



*Routledge Studies on Challenges, Crises and
Dissent in World Politics*

ULTIMATE ECONOMIC CONFLICT BETWEEN CHINA AND DEMOCRATIC COUNTRIES

AN INSTITUTIONAL ANALYSIS

C.Y.C. Chu, P.C. Lee, C.C. Lin and C.F. Lo



'This book by Cyrus Chu and his colleagues at Academia Sinica is a timely book with deep insight from both an academic and practitioner perspectives on the deep rift between the U.S. and China with a focus on their economic relationship. Contrary to expectation among many policymakers and scholars over the last few decades, gaining entry into the WTO and other multilateral institutions did not bring China to become more aligned with the U.S. and its allies in terms of common values. Instead, China has become more prosperous while attempting to change the liberal economic order to the benefit of the Chinese government. This is an important work of research with a clear framework supported by excellent evidence to understand the rivalry between the two superpowers is founded on different principles which will be hard to bridge.'

Syaru Shirley Lin, *University of Virginia, Chinese University of Hong Kong and Brookings Institution*

'This eye-opening book transformed the way I think about China's relationships with the advanced market economies, and I strongly recommend it to those seeking to understand the dilemma that so much of the world now faces: the desire to access the large and growing Chinese market, and yet the understandable concern with the serious risks and challenges they will encounter in doing so.'

Harry Harding, *University of Virginia, USA*

'This is a timely book providing a unique and in-depth analysis of the economic conflicts between market economies and state capitalism in general and between U.S. and China in particular. Each main chapter of the book addresses an important and fundamental difference between the two systems arising from the institutional differences. The discussion is clear and informative with empirical support. It helps readers understand better the current economic conflicts between the U.S. and China and points out the difficulties involved in resolving these conflicts under the exiting WTO rules and legal systems. As stated by the authors, the book is not about which system is better, but about understanding the root cause of the conflicts. Only by understanding the real source of the conflicts, we can find ways to overcome them. Thus, I strongly recommend the book to all policymakers in business and economics and to everyone who cares about the relationship between U.S. and China.'

Ruey Tsay, *H.G.B. Alexander Professor of Econometrics and Statistics, Booth School of Business, University of Chicago, USA*

‘In *Ultimate Economic Conflict between China and Democratic Countries: An Institutional Analysis*, C. Y. C. Chu and co-authors offer careful research and richly elaborated review of the unresolved questions surrounding China’s unique system of state capitalism. In recent years many policymakers set aside scholarly efforts to inform decision-making, instead proceeding based on instinct. Policymaking uninformed by research will fail to protect liberal economy interests. With a thorough reexamination of economic literature and great attention to the most salient examples of tensions between China’s approaches and market expectations, the authors refresh the debate and bring it up to date. This is extremely timely, as major economies move beyond the ill-informed unilateralism of recent years and commence serious work together to plan an agenda for an era in which convergence with market principles is no longer the base case.’

Daniel Rosen, *Partner at Rhodium Group, USA*

Ultimate Economic Conflict between China and Democratic Countries

This book investigates various dimensions of the economic conflicts between the US – and other democratic market-economy countries – and state-capitalist communist China in the past decade, examining how differences in institutions and ideology bring these about.

Through the lens of institutional analysis, the book elaborates and explains the underlying institutional designs and reasons behind the disputes, highlighting how such variances are embedded and reflect fundamental value divergences between China and other democratic countries.

This book will be of key interest to scholars, students, and practitioners in law, economics, political sciences, international relations, international organisations and global governance.

C.Y.C. Chu is a Distinguished Research Fellow at the Institute of Economics and an elected Academician (2008) of Academia Sinica, Taiwan.

P.C. Lee is a practicing lawyer in WTO and trade remedy laws in Taipei, Taiwan.

C.C. Lin is an Associate Research Professor of the Institutum Iurisprudentiae of Academia Sinica, Taiwan.

C.F. Lo is Taiwan's Permanent Representative to the WTO.

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An Institutional Analysis

**C.Y.C. Chu, P.C. Lee, C.C. Lin and
C.F. Lo**

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Contents

<i>List of Illustrations</i>	viii
<i>Preface by Jerome A. Cohen</i>	ix
<i>Authors' Preface</i>	xi
<i>List of Abbreviations</i>	xiii
1 Introduction: When the 'Socialist Market Economy' Meets the Globe	1
2 Made in China 2025: The Issue of Government Subsidies	18
3 E-commerce Mercantilism: The Issue of Internet Blocking	41
4 Banning TikTok and WeChat: The Issue of Privacy Protection and National Security	62
5 Enacting HFCAA: The Issue of Listed Company Accountability	87
6 Antitrust Laws in the Globalisation Era: The Issue of Competitive Neutrality	106
7 Cross-border M&A Control: The Issue of State Capitalism	124
8 Forced Technology Transfer: The Issue of Insufficient Remedies	151
9 How to Dance with the Dragon?	173
<i>Index</i>	186

Illustrations

Figures

2.1	The time line from laboratory to factory and market	30
3.1	Conceptual trade stages	46
6.1	Horizontal and vertical ICT map of Huawei	117

Tables

3.1	Country-wise e-commerce, GDP and HFCE data	43
3.2	Per-click waiting time: major retailing platforms	48
3.3	Per-click waiting time: major retailing platforms, two-way tests	49
3.4	Per-click waiting time: typical SME websites	51
3.5	Per-click waiting time: SME websites, two-way test	52

Preface

Before reading this informative and stimulating book, I was puzzled by the authors' request that I write its preface. Like many observers of the increasingly fraught US-China relationship, although worried about the various economic conflicts that daily threaten to make matters worse, I am not an economist and strain to grasp the details and implications of each of the Sino-American trade, investment and technology transfer disputes featured in the media. I am happy to report that this book is not written solely for economic specialists but also for the broad community concerned with the comprehensive interactions of the world's two superpowers. Indeed, as its title suggests, its analysis reveals the profound obstacles that confront cooperation and understanding between authoritarian and democratic societies generally.

China, of course, is the leading contemporary example of authoritarian, if not totalitarian, government. The authors persuasively demonstrate that the Chinese Communist Party-state is very different from the governments of those countries that can meaningfully be deemed democratic, and that this difference has a decisive impact on each of the often-disparate economic disputes that have arisen.

They characterize this difference as essentially between competing ideologies, even between rival value systems, and not merely a clash between economies. They spell out the major manifestations of this difference in terms of the distinctive goals, organization, institutions and practices of the People's Republic of China (PRC). Moreover, the authors emphasize how the Chinese Communist Party has monopolized and mobilized its unique arrangements in order to maximize opportunities for the enhancement of national power that recent decades of remarkable technological innovation have made possible.

It is against this unique background that individual economic conflicts with the PRC, its companies and its representatives have to be understood. As this book stresses, negotiating with the PRC and its emanations is not like negotiating with counterparts from the British Commonwealth, Western Europe, Japan or India. Whatever the nominal status of the PRC people across the table, they are in fact usually minions of the Party and, necessarily, obedient servants of its will. Similarly, in the search for remedies against perceived commercial grievances, foreigners seeking satisfactory modes of dispute

resolution, whether in international fora or before PRC legal institutions, often confront unusual political barriers.

The authors illustrate this theme in a range of contexts. In view of their long-established expertise in WTO matters, it is not surprising that, although complex, their discussion of the consequences of the PRC's entry into that organization is highly illuminating. I also find very helpful their explanation of why, in the new era of Artificial Intelligence and Big Data, companies as seemingly innocuous as TikTok, for example, might be considered detrimental to the national security of the United States and other free countries. Many other controversies of current interest benefit from this book's analysis, including the extraterritorial reach of the PRC's Anti-Monopoly Law, the bitter dispute between Huawei and Qualcomm, and the belated enforcement of the requirement that Chinese companies listed on American stock exchanges comply with the same transparency obligations imposed on all other companies.

Of especial interest to many readers will be the authors' portrayal of today's PRC legal system and its contribution to settling economic conflicts. They rightly emphasize the blatant contradiction between, on the one hand, new laws and tribunals created in explicit efforts to establish in China a more attractive international dispute resolution environment and, on the other hand, the incessant, open confirmation of the Communist Party's total control of the courts and the legal profession. As the authors note, in China the many relevant laws that now exist are often vague and susceptible to expedient interpretation, and indeed often ignored in practice.

Finally, also worthy of comment is the Taiwan origin of all four of the very able authors, as well as the professional eminence of the two senior scholars, Professor Chu and Professor Lo, in their respective areas of economics and law. Both continue to play prominent roles in Taiwan's impressive modernization, in government as well as in academe. Their backgrounds and experiences have also plainly enhanced their capacity for analyzing the PRC and its relations with the world instead of distorting their perceptions and judgments of the regime across the Strait.

While many Taiwan experts are understandably preoccupied with the island's stunning political, economic and legal progress and its significant remaining challenges, some have already done distinguished work relating to the PRC. I hope that many more will emulate the authors of this important volume and devote their energies to studying developments concerning the PRC as well.

Jerome A. Cohen (May 14th, 2021)

Founding Faculty Director Emeritus,
US-Asia Law Institute, New York University Law School;
Adjunct Senior Fellow for Asia,
Council on Foreign Relations.

Authors' Preface

This is a book about the US–China trade and technology conflicts, which started roughly ten years ago and, as this book goes to print, are expected to last for some years to come. We should thus explain why these conflicts have caught our attention, and why we are able to provide a comprehensive analysis.

Most students who have majored in economics have read the classic book *The General Theory of Employment, Interest and Money* (1936) by John Maynard Keynes. The idea in that book was motivated by Keynes's observations during the period of the Great Depression. However, for economists born after the Second World War, if they had not witnessed the 2008 financial tsunami, they might not have been able to imagine the impact of a global financial crisis, and the important role played by government spending in helping facilitate the recovery of the economy. Similarly, most young students of international relations have read the stories of the trade embargoes and retaliations in the 1920s. However, if it had not been for the unprecedented US–China trade war during the Trump administration, many of them might not have really understood what a global trade war looks like.

Being either lucky or unlucky, we had the opportunity to sit on the front row of the WTO, to observe the drastic change in the WTO atmosphere, to observe the paralysis that had gripped the WTO's dispute settlement system, and to see how frustrated WTO Members had become during this trade-war period. This unique 'war front' observation became our first motivation to write this book, as we sought to study and find the institutional reasons behind this huge conflict between the world's two largest countries. If the US believed that it had made a mistake to let China enter the WTO, why was it? If there were some flaws in the WTO rules that are behind the conflicts, what are they? If there are such flaws, why has the WTO been unable to amend them? Finally, how will such conflicts eventually end?

Other than the above-mentioned trade conflicts, the Trump administration also brought a new dimension to the US–China confrontation: technology embargoes, using executive measures such as an 'entity list'. From the perspectives of trade and economics, these embargos can be treated as a 'non-tariff trade barrier', which may be even more powerful than a tariff as

a traditional trade barrier. Indeed, as many commentators have brought in a new term referred to as 'technology decoupling', it appears that a technology embargo may even become a new and more effective weapon of confrontation. Furthermore, if technologies are to be decoupled, how will global trade develop in the future? Our second motivation in writing this book is to understand the interface between the trade and the technology aspects. Specifically, if there is an institutional reason behind the trade war, we would like to know how the technology side contributes to and becomes a part of the trade issue.

Looking ahead, if technologies are indeed to be decoupled, we cannot be optimistic about the revival of a unified rule-based global system, such as the old WTO. In particular, since many national security networks and weapons systems use the Internet as their backbone infrastructure, future Internet standards (such as 6G) are likely to be decoupled, or even dichotomised, for national security reasons. If future trades in services are mostly built on the infrastructure of the Internet, we are becoming increasingly sceptical as to whether a unified global system of trade in services could be sustained.

The above-mentioned US–China conflicts involve various aspects of knowledge: economics, international relations, comparative legal systems, WTO laws, and science and technology policies. It is such knowledge that has brought the authors together.

Some of the content in Chapters 2 and 3 is abstracted and rewritten from our previous publications.¹ Moreover, we would like to thank many colleagues and friends of the authors' various affiliations for their valuable comments and feedbacks to our earlier drafts. In particular, colleagues of many Missions of the WTO have made significant contributions to formalise the structure of this book. We are grateful to the publishers for allowing us to use these contents here. We would also like to thank the editor of this series, Andrew Taylor, for his efficient handling.

C.Y.C. Chu, C.C. Lin
P.C. Lee, and C.F. Lo
August 2021

Note

- 1 Chapter 2 is rewritten from C.Y. Cyrus Chu and Po-Ching Lee (2019) 'Three Changes Not Foreseen by WTO Rules Framers Twenty-Five Years Ago', *Journal of World Trade*, 53(6): 895–922. Available at <https://kluwerlawonline.com/journal/article/Journal+of+World+Trade/53.6/TRAD2019036>. Chapter 3 is rewritten from C.Y. Cyrus Chu and Po-Ching Lee (2020) 'E-commerce Mercantilism-Practices and Causes', *Journal of International Trade Law and Policy*, Vol. 19, No. 1: 51–66. Available at www.emerald.com/insight/content/doi/10.1108/JITLP-08-2019-0054/full/html.

Abbreviations

ACCC	Australian Competition and Consumer Commission
ACGA	Asian Corporate Governance Association
AI	artificial intelligence
AMC	Anti-monopoly Commission
AML	Anti-monopoly Law of China
CCMCs	Communist Chinese Military Companies
CCP	Chinese Communist Party
CFIUS	Committee on Foreign Investment in the United States
CMA	Competition and Markets Authority of the United Kingdom
COMECON	Council for Mutual Economic Assistance
CPC	Central Product Classification
CPTPP	Comprehensive and Progressive Agreement for Trans-Pacific Partnership
CSLI	cell-site location information
CSRC	China Securities Regulatory Commission
CTBs	cyberspace trade barriers
DARPA	Defense Advanced Research Projects Agency
DSB	Dispute Settlement Body of WTO
E-WTO	E-world Trade Organization
EC	European Commission
F-WTO	fair-competition WTO
FDA	United States Food and Drug Administration
FDI	foreign direct investment
FIRRMA	Foreign Investment Risk Review Modernization Act of the United States
FTTs	forced technology transfers
GAAP	generally accepted accounting principles
GAFA	Google, Apple, Facebook, Amazon
GATS	General Agreement on Trade in Services
GATT	General Agreement on Tariffs and Trade
GDPR	General Data Protection Regulation
GFW	Great Firewall of China

HFCAA	Holding Foreign Companies Accountable Act of the United States
HFCE	household final consumption expenditure
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
IP	intellectual property
JSI	Joint Statement Initiative
M&As	mergers and acquisitions
MNCs	multinational companies
NASSP	National Administration of State Secrets Protection of China
NATO	North Atlantic Treaty Organization
NIH	National Institutes of Health of the United States
NME	non-market economy
NSF	National Science Foundation of the United States
NYSE	New York Stock Exchange
OSC	Orbital Sciences Corporation
PCAOB	Public Company Accounting Oversight Board of the United States
PPF	production possibility frontier
PWG	President's Working Group on Financial Markets
SASAC	State-owned Assets Supervision and Administration Commission of China
SBIR	Small Business Innovation Research Program
SCM	Agreement on Subsidies and Countervailing Measures
SEC	Securities and Exchange Commission of the United States
SOEs	state-owned enterprises
SPC	Supreme People's Court of China
STTR	Small Business Technology Transfer
TFEU	Treaty on the Functioning of the European Union
TIER	Technology Import and Export Regulations of China
TiSA	Trade in Services Agreement
TRIMs	Agreement on Trade-Related Investment Measures
TRIPS	Trade-related Aspects of Intellectual Property Rights
UNSD	United Nations Statistics Division
US BRAIN Initiative	Brain Research Through Advancing Innovative Neurotechnologies
USMCA	United States–Mexico–Canada Agreement
VPN	virtual private network
WFOE	wholly foreign-owned enterprise
WTO	World Trade Organization

1 Introduction: When the ‘Socialist Market Economy’ Meets the Globe

The US–China Escalating Conflicts

It may be fair to say that, between 2017 and 2020, during the Trump administration, the most important event in international relations was the escalation in conflicts between the US and China. This is certainly not the first time in history that the US had trade conflicts with other countries. One of the most notable cases might be the US–Japan trade conflict in the 1980s. Nonetheless, the ongoing US–China trade war is critically different from that between the US and Japan. In the latter case, although there were some institutional differences between the US and Japan, economic decisions in both countries still generally followed the rationale of the market.¹ This was different from, say, socialist economies, in which ‘state ownership of the means of production, state planning, and bureaucratic goal-setting are not rational means to a developmental goal ... they are fundamental values in themselves, not to be challenged by evidence of either inefficiency or ineffectiveness.’² From this perspective, it is not surprising that the conflicting issues between the US and China, which were originally to do with unfair trade, were soon extended to encompass technology and even finance. Bolton (2020) argued that the trade issue is just the tip of the iceberg, and it is the structural problems between the US and China that are the true factors contributing to the US–China conflicts. This is certainly not Bolton’s individual hawkish view; in fact, many have made similar statements.³ The following paragraphs provide a sketchy description of the above-mentioned structural problems, as well as the evolution of this conflict. Detailed references will be provided in later chapters, and hence we will skip them in this introduction.

On 22 March 2018, the Trump administration started to impose tariffs on \$60 billion of Chinese products imported into the US. This pure trade measure was soon expanded to cover the technology aspect, when the US announced on 16 April 2018 that it would ban US exports of components to ZTE, a Chinese telecommunications manufacturer. Then, on 1 December 2018, at the request of the US Justice Department, the Canadian government arrested Meng Wanzhou, the deputy Chair and CFO of Huawei, the leading telecommunications company in China. The charge was that Huawei had

2 The 'Socialist Market Economy' Goes Global

used some US components in its products and had exported such products to Iran and North Korea, violating the US sanctions imposed on these two countries. Because Huawei was the Chinese leader in the development of the fifth-generation telecommunications (5G), it was believed that arresting Meng reflected the US's attitude toward the 5G competition between the US and China.⁴

On 6 August 2020, the Trump administration ordered that a number of companies listed on the New York Stock Exchange and the Nasdaq be subjected to US auditing, or they would be delisted by 2022. Although this administrative order did not refer to specific company names, all of the news reports pointed out that the 283 listed companies affiliated with China and Hong Kong were the target. This was the first time the US–China conflicts had spread to the financial dimension. Moreover, at various times in 2020 the US government gradually increased the number of sanctioned items against China by including more and more Chinese companies on the *Entity List*.⁵

This book's goal is to analyse various dimensions of the above-mentioned economic conflicts between the US and China. To analyse the fundamental reasons why such US–China conflicts arise, we review the institutional and ideological differences between China with its authoritarian communism and the democratic capitalist countries such as the US, Japan, Germany, the UK and Australia. In our book, we use 'market economy' and 'state capitalism' to refer to the economic models in Western liberal democracies and in China respectively. According to Musacchio and Lazzarini (2014), state capitalism refers to 'the widespread influence of the government in the economy, either by owning majority or minority equity positions in companies or by providing subsidized credit and/or other privileges to private companies'. By contrast, a market economy is an economic system in which the role of governments is limited, and economic decisions are driven and influenced mainly by supply and demand. Notably, we are fully aware that these two models are just ideal types. Namely, they cannot precisely describe the economic behaviours in either the West or in China. Also, there are many intermediate or hybrid models that combine elements of both ideal types. With this limitation in mind, we believe the two concepts, albeit imprecise, could still serve as useful analytical tools for the purpose of our book.

As we compare the ideological differences behind state capitalism and market economy, our focus is not on the often-mentioned differences in values such as freedom or liberty per se, but rather on the economic implications of such differences in values and institutions. To understand why differences in values will bring about ultimate economic conflicts, we begin by discussing the well-known critique made by the Trump administration: that allowing China to enter the WTO was a mistake made by the US.⁶ What does it mean to say that the US government *made a mistake*? If China's economic practices did not turn out to be as the US and other democratic countries expected, why was that? If China retains its current political regime, what economic conflicts can we predict and expect?⁷ Furthermore, and most importantly,

how are such conflicts related to the institutional differences between authoritarian communism and democracy?

Will Economic Development Lead to Democracy?

As to the relationship between economic development and political regime changes, when China was admitted to the WTO two decades ago, the orthodox beliefs were that bringing China into the global market-economy system would eventually change its political institutions, and that it was likely it would move toward democracy, at least to some extent or in some domains. It turned out that instead of living up to that expectation, China has abused its exponential economic growth to further its dictatorial goals since joining the WTO. In other words, the problem is not only that China has not democratised, but that China has become even more totalitarian than it used to be, in parallel with its economic success. The above-mentioned orthodox belief, now realised to be purely unrealistic wishful thinking, was actually backed by some theoretical analysis in the literature on political science. We will first review such a discussion, before providing our own interpretation.

The interaction between democracy and economic development has long been discussed by social scientists. The pioneer was Lipset (1959), followed by Diamond and Linz (1989), Inglehart and Welzel (2010), Acemoglu and Robinson (2012, 2019), and many others. Lipset (1959) argued that economic development may lead to democracy because it produces certain socio-cultural changes that shape human actions. Diamond and Linz (1989) referred to three possible channels for such interactions, which we discuss one by one below.

First, it was argued that poor economic performance is likely to raise public anxiety and dissatisfaction. An increasingly anxious social atmosphere usually welcomes strong-arm leadership, which is not compatible with the development of democracy. Thus, a healthy economy is a necessary condition for a sustainable democracy. We note, however, two things: 1) this argument does not suggest that economic development is a sufficient condition for democracy, and 2) this argument is about the *sustainability* of democracy, and not about *switching* from an authoritarian regime to democracy.

Second, researchers found that economic development often goes hand in hand with urbanisation, increasing literacy, an expansion of educational opportunities, and more intensive interactions with people in democratic countries. These all facilitate human rights perceptions and the promotion of democratic ideas. However, again, this argument is only based on the simultaneity or correlation between economic development and a social environment that is compatible with democracy. There is no causal implication that economic development will lead to democracy.

Third, we observe that economic development also helps promote the rise of the middle class and related social organisations, which in turn help generate pressure and impose checks and balances on government policies.

4 *The 'Socialist Market Economy' Goes Global*

These checks and balances are of course coherent with democratisation. Once again, the argument here is not rigorous: whether the rising middle class will indeed drive the regime toward democracy is a presumption, and there are many factors that may lead the regime in other directions.

To sum up, the above-mentioned arguments provide conceptual support for the *correlation* between democracy and economic development, but do not constitute a rigorous *causal* argument. Researchers such as Inglehart and Welzel (2010) have provided numerous examples that economic development has not led to democracy. Even with the help of data from large-scale cross-country surveys, scholars have still been unable to identify a clear-cut pattern of causality from economic success to democracy. Furthermore, since the speed of political regime change is slow, it is difficult for researchers to observe the full picture of political regime stabilisation, and to draw rigorous inferences from the limited observations. Finally, the case of Singapore also characterises a grey area: It is neither democratic nor completely authoritarian, but it has had one of the fastest-growing economies in the world for half a century.

The most disturbing example in the past three decades is, of course, China. In the period from 1990 to 2020, China experienced unprecedentedly high economic growth, which has increased its GDP roughly 40 times in 30 years. China tends to interpret its economic success as a role model of a *socialist market economy*, or a market economy with socialistic characteristics. Leaving aside what exactly a socialist market economy means, if this role-model story is valid, it evidently refutes the prediction that economic development will lead to political liberalisation. Regardless of the measures used, in 2021 China is perhaps one of the most authoritarian nations in the world.

So, if the prediction that economic development will lead to political liberalisation is not valid, what will happen to the authoritarian nations next? In this period of irreversible globalisation, how will the world cope with two economic regimes, one a democratic market economy, and the other the authoritarian socialist market-economic structure represented by China, and perhaps Russia as well? Should we expect to see a world economic order somehow harmonising these two regimes? Alternatively, are there fundamental institutional differences between these regimes which mean that future conflicts are unavoidable? This book tries to provide answers to these questions.

The Peculiar Role of Chinese Businesses in the World Economy

The economic success story of China is also related to the US–China conflicts that we mentioned at the beginning of this chapter. In various WTO documents,⁸ the United States Trade Representative (USTR) has argued that China's economic success was (at least partly) due to its forced technology transfers, intellectual property theft, substantive government subsidies, exchange rate manipulation and many other measures that either violated the existing WTO rules, or took advantage of the loopholes. According to

Chu and Lee (2019), the US's accusations are valid in the sense that there are indeed loopholes in the Marrakech Agreement enacted in 1994. The world today is very different from what it was in 1994, and China did take advantage of the old WTO rules enacted long ago.

Chu and Lee (2019) pointed out three environmental discrepancies between now and the time when the Marrakesh Agreement was framed. Here we only briefly mention two of them; more details will be provided as we proceed in later chapters. The first discrepancy between the situation of the present time and that of 1994 is the rise of the knowledge-based economy. By 1994, Moore's law was only a conjecture; the human genome initiative had not yet been started; the R&D-based high-tech industries and venture capital firms were not yet popular, and many high-tech giants today had not even existed at that time. The General Agreement on Tariffs and Trade (GATT) rules enacted in 1994 were mainly concerned with traditional industries. To ensure fair international competition in these traditional sectors, the trade rules mainly regulated government subsidies *between the factory and the market*. For high-tech industries, however, a new and perhaps more important subsidy arose: *from the R&D laboratories to the factories*. In China, many government subsidies were provided at this early stage, but the current WTO rules cannot regulate these subsidies effectively.

The second aspect not foreseen in 1994 relates to the consequences of the rapid marketisation of the Soviet Union and China. The Soviet Union began to disintegrate in 1991, and Deng Xiaoping's southern-tour speech, which fully launched China's marketisation, was delivered in 1992. At the time when the Marrakesh Agreement was drafted, the general assumption was based on the orthodox Western market-economy background, namely, that businesses and governments are separate entities. However, in China, the situation is totally different.

In China, the rapid privatisation of the previously wholly government/party-owned enterprises created a large percentage of pseudo-private businesses, in most cases owned or controlled by individuals who had good connections with the government's or party's high officials. This bizarre economic situation can be referred to as *state-party crony capitalism*. Thus, in China and perhaps Russia as well, business and government entities actually overlap considerably in terms of their institutional designs and key personnel.

According to McGregor (2010), in terms of decision making, there are no real 'private' businesses in China; all businesses, especially large ones, are strictly controlled by the Chinese Communist Party (CCP). McGregor (p. 199) cited others and concluded that only 20–30 per cent of Chinese GDP is contributed by truly private businesses. Many CEOs of large Chinese businesses have a 'red telephone' in their office (p. 29), which is a within-CCP communication line, and which allows these CEOs to be instructed by the CCP about the direction of business when necessary. Most businesses do not regard this red telephone as a nuisance. Rather, it is a symbol of their close ties with the CCP, which can facilitate the business's operations in critical times.

Although nothing is written explicitly in law, according to McGregor (2010), the CCP can instruct the company regarding its policies, choose company board members and business CEOs, ignore or bypass all business-related laws, and suppress all labour union movements within any company. Even for listed companies, the published documents never describe who are the foremost decision makers (p. 74). In private negotiations, many foreign companies have complained that they are actually negotiating with a nation, not a company, although on the surface it appears as a business-to-business bilateral deal (p. 81). To sum up, we believe that the innocent term *socialist market economy* cannot fully capture the reality of Chinese business practices any more.

This has serious implications. For instance, the WTO obligations are applicable to Member governments with an eye on regulating government actions over commercial activities. However, when the businesses are actually run by the government/party, the above-mentioned regulation is clearly not as effective as it should be.

The above analysis highlights the fact that the current WTO rules established 27 years ago are somewhat obsolete when it comes to dealing with modern-day international trade in 2021. Moreover, since rule revisions need a consensus in the WTO, those countries that have benefited from the current rule loopholes are unlikely to give up their advantages, and therefore old rules cannot be amended toward a reasonable direction. Thus, the present situation encourages countries to exploit the current WTO rules' loopholes. This has in large part been the accusation by the US.

The Intrinsically Unfair Economic Competition

Although the existing WTO rules have loopholes as we mentioned above, the opportunities arising from such loopholes are not the same for all countries, with some systematically having more of an advantage than others. In the following chapters of this book, we argue that in terms of the competition between businesses in authoritarian countries and those in democratic ones, those in the latter are intrinsically at a disadvantage.

We should emphasise two things at the outset. First, although in the above discussion we have mentioned that there are many obsolete WTO rules, the unfair competition is *not* restricted to those cases where the WTO rules apply. For instance, in Chapter 6, we will demonstrate how antitrust laws in democratic countries result in unfair competition, even though the WTO rules do not touch upon antitrust. The conflicts between authoritarian states and democratic capitalism arise in many different areas beyond the WTO rules.

Second, when businesses in democratic countries are at a disadvantage in their competition with those in authoritarian states, such disadvantages do not have anything to do with competitive efficiency per se; instead they are due to other implicit or explicit democratic values that are sometimes enacted in laws. The situation is just like that for labour rights: in countries that respect

human dignity and honour labour rights, businesses are constrained by various labour recruiting regulations, including a working hours upper limit, minimum paid vacations, a minimum juvenile labour age, a mandatory probation period for layoffs, etc. And hence as they compete with businesses in countries that offer little labour protection, they are bound to have higher labour-force expenditures. We understand that democratic countries may differ in terms of their labour rights protections, which then results in different business costs; but the distinctions between authoritarian and democratic countries are so drastic that the unfairness in relation to competition is something that cannot be ignored.

In the following chapters of this book, we will illustrate various dimensions of such constrained inefficiency in democratic countries, as compared with authoritarian ones. We should emphasise, however, that our discussion does not have any implications concerning whether an authoritarian state (such as China), or a semi-authoritarian state (such as Singapore), or a democratic one is most efficient. As we have mentioned, the competition constraints arise from fundamental differences in constitutions, which have moral values that transcend efficiency. It may well be the case that it is more efficient to use slaves to plant cotton fields, but today all democratic countries have chosen to abolish the slavery system and prohibit it by law. When two countries, one using slaves and the other not, compete in the cotton industry, we should understand the underlying unfairness of the competition, rather than jump to conclusions as to which one is more efficient.

In short, we believe that the ultimate economic conflicts between authoritarian China and democratic countries are embedded in their *ideological and value differences*. Democratic countries respect the rule of law, so that their administrations simply cannot block the Internet, as China does. Democratic administrations have a long tradition of providing checks and balances, so that the administrative entities simply cannot influence the court rulings in favour of domestic businesses, as China does. Officials in democratic governments simply cannot provide subsidies to specific private companies, as China has done and has announced publicly. Furthermore, all listed companies in democratic countries are accountable to the general public, unlike Chinese companies which are accountable only to the CCP. These differences are in fact the key behind all unfair competition. Because such differences are embodied in the fundamental distinction between democratic countries and authoritarian regimes, it is difficult to imagine how the ensuing economic conflicts can be avoided. This is why we refer to these conflicts as 'ultimate' economic conflicts.

Are Political Institutions Malleable?

Another criticism of the 'economic development leads to democracy' hypothesis concerns the operation of politics in reality. When we discuss how economic development may lead to democracy, an implicit assumption seems to

be that the political institution is changeable or malleable, something that can be influenced by education, literacy and the economic environment. However, this is not always the case in reality. Authoritarian ruling elites are never willing to give up their vested benefits; they always tend to resist changes that may compromise their interests. Furthermore, the assumption of malleable political institutions is definitely more questionable in Leninist party-state countries such as China. In such Leninist authoritarian countries, according to their constitutions, the Communist Party/State is the ultimate decider of almost everything. It is also the political institution that determines the flexibility of the market mechanism, and not the other way around.

Admittedly, every political system, be it democratic or not, affects or modifies the operation of an economy to some extent, which is why we have antitrust law, among other things. Nonetheless, the situation in China is different in terms of both the type and the degree. In Western societies, the political system usually intervenes near the margins when it tries to nudge or regulate the market, with an eye to facilitating business operation and preventing market failure. By contrast, in China, the political system firmly controls and comprehensively monitors the operation of the economy. The purpose is to ensure that the economy's operation serves the interest of the Party. Indeed, when China asserts that it has a 'socialist market economy' (or market socialism), it means that the CCP has transformed the Western market economy into a different species.

For instance, in China, in order to maintain its control, the CCP has built the most severe Internet censoring mechanism, referred to as the 'Great Firewall'. According to Freedom House,⁹ of the 1,000 most-visited websites in the world, China has blocked 171 of them, not to mention more than 10,000 commercial websites that are also blocked.¹⁰ This Internet blocking is based on an upper-tier political decision, and all economic activities are regulated and fine-tuned according to this upper-tier constraint. Given this Great Firewall, the Chinese can access foreign websites only if they use a VPN (virtual private network) to 'climb over the wall'. However, this is illegal, and the government's monitoring of such activities has become increasingly strict since 2017.¹¹

As mentioned above, all economic activities need to be adjusted in accordance with the Internet blocking. For instance, it is very inconvenient for the Chinese public to access the website of the US platform Amazon (to be discussed in detail in Chapter 3); as a result, if Amazon wants to be more conveniently accessed in China, the company must set up a local version of its website for potential Chinese customers, despite it being inefficient for Amazon to do so. This Amazon Chinese website is connected to a local server, and is subject to Chinese government scrutiny. For example, if one accesses the Amazon website in the US and searches for Tiananmen Square, hundreds of books and related commodities will pop up on the screen. However, on the Chinese website of Amazon, less than five such items will appear, with all

others having been censored and deleted. Thus, Amazon's global business is adjusted in accordance with Chinese politics.

Another example is that in China, any organisation having more than three Communist Party members needs to have a party organisation, and all listed companies have a party secretary sitting on the company's board.¹² Because the CCP is the highest-ranking party in the country, all company policies must be subjected to the party's scrutiny. In addition, because the operation of the CCP is not transparent, this internal control mechanism will of course compromise the company's accountability to its stock- and debt-holders. So again, it is the political regime that modifies the economic rules and activities, and not the economic activities that affect the political regime.

The above examples tell us that, in authoritarian regimes in general, and in Leninist Communist countries in particular, it is the economic activities that are subject to modification, and not the political institution. Thus, in this era of globalisation, businesses in democratic countries are facing economic competition from two different sources: Some businesses are from countries with similar democratic institutions, and others are from authoritarian states in which there are totally different political ideologies. The natural questions which then arise are: is this competition fair? Which side is intrinsically favoured in this competition? These are the problems to be analysed in future chapters.

The Unjust Advantages of Authoritarian Rulers in the 21st Century

Another aspect that influences the economic/political interaction concerns the recent developments in technologies. In both Acemoglu and Robinson (2012, 2019) and Rajan (2019), the authors emphasise the power of 'society' or 'community' that can check and balance the power of the state. Acemoglu and Robinson have argued that only in a very *narrow corridor* can democracy and economic growth be stably developed and sustained. We are of the view that recent technological developments are likely to make this narrow corridor even more constricted, so that the authoritarian states can more easily maintain the stability of their operations.

We understand that switching from authoritarian control to democracy must involve some kind of 'social movement', either in the form of a harsh revolution or a series of peaceful protests, such as the famous gatherings during the Arab Spring in 2010–12. After the Arab Spring, many social movement advocates began to realise that social media was an efficient means of mobilising social gatherings and protests. On the other hand, the state powers also realised that anti-government movements are often facilitated by communication on the Internet. This triggers the state to impose more effective control over the Internet. Thus, both sides seek to make the best use of the Internet, one for mobilising social power and the other for containing it. The question is: which side is able to benefit more?

We are of the view that the state power has more of an advantage in the Internet era. The key technological changes that lead to our conclusion are the emergence of big data and artificial intelligence (AI). As for big data, the government naturally has data from demographic records, family relationship booklets, tax filings, real estate registrations, traffic monitoring, credit records in banks, public transportation cameras, publicly owned telecommunications footprints, satellite GPS, and so on. By contrast, social groups or organisations can hardly access any of these. If the government wants to use such big data to monitor civil society's opposition forces, then it has the potential to know every move of every person within its territory, and even beyond.

Given that the above-mentioned big data are in the hands of the authoritarian state, the government has one more advantage in using such data: artificial intelligence. It should be noted that the use of big data usually involves complicated parallel computing algorithms and a very fast computing speed. This is a facility that is very expensive for social groups, but can be easily acquired by the authoritarian state. Thus, big data plus AI disproportionately increases the government's ability to monitor social groups, to know their networks, to pre-empt their gatherings, and even to tear apart the unification of these groups.

The most well-known case, of course, is China, which has publicly bragged about its social credit system, which records every individual's previous behaviour and statements, and gives 'scores' to each of them. These scores are then used to determine whether a person is allowed to buy a high-speed rail ticket, whether their children can be admitted to good public schools, whether they can get a job they've applied for, and whether they will be charged with certain crimes.¹³ With all such monitoring using big data and sophisticated AI algorithms, it is expected that social powers will be severely constrained, and the transformation toward democracy will become even more difficult.

In democratic countries, however, big data and AI do not provide sufficient cause for concern. The largest social media entities in democratic countries are perhaps Facebook, Google and Amazon, with each being only able to capture a slice of users' private information. Democratic countries also have more comprehensive privacy protection laws, and it is not possible for their constitutions to allow the merger and use of these separate entities' 'clouds'.

To sum up, we believe that in this digital era of big data and AI, the authoritarian states have more of an advantage in oppressing social forces, and hence they are more stable than they were before. Therefore, we expect to observe a prolonged confrontation between democratic countries and authoritarian states. If there are any fundamentally unfair competition scenarios between businesses in these two regimes, we expect that these scenarios will persist for quite some time. Our purpose in this book is to identify the various dimensions of such unfair competition.

Is Smooth Competition Possible?

We would like to point out that some of the above-mentioned unfair competition dimensions, such as the unfair government subsidies and market access (Chapters 2 and 3), have already come up in the US–China conflicts; but others, such as the antitrust constraints (Chapter 6), have not yet been raised. Even for those aspects that have already been brought up, politicians have only pointed out the facts of unfair competition, but have been unable to clarify the theoretical structure of such conflicts. In this book we try to provide a systematic analysis of all such unfair competition dimensions. We point out clearly why democracy imposes some natural restrictions on the business operating environment, and why authoritarian countries do not have such restrictions. Because these restrictions are built into the democratic system, the unfairness is unavoidable, and the authoritarian states can always take advantage of such situations.

In the above discussion, we mentioned that democracy involves a regime with checks and balances, and that economic activities are constrained by values such as human rights and individual freedom, which usually reduce efficiency. Authoritarian states are not constrained by such values, and may therefore be more efficient. Other than such inefficiencies due to built-in democratic values, are there any aspects that are advantageous to democracy? The answer, according to Acemoglu and Robinson (2012), is that there are. Although this is not the focus of this book, it is worthwhile providing a very brief explanation at the beginning.

Acemoglu and Robinson (2012) classified economic systems into two broad categories: *inclusive* and *extractive*. The inclusive system is one in which property rights are well specified, the majority of individuals can share the fruits of economic growth, and there are constitutional designs to warrant such rules on profit-sharing. By contrast, an extractive system is one in which most of the rewards of economic growth go to a small group of ruling elites, property rights are not well protected, and institutions are designed to match this extraction. Because of these extractive institutional designs, people have few incentives to become involved in economic activities, so that their economic performance under the elites' authoritarian controls is relatively poor. Typical examples of inclusive regimes are stable Western democratic nations, and typical examples of extractive states are the Philippines, Indonesia and some Latin American countries of the 1960s, with crony capitalism associated with a few ruling elites.

According to Acemoglu and Robinson (2012, Chapters 3 and 15), despite China's rapid economic growth rate over the past thirty years, it is believed that this was due to the fact that China was in its early stage of economic development. Using the terms referred to in Acs et al. (2000), China has had rapid economic growth because it has been in the factor-driven and efficiency-driven stage of economic development. The growth in these stages is akin to moving from inside the production possibility frontier (PPF) toward its

edge, which can at times be very rapid. When the economy is already on or close to the PPF, any further growth must be driven by innovations. Once the economy is in this innovation-driven stage, Acemoglu and Robinson claim that China's extractive system will slow the economy down.

If, as Acemoglu and Robinson proposed, the democratic inclusive regime is better than the authoritarian extractive regime in the era of innovation-driven growth, does it mean that we do not have to worry about the unfairness in competition in the current stage? Can we ignore it and expect there to be eventual smooth competition between democratic and authoritarian regimes? The answer is that we cannot, and the arguments are fourfold, as follows.

First, when we say that system A is more efficient than system B, we do not take into consideration the *interactions* between A and B. However, when the two systems do interact in the globalised trading system, then the story becomes more complicated. For instance, high-tech industries with increasing returns or network economies have the property that a frontrunner today is likely to keep or even increase its advantage tomorrow. As we will argue in Chapter 2, this increasing-return property gives the government an extra incentive to subsidise its high-tech industries now, in order to gain an advantage in the future. Thus, if US-based Qualcomm loses the competition for 5G or 6G to Huawei due to China's massive state subsidies, given the increasing returns, it is extremely difficult for Qualcomm to reverse the game in future generations of telecommunications competition. This implies that the unfairness created today may be long-lasting. The above-mentioned innovation advantage of an inclusive system may not have any chance of being effective.

Second, there are some highly regulated industries that have 'standards', and whoever's product is the first on the market may become the market standard. Again, this invites authoritarian governments to provide subsidies, and companies in other democratic countries may have to terminate their operations after the standard is established, despite the possibility that the latter's products might have been better.

Third, when we say that system A is more efficient than system B, it is a *ceteris paribus* comparison, meaning 'other things being equal'. Government support comes in as an *extra* force that disturbs the original competition between two businesses in different countries, and therefore the other elements are not equal anymore. If we allow all governments to subsidise their businesses to compete with foreign ones, then the competition is between governments, and not between businesses, and the whole context of international trade becomes different.

Fourth, there are some commodities or services that may be directly related to national security. For example, 6G is essential to precise geographic positioning, which is pivotal to satellite, GPS and high-tech wars; TikTok or any Apps that use AI as tools for their algorithm design may collect extremely useful individual data for the ability to precisely spread information in the future. Evidently, both of these are likely to compromise national security. If private businesses such as Facebook or Google are involved in technologies

related to national security, democratic countries may be less concerned. However, if a company run by the CCP is involved, then many countries will become extremely worried. Thus, in industries where national security is involved, competition between democratic and authoritarian countries is unavoidable. This is an area where smooth competition is unlikely.

Why Do We Focus On Economic Conflicts?

We mentioned previously that the conflicts between China and the US and other democratic countries are embedded in the fundamental value differences between democracy and authoritarianism, and we also pointed out that in many countries these values transcend the sheer efficiency concerns of economic operations. For instance, China imposes strict controls on the use of the Internet, which evidently violates the democratic values of the freedom of expression, the free flow of information, the freedom of the press, and so on. Why, then, do we wish to study the *economic impacts* of such an oppression of freedom, rather than the freedom itself?

Indeed, our analytical approach may have been compromised by practical international politics. In reality, whenever an authoritarian country is accused of some human rights violation in terms of persecuting a group of people or banning some particular activities, the common defence by the authoritarian state is 'This is a domestic matter; these people and their activities have caused our national security concerns, and we have dealt with it according to our domestic laws.' As to the regulation of the Internet, for instance, every country in the world imposes some restrictions on the dissemination of pornographic content or videos, but the restrictions may differ from one country to another, and sometimes across different local religious groups. It is thus difficult to draw a line between the accusation of human rights violations and interfering with another country's domestic affairs.

Economic conflicts, however, leave us with an alternative angle from which to approach this. In the era of globalisation with its widespread international competition, a fair competition environment or a level playing field is a concept that almost every country in the world can accept. While country A may not have a legitimate reason to inquire about the extent of country B's Internet regulation policies, if such policies systematically cause market access problems, and form an unfair competition environment for e-commerce platforms in country A, then country A does have a strong reason to challenge such an unfair trade practice. Conceptually, therefore, it sometimes does not pay to criticise an unreasonable policy itself, but it does pay to criticise the *implications* of such a policy. The strategic focus of this book is indeed on the economic implications of the value differences between democratic and authoritarian countries. We believe that this strategy may be more successful to avoid using the excuse of 'it is our domestic matter.'

Since the enactment of the Marrakesh Agreement in 1994, the world has had a set of common rules on trade and economic activities. Agreements

such as the GATT, GATS (General Agreement on Trade in Services), TRIPS (Agreement on Trade-related Aspects of Intellectual Property Rights) and SCM (Agreements on Subsidies and Countervailing Measures), etc. have become commonly accepted doctrines that most countries in the world accept as the bases and rules of economic interactions. At the end of 2020, there were 164 Members in the WTO, in which the above-mentioned trade rules have been adopted. The WTO also has a dispute settlement system, through which binding decisions can be made when Members file complaints about other Members' violations of such rules. In short, unlike freedom or human rights issues in which countries may have wildly different opinions, many economic conflicts can be judged relatively more easily under the current WTO system.

As far as private entities are concerned, a fair business competition rule and a level playing field are also what individual businesses demand. Businesspeople may not know much about the human rights violations of country A's Internet blocking, but if such blocking results in the citizens of country A being unable to access the e-commerce platform of country B, then this unfair economic impact can be easily perceived. As such, if we focus on the *economic* consequences of human rights violations, very often we will receive more attention and support than if we focus on the human rights aspects per se. If the support and attention we receive are sufficiently strong, we may be able to rectify the human rights issue indirectly, through the challenge of the economic conflicts.

It is certainly correct that not all ideological differences have economic consequences, and therefore our economic-conflict approach always has its limits. Indeed, we are humble about the range of problems that we are able to resolve. Our efforts may not be able to create an ideal world, but we do hope that we can help make it better.

Organisation of This Book

The remaining chapters of this book are organised as follows. Chapter 2 starts with an analysis of the *Made in China 2025* project, which provoked severe reactions from the US, the EU and Japan in 2018. We explain the key controversy behind this project, and why democratic countries were not happy. Chapter 3 examines the economic consequences of authoritarian countries' Internet control, and explains why this creates new and unfair trade barriers for e-commerce. In Chapter 4, we address another dimension of unfair competition in the Internet era: privacy and national security. This is related to the recent development of artificial intelligence. Because privacy is a fundamental value of democratic societies, we believe that it is difficult to find a compromise solution here.

Chapter 5 discusses the accountability of publicly listed companies, and explains the fundamental differences between laws in China and those in democratic countries. These laws generate the unfair competition environment between these two systems. Chapter 6 touches on an area not previously

mentioned in the literature: antitrust laws. The fundamental idea behind the antitrust laws in democratic countries is that they provide a system of checks and balances, which in authoritarian countries is a ridiculous concept. We show how unfair competition may arise from this conceptual difference. Chapter 7 is concerned with the national security issue. Because the CCP can impose various top-down national objectives on businesses, it causes other countries' national security systems to check on Chinese foreign direct investment (FDI). In recent years these 'responsive' national security checks gradually become a priority rather than an exception. This of course makes economic activities more inefficient. Chapter 8 is about judicial remedies. In democratic countries, the judicial system is independent, and all international businesses can pursue reliable judicial remedies if they face unfair treatment in relation to competition in other countries. However, in authoritarian states such as China, there is no such thing as judicial independence, and therefore foreign businesses often face unfair treatment when they try to find judicial remedies. Finally, Chapter 9 draws together our findings and makes some conclusions.

The above-mentioned institutional differences between democratic and authoritarian countries are often concretised in various laws. As such, when we analyse the conflicts between countries, we often pin them down to the point where 'the conflicts can be avoided if the laws in democratic or authoritarian countries are amended in certain directions.' However, very often we find that amending laws as suggested would violate the fundamental value of democracy or the fundamental definition of an authoritarian state. This is why we believe that such conflicts are ultimately unavoidable. Because we analyse various laws to establish our propositions, this book can be treated as an analysis of law and legal institutions. Whenever applicable, we will match our analysis with the existing rules of the WTO.

In sum, this is not a book that aims to help the reader to understand the operation of China's socialist market economy; rather, we focus on the *interaction* between China and other market economies in democratic countries.¹⁴ Nor do we blame any single person, such as Xi Jinping or Donald Trump, for the US–China conflicts; instead, we try to point out the incongruence between the Chinese system and the Western democracies. Fundamentally, it is the institution that matters, not the individuals who formulate and abuse the institution. We believe that the trade war and other economic conflicts between the US and China are merely superficial phenomena, and that underneath there are deep institutional differences, or to put it simply, the difference between democracy and authoritarianism.

Some people tend to refer to the US–China conflicts as a 'Thucydides trap', describing the tendency for there to be confrontation when a rising power encounters the incumbent power. We, however, do not tend to think of the conflicts this way. If it is just a Thucydides trap between any two entities, there is only a power struggle, and no values are involved. However, between democracy and authoritarianism, the differences in values are the key. The

confrontations are not like Thucydides traps that drive both sides to fall in, but embody deep value differences that push both sides apart. The ultimate economic conflicts are between two parties with very different ideologies.

Notes

- 1 Chalmers Johnson, *MITI and the Japanese Miracle: The Growth of Industrial Policy, 1925–1975* (Stanford University Press 1982).
- 2 Ibid., 18.
- 3 Aaron L. Friedberg, 'Competing with China' (2018) 60 *Survival* 7; Hal Brands, 'Democracy vs Authoritarianism: How Ideology Shapes Great-power Conflict' 60(5) *Survival*, 61–114 (2018).
- 4 According to BBC News on 25 September 2021, Meng was finally released after reaching a plea deal with US prosecutors.
- 5 The US Bureau of Industry and Security (BIS) controls the export of commodities and related technologies whose widespread release could compromise US national security. This *Entity List* can be found in Supplement No. 4 to Part 744 of the Export Administration Regulations. It includes the names of foreign persons, government organisations and companies that are subject to specific licensing requirements for the export of certain items. These specific licensing requirements are in addition to those requirements outlined within the Export Administration Regulations. If any item is included in the *Entity List*, its export needs to be approved. Violating this regulation may result in criminal charges.
- 6 See Shawn Donnan, 'US Says China WTO Membership was a Mistake' *Financial Times* (20 January 2018), www.ft.com/content/edb346ec-fd3a-11e7-9b32-d7d59aace167 accessed 15 September 2020.
- 7 Another explanation for the US–China conflicts was the Thucydides trap (Allison 2017), which says that two giants are bound to face conflicts when the originally smaller player (China) gradually becomes a serious challenger to the incumbent larger player. Our analysis can be treated as a complement to this argument. Even if two giants do have the instinctive motivation to confront each other, our analysis provides an objective analysis as to why such conflicts are unavoidable.
- 8 See, for instance, USTR, *2017 Report to Congress on China's WTO Compliance* (2018), <https://ustr.gov/sites/default/files/files/Press/Reports/China%202017%20WTO%20Report.pdf>; USTR, *2016 National Trade Estimate Report* <https://ustr.gov/sites/default/files/2016-NTEReport-FINAL.pdf>, 91; USTR, *2018 National Trade Estimate Report on Foreign Trade Barriers* (2018), <https://ustr.gov/sites/default/files/files/Press/Reports/2018%20National%20Trade%20Estimate%20Report.pdf>, 104, and 'Joint Statement of the Trilateral Meeting of the Trade Ministers of the United States, Japan, and the European Union' (31 May 2018), <https://ustr.gov/about-us/policy-offices/press-office/press-releases/2018/may/joint-statement-trilateral-meeting>.
- 9 Freedom House, *Freedom on the Net 2020: China Country Profile*, <https://freedomhouse.org/country/china/freedom-net/2020>.
- 10 See USTR, *2020 National Trade Estimate Report on Foreign Trade Barriers*, https://ustr.gov/sites/default/files/2020_National_Trade_Estimate_Report.pdf, 119.
- 11 'China Tells Carriers to Block Access to Personal VPNs by February', *Bloomberg News* (11 July 2017), www.bloomberg.com/news/articles/2017-07-10/china-is-said-to-order-carriers-to-bar-personal-vpns-by-february.

- 12 See Michael Martina, 'Exclusive: In China, the Party's Push for Influence Inside Foreign Firms Stirs Fears', *Reuters* (24 August 2017), www.reuters.com/article/us-china-congress-companies-idUSKCN1B40JU. See also the rules announced by the China Securities Regulatory Commission on 16 June 2018.
- 13 Stanley Lubman, 'The Unprecedented Reach of China's Surveillance State', *The China File* (15 September 2017), www.chinafile.com/reporting-opinion/viewpoint/unprecedented-reach-of-chinas-surveillance-state?utm_source=feedburner&utm_medium=feed&utm_campaign=Feed%3A+chinafile%2FAll+%28ChinaFile%29; Josh Chin and Gillian Wong, 'China's New Tool for Social Control: A Credit Rating for Everything', *Wall Street Journal* (28 November 2016), www.wsj.com/articles/chinas-new-tool-for-social-control-a-credit-rating-for-everything-1480351590?mg=prod/accounts-wsj.
- 14 For a general introduction regarding China's political system, see for example, William A. Joseph (ed.), *Politics in China: An Introduction* (Oxford University Press 2019); June Teufel Dreyer, *China's Political System: Modernization and Tradition* (Routledge 2019).

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2 Made in China 2025

The Issue of Government Subsidies

Background – Imposing a Future Tariff?

In 2018, then US President Donald Trump began to impose tariffs and other measures on China, with the goal of forcing China to make changes to respond to the US's 'unfair trade practices'. On 11 July 2018, the US submitted a communication to the WTO General Council entitled 'China's Trade-Disruptive Economic Model', which can usefully serve as a snapshot of US accusations against China in the two countries' trade war. In this communication, the US accused China of having adopted a disruptive economic model of party-state authoritarian control, exercised forced technology transfers (FTTs), facilitated the cyber theft of intellectual property (IP), imposed overly broad cyber security measures, and applied inappropriate industrial policies, such as in its industrial plan *Made in China 2025*, to support its industries.¹ The US also asserted that China was benefiting from its WTO membership at the expense of other Members, and concluded that 'As long as China remains on this path, the implications for WTO would be decidedly negative.' China denied all such allegations.²

Around the same time, on 6 July 2018, the initial action of the US Section 301 investigation, that is, an additional *ad valorem* duty of 25 per cent on 818 items of Chinese origin, came into effect.³ According to the Office of the US Trade Representative (USTR), the list of goods takes into account 'products that benefit from Chinese industrial policies, including *Made in China 2025*'.⁴ While the traditional tariffs are collected from the current imports, the US actually promises a future measure on products not yet existent today. Typical examples are 'information technology (5G and 6G)' and 'new (nano) materials', which are items included in *Made in China 2025*, but were not yet on the market in 2018. Announcing the imposition of a tariff on a 'next generation' product before it comes into existence is indeed unprecedented.

Several questions could be asked here. Why is the US targeting China's future industries? Why is *Made in China 2025* of so much concern? Why does the US assert that the WTO serves to benefit China rather than regulate its trade disruptive actions? How could 'overly broad cyber-security measures' constitute trade unfairness?

The purpose of this chapter and the next (Chapter 3) is twofold. First, we intend to show that the answers to the above questions are all related to three major structural changes in the global economic and trade landscape that have occurred during the 27 years after the Marrakesh Agreement was enacted (1994) – namely, the rise of the knowledge-based economy with its increasing returns, the emergence of economies with a large proportion of state-owned enterprises (SOEs) or quasi-SOEs, and the digital revolution. Because the framers of the GATT, and even those of the WTO, could not foresee these trends and their impacts on the patterns of economic operation, the WTO rules existing today contain loopholes when applied to the modern international trading environment. Moreover, such regulatory deficiencies have become worse, due to the stagnation of the WTO negotiations.

On the other hand, the authoritarian party-state regimes of some countries have been making the most of exploiting these loopholes and have benefited greatly. This lopsided and unfair competition is at the root of the US–China trade war. The long-term solutions therefore lie in the possible amendment of the WTO rules to address the needs of modern international trade.

In this chapter, we will discuss in more detail the current WTO rules on government subsidies, and explain why they cannot deal with cases such as *Made in China 2025*. This subsidy issue, as we will explain, is related to two of the three changes mentioned above.

The remainder of this chapter is arranged as follows. First, we provide a brief summary of the trade war between the US and China in 2018. We then elaborate on the reasons for and impacts of the three structural changes one by one, but with more emphasis on the first and the second one. The impact of the internet revolution will be further elaborated later in Chapter 3. For each change, we also examine and assess the applicability of the existing WTO rules. We finally conclude that further WTO reforms must at minimum address these three changes if a fair and open global international trading environment is to be maintained.

The Big Picture – The US–China Trade War

In January 2018, the USTR published its 16th report to the US Congress regarding China’s WTO compliance, the first such report since President Donald Trump took office in January 2017.⁵ Unlike the previous 15 reports, the 2018 USTR report contained no positive accounts of US-China trade, and delivered the blunt conclusion that ‘it seems clear that the United States erred in supporting China’s entry into the WTO on terms that have proven to be ineffective in securing China’s embrace of an open, market-oriented trade regime.’ The USTR went on to say that ‘it is now clear that the WTO rules are not sufficient to constrain China’s market-distorting behaviour’, and that the WTO dispute settlement mechanism is ‘not effective in addressing a trade regime that broadly conflicts with the fundamental underpinnings of the WTO system’. In this regard, the US pledged to ‘take all other steps

necessary to rein in harmful state-led, mercantilist policies and practices pursued by China, even when they did not fall squarely within WTO disciplines, as evidenced by the USTR's ongoing investigation of Chinese technology transfer policies and practices pursuant to Section 301 of the Trade Act of 1974, as amended'. Moreover, China's industrial plan *Made in China 2025*, which was touched on slightly in the 2017 report, was ranked at the top of the US's list of priority issues in 2018.

In hindsight, this report had clearly pictured the events that had happened in the following months. The White House announced in March 2018 an increase in tariffs on approximately US\$50 billion worth of Chinese imports as a result of the US Section 301 investigation.⁶ Although a few seemingly positive bilateral talks were held in May,⁷ the ultimate 25 per cent tariff increase on more than 2,000 goods items among imports from China was imposed in two tranches, in July and August. A number of denouncements about China's party-state regime and trade-distortive policies, such as the communication mentioned at the beginning of this chapter, were delivered at the WTO around the same time. On 18 September 2018, the USTR further imposed a 10 per cent tariff increase (which was scheduled to move up to 25 per cent on 1 January 2019) on an additional US\$200 billion worth of Chinese imports.⁸

From the US's descriptions of the trade war, we can abstract two distinct lines of discussion. The first relates to the party-state disruptive measures being adopted by China. The second is the ineffectiveness of the WTO rules in treating such measures.

The party-state situation in China, however, is not new. The Chinese Communist regime actually follows the Leninist doctrine, which existed even in 1947 when the GATT was first framed. So why does it pay China to exercise its government intervention in industries today, and why did it not pay the socialist GATT contracting parties in the Cold War period? Is it simply because China is different from other socialist states? Or is it because the environment today is different from that of the 1950s? If there are changes in the environment, then what are they? Furthermore, what is the connection between a party-state system and the *Made in China 2025* project?

Three Structural Changes Since the Marrakesh Agreement

We argue here that there are three major changes that have taken place since the Marrakesh Agreement enacted in 1994, which made the current WTO rules incomplete and inadequate. The first is the rise of the knowledge-based economy, which emphasises the importance of knowledge as a production input, thereby generating the associated property of increasing returns. The second change is associated with the accessions of China and Russia to the WTO. In addition to the large numbers of SOEs, the close ties between government and business in China pose particular challenges to WTO rules. These first two changes are both related to government subsidies, which is the

major focus of this chapter. The third change is the prevalence of the Internet, or more broadly speaking, the changing environment brought about by the digital revolution.

Regarding the first change, the property of increasing returns generated by the knowledge-based economy alters the competition strategy of businesses (Romer 1990). It also provides a strong incentive for governments to use subsidies to help their businesses become the front-runners in their particular field. In order to take full advantage of the changes mentioned above, a government needs to be in a position of extreme power, especially when it comes to allocating resources, and this is where the party-state and its highly centralised control regime can play a critical role.

As for the second change, the commonly held perception of the GATT was that it was the club of market economies, in stark contrast to the then Soviet Union-led Council for Mutual Economic Assistance (COMECON). Based on the presumption that all, or almost all, of the participating country governments remain at an arm's-length distance from the markets, neither the GATT nor the WTO specifically regulates a government's direct control over its economic activities. Detailed discussions of the original design can be found in Thorstensen et al. (2013) and Irwin et al. (2008).

China's accession to the WTO manifests this loophole. While continuing to act consistently within the WTO rules, the 'socialist market-economy' governments can put international competitors at a disadvantage by directly participating in, rather than regulating, the playground. This second change is also related to the first one: in a party-state system such as China, it is difficult to tell whether a business of which 45 per cent of the stock shares are owned by the state receives 'government subsidies' or not.

The third change is the digital revolution that has created the notion of cyberspace, which is entirely different from the physical space where the WTO/GATT rules mainly apply. In the digital world, online regulatory measures can be transformed into various types of cyberspace trade barriers (CTBs), to the advantage of domestic businesses. We will briefly discuss this toward the end of this chapter, and elaborate more in Chapter 3.

First, let us go through these three changes one by one. More discussion regarding the impact of the changes brought about by the Internet on privacy intrusion will be provided in Chapter 4.

First Change: Prevalence of Increasing Returns

In evaluating the US's accusations against China, we start by exploring the various elements of China's *Made in China 2025* project.⁹ We find that the key feature of this huge industrial policy project is that the ten industries targeted are all high-tech ones. As we all know, there is hardly any advantage in being a front-runner in traditional sectors, such as retailing, clothes manufacturing, or agriculture. As long as technologies and resources are free to move, due to the law of diminishing returns, the latecomers are likely to eventually catch up

with the front-runners. In macroeconomics, this is referred to as ‘the convergence hypothesis’. A further discussion can be found in Durlauf (1996) and the references therein.

As *Made in China 2025* is designed to help Chinese businesses establish themselves as front-runners in these ten high-tech industries, one question that obviously springs to mind is: what is the key difference between subsidising the agricultural sector (which many countries are doing) and subsidising high-tech industries?

The answer to this question is simple: increasing returns. In the traditional manufacturing industries or agriculture, it is the law of diminishing returns that is most likely to apply. In these traditional sectors, even being the front-runner or having a large market share is still unlikely to pay dividends. Diminishing returns will soon suppress the rewards down to the same levels as the latecomers or smaller competitors. By contrast, for industries characterised by increasing returns, the front-runners’ advantage may be sustained and even accelerate. In other words, there is a greater chance of the strong ones of today becoming even stronger tomorrow, thus making the efforts needed to secure that leading position much more worthwhile.

Why Do High-tech Industries Have Increasing Returns?

It is important to note that the increasing-returns advantage of the high-tech industries is not just due to the property of decreasing average cost that is often discussed on the technology side. According to Brian Arthur (1996), several features of the high-tech industries explain why they tend to generate increasing returns.

First, high-tech manufacturing or production often involves large up-front research and development costs. Once the manufacturer becomes a front-runner and grabs a large market share, the average cost of providing goods is low, but for later challengers the up-front threshold is still there. This makes it difficult for latecomers to compete with the front-runner, thereby helping the incumbent leader to sustain its position. In short, this refers to the cost-side increasing returns, meaning that the larger leader’s average cost is below that of smaller latecomers. Furthermore, due to reasons such as learning-by-doing, the marginal costs of making additional improvements may be lower for front-runners.

Second, there is the network effect. For instance, the success of Amazon enables it to carry more commodity items than its competitors, and hence gather more website browsers. This in turn makes it more attractive to new firms looking for a platform for their products. Furthermore, as more new firms decide to choose Amazon as their sales platform, Amazon’s own sales and customer base become larger, making it even more successful than its competitors. This network effect refers to the demand-side increasing returns, meaning that successful front-runners are more attractive to individuals – hence the demand curve that the front-runners face shifts upward.

Third, according to Arthur (1989), there is also the customer lock-in effect benefiting the front-runner. For instance, e-commerce consumers may be used to the algorithm, payment system, and search logic of the front-runner's website, and therefore unwilling to learn about and switch to competitors.

Also worthy of mention is that you do not have to be truly efficient to be a front-runner in the high-tech industries. As Arthur (1989, p. 119) points out, front-runners may well be inefficient. Some make it to the front just by luck or because they happened to be early entrants in the market. What is evident, though, is that competing from the front is more advantageous, even without substantive efficiencies. In addition, the fact that being a front-runner helps a company maintain its market advantage certainly gives the government much of an incentive to help its high-tech industries to fight to get to the front line.

In short, in high-tech industries, the target of government subsidies is not to help a firm to cut its costs or to lower its prices, but to help it gain a front-runner's position.¹⁰

Another typical means of achieving front-runner success in the high-tech competition is to win the 'standard'. For instance, at present the standard for future telecommunications (6G) is yet to be determined. Whoever wins this standard competition will gain huge industrial profits. In pharmaceuticals, as another example, the government's regulation and approval (such as by the US Food and Drug Administration (FDA), or the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH)) have also become the 'standard', which precludes other unapproved entities from competing in the drug market. High-tech industries with the feature of the 'standard' have an even greater incentive to become front-runners.

What About the US Human Genome Project?

A question that one may ask is what is the difference between the *Made in China 2025* project and the US Science projects, which also subsidise R&D with government money? We should first note that most of the US national research programs are science-based, and not technology-based. The Human Genome Project includes gene sequencing, genomics, proteomics, etc., and is designed to explore scientific knowledge not directly related to any specific application objectives. Similarly, the US BRAIN Initiative (Brain Research Through Advancing Innovative Neurotechnologies) sets out to facilitate our understanding of the working and operation of memory and learning.¹¹ It may, at the same time, be remotely related to a cure for autism, amnesia and Parkinson's disease, but the research itself has little to do with any specific industrial values or business revenues.

By contrast, the *Made in China 2025* project has some very specific and concrete industrial targets. It encompasses telecommunications, robotics, aeronautical and astronomical equipment, maritime engineering, rail equipment, green energy, electric cars, nano materials, and pharmaceuticals. These are

specific industries, as opposed to the more abstract description of the US's BRAIN or Human Genome science initiatives. For each of the ten *Made in China 2025* industries, there may be a technology breakthrough that is critical to its development. Facilitating such technology breakthroughs is a key element of *Made in China 2025*, with the ultimate aim being for China to play a leading role in these industries in the world.

What About Germany's Industrie 4.0 Project?

Germany describes its Industrie 4.0 project as being more application- and engineering-oriented than just pure scientific research. However, it still only contains concepts, such as information and communications technology, ICT-based manufacturing, smart factories, individualised product specifications, computerised manufacturing, the Internet of things, the Internet of persons, self-diagnosis, self-configuration, self-optimisation, and so on. It is more like a description of future technology/production roadmaps which German scientists and engineers need to explore jointly, rather than any specific industrial subsidy project that has direct revenue flows.¹²

By pointing to these future directions as being keys to the fourth industrial revolution, the German government seems to be lighting the way for their own (and other countries') R&D orientations. In fact, the Industrie 4.0 project has been much criticised as being too diffuse and somewhat impractical to implement. *Made in China 2025*, by contrast, lays out a set of top-down unified doctrines.

Can Democratic Governments Subsidise Particular Businesses?

Even if the US and Germany were to announce their priority national industrial projects (instead of science initiatives or roadmaps), neither would suggest any allocation of government resources to specific industries, sectors, or businesses. In fact, this is not part of the mechanism of any democratic government.

The major US science funding agencies or mechanisms are the National Science Foundation (NSF), the National Institutes of Health (NIH), the various National Laboratories of the Energy Department, and the Defense Advanced Research Projects Agency (DARPA) of the Defense Department. These funding agencies may have some projects that are more application-oriented, but they are not industrial projects and cannot be directly funded to support specific large companies.

The German case is slightly different: its public funding involves both scientific research conducted by universities and institutes such as Max Planck, and applied projects such as those executed by the Fraunhofer Institute, which focuses on bridging knowledge and applications. Successful bridging will generate new businesses and profits, which becomes an important budget source of the Fraunhofer Institute (70 per cent). Again, this is not a direct subsidy to specific companies, because the 'successful' ones are unknown in advance

in the bridging stage. In general, there is a clear distinction between incubating potential new firms and subsidising already existing firms; the latter is very much restricted in democratic regimes, which we will discuss below in the section ‘The Impact of Prevalent State-owned Enterprises’.

The mechanism for bridging knowledge and applications is generally in the form of feasibility analysis, animal experimentation, the testing of market acceptability and various other forms often seen in the US’s Silicon Valley. The German Fraunhofer Institute may replace Silicon Valley’s market role in some respects, but the German case does not involve any subsidy to large existing industries and firms, either.

The EU-level project follows the same pattern. For example, the EU’s Horizon 2020 funds future and emerging technologies (FET) based on ‘new ideas for radically new future technologies, at an early stage when there are few researchers working on a project topic’. The project also ‘supports areas that are not yet ready for inclusion in industry research roadmaps with the aim of building up and structuring new interdisciplinary research communities’. FET Flagships specifically sets a €1 billion budget to unite European research forces in solving ambitious scientific and technological challenges, like understanding the human brain or developing the new materials of the future, such as graphene.¹³

In sum, the US’s, Germany’s, and the European Union’s subsidies are either entirely science-based or focused on bridging the gap between knowledge and applications. This kind of bridging diversifies the individual technology investor’s risk at the very early stage, and thus together the country enjoys a higher than expected return, due to the law of large numbers. By contrast, *Made in China 2025* is targeted at specific industries and firms, and its purpose as described is not to diversify the risk, but to attain the front-runner’s advantages in industries with increasing returns.¹⁴

Few Exceptions in Democratic Countries

In democratic countries, there are a few cases where government subsidies are allowed. For example, in Germany, the Federal Ministry of Education and Research has a KMU program which subsidises small- and medium-sized enterprises to try out cutting-edge research. In addition, the Federal Ministry of Economic Affairs and Energy has a ZIM program, which also seeks to encourage small- and medium-sized enterprises to adopt market-oriented innovation development.

The US Seed Fund under the NSF has an STTR (Small Business Technology Transfers) program to support technology transfers. The US DARPA has an SBIR (Small Business Innovation Research) program which helps small businesses with their innovation. The Massachusetts Institute of Technology, collaborating with the US Air Force, also has an STTR program.

Similar programs exist in Japan, Taiwan and many other democratic countries, and we will not repeat them here. The general common feature of these

projects is that they focus on small- or at most medium-sized enterprises. Because the target of such subsidies are SMEs, international spillovers are not involved, and hence the fairness of international competition is not affected.

In particular, on 8 June 2021, the US Senate passed the United States Innovation and Competition Act (USICA) to provide more than \$250 billion over the next five years to boost American industries in several critical areas, including semiconductors, 5G communications and artificial intelligence. At the time of writing (December 2021), the bill was on its way to the House and expected to pass.

As the ‘the largest investment in scientific research and technological innovation in generations’ according to Senator Chuck Schumer, the USICA intends to limit government spending to science R&D and gap bridging. The core part of the USICA, the Endless Frontier Act (EFA), aims to appropriate tens of billions for the National Science Foundation and to establish a Directorate for Technology and Innovation within the Fund to ‘direct basic and applied research, advanced technology development, and commercialization support in the key technology focus areas’. The approaches include establishing innovation centres, providing scholarships and R&D awards, facilitating academic technology transfer and establishing testbeds. Again, the subsidies provided here focus on basic R&D and are remote from specific industrial benefits.

A more controversial feature of the USICA is that it will further provide \$52 billion to fund the semiconductor manufacturing programs stipulated under the US Fiscal Year 2021 National Defense Authorization Act (NDAA). An essential program under Section 9902 of the NDAA is that it plans to provide ‘financial assistance to covered entities to incentivize investment in facilities and equipment in the United States for semiconductor fabrication, assembly, testing, advanced packaging, or research and development’. The USICA also includes \$1.5 billion for the Public Wireless Supply Chain Innovation Fund to spur mobile broadband technologies.

Although we are of the view that the inherent checks and balances under the democratic system should very much restrain the market-distorting effects of USICA-like subsidisation (for example, it could never directly designate all classes of state resources to a couple of enterprises), we still see the concern. The USICA has been drafted with the express purpose of competing with China. It may thus be seen as a sign of a downward spiral, in that liberal democracies are adopting state capitalism to fight against state capitalism, with market competition becoming more like competition between states. The orthodox liberal economic theory is looming.

The Crowding-out Effect of the Made in China 2025 Project

The ten industries specified in *Made in China 2025* may not yet exist, but some rough sketches have been drawn. From the Chinese official documents (cited in note 9), we can see that the government measures include financial support,

tax waivers, expertise nurturing and experimental site advantages, etc. These are exactly the kinds of government subsidies that would help to establish front-runners at the head of the ten targeted industries by 2025.

When a high-tech or knowledge-based industry is characterised by increasing returns, it is usually the case that very few players remain in the market equilibrium. For instance, in pharmaceuticals, only FDA- or ICH-approved firms can produce and sell new drugs and medical devices. In semiconductor wafer manufacturing, around the world there are really only Intel, TSMC and Samsung. In high-tech industries, there may be some small firms focusing on specific customers, but the niche is very limited and their market shares are usually extremely small. Increasing returns will reinforce the advantage of the already strong, and gradually drive away the weak and smaller ones. In short, as far as high-tech industries are concerned, very few businesses can possibly survive in the final equilibrium under competition.

According to Arthur (1989), which business will end up in the high-tech equilibrium is a probabilistic event, and not a deterministic one. However, a dedicated effort can increase the chances of success, and the *Made in China 2025* project sets out to achieve exactly that: to increase the probability that a Chinese firm will end up in the equilibrium. Moreover, because very few firms can survive in the equilibrium, the success of *Made in China 2025* must imply the failure of some competitors from other countries, who might have been more successful had there been no Chinese governmental support. The battle is not for market share, but for the future leading position. This crowding-out of market shares, however, is not different from the battle to win seats.

In the case of pharmaceuticals, the crowding-out may be more serious in terms of FDA approval. Once the first drug is approved by the FDA, its age-specific doses, treatments and side-effect avoidance can be better improved after receiving the patients' feedback, which further narrows the later-approved drugs' market. As for telecommunications chips, there are also 'standards' to be accommodated, to make the communication between any two agents compatible. Once a chip standard is adopted, any other chip design has almost zero market demand. This is why large telecommunications firms all try to win the 'standard wars'.

Flaws in the WTO's Government Subsidies Rules

Our next question is: if the Chinese government does use various measures to provide support to its ten industries as described in the *Made in China 2025* project, is China in violation of any WTO rules? Why cannot other countries file a complaint through the multilateral dispute settlement system in the WTO?

The USTR's report on China's WTO compliance indicates that certain detrimental actions of the Chinese government do not appear to implicate specific WTO obligations. The WTO does have rules on government subsidies, although they were framed decades ago, when the increasing return property

of knowledge-based economies and high-tech industries was hardly known. More general economic analysis of the WTO subsidy rules can be found in Bagwell and Staiger (2006). Here we focus on why high-tech industries with increasing returns cannot fit in the existing WTO rules framework.

The WTO's Agreement on Subsidies and Countervailing Measures (SCM Agreement) prohibits a government from providing financial support that can benefit specific businesses if its grant is contingent upon the businesses' export performance or the use of domestic over imported goods (Article 3.1).

Government subsidies other than the above 'prohibited subsidies', but shown to cause adverse effects to another WTO Member, shall also be modified or removed ('actionable subsidies') (Article 5 of SCM). The 'adverse effects' of subsidies include: (1) injury to a domestic industry of another Member caused by imports of the subsidised product (i.e., export expansion); (2) displacement of another Member's exports by the subsidised product in either the subsidising Member's or in a third-country's market (i.e., import substitution), and (3) nullification or impairment of benefits accruing under the GATT 1994 – in general the tariff concessions' market access being undermined by the subsidies (Article 6 of SCM).

The Member granting or maintaining actionable subsidies shall take appropriate steps to remove the adverse effects or shall withdraw the subsidy (Article 7.8). In the case of injury to a domestic industry, the importing Member may impose countervailing duties to offset the effect of a subsidy.

In sum, the existing WTO subsidy rules in general focus on the subsidy's effect of reducing the production and sales costs of some products and therefore distorting fair competition with like products from other countries. However, in high-tech industries, such rules suffer from the following logical flaws. Take *Made in China 2025* as an example: the industry for nano materials or next-generation telecommunications evidently does not exist in either China or any other countries today, and therefore no exports or domestic sales of the products have taken place which we can relate to China's subsidies. Moreover, because the industries do not yet exist, other WTO Members may find it hard to claim any injury or adverse effects with respect to China's subsidies. In short, the present rules do not allow WTO Members to file complaints against the government subsidies aimed at competing for the winning seat among the future high-tech industries.

We believe that this flaw should be eradicated. As mentioned above, there are only a very few firms that could possibly survive in the final equilibrium of high-tech competition. Crowding out the seat of another WTO Member should surely be equivalent to an 'adverse effect' in the *ex ante* sense.¹⁵

Summary and SCM Revision Directions

Summarising the discussion above, we believe that the theoretical argument behind the US's imposition of tariffs on some future products of the *Made in China 2025* project may be as follows: (1) One Member subsidises its high-tech

businesses to increase their chances of winning an advantageous position in the final equilibrium. (2) Because of the property of increasing returns, a business that gains an advantageous position today is likely to maintain it tomorrow. (3) From this advantageous position, the Member may generate a future flow of profits or economic rents. (4) With seats in the equilibrium being limited, the Member's ability to extract such economic rents leads to other Members being expelled, hence losing their economic rents. (5) In probabilistic terms, if there were no governmental subsidies by this Member, then other competing Members' benefits might be different. This, in short, is an adverse effect. The US may believe that this issue must be brought up now, because it is difficult, if not impossible, to establish the causality between subsidies years ago and industry success later on.

Those who are familiar with the WTO rules may not feel comfortable with the above-mentioned 'probabilistic' argument. All the GATT, GATS and other WTO rules are dealing with deterministic, realised damages or infringements. While this approach to designing the rules was fine in the past, it may not be sufficient in today's era of the knowledge-based economy.

In 1939, Economics Nobel Laureate John Hicks (1939, p. 39) warned that admitting increasing returns would lead to 'the wreckage of the greater part of economic theory'. The economic theory Hicks had in mind was that of diminishing returns, which is appropriate when describing traditional production or manufacturing processes. This process involves repetitive production, marginally improving quality, and reducing costs.

Starting from the 1970s, a new production process has emerged, which is referred to today as 'knowledge-based'. This new process now operates in parallel with the old one, and does not replace it. The key features of this new process are innovation and IP, which are elements of the so-called 'high-tech' industries. The traditional wisdom of diminishing returns does not apply to the high-tech businesses. The dominating factors in such a knowledge-based economy are high R&D costs, the network effect, and the lock-in effect that applies to currently advantageous businesses. Because the overall logic of the new process is different, it is hardly surprising that the old rules of the WTO are insufficient to guarantee fair competition in the knowledge-based economic sectors.

As already mentioned in this chapter, a key competition strategy of the high-tech industries is to try to become a front-runner as quickly as possible. This creates a huge incentive for the government to devote resources to that goal. However, because the future industry in question does not yet exist, the government support does not have any directly observable adverse effects, and hence no dispute settlement cases can possibly be established according to the present WTO rules.

A simple way of illustrating the direction of the WTO rules revision is shown in Figure 2.1. Point A is where the production is taking place. The traditional WTO rules regulate government interventions at or after point A, so that government subsidies to reduce production costs or to increase export

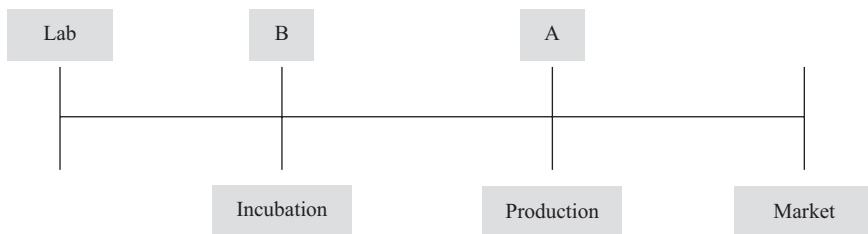


Figure 2.1 The time line from laboratory to factory and market

competitiveness, thus causing damage to companies in other Member countries, are prohibited. However, for high-tech industries, we should extend the time horizon leftwards, towards the laboratory, where the technology idea was first invented but not yet made feasible in the real world. Because of the property of increasing returns, regulating subsidies after point A comes far too late. In order to maintain a level competitive playing-field, government subsidies should be prohibited at a much earlier stage, say, after point B, which is located somewhere between the lab and the production point A.

For instance, in the pharmaceutical business, the government can only support up to the stage of incubation or germination. In other high-tech engineering or electronic industries, the subsidy may be allowed up to the new-product prototyping. The details may require an elaborate discussion by WTO Members. For small businesses which do not have international antennae, the regulation may be looser.

Second Change: The Impact of Prevalent State-owned Enterprises

The second major change in the 27 years following the Marrakesh Agreement is the prevalence of state- (or party-) owned enterprises (SOEs). While SOEs have long been an issue in the context of economic liberalism, it was not until the 21st century that the tension between SOEs and the international trading system came under the spotlight. The discussions may have been triggered somewhat by the global overcapacity in the steel industry, as Chinese state-owned steelmakers have been blamed for most of the capacity increase since 2000.¹⁶ Since then, as expressed by the US in its WTO communication, the criticism keeps on being amplified continuously.

As mentioned earlier, the US contends that the existing WTO rules cannot adequately address the SOE problem. In academia, putting this in a more moderate way, most commentators agree that tensions exist between the large-scale presence of SOEs and the WTO rules.¹⁷ Among WTO diplomats, reform initiatives have been proposed. For example, in their Joint Statement announced in May 2018, the US, Japan and the EU agreed to discuss, among other issues, ‘additional obligations and rules for public bodies and SOEs’.¹⁸

We agree that the large scale of SOEs poses challenges to the existing international trading system and that the WTO rules should be strengthened accordingly. However, we would submit that the root of the problem may be hidden more deeply and broadly underground. In particular, with the prevalence of state-controlled enterprises or superficially private enterprises, the true government subsidies are not transparent. As such, WTO rules on subsidies cannot be implemented meaningfully. Let us start to explain this by providing some historical background.

Chinese Economic Reforms – The Origin of the Prevalence of SOEs

China's economic reform started in 1979, but the beginning of the process was not smooth, and it was not until 1992, when the late leader Deng Xiaoping disclosed his 'Southern Inspection Tour Speeches', that the opposition to economic reform receded and the tone for reform was set. In Europe, the Berlin Wall fell in 1989, closely followed by the disintegration and economic reform of the Soviet Union in 1990–91. In 2001 and 2012, respectively, China and Russia joined the WTO. Thus, it is only very recently that democratic countries began to face these two previously communist countries. Admittedly, some of the previous Soviet Union members are now Member States of the EU. These countries, however, are not sufficiently large to alter the market economy structure of the EU. So, what is so special about China's joining the WTO?

In the context of this chapter's discussion, it is the large scale of privatisation on the surface, and the state-control-in-essence economic system underneath. Before their respective economic reforms, businesses in China were mostly (centrally or locally) government-owned, or party-owned, or jointly-owned. After the reform, or the dissolution of the original central-control system, a large number of originally public entities were 'privatised'.

More often than not, such privatisation processes were not fair and transparent, and as a result, many of the 'privatised' businesses went into the hands of government-related friends. A detailed explanation can be found in Bernstein (2017). Very possibly, the party or government still holds the largest percentage of shares and control, but on the face of it the business is *de jure* private, which is our reason here for putting the description 'private enterprises' in quotation marks. Only for the record are they private. Wu (2016) describes the complex network between the public and private sectors in China as 'China Inc.'. We should add, though, that this is not something unprecedented in human history. Whenever a large public institution is dismantled, it is always close friends of the authorities that end up gaining the economic rents. The term 'crony capitalism' was indeed used to describe the practices whereby a large percentage of businesses originally owned by a colonial regime went to friends and relatives of the new leader after the colony's independence.

This majority of public enterprises behind the ‘private’ veil is something that the WTO has never faced before. In a sense, this is broadly speaking a transparency problem. The economy in question is not sufficiently transparent to fully convince other Members that they are dealing with a private business, when they are actually dealing with a state-controlled one. When WTO Members accused China of operating a non-market economy (NME), they were actually accusing it of disclosing information that was not transparent enough, and outsiders could not easily see whether and to what extent state intervention or subsidies were in operation behind the scenes.

Another kind of transparency problem is that many ‘public-in-essence’ enterprises end up being registered in certain tax havens with disguised ultimate beneficiaries. For instance, according to Bernstein (2017), many companies which appear to be private in authoritarian countries are actually owned by close confidants of the dictator or corrupt officials in authoritarian countries. When this is the case, one can hardly claim that there is no state intervention.

Regulation Deficiencies

When public-in-essence but superficially private enterprises are prevalent, many of the WTO rules are void. For instance, a Member can subsidise an export manufacturing company by ordering one of its ‘private’ businesses to sell inputs to that company at an unreasonably low price. On the face of it, in this example, to establish a case of government subsidy may not be easy. Similarly, the Chinese government with its party-state control can easily intervene in policy by ordering the prevalent party secretaries in many ‘private’ companies to do certain things, and yet it is difficult to verify such an order. To visualise an extreme example, suppose that a WTO Member has been identified to be treated as an NME, suggesting that government intervention in the market is serious. Then the Member’s government can easily refute its NME status by re-registering all its state- or party-controlled entities in the Virgin Islands, thereby hiding the true ultimate beneficiaries, and still doing the same thing.

To illustrate the transparency issue, we may consider the problem from another angle. In their article about China’s SOE reform, in response to the criticism that the Appellate Body’s ‘authority approach’ would make it difficult to classify a SOE as a public body, Zhou et al. (2018) contended that the Appellate Body leaves ample opportunities for an investigating authority to present evidence of the existence and exercise of government control over the SOE so as to prove that the SOE is exercising governmental authority. This argument is really misleading; it simply ignores the big picture and the practical reality.

In our view, the argument of Zhou et al. could be correct only if the government and the SOEs’ operations are fully transparent. Under the current situation, we do not think that it is easy for any individual investigating

authority or a petitioning enterprise to present concrete and comprehensive evidence of government control over a Chinese SOE. In the case of public-in-essence enterprises as we have discussed here, the chances of providing such evidence are even fewer. The result is the unfair competition between businesses in democratic countries, which do not have government subsidies, and public-in-essence ‘private’ businesses in China, which have all kinds of hidden subsidies. Indeed, a major problem of the ‘socialist market economy’ is that the *market* is mingled with the *state*. For these countries, any WTO rule regarding government subsidies to public-in-essence private businesses is doomed to be in vain.

Rule References in Regional Agreements

Are there examples in other international treaties that have dealt with this hidden-SOE issue? The SOE chapter (Chapter 17) of the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP), a trade deal signed by 11 Pacific Rim countries, should represent the greatest efforts by far in tackling this issue at the plurilateral level.¹⁹ The CPTPP lays down a number of substantive obligations regarding the operation of SOEs, requiring the Parties to ensure that their SOEs follow non-discriminatory principles and commercial considerations (Article 17.4). A CPTPP Party shall also ensure that the non-commercial assistance provided by the government to an SOE, or by an SOE to another SOE, will not cause adverse effects to another Party or injure another Party’s industries (Article 17.6). Moreover, Article 17.10 introduces a transparency mechanism: each Party should provide a list of its SOEs and renew the list annually; upon request, a Party must provide information about a specific SOE, including the government ownership and any government official serving as an officer or member of the board, and information on any policy for non-commercial assistance.

The CPTPP’s SOE chapter indeed expands the traditional doctrine. For example, CPTPP rules on non-commercial assistance cover both services and investments, which are outside the realm of the WTO SCM Agreement. In addition, the disciplines are applicable to non-commercial assistance provided by an SOE to another SOE, which is, in a way, setting a broader scope than the term ‘public body’ under the SCM Agreement.

In the above expansion in rules, however, the definition of an SOE under the CPTPP is very narrow. The term SOE refers only to a commercial enterprise in which a Party owns more than 50 per cent of the share capital; controls through ownership interests the exercise of more than 50 per cent of the voting rights; or holds the power to appoint a majority of members of the board of directors (Article 17.1). With this definition, as pointed out by Kawase and Ambashi (2018), the above-mentioned public-in-essence enterprises in China cannot be captured by the CPTPP.

It should be noted that although ‘50 per cent of ownership or voting rights’ is a usual threshold in determining an SOE, it is not a universal norm. As

Kawase and Ambashi (2018) highlight, in the US-Singapore FTA, the concept ‘effective influence’ is used to identify government enterprises. It includes enterprises in which the government owns 50 per cent or less of the voting rights if it has the ability to exercise substantial influence over executive appointments or other important management decisions. If the government owns more than 20 per cent of the voting rights that constitute the largest block thereof, a rebuttable presumption of effective influence exists.²⁰

Suggestions for WTO Reforms

Our goal here is to figure out ways of revising the WTO rules so that such prevalent public-in-essence enterprises, in whatever form, can be suppressed. Enhanced transparency could be the most useful first step. A notification and review mechanism should be established, requiring WTO Members to notify their SOE information. In addition to *de jure* SOEs and the relevant laws and regulations, the mechanism should also cover *de facto* parts of SOEs, such as those ‘public-in-essence’ enterprises, as well as any relevant unwritten norms, practices, or operations. Taking into account the administrative burdens that this could cause, perhaps a Member should notify the *de facto* information only if another Member so requests.

To more effectively unveil those *de facto* parts of SOEs, Members should consider bringing in the ‘counter-notification’ model. In the event that a Member fails to notify its SOE information as requested, the requesting Member/s themselves may present a counter-notification to the WTO. The Member to which the original request was made should have a reasonable period of time to clarify the information contained in the counter-notification. Individual WTO Members’ domestic measures based on the counter-notification along with the clarification information should be presumed to be WTO-consistent, for example, in the case of the determination of a ‘public body’ in countervailing investigations, or the determination of an NME or a particular market situation in anti-dumping investigations.

WTO Members may consider more comprehensive and stringent schemes to monitor governments’ direct economic interventions, and ways to offset the negative impacts thereof. The information and experiences obtained through the counter-notification mechanism suggested above should serve as a basis for more ambitious discussions.

Third Change: Challenge from the Internet

Nicolas Negroponte of the MIT Media Lab described in 1994 the digital revolution as consisting of two major transformations: an atom/bits transformation and a static/mobile switch. As for the former, atoms are the basic units of physical products, whereas bits are the basic units of digital content. Because the shipment of atoms is completely inefficient compared with the transmission of bits, in the digital era, all the physical commodities that can

be transformed into bits will be shipped in the format of bits. A discussion of the static/mobile switch will be provided later in Chapter 4.

The Digital Revolution and Unfair Trade

E-commerce is not a new concern to the WTO; it has been discussed since the second Ministerial Conference in May 1998, at which ministers adopted the Declaration on Global Electronic Commerce, calling for a work program on this issue.²¹ The issues being discussed in the WTO vary widely. In general, the focus is largely on how to form a regulatory framework to facilitate and secure online transactions. Issues of such kinds include the protection of privacy and public morals, prevention of fraud, protection and enforcement of IP rights, rules on e-signatures and e-payments, data localisation, and so on. The issue regarding the participation of developing countries in the e-commerce market-place is also highlighted in some forums.²² Indeed, all the topics are critical to shaping a modern regulatory regime. However, the very fundamental change brought by the Internet to international trade that has been missing in the rise of e-commerce is the 'atom/bit transformation'. Digital transmission is playing an increasingly important role in modern transactions.

It should be stressed that e-commerce here does not refer solely to the completion cycle of online transactions, that is, the ordering, delivery and receipt of goods or services between buyers and sellers. It also covers the preliminary stages prior to the actual ordering and delivery. For instance, the matching of potential sellers and buyers involves activities such as searching, comparing items on competing websites, checking commodity specifications, reading public comments and ratings, as well as researching government regulations. In addition, in order to complete a deal, activities such as electronic payments, delivery tracking, customer complaints, and other logistics or checks are required. Thus, e-commerce actually involves Internet communications at all of the above-mentioned transaction stages. If communication obstacles occur, the trade opportunities of e-commerce will be eroded. This is where cyberspace trade barriers come into play, and this is also why the US communication WT/GC/ W/745 specifically pointed out the disruption of e-commerce trade barriers.²³

In the context of trade in goods, traditional trade barriers, which take the form of import taxes, quotas, and regulatory or technical restrictions on trade, are mostly imposed at the customs border. These are the focus of the existing WTO rules. CTBs can occur, however, at any or all stages of an online transaction. Traditional barriers obstruct the physical flow of atoms (products), while CTBs interfere with the transmission of 'bits' (electronic signals).

Like traditional trade barriers, the existence of CTBs causes unfairness in international trade. For instance, Country A maintains an open Internet, therefore buyers in Country A can reach and buy from sellers all around the Internet, including those having their servers in Country B. However, Country B blocks its people from accessing overseas websites. Thus, the

sellers in Country A cannot be equally reached online by the potential buyers in Country B; the online supply demand match thus becomes impossible. Obviously, this kind of accessibility asymmetry causes a one-sided trading deficit that favours the sales of Internet-blocking Country B.

The same logic also applies to trade in services. It should be noted that a CTB may not only hinder services delivered across borders through electronic means (i.e., mode-1 under GATT), but it may also hinder services delivered in person. For example, in the case where the reservation website of a hotel located in Country A is blocked by the government of Country B, no Country B travellers can book rooms in that hotel online, and thus will not make use of its services in Country A. Sometimes the travel plan may be cancelled altogether. In this sense, Country A's hotel services (supplied in mode 2, consumption abroad), are therefore blocked.

Website-blocking and Other Business Interference

In its 2016 National Trade Estimate report, the US for the first time listed China's GFW as a trade barrier.²⁴ In its 2018 report, the USTR continued marking China as the most serious Internet-censoring country, where a total of 13 out of the 30 most world-trafficked websites are blocked.²⁵ In addition, according to Freedom House's report, citing the watchdog website Great Fire China, some globally popular news and social media websites, such as YouTube, Google, Facebook, Flickr, SoundCloud and WordPress, are all blocked in China.²⁶ As long as foreign businesses or platforms use such a website as their connecting window, then they cannot be accessed by as many as 1.4 billion Chinese users, and possible derived transactions can never be made.

Some may question the impact of asymmetric access, because the blocking of access to a foreign website prevents both domestic buyers and sellers from reaching their counterparts abroad. However, it should be noted that, in general, bulky commodities such as steel or aircraft are not traded through the Internet. On the other hand, for retailing commodities that e-commerce usually covers, it is always consumers who search for supply sources, and not sellers who search for millions of potential buyers. Thus, access asymmetry does benefit the Internet-blocking country, as we described above. More discussion about the Internet revolution in e-commerce will be presented in Chapter 3.

Conclusions

In this chapter, we argue that three unforeseeable structural changes to the global trading economy have served to outdate the existing WTO rules and at least partially contribute to the US–China trade war.

The first change is the increasing prevalence of increasing returns in knowledge-based industries. Given the fact in today's commercial environment that a front-runner in an emerging industry is most likely to obtain

an unbeatable advantage through increasing returns, certain authoritarian governments are motivated to provide subsidies to push their own businesses to the front line. Democratic countries by contrast are constrained by laws and are unable to implement such subsidies. Because the subsidies are directed to not-yet-existing industries, the WTO's existing subsidy rules are unable to effectively regulate these subsidies.

The second change is the presence of large, previously communist economies, in which the distinction between public and private sectors is blurred. The large-scale prevalence of SOEs shakes the free-market foundations of the WTO. Moreover, due to the lack of transparency, superficially private companies in those economies may still maintain strong government ties, which makes the definition of government subsidies meaningless. The problem worsens in the case where companies with hidden ultimate beneficiaries in tax havens place a veil over the governments' interference.

The third change is the prevalence of the Internet, which has led to the rapid growth of international e-commerce. Due to the lack of proper and relevant WTO rules in this field, countries that are capable of and that desire to exert strict control over the Internet can close out international competitors from their online market, yet help their own businesses to reach out to foreign customers through the Internet. We will provide more discussion of this in the following chapter.

The WTO website states that 'The WTO is sometimes described as a free trade institution, but that is not entirely accurate ... More accurately, it is a system of rules dedicated to open, fair and undistorted competition.'²⁷ We submit, therefore, that the WTO should never allow itself to become a system that embraces unfair trade and distorted competition, or that facilitates trade in those products whose market competition has been unfairly distorted by cyberspace measures, intensive government support for some high-tech industries, or invisible SOEs. Instead, a revitalised WTO must actively confront the specific issue of unfair competition created as a direct result of interference in products, trade and markets, especially in areas that have undergone significant changes in recent decades.

Notes

- 1 *China's Trade-Disruptive Economic Model: Communication from the United States*, WT/GC/ W/745 (16 July 2018).
- 2 *China and the World Trade Organization: Communication from China*, WT/GC/W/ 749 (23 July 2018).
- 3 China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property, and Innovation, 83 Fed. Reg. 28,710 (20 June 2018).
- 4 United States Standards for Grades of Almonds in the Shell, 78 Fed. Reg. 14,907 (6 April 2018).
- 5 USTR, *2017 Report to Congress on China's WTO Compliance*, [https://ustr.gov/sites/default/files/files/ Press/Reports/China%202017%20WTO%20Report.pdf](https://ustr.gov/sites/default/files/files/Press/Reports/China%202017%20WTO%20Report.pdf) accessed 24 October 2018.

- 6 Actions by the United States Related to the Section 301 Investigation of China's Laws, Policies, Practices, or Actions Related to Technology Transfer, Intellectual Property, and Innovation, 83 Fed. Reg. 13,099 (22 March 2018).
- 7 See for example, the report by Lindsay Dunsmuir and Howard Schneider, 'U.S., China Putting Trade War on Hold, Treasury's Mnuchin Says', *Reuters* (20 May 2018), www.reuters.com/article/us-usa-trade-mnuchin/u-s-china-putting-trade-war-on-hold-treasurys-mnuchin-says-idUSKCN1IL0JG.
- 8 Notice of Modification of Section 301 Action: China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property, and Innovation, 83 Fed. Reg. 47,974 (21 September 2018).
- 9 See The State Council of China, *Notice Issued by the State Council Regarding Made in China 2025*, Guo-Fa (2015) no. 28 (19 May 2015), www.gov.cn/zhengce/content/2015-05/19/content_9784.htm accessed 24 October 2018. The ten priority industries are: 1) New advanced information technology; 2) Automated machine tools and robotics; 3) Aerospace and aeronautical equipment; 4) Maritime equipment and high-tech shipping; 5) Modern rail transport equipment; 6) New-energy vehicles and equipment; 7) Power equipment; 8) Agricultural equipment; 9) New materials, and 10) Biopharma and advanced medical products.
- 10 Regarding the general economic analysis of the effects of strategic R&D subsidies, see Bagwell and Staiger (1994).
- 11 National Human Genome Research Institute, 'An Overview of the Human Genome Project', www.genome.gov/12011238/an-overview-of-the-human-genome-project/; Brain Initiative, 'The BRAIN Initiative Mission', www.braininitiative.org/mission/.
- 12 Federal Ministry for Economic Affairs and Energy of Germany, 'What Is Industrie 4.0?', www.plattform-i40.de/I40/Navigation/EN/Industrie40/WhatIsIndustrie40/what-is-industrie40.html.
- 13 See the European Commission. 'Future and Emerging Technologies', <https://ec.europa.eu/programmes/horizon2020/en/h2020-section/future-and-emerging-technologies>.
- 14 See for example, James McBride, 'Is "Made in China 2025" a Threat to Global Trade?' (Council on Foreign Relations, 2 August 2018), www.cfr.org/backgrounder/made-china-2025-threat-global-trade.
- 15 We think it is important, at this point, to highlight the difference between the issue we are discussing here and the issue of subsidies and new markets that was addressed in the WTO dispute regarding *Canada–FIT*. In *Canada–FIT*, the Appellate Body drew a distinction between 'government interventions that create markets that would otherwise not exist' and 'government interventions in support of certain players in markets that already exist', and indicated that in the former scenario 'it cannot be said that the government intervention distorts the market.' See *Canada – Measures Relating to the Feed-in Tariff Program ('Canada – FIT')*, WT/DS412/AB/R, adopted 24 May 2013, para. 5.188. This distinction is needed in order for there to be a proper standard by which to determine whether the renewable energy industries receive benefits under Article 1.1(b) of the SCM Agreement. See also Cosbey and Mavroidis (2014).

In our view, however, the essence of the issue in *Canada–FIT* is that the government of the Province of Ontario based its guaranteed electricity prices on the generator's use of local goods. The beneficiary of this arguable subsidy program

is thus the Canadian renewable energy goods producers (for example, solar cells). The relevant industries might not exist in Ontario then, but they have already existed in other corners of the world. Whether or not these industries have the property of increasing returns is never the issue in this case. Therefore, by its very nature, that dispute is within the realm of the existing SCM Agreement. By contrast, the ‘new industry’ issue that we are discussing here is about an industry not yet existing at a global level, yet already firmly expected to be characterised by increasing returns.

- 16 See OECD, ‘Excess Capacity in the Global Steel Industry and the Implications of New Investment Projects’ (2015) ECD Science, Technology and Industry Policy Papers, No. 18, 5, <https://doi.org/10.1787/5js65x46nxhj-en>.
- 17 See for instance Mavroidis and Janow (2017).
- 18 See ‘*Joint Statement on Trilateral Meeting of the Trade Ministers of the United States, Japan, and the European Union*’, <https://ustr.gov/about-us/policy-offices/press-office/press-releases/2018/may/joint-statement-trilateral-meeting>.
- 19 The full text of Chapter 17 of the CPTPP is available at www.mfat.govt.nz/assets/Trans-Pacific-Partnership/Text/17.-State-Owned-Enterprises-and-Designated-Monopolies-Chapter.pdf.
- 20 Article 12.8.5 of the United States – Singapore Free Trade Agreement: ‘5. effective influence exists where the government and its government enterprises, alone or in combination: (a) own more than 50% of the voting rights of an entity; or have the ability to exercise substantial influence over the composition of the board of directors or any other managing body of an entity, to determine the outcome of decisions on the strategic, financial, or operating policies or plans of an entity, or otherwise to exercise substantial influence over the management or operation of an entity. Where the government and its government enterprises, alone or in combination, own 50% or less, but more than 20%, of the voting securities of the entity and own the largest block of voting rights of such entity, there is a rebuttable presumption that effective influence exists. Annex 12A provides an illustration of how the analysis of effective influence should proceed.’
- 21 *The Geneva Ministerial Declaration on global electronic commerce*, WT/MIN (98)/DEC/2 (20 May 1998).
- 22 See WTO Secretariat, ‘Electronic Commerce’, www.wto.org/english/thewto_e/minist_e/mc11_e/briefing_notes_e/bfecom_e.htm.
- 23 See also *Removing Cyberspace Trade Barriers: Towards a Digital Trade Environment with Reciprocally Equal Access: Non-Paper from the Separate Customs Territory of Taiwan, Penghu, Kinmen and Matsu*, JOB/GC/170 (16 February 2018).
- 24 USTR, *2016 National Trade Estimate Report*, 91, <https://ustr.gov/sites/default/files/2016-NTE-Report-FINAL.pdf> accessed 24 October 2018.
- 25 USTR, *2018 National Trade Estimate Report on Foreign Trade Barriers*, 104, <https://ustr.gov/sites/default/files/files/Press/Reports/2018%20National%20Trade%20Estimate%20Report.pdf> accessed 24 October 2018.
- 26 Freedom House, ‘Freedom on the Net 2017: China Country Profile’, <https://freedomhouse.org/report/freedom-net/2017/china> accessed 24 October 2018.
- 27 World Trade Organization, ‘Principles of the Trading System’, www.wto.org/english/thewto_e/whatis_e/tif_e/fact2_e.htm#:~:text=The%20WTO%20is%20sometimes%20described,open%2C%20fair%20and%20undistorted%20competition.

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3 E-commerce Mercantilism

The Issue of Internet Blocking

Background – China’s Internet Blocking and the Great Fire Wall

The Internet has created a whole new marketplace, but it has brought with it a whole new set of mercantile challenges for international trade as well. The rapidly growing prevalence of the Internet and the anticipated boom in e-commerce may well induce and fertilise a mercantilist mindset in governments. A government may thus be incentivised to shield its e-commerce businesses from foreign competition, while allowing its own industries to prey on the more liberal e-commerce market access of other countries at the same time. Unlike traditional trade barriers, which are generally applied directly to the object of the transaction (that is, goods or services), the protectionist measures used in the digital era of today often come in the form of government interventions in the cross-border data flow by, for example, the imposition of Internet censorship or cross-border data-flow restrictions. China is the most prominent exponent of this so-called ‘digital mercantilism’. As observed by *The Economist* magazine (6 August 2016):

China is a sort of technological Galapagos Islands; a distinct and isolated environment in which local firms flourish. Chinese firms are protected from external competition by government regulation and the Great Firewall. And that protection means that they need not innovate but can thrive by copying business models developed in the West.

For some time now, China has been the best subject for case study in the relevant research (Wu, 2006; Aaronson, 2019).

Rather than attempting here to give an overview of the full range of China’s digital mercantilist policies, this chapter will focus on one greatly influential constituent that deters e-commerce access – Internet blocks and connection speeds. Conceptually and for the purpose of the discussion in this chapter, a blocked access refers to the case where the connection waiting time is infinitely long.

There is no doubt that a long webpage loading time drives away visitors. For example, according to a Google survey, as many as 53 per cent of visits are

abandoned if a mobile site takes more than 3 seconds (hereafter abbreviated as 3s) to load (Google, 2016). Other data show that when the page-loading time goes up from 1s to 10s, the probability of a mobile site visitor ‘bouncing’ (entering the site and then leaving) increases by 123 per cent (An, 2017).

Our empirical data show that, as a general phenomenon, Chinese consumers need to wait for a much longer time to connect to foreign websites (if the connections are not simply blocked) than foreign consumers to connect to Chinese ones from abroad. These two-way asymmetrical Internet speeds are caused by China’s censorship of cross-border data flows, also known as ‘the Great Firewall’ (GFW). Very little of the literature on the subject discusses China’s low Internet speeds in the context of trade. In our view, this is additional firm evidence of Chinese digital mercantilist practices.

Following this introduction, we proceed to present the current and projected market data that demonstrate the significant advantages already gained by China in the area of e-commerce. The next section contains a short description of how China’s digital mercantilist policy works. This is followed by empirical data on the asymmetrical Internet connection speeds within China, showing that even when *not* being blocked, foreign e-commerce platforms face very unfair market access (compared with the Chinese ones), and hence they can hardly compete in the Chinese market. Finally, we explain why the existing WTO rules cannot effectively address these unfair practices. We conclude by adding our observations of the ongoing WTO plurilateral e-commerce negotiations and by suggesting that a provision guaranteeing the free flow of data should be included in the outcome agreement.

Before going further, it might be useful to clarify in advance some of the terminology that we use in the chapter. First, we use the terms ‘e-commerce’ and ‘digital trade’ interchangeably. Both terms refer to their broadest meaning – ‘any transaction occurring online’ – and their scope includes transactions in goods, services and digital content. The same logic applies to the terms ‘information’ and ‘data’. Although some authors tend to consider that each of them refers to different concepts (Aaronson 2019, pp. 2–3), we do not make such a distinction here. Also, in this chapter, ‘data flow’ and ‘transmission of information’ both refer to the ‘transmission of electronic signals via the Internet’. This includes any online communication and transfer of transactional or personal data, although the WTO is still debating what should and what should not be included.¹

China’s E-commerce Advantages

We start with some market data to help picture the global landscape of e-commerce today. The e-commerce retail sales data shown here are collected and compiled by Statista, a Hamburg-based open-data source containing country-wide e-commerce sales amounts and predicted growth rates. The gross domestic product (GDP) and the growth-rate data are from the open database provided by the International Monetary Fund. The data on household

final consumption expenditure (HFCE) are from the World Bank. Given the fact that various data become available at different times, the e-commerce and GDP numbers are based on data collected in 2018, whereas the HFCE data are from 2017.

The Disproportionate Prosperity of China's E-commerce

As shown in Table 3.1, in the e-commerce business-to-consumer (B2C) retailing sector, China is by far the greatest beneficiary of e-commerce revenue and prosperity. In 2018, China accounted for 40.57 per cent of the total revenue of the top ten e-commerce operating countries, a much bigger share than the No. 2 country, the US, which accounted for 32.18 per cent. At the same time, China's e-commerce businesses are expected to grow at a much higher rate (70.7 per cent) than any of its foreign counterparts, which means

Table 3.1 Country-wise e-commerce, GDP and HFCE data

	<i>E-commerce retail revenue (US\$billion)^a</i>			<i>GDP (US\$billion)^b</i>		<i>HFCE (US\$billion)^c</i>
	<i>2018 (% of 10)</i>	<i>2023 (% of 10)</i>	<i>Growth (%)</i>	<i>2018 (% of 10)</i>	<i>Growth (%)</i>	<i>2017 (% of 10)</i>
China	636.1 (40.57)	1,086.1 (45.38)	70.7	13,407.4 (24.44)	6.6	4,704.44 (16.17)
US	504.6 (32.18)	735.4 (30.73)	45.7	20,494.1 (37.36)	2.9	13,321.41 (45.80)
Japan	81.7 (5.21)	103.6 (4.38)	26.8	4,971.9 (9.06)	0.8	2,696.81 (9.27)
Germany	70.3 (4.48)	95.3 (3.98)	35.6	4,000.4 (7.29)	1.5	1,951.97 (6.71)
UK	86.5 (5.52)	113.6 (4.75)	31.3	2,828.6 (5.16)	1.4	1,733.55 (5.96)
France	49.4 (3.15)	71.9 (3.00)	45.6	2,775.3 (5.06)	1.5	1,396.80 (4.80)
Canada	39.9 (2.54)	55.4 (2.31)	38.8	1,711.4 (3.12)	1.8	958.55 (3.30)
Russia	17.2 (1.10)	24.8 (1.04)	44.2	1,630.7 (2.97)	2.3	831.57 (2.86)
Korea	63.7 (4.06)	80.2 (3.35)	25.9	1,619.4 (2.95)	2.7	736.21 (2.53)
Australia	18.6 (1.19)	26.9 (1.12)	44.6	1,418.3 (2.59)	2.8	755.11 (2.60)

Notes: ^a Statista, as cited in Orendorff, 2019 at www.shopify.com/enterprise/global-ecommerce-statistics#2; ^b The International Monetary Fund DataMapper, *World Economic Outlook, Gross Domestic Product* at: www.imf.org/external/datamapper/datasets/WEO/1; ^c World Bank, *Households and NPISHs Final Consumption Expenditure*, at <https://data.worldbank.org/indicator/NE.CON.PRVT.CD>.

it should account for 45.38 per cent of the total revenue of the top ten most active e-commerce countries by 2023.

China's advantage in e-commerce appears to be disproportionate to its share of the global economy. As Table 3.1 also shows, China's GDP accounts for just 24.44 per cent of the above-mentioned ten-country group, a much smaller share of the group total than that of its e-commerce. If we compare the e-commerce figures to the HFCE numbers (that is, the consumer market sizes), the disproportionality is even more obvious, with China's HFCE share accounting for only 16.17 per cent of the group total. This may be partly explained by the fact that Chinese consumers shop online more often than they do offline, but the huge discrepancies mentioned above appear to go well beyond what might normally be expected in response to simple differences in consumer behaviour.

Indeed, GDP and HFCE include both the goods sector and the services sector. In general, developed economies, such as the US and Japan, have relatively larger services sectors compared with those of the developing economies, such as China. Some may therefore doubt whether it is fair to compare GDP and HFCE with e-commerce retailing sales revenues. It should be noted, however, that the sales revenues from e-commerce retailing include not only the revenues stemming from the sale of goods, but also the revenues from the provision of retailing and other associated services (for example, logistics and payment). Therefore, at the B2C level at least, the e-commerce retailing sales revenues should serve as a meaningful reference for estimating the overall e-commerce activities within an economy and be comparable with GDP and HFCE.

For business-to-business (B2B) e-commerce, such as system engineering services in the green energy sectors, and international finance and banking, Internet blocking per se may not constitute a trade barrier as effectively as that for B2C e-commerce. The reason for this is that B2B foreign businesses are sometimes able to set up a branch in the target country, and provide the services in mode-3 (services through foreign affiliates). We will include a brief discussion of the case of Amazon later in this chapter. This mode-3 choice, however, is not always possible. For instance, services in the 'social media' and 'search engine' categories have never been allowed to compete freely and fairly in China. Further discussion of this follows in the next section.

The question one might ask is as follows: does it not seem intriguing that China has managed to secure such a commanding position in e-commerce business when its GDP accounts for just over 24 per cent of the total of the same ten countries? Is it because Chinese businesses are so much more competitive in the digital world?

Unfair E-commerce – Collateral Damage of Cross-border Data-flow Blocking

We argue that China's leading position in B2C e-commerce largely stems from its widespread and strict regulation of the cross-border online data flow, or the

GFW as it has come to be known. Such a regulation was originally intended for political purposes, that is, preventing the Chinese people not just from producing and disseminating, but also from searching and reviewing ‘harmful information’, on subjects such as the June 4 Tiananmen Square uprising, the Tibet riots, the Uyghur ‘re-education’ camp and so on, to oppress voices of democracy and to prevent any incidents like the ‘Arab Spring’ (*The Economist*, 6 April 2013). One economic consequence of Internet blocking, unavoidably, has been to make it impossible, or at least very difficult, for Chinese people to access foreign websites.

To give the reader some idea of how broad is the definition of ‘harmful information’, we cite the following regulation. Article 5 of Order No. 33 of the Ministry of Public Security of China, entitled ‘Measures for Security Protection Administration of the International Networking of Computer Information Networks’, provides that:

No unit or individual shall use international networking to produce, duplicate, search and disseminate the following information: 1) information that instigates the resistance to and disruption of the implementation of the Constitution, laws and administrative regulations; 2) information that instigates the subversion of the state political power and overthrow of the socialist system; 3) information that instigates the splitting up of the country and sabotage of national unity; 4) information that instigates hatred and discrimination among nationalities and sabotages solidarity among nationalities; 5) information that fabricates or distorts facts, spreads rumors and disrupts social order; 6) information that propagates feudalistic superstitions, obscenity, pornography, gambling, violence, murder and terror and instigates crimes; 7) information that openly insults others or fabricates facts to slander others; 8) information that damages the reputation of state organs; and 9) other information that violates the Constitution, laws and administrative regulations.²

Because the prohibited target is ‘information’, it goes without saying that anything possibly printing or carrying such information is not allowed to be distributed. Thus, any T-shirt on which a picture of the ‘Tank Man’ standing in the Tiananmen Square is printed cannot be sold in China, not to mention books, videos and news reports. In addition, item 8) is ridiculously vague and broad, and hence the Internet content to be screened is also extremely broad. Furthermore, logically it makes sense to prohibit duplication and dissemination, but it does not make sense to block ‘search’, for without searching how could anyone possibly know whether the website contains ‘harmful’ information? In short, in order to guarantee that all such prohibitions are implemented, the Internet checks have to be outrageously severe.

At the same time, however, foreigners are not prevented from accessing Chinese websites. This naturally results in market-access asymmetry: it makes it less likely, if not impossible, for Chinese consumers to buy from the rest of

the world, despite the fact that it is still convenient for the rest of the world to buy from Chinese platforms. The playing field is thus skewed to the advantage of Chinese e-commerce operators, resulting in a whole new kind of mercantilism in the digital era.

How Does Digital Mercantilism Work?

Conceptually, the traditional international trade in goods can be divided into a series of different stages, as shown in Figure 3.1. The first stage is that of a potential buyer acknowledging the existence of a commodity; in the second step, the potential buyer compares the target commodity with other competitive ones; the third step is the placement of orders; the fourth is the shipment of the commodity; the fifth step is the customs check and taxation; the sixth step is domestic (or local) logistics or marketing; and finally, in the seventh step, the buyer gets the commodity.

In the context of traditional trade in goods, barriers to trade are most often applied in Stages 4 to 6. These usually take the form of high customs taxes, non-tariff barriers, or other harassments. In digital trade, however, the barriers are different. Most of these barriers are implemented in Stages 1–3, and they usually take the form of regulatory measures related to the cross-border data flow.

For example, the Chinese government intervenes in Internet traffic, either directly or with the assistance of the Internet service providers, to filter online requests and to block access to foreign e-commerce platforms. Although the same censoring requirements may also be imposed on domestic industries, such interventions mostly result in higher marketing, transaction and administration costs for the foreign e-commerce operators. This is because foreign websites are more likely to sell products that contain ‘harmful’ information

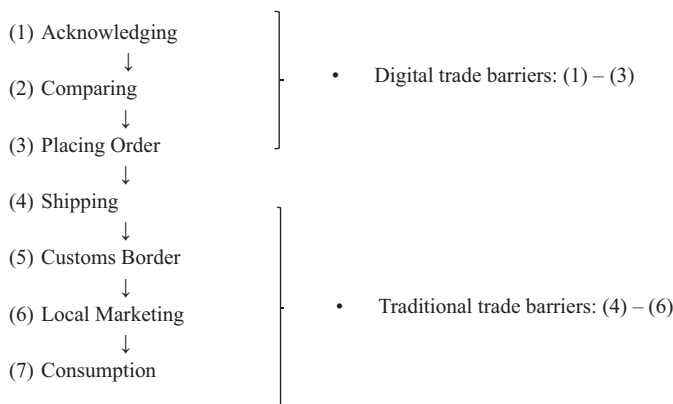


Figure 3.1 Conceptual trade stages

than obedient domestic ones; hence, the access to foreign websites is subject to more censorship and securitisation. In extreme cases, the foreign operators can be completely blocked out from the Chinese consumers.

Chinese e-commerce businesses, however, face no similar hurdles in most of the foreign markets. These businesses can thus keep benefiting from the expansion in global e-commerce, and with their protected domestic markets, they will always have an edge on the forthcoming competition.

How Broad Is China's Internet Blocking?

We are not the first to put forward this argument. China's strangulating regulation of the Internet is a well-known fact. In 2020, it was ranked as the country with the least Internet freedom by the US-based nongovernmental organisation, Freedom House, for the sixth consecutive year.³ According to Freedom House's report in 2018, citing the watchdog website GreatFire.org, some world-popular news and social media websites, such as Google, Facebook, WhatsApp, YouTube, Flickr, Tumblr, Dropbox, Instagram, SoundCloud, WordPress and Pinterest are all blocked by the Chinese government.⁴ As a matter of fact, many small and medium-sized enterprises (SMEs) use these platforms (only sometimes) as their connecting window for international sales.⁵ In effect, therefore, the blocking of these platforms essentially blocks access to such businesses by all of China's 1.4 billion potential consumers. According to the USTR 2020 report to Congress, China has blocked more than 10,000 websites worldwide,⁶ and hence they are not accessible to people in China.

Commercial advertising is another important element in the whole process. Blocking Facebook would definitely prevent Chinese consumers from accessing the advertisements published there. Moreover, because of China's blockage of foreign platforms such as Facebook, foreign businesses seeking exposure in the Chinese market have no choice but to register themselves on the Chinese platforms that are similar to Facebook, such as Weibo. However, Chinese businesses do not necessarily need to choose to advertise on Facebook to market their goods abroad, because international consumers are free to access the Chinese platforms. China's blocking thus grants additional competitive advantages (such as advertisements) to its local platform industries vis-à-vis foreign ones.

China's trading partners share the same concerns. In 2016, the US first listed China's GFW as a trade barrier (USTR, 2016). In its released trade estimate report, the US places emphasis on the mercantilist dimension of the Chinese Internet regulatory regime, accusing Chinese companies of dominating the domestic market 'due in large part to restrictions imposed on foreign companies by the Chinese government' (USTR, 2019).

E-commerce Implications of Internet Blocking

To verify the argument of digital mercantilism, we have implemented an extensive on-site experiment to gather data and reflect the general experiences of

consumers connecting from China to e-commerce platform websites in other countries, and vice versa – that is, the experiences of consumers connecting from other countries to China's e-commerce platforms. Instead of checking only whether access to specific websites was being blocked, we measured the average waiting time before access was achieved (if achieved at all). We also checked the two-way traffic between the two countries to make sure that any delays experienced in trying to connect were not caused by any general infrastructure deficiency. In sum, our findings were that Chinese consumers usually need to wait for a much longer time period to connect to foreign websites than foreign consumers do to access Chinese sites. In fact, for the latter, there is a much shorter waiting time.

Tables 3.2 and 3.3 demonstrate the outcomes of our on-site surveys. In both tables, the waiting time is calculated using a standard method. We asked our research investigators to use a standard personal computer and to connect with the local Wi-Fi. The calculation of the 'per click waiting time' starts from the pressing of the 'Enter' key and monitors the time up to when the full screen is shown. Multiple trials were made, and a simple average value was used if discrepancies existed. We also specified the capacity range of the personal computer facilities used by each agent, to make sure that the waiting time would not be affected by computer capacity differences. The testing

Table 3.2 Per-click waiting time: major retailing platforms

<i>Time Per Click (seconds)</i> <i>Destination Platform</i>		<i>Access from</i>			
		<i>Shanghai</i>	<i>Shenzhen</i>	<i>Beijing</i>	<i>New York</i>
Australia	Big W	113.72	111.18	–	–
Brazil	Americanas	>120.00	>120.00	–	–
Canada	Walmart	14.11	30.35	–	–
France	Cdiscount	18.40	17.53	61.87	0.96
Germany	Otto	15.22	42.08	21.39	1.52
India	Flipkart	37.45	11.45	>120.00	–
Indonesia	Tokopedia	20.80	61.42	>120.00	–
Japan	Rakuten	>120.00	>120.00	31.19	2.57
Korea	G market	5.33	6.47	0.24	0.68
Malaysia	Mudah	4.54	51.65	9.07	4.12
Mexico	Mercado Libre	109.55	>120.00	–	–
Singapore	Qoo10	5.22	7.66	6.65	1.43
Switzerland	Richardo	54.52	43.94	–	–
Taiwan	PChome	>120.00	>120.00	>120.00	0.88
Turkey	Akakce	63.97	51.75	–	–
UK	ASOS	7.61	11.92	14.90	0.75
US	eBay	>120.00	>120.00	11.74	0.62
US	Amazon	15.06	–	14.57	5.34

Notes: Waiting time calculated from pressing the 'Enter' button to when the full screen shows. '–' means no test is executed. '>120.00' means waiting time is longer than 120s. Selling platform chosen is that with the highest traffic in each country

Table 3.3 Per-click waiting time: major retailing platforms, two-way tests

<i>Time Per Click (seconds)</i>		<i>From others to Taobao</i>	<i>From China to others (simple average of Table 2)</i>
Australia	Sydney	7.51	112.45
Brazil	São Paulo	10.14	>120.00
Canada	Ottawa	2.17	22.18
France	Paris	1.54	32.60
Germany	Berlin	5.26	26.23
India	New Delhi	13.21	24.45
Indonesia	Jakarta	8.35	41.11
Japan	Tokyo	1.97	>120.00
Korea	Busan	2.56	4.01
Malaysia	Kuala Lumpur	4.22	21.73
Mexico	Mexico City	8.55	114.78
Singapore	Singapore	2.15	6.51
Switzerland	Geneva	3.34	49.23
Taiwan	Taipei	4.95	>120.00
Turkey	Ankara	3.87	57.86
UK	London	7.32	11.48
US	New York	0.87	14.57

took place from 15 to 21 April 2019, during the local time period of 8:00 to 10:00 p.m. each day. A list of the uniform resource locators (URLs) used in the survey is annexed at the end of this chapter.

Per-click Waiting Time Statistics – Cross Country Tests

In Table 3.2, we ran the on-site test from three major cities in China (Shanghai, Shenzhen and Beijing) to five countries in Europe (France, Germany, Switzerland, Turkey and the UK), four countries in Pan-America (Brazil, Canada, Mexico and the US), and eight countries in the Asia-Pacific region (Australia, Japan, Korea, India, Indonesia, Malaysia, Singapore and Taiwan). We also conducted remote tests on the websites GreatFire.org, WebSitePulse.com and WebPageTest.org, where a user can measure the connection speed to a specific URL from China or other locations. The remote test and on-site test outcomes are consistent with each other; therefore, in what follows we only present our on-site test outcomes.

Table 3.2 presents a comparison of the per click waiting time for connecting from a Chinese city to some of the largest e-commerce retailing platforms in other countries. It is obvious that it usually takes an intolerably long time for a Chinese consumer to access those foreign websites. For instance, it takes 62s for a click to connect from Beijing to Cdiscount of France. If a transaction takes ten clicks to complete (from the home page to check out and payment), 62s per click adds up to a total of more than ten minutes to complete a transaction. This exceeds the usual tolerance levels of almost all users.

From Table 3.2, we can summarise the following conclusions. Despite the fact that the exact numbers of waiting-time seconds vary across Chinese cities and across destination websites, the prolonged waiting-time pattern is the same: in all three cities, it takes a prohibitively long time for Chinese consumers to buy products from e-commerce websites worldwide. The only exception is access to G Market of Korea from Beijing, which takes just 0.24s per click, for reasons unknown. Moreover, if we compare such waiting times with those recorded in New York City, as shown in Column 4 of Table 3.2, it is obvious that the US consumers can buy goods from foreign e-commerce operators much more conveniently than Chinese consumers, even though the US is geographically farther away from, say, Malaysia, than is China.

Per-click Waiting-time Statistics – Reverse Tests

There may be some suspicion that it is the low fibre capacity that connects China and France which results in the long connection waiting time from China to Cdiscount in France, compared with the same access trial from New York.

To verify such a possible explanation, we present, in Table 3.3, a ‘reverse test’. Here, we calculate the waiting time per click from other countries to China’s Taobao website, the largest retailing website owned by Alibaba. Internet communications are like two-way highways, so it is not possible systematically for traffic travelling one way to take longer to connect than traffic travelling in the other direction. As we can see from Table 3.3, the waiting times are repeatedly longer for connections from China to other countries, than the other way around. This evidence shows that the long waiting time in China is because of China’s GFW.

Mode-3 Alternative for a Few Large Businesses

Hardly any company from another country can avoid China’s GFW harassments, unless it has the capacity to provide a ‘special service’ solely for Chinese. Amazon provides a Chinese-access webpage, which does not have to be accessed through the GFW, mainly because it has been censored already.

There is no way one can see the comprehensive and detailed differences between the Chinese Amazon webpage and the worldwide Amazon webpage. However, our test shows that at least books and t-shirts with the keyword ‘Tiananmen Square’ cannot be found on the Chinese Amazon website, although there are hundreds of items of this type on Amazon.co.uk. Thus, the commodity spectrum limited by the Chinese government still exhibits access asymmetry. Furthermore, most businesses and SMEs do not have the capacity to access Chinese consumers in this way.

Per-click Waiting-time Statistics – SME Tests

Tables 3.2 and 3.3 summarise the results of accessing large e-commerce platforms. Table 3.4 shows the on-site experiment results of accessing typical SMEs in various countries. We refer to Google and the country-specific tour websites, picking out the most typical kind of food of the country and choosing the most famous or most recommended food store's website. Our understanding is that it is not usual for cross-border food to be ordered from the website, but food seems to be a featured product of every country, and hence a comparison is easier to make. We should also emphasise that our focus is entirely on the task of testing the comparative Internet accessibility and waiting times, and not on ordering a meal. Obviously, food can hardly contain 'harmful information'; if it is another general product that is picked out to perform the comparison, it is believed that the result will reveal more access asymmetry.

Indeed, it is intrinsically difficult to find SMEs in different countries selling the same product in similar business models (otherwise they would probably not be SMEs in the first place). We understand that a strict comparison of one SME with another at the global level could be difficult, if not impossible. We did not intend to make an inter-SME comparison. Instead, we demonstrate that access to the Chinese online market is foreclosed by the intended slow-down in the connection times to foreign websites.

Table 3.4 Per-click waiting time: typical SME websites

<i>Time Per Click (seconds)</i>		<i>Access from</i>	
<i>Destination Website</i>		<i>Shanghai</i>	<i>Shenzhen</i>
Australia	Vegemite	32.97	>120.00
Brazil	B.Lem Bakery	35.00	45.21
Canada	Dare Foods	>120.00	>120.00
France	La Mère Poulard	25.83	29.63
Germany	Café Schäfer Triberg	>120.00	6.12
India	Haldiram	54.41	50.02
Indonesia	Lapislapis Kue Lapis	9.70	21.30
Japan	Nagomi-Yoneya	117.59	39.87
Korea	Ottogi Ramen	14.95	16.3
Malaysia	Old Town White Coffee	116.10	30.41
Mexico	La Churreria El Moro	111.38	68.91
Singapore	Bee Cheng Hiang	>120.00	14.67
Switzerland	Tristan Chocolatier	70.46	36.01
Taiwan	Yu Jen Jai	26.20	43.78
Turkey	Haci Bekir	7.21	12.73
UK	The Astronomer	>120.00	25.43
US	Federal Donuts	70.51	>120.00

Table 3.5 Per-click waiting time: SME websites, two-way test

<i>Time Per Click (seconds)</i>		<i>From others to Shaoxing Liquor Company</i>
Australia	Sydney	9.13
Brazil	São Paulo	8.17
Canada	Ottawa	4.05
France	Paris	1.73
Germany	Berlin	6.22
India	New Delhi	23.03
Indonesia	Jakarta	7.87
Japan	Tokyo	2.81
Korea	Busan	2.71
Malaysia	Kuala Lumpur	2.54
Mexico	Mexico City	7.12
Singapore	Singapore	3.09
Switzerland	Geneva	2.54
Taiwan	Taipei	6.52
Turkey	Ankara	11.03
UK	London	4.15
US	New York	1.29

From Table 3.4, we can see that, with only a few exceptions, accessing the websites of foreign SMEs from China is even more difficult than accessing the big e-commerce platforms, and the country-wide waiting-time pattern is consistent with that shown in Tables 3.2 and 3.3. The general message is clear: If it is difficult to access e-commerce platforms, the same applies in the case of SMEs.

In Table 3.5, we show the waiting time to access SMEs' websites experienced in the 'reverse test', which refers to the time needed from various countries to access the Shaoxing Liquor Company, a famous SME wine company in China. Again, the length of time needed to access China's Shaoxing Liquor Company is much shorter than the time needed from China to access the sample of SMEs in the rest of the world. So, the same access asymmetry exists with SMEs as well.

It should be noted here that we are not contending that, because of the asymmetric internet speeds, China's Shaoxing Liquor Company makes, or is able to make, more online sales than, for example, Federal Donuts, either in the Chinese or the international markets. Instead, we try to argue that the asymmetry in Internet speeds constitutes a barrier for Chinese consumers, when it comes to accessing the foreign online market in general, regardless of origin, business size (Amazon or Federal Donuts), or product type (whether a wide range of groceries or single food items). In terms of the WTO rules, this is a 'market access' rather than a 'national treatment' issue. Other countries do not maintain the same online market access barrier against Chinese companies and their products. Therefore, what we have here is *prima facie* evidence of the system being skewed solely to the interests of China.

In sum, because of the GFW, the Chinese consumers face a serious barrier when attempting to access foreign e-commerce platforms. It may not be China's direct intention to prevent its citizens from buying from foreign suppliers, but it results from access asymmetry, which can be described as e-commerce mercantilism. The simple fact is that it is much more difficult for Chinese potential buyers to access foreign suppliers than it is for foreign buyers to access Chinese suppliers. This gives China an unnatural and unfair advantage in the sale of its goods to the rest of the world. This probably explains why China has a disproportionate percentage of e-commerce sales compared with other countries (Table 3.1), despite the fact that China's GDP or HFCE share is comparatively much smaller.

Can the WTO Deal With Digital Mercantilism?

Our next question is: if digital mercantilism causes market-access unfairness in global e-commerce, can the WTO rules, and the GATS in particular, address this problem?

Weak Commitments of the GATS

Given that the GATS makes no distinction between services supplied online and those supplied offline, its rules do indeed apply to government measures related to e-commerce. The so-called 'technical neutrality' principle has been confirmed by WTO Members as well as in a number of WTO dispute cases.⁷ For example, the WTO dispute settlement Panel on US Gambling rules concludes that the US's commitment to liberalise 'Other Recreational Services' covers online gambling services and, thus, the US's ban on online gambling services violates its obligation under the GATS. In the words of the Panel:

[W]hen a Member inscribes the word 'None' in the market access column of its schedule for mode 1, it commits itself not to maintain measures which prohibit the use of one, several or all means of delivery under mode 1 in a committed sector or sub-sector. This is especially so in sectors and sub-sectors where cross-border supply is effected essentially if not exclusively through the Internet.⁸

In *China – Publications and Audiovisual Products*, the Panel also found that the scope of China's commitment in its GATS Schedule on 'sound recording distribution services' extends to sound recordings distributed in non-physical form through technologies such as the Internet.⁹

Beyond individual disputes, however, the structure of the GATS limits its capability to provide a systemic solution to the digital mercantilism problem. In the GATS, the provisions related to substantive obligations, such as Articles XVI (Market Access), XVII (National Treatment) and VI (Domestic

Regulation), apply only to those service sectors and models under which WTO Members make specific commitments. If, for instance, a Member has not inscribed commitments for payments and money transaction services in its schedule, then the above GATS obligations do not apply. That Member is then free to impose restrictions on electronic payment services without having to worry about the possibility of later disputes.

Most of the existing GATS commitments were the outcome of the Uruguay Round negotiations during 1986–93. For countries that joined the WTO after 1995, the commitments are set through individual accession negotiations. The WTO has failed to achieve another round of large-scale services trade negotiation. In this regard, the level of liberalisation under the GATS compared with today's e-commerce coverage is very limited. In the case of China, its e-commerce commitments are few.

Besides, the classification of services adds another restraint to the GATS' applicability to e-commerce (Ahmed, 2019). In practice, almost all WTO Members' Schedules of Commitments under the GATS are based on classifications contained in the provisional version of the United Nations Statistics Division (UNSD)'s Central Product Classification (CPC) system, which was published in 1991.¹⁰ Although the UNSD keeps updating the CPC,¹¹ WTO Members have not modified their Schedules accordingly. Therefore, even some common Internet services today, such as search engine services, require special efforts if they are to even barely fit into one Member's Schedule, let alone the newly emerging business models (Wu, 2006).

The asymmetry in Internet speed in China raises another question here: can the six market access barriers defined in Article XVI of the GATS cover all digital protectionist measures?

Could the GATS and GATT Articles Apply?

Indeed, the overall blockage of a foreign website may be regarded as being a 'zero quota' and thus constitute the 'limitation on the number of service suppliers' under Article XVI:2 (a) of the GATS. It is questionable, however, whether a measure causing very slow connections to foreign websites falls within the meaning of any sub-paragraph of Article XVI, such as the 'limitations on ... the total quantity of service output expressed in terms of designated numerical units in the form of quotas or the requirement of an economic needs test', as provided for in Article XVI:2 (c) of the GATS.

Beyond the GATS, there is little discussion concerning digital mercantilism under the GATT, which applies only to trade in goods. In the case where an internet regulatory measure makes it difficult for foreign e-commerce operators to reach the domestic consumers and thus to supply the goods, Article XI of the GATT may come into play as a provision that prohibits any trade restrictions of either a *de jure* or *de facto* nature.

This argument may have a chance of becoming more viable under the broad interpretation of Article XI of the GATT. In practice, however, a foreseeable challenge for the complainants concerns how to answer some difficult hypothetical questions. For instance, would trade still be possible (via physical channels) even if the Internet measure were not in place? What would the trade volume be if there were no such measures? Logically, answering such ‘what if’ questions is needed in order to establish a comparison between the trade volume differences with and without the Internet blocking. This comparison is necessary to establish how an Internet blocking measure constitutes ‘prohibitions or restrictions’ on the importation of products under Article XI of the GATT 1994, thereby a possible dispute settlement case in WTO. The evidentiary threshold associated with the above ‘what if’ questions is likely to be so high as to be almost impossible for the WTO Member to overcome.

Could Other Countries Counteract?

Faced with the e-commerce mercantilism in China, and given that the WTO rules do not explicitly prohibit its practices, one reasonable question to be asked is whether democratic countries are able to retaliate. Note that the most serious CTBs, such as Internet blocking and control, are only possible under an authoritarian regime. Government interventions by restricting information flow in essence are at odds with liberal democratic values. Technically speaking, to effectively regulate in the borderless cyberspace, a government needs a big policy tool-kit and extreme, even unlimited, power.

Indeed, Internet control requires the extensive cooperation of all kinds of service providers. For instance, to be able to block some websites successfully, the servers must be controlled by operators that are willing to follow government orders. To screen out sensitive word-searches, the search engines must comply with the required search algorithm design, and, to avoid any risk of the public challenging the government, social media operators must monitor sites regularly and shut down the network as instructed by the government whenever necessary. This scale of order-and-obedience is possible only under an authoritarian regime.

In practice, it may be that the mandatory order may not work as efficiently as the incentive plan. To encourage Internet operators to cooperate with the regulator, an authoritarian government often grants network-related monopoly status to those that comply. Such grants are reviewed and renewed periodically to ensure the operators’ continuous compliance. In democratic countries, such businesses are usually not franchised by the government. Furthermore, even if they are franchised, the franchise-review system in democratic countries can hardly be manipulated by officials.

It should be noted that we are talking here about government actions which are enforced with the cooperation of Internet operators. They should be differentiated from the case where an Internet operator, say, a search

engine, acts on its own initiative to remove its competitors' content from the search result. In the latter case, competition law instead of international trade law should apply.

Other Issues Related to China's Internet Control

The combination of an authoritarian regime and Internet control is more likely today to lead to an outcome which is worse than the simple CTBs and asymmetric market access: data abuse. One feature of the digital revolution that we mentioned in Chapter 2 is the static/mobile switch à la Negroponte (1994). Unlike in the 1990s when devices were mostly static and data were mainly transmitted through mobile antennae, today versatile data can be collected by mobile devices anywhere, and transmitted efficiently by static fibres. As a result of this static/mobile transformation, most data are now generated and collected by local branches or local companies. If governments can access the data from international terminals (though search engines or online shopping, for example),¹² and if such data are further profiled and applied to future business screening, then the unfairness will be further badly deepened.

For instance, it has been reported that China has collected various personal data and established 'scores' for all individuals. Leaving aside the privacy issue that we will discuss in Chapter 4, this long-unaddressed and unfair situation places some Internet-censoring countries in an advantageous position, right at the time of a boom in e-commerce. Because such electronically collected data cover buyers of all nationalities, these data can be used by the Chinese government to do further profiling for accurate marketing. When the personal data of foreign visitors in China (or foreigners using a Chinese search engine, or shopping on the Chinese Alibaba website) are collected, profiled and used for business purposes, no democratic government may know about it or be able to do anything about it. This is another dimension of efficiency in China's unjustified favour.

The regulatory deficiency described above partly explains why no issues of a CTB or data intrusion nature have ever been brought to the WTO's dispute settlement mechanism. As a consequence, this allows these countries' e-commerce operators to profit hugely by preying on the freedom of the Internet, while being shielded from international competition in their domestic markets.

Future WTO reforms thus need to properly address the fact that a revolutionary new way of conducting trade internationally is indeed affecting global trading patterns and the dynamics of world economic power. With regard to the CTB issue, WTO Members should confirm their recognition of the undesirable effects of CTB measures and commit to ensuring and securing the free cross-border transmission of information. New rules should also contain a description of the legitimate policy objectives that Members may pursue by applying CTB measures, while at the same time specifying the exceptionality

of the circumstances under which CTB measures are allowed. Members should also assess the impact of data abuse on international trade.

Conclusion – Unpromising WTO E-commerce Reform

In our view, in order to prevent digital mercantilism, a universal norm of prohibiting any restrictive measures on cross-border online data flow should be established at the international level. Certain giant free trade agreements in recent years, such as the CPTPP and the United States-Mexico-Canada Agreement (USMCA), have already incorporated rules to this effect.

Article 14.11 of the CPTPP provides that:

- 1) The Parties recognize that each Party may have its own regulatory requirements concerning the transfer of information by electronic means.
- 2) Each Party shall allow the cross-border transfer of information by electronic means, including personal information, when this activity is for the conduct of the business of a covered person.
- 3) Nothing in this Article shall prevent a Party from adopting or maintaining measures inconsistent with paragraph 2 to achieve a legitimate public policy objective, provided that the measure:
 - is not applied in a manner which would constitute a means of arbitrary or unjustifiable discrimination or a disguised restriction on trade; and
 - does not impose restrictions on transfers of information greater than are required to achieve the objective.¹³

Article 19.11 of the USMCA provides that:

- 1) No Party shall prohibit or restrict the cross-border transfer of information, including personal information, by electronic means if this activity is for the conduct of the business of a covered person.
- 2) This Article does not prevent a Party from adopting or maintaining a measure inconsistent with paragraph 1 that is necessary to achieve a legitimate public policy objective, provided that the measure:
 - is not applied in a manner which would constitute a means of arbitrary or unjustifiable discrimination or a disguised restriction on trade; and
 - does not impose restrictions on transfers of information greater than are necessary to achieve the objective. [footnote omitted]¹⁴

The WTO, however, has failed to deliver any meaningful outcome under its Work Program on Electronic Commerce, which was initiated in 1998.¹⁵ The existing focus shifts to the negotiation process, initiated by 76 WTO Members,

including the US and China, which formulated a Joint Statement Initiative (JSI) in January 2019, without participation from India and South Africa.¹⁶

To date, many proposals have been submitted, and the negotiations are intensively ongoing. According to information revealed, the US's proposal is in general copied from the USMCA, asserting strict rules on digital mercantilist measures. It contains a provision requiring the signatories not to 'prohibit or restrict the cross-border transfer of information ... by electronic means' unless the measure passes the 'legitimate objective test' and the 'necessity test' stipulated therein. Discrimination between within-territory information transfer and cross-border information transfer is deemed a particularly condemnable model, which cannot be justified (Manak, 2019). On the other hand, China's negotiation focus is on the 'cross-border trade in goods enabled by the internet'.¹⁷

According to media, in its latest textual proposal, China suggests that the JSI participants strive for 'enhanced trade facilitation' to facilitate 'cross-border e-commerce where products are sold online and later physically delivered to consumers or buyers within their territories through measures including implementation of the Trade Facilitation Agreement'.¹⁸ China also called for negotiations on 'trade facilitating and supportive services', such as customs clearance, logistics, payment and so on (Third World Network, 2019). China continues to call the data-flow issue complex and sensitive, and thus any data flow should be 'subject to the precondition of security, which concerns each and every Member's core interests', and be 'in compliance with Members' respective laws and regulations'.¹⁹

Apparently, in the ongoing JSI negotiations, the US's emphasis is on the freedom of 'data/information' flow, for the purpose of ensuring a level playing field, while China's emphasis is on the freedom of 'physical goods' flow, for the purpose of continuing its originally unfairly acquired e-commerce advantages. As we explained above and in Figure 3.1, digital trade barriers are usually imposed in the form of barriers to data flow. We therefore consider that the US approach is the correct and most effective one to tackle e-commerce mercantilism issues. Pursuant to the provision on the cross-border transfer of information proposed by the US, the asymmetry of connection speeds caused by the GFW in China in itself constitutes a prohibitive restriction on cross-border data flow. Furthermore, because China accords discriminatory treatment to data flow across the border of China, it may not be easy to find consensus.

China, by contrast, concentrates on facilitating the trade in goods usually dealt with online (for example, small-amount transactions where goods are delivered in parcels), which overlaps or comes on top of WTO efforts already being made in this regard, such as in the Trade Facilitation Agreement. It is patently evident that China's approach is aimed at strengthening its own advantages in the e-commerce retail sector, while at the same time, refusing

any compromise on the data-flow issue. Under such circumstances, the GFW and the access asymmetry are expected to remain. Given that any JSI outcome depends on the consensus of all the participants, we can hardly expect the plurilateral process to lead to legal instruments that can effectively curtail digital mercantilism.

Appendix: URLs used in the survey

<i>Country</i>	<i>Website</i>	<i>URL Link</i>
Australia	Big W	www.bigw.com.au/
Australia	Vegemite	https://vegemite.com.au/
Brazil	Americana	www.americanas.com.br/
Brazil	B.Lem Bakery	www.blembakery.com/
Canada	Walmart	www.walmart.ca/en
Canada	Dare Foods	www.darefoods.com/ca_en
China	Taobao	www.taobao.com/
China	Shaoxing Liquor Company	www.nuerhong.com.cn/blank1.html
France	Cdiscount	www.cdiseount.com/
France	La Mère Poulard	https://lamerepoulard.com/en/home2/
Germany	Otto	www.otto.de/
Germany	Café Schäfer Triberg	www.cafe-schaefer-triberg.de/
India	Flipkart	www.flipkart.com/
India	Haldiram	www.haldirams.com/
Indonesia	Tokopedia	www.tokopedia.com/
Indonesia	Lapislapis Kue Lapis	http://lapislapis.co.id/story/
Japan	Rakuten	www.rakuten.com/
Japan	Nagomi-Yoneya	www.nagomi-yoneya.co.jp/en/products/index.html
Korea	G market	http://gmarket.co.kr/
Korea	Ottogi Ramen	www.ottogi.co.kr/eng/main/main.asp
Malaysia	Mudah	www.mudah.my/
Malaysia	Old Town White Coffee	www.oldtown.com.my/
Mexico	Mercado Libre	www.mercadolibre.com.mx/
Mexico	La Churreria El Moro	elmoro.mx/
Singapore	Qoo10	www.qoo10.com/
Singapore	Bee Cheng Hiang	www.beechenghiang.com.sg/
Switzerland	Richardo	Richardo.ch
Switzerland	Tristan Chocolatier	www.chocolatier-tristan.ch/
Taiwan	PChome	shopping.pchome.com.tw/
Taiwan	Yu Jen Jai	www.1877.com.tw/
Turkey	Akakce	www.akakce.com/
Turkey	Haci Bekir	www.hacibekir.com/
UK	ASOS	www.asos.com/
UK	The Astronomer	www.theastronomerpub.co.uk/food
US	eBay	www.ebay.com/
US	Amazon	www.amazon.com
US	Federal Donuts	https://www.federaldonuts.com/

Notes

- 1 See *Work Program on Electronic Commerce*, WT/L/274 (25 September 1998). Debates are ongoing about the characterisation of digital content/products, such as computer programs, text, videos, images and sound recordings. We do not intend to address this issue here.
- 2 Ministry of Public Security Order No 33 (16 December 1997), www.dffy.com/faguixiazai/xzf/201003/20100325213033.htm. Similar standards are adopted by the Internet Illegal Information Reporting Center under the Office of the Central Cyberspace Affairs Commission of China. The Internet Illegal Information Reporting Center, 'Guidance on Reporting' (5 February 2018), www.12377.cn/txt/2018-02/05/content_40215072.htm.
- 3 Adrian Shahbaz and Allie Funk, *Freedom on the Net 2020: The Pandemic's Digital Shadow* (Freedom House 2020), <https://freedomhouse.org/report/freedom-net/2020/pandemics-digital-shadow>.
- 4 Adrian Shahbaz, 'Freedom on the Net 2018: The Rise of Digital Authoritarianism' (2018), <https://freedomhouse.org/report/freedom-net/2018/rise-digital-authoritarianism>.
- 5 See, for example, business case studies at Facebook, 'Case studies', www.facebook.com/business/success.
- 6 See USTR, *2020 National Trade Estimate Report on Foreign Trade Barriers*, https://ustr.gov/sites/default/files/2020_National_Trade_Estimate_Report.pdf.
- 7 See *Work Program on Electronic Commerce – Progress Report to the General Council*, S/ L/74 adopted on 27 July 1999, para. 4.
- 8 *United States – Measures Affecting the Cross-Border Supply of Gambling and Betting Services*, WT/DS285/R, adopted on 20 April 2005, para 6.287.
- 9 *China – Measures Affecting Trading Rights and Distribution Services for Certain Publications and Audiovisual Entertainment Products*, WT/DS363/R, adopted 19 January 2010, para 7.1218.
- 10 *Services Sectoral Classification List*, MTN.GNS/W/120 (10 July 1991).
- 11 At the time of writing, the latest version of CPC (version 2.1) was published in 2015, and is available at <https://unstats.un.org/unsd/classifications/unsdclassifications/cpcv21.pdf> (accessed 29 July 2019).
- 12 *The Economist*, 'China Invents the Digital Totalitarian State' (17 December 2016), www.economist.com/briefing/2016/12/17/china-invents-the-digital-totalitarian-state.
- 13 Available at www.mfat.govt.nz/assets/Trade-agreements/TPP/Text-ENGLISH/14.-Electronic-Commerce-Chapter.pdf.
- 14 Available at <https://ustr.gov/sites/default/files/files/agreements/FTA/USMCA/Text/19-Digital-Trade.pdf>.
- 15 See *Work Program on Electronic Commerce*, WT/L/274 (25 September 1998).
- 16 See *Joint Statement on Electronic Commerce*, WT/L/1056 (25 January 2019).
- 17 See *Joint Statement on Electronic Commerce – Communication from China*, INF/ ECOM/19 (24 April 2019), para. 2.5.
- 18 See D.R. Kanth, 'JSI members seek market access outcomes in e-compluri-talk', *Third World Network* (25 September 2019).
- 19 See *Joint Statement on Electronic Commerce – Communication from China*, INF/ ECOM/19 (24 April 2019), para. 4.3.

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4 Banning TikTok and WeChat

The Issue of Privacy Protection and National Security

Introduction

It is not difficult to see that the unfair competition in e-commerce that we mentioned in Chapter 3 affects both the market share and the data that can be accessed, collected and processed. Jack Ma, a Chinese tech billionaire, has maintained that the data resource is the oil and water of the next generation.¹ Being able to access as much data as possible has therefore become a strategically important task. By taking advantage of their unfair competitive advantage, Chinese tech companies have been able to acquire trillions of bits of data from democratic countries every day, which may be used to further sharpen their algorithm and reshape the economic order in an age of surveillance capitalism (Zuboff, 2019). In addition, it may cause a serious threat not only to personal privacy both inside and outside China but also to the national security of other countries. Aware of this worrying situation, several countries have taken actions in response. The controversy over TikTok and WeChat in the United States is one of the latest examples. Let us start by briefly reviewing the background to and controversy over TikTok.

In 2017, ByteDance, a Chinese AI tech company engaged in content distribution, acquired Musical.ly and later renamed it TikTok, a short-form video app that became popular around the globe, except in China. Within China, the market is dominated by Douyin, which is the Chinese version of TikTok and is also owned by ByteDance. Because TikTok is unavailable in China, most of the footage on TikTok's platform is from Western and Indian users, at least before India banned TikTok along with other Chinese mobile apps in June 2020.

In February 2019, the US government fined TikTok US\$5.7 million for illegally collecting children's data, including their names, email addresses, pictures and locations, a practice that violated the US Children's Online Privacy Protection Act. Moreover, it was reported that videos criticising China, such as those to do with the Hong Kong protests, could not be searched for unknown reasons. Given the company's possible direct or indirect connection with China, the concern over the possible violation of Americans' privacy

quickly snowballed into a comprehensive investigation over national security and unfair competition between the US and China. To fully understand the implications of the TikTok saga, and how it would affect the struggle between the US and China, we need to weave together several threads of law, economics and national security.

Before analysing the TikTok event and its implications, this chapter will first introduce privacy law in the US and in the EU. By doing so, we may understand why the privacy concern of American users on TikTok has gradually escalated into a threat to national security and economic development. This does not mean there is no privacy law in China. However, the gap between laws in action and laws in books is wider in China compared with that in the US and in the EU. More crucially, the dividing line between the state/CCP and the private sector in China is thin under some circumstances, if it exists at all. This is pivotal to understanding why the data collection of an ostensibly innocuous social media app is becoming a national security problem between the two most powerful countries in the world.

Admittedly, privacy and national security concerns are not new issues. However, thanks to globalisation and the Internet revolution, old problems have now been exacerbated in a dramatic way. Specifically, globalisation means the interdependence of production lines, the division of labour, the wide extension of markets, and the increased interconnectedness through cyberspace. In the past, national security concerns usually arose because of military competition or terrorism. Nowadays, they can come from any innocuous section of the economy. Computer hacking directed at high-tech companies can be accomplished within a few clicks, and we may not even be aware of any leakage.

Furthermore, the Internet revolution also involves the static/mobile transformation mentioned by Negroponte (1994). This means that terminal devices such as smart phones or pad computers would become mobile, unlike the traditional immobile bulky television, whereas the transmission of information takes place through fibre stably buried underground. Because mobile devices collect all kinds of information from everyone in all places at all times, the possibility of an intrusion of privacy increases. In addition, since personal data collected from versatile sources can be used for accurate profiling using AI techniques, behavioural patterns and trends may be discerned using AI. If such behavioural patterns are related to soldiers, satellites, officials, vaccine production, weapon transportation, voting interferences, fake news in elections, bank accounts, etc., a potential threat to national security is imaginable.

In short, the development of information technology and globalisation has fundamentally and irreversibly changed the concept of a threat to our national security and privacy. As such, the following discussion on the privacy and national security concerns should be understood against the background of economic globalisation and technological development.²

Privacy Protection in the US

In the United States, the right to privacy is not explicitly enshrined in the Constitution. Nonetheless, it is now generally well-protected and its scope is continually expanding, paralleling the development of information technology. Indeed, the evolution of the concept of privacy has always been deeply intertwined, and most of the time, in tension with technological innovation.

The concept of a right to privacy dates back to the late 19th century, when Samuel Warren and Louis Brandeis (1890) published their seminal paper ‘The Right to Privacy’. Thanks to the invention of the Kodak camera, which facilitated the emergence of the tabloid press, the importance of privacy had surfaced and had become an increasingly urgent issue. Since then, the concept has been expanded and refined whenever new technologies have threatened the protection of privacy. For instance, William Prosser (1960) further distinguished four kinds of violations to privacy, which included intrusion, public disclosure of private facts, placing in a false light in the public eye, and (mis)appropriation.

Historical Development

The concept of privacy is by no means of academic interest only. In 1965, the United States Supreme Court officially recognised the right to privacy in its landmark decision *Griswold v. Connecticut*. In this decision, Justice William O. Douglas, the author of the majority opinion, predicated his conclusion on the so-called ‘penumbras theory’, arguing that ‘The [precedents] suggest that specific guarantees in the Bill of Rights have penumbras, formed by emanations from those guarantees that help give them life and substance. Various guarantees create zones of privacy.’³ Notably, the privacy right in *Griswold* involved the privacy of married couples to buy and use contraceptives without government restriction. The most important guarantee Douglas had mentioned was the Fourth Amendment, which prescribes ‘The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no Warrants shall issue, but upon probable cause, supported by Oath or affirmation, and particularly describing the place to be searched, and the persons or things to be seized.’ That is to say, this Amendment protects American people from unreasonable searches and seizures; but what are ‘searches and seizures’?

In the past, the US Supreme Court adopted a narrow understanding of ‘a search’: only physical intrusion of someone’s property counts. This interpretation resulted in many cases of eavesdropping, spying, or recording not constituting a search.⁴ As a corollary, the Fourth Amendment could not be invoked to protect someone’s privacy in many circumstances. This is generally known as the property doctrine or the trespass doctrine in the context of the Fourth Amendment.

The property doctrine was vehemently criticised, along with the development of technology, because it was too wooden a standard to protect fundamental rights. In *Katz v. United States*,⁵ the Supreme Court finally switched its stance and expanded its originally narrow interpretation, arguing that the Fourth Amendment protected ‘reasonable expectations of privacy’, even though one’s property was not physically searched or intruded on. The majority opinion articulated that a government action constituted a search as long as 1) a person has an expectation of privacy, and 2) that expectation is recognised as reasonable by society. Based on this criterion, therefore, a wiretapping of Katz’s conversation in a public phone booth still constituted a search, even though the FBI agents did not physically enter the booth. By contrast, Katz’s movements into the booth would not be protected because it was publicly visible to anyone nearby. This is known as the public observation doctrine.

The public observation doctrine, however, excludes many of the modern investigative techniques from the scope of the Fourth Amendment. For instance, the movement of a car on a street is publicly visible to other people. Furthermore, the *Katz v. United States* case also planted the seed of the third-party doctrine, which suggests that ‘Telephone users, in sum, typically know that they must convey numerical information to the phone company; that the phone company has facilities for recording this information ... for a variety of legitimate business purposes ... it is too much to believe that telephone subscribers ... harbor any general expectation that the numbers they dial will remain secret.’⁶ Thus, individuals could not reasonably have expectation of privacy once they agreed to submit their information to a third party, be it a natural person or a telecommunications company.

The combination of the third-party doctrine and the public observation doctrine makes the right to privacy more vulnerable to modern surveillance techniques, techniques that make every movement trackable without physically intruding on an individual’s property. In the contemporary world, it is simply unthinkable that one could live a normal life without revealing some personal information to a third party, be they banks, e-commerce companies, or social media. From this perspective, it is obvious that the development of technology has left the evolution of privacy law far behind.

Changes After the 1980s

The situation changed in *United States v. Jones*,⁷ where the unanimous opinion held that the installation of a GPS tracking device on a person’s vehicle without a warrant constituted an unlawful search under the Fourth Amendment. What is noteworthy in this case is the ‘mosaic theory’ alluded to in Justice Alito’s and Justice Sotomayor’s concurring opinions. The mosaic theory suggests that ‘disparate items of information, though individually of limited or no utility to their possessor, can take on added significance when combined with other items of information’ (Pozen, 2005; Kerr, 2012;

Schlabach, 2015). The key idea was that ‘each individual piece of intelligence information, much like a piece of jigsaw puzzle, may aid in piecing together other bits of information even when the individual piece is not of obvious importance in itself.’⁸

In the context of informational privacy, the mosaic theory suggests that even though some ostensibly non-sensitive information, such as our favourite restaurant or baseball team, does not reveal too much, the collection of each piece may provide a full profile of who we are and constitute a violation of privacy. This is an idea strongly related to AI. It remains to be seen whether the Supreme Court will further apply the mosaic theory in this domain.

Of particular note, in a report issued in May 2014, the US President’s Council of Advisors mentioned ‘data fusion’, which ‘occurs when data from different sources are brought into contact and new facts emerge’ (President’s Council of Advisors on Science and Technology, 2014). Data analysts—and their bosses, tech companies—may identify individual people by creating personal profiles and uncovering behaviour patterns. As a corollary, data fusion could easily result in a comprehensive profiling that allows the government to identify any individual. Because of this possibility, several concerns arise immediately, depending on the identity of that person: first, the privacy of that identified person would be infringed upon; second, business secrets could be probed and lead to millions of dollars of losses, and finally, in the case where the person identified is a high-ranking official, national security is at stake.

Prevalent Surveillance After 9/11

Following the 9/11 event, as authorised by the Foreign Intelligence Surveillance Act, the US government conducted mass surveillance upon a broad swath of people, a counter-terrorist measure that gave birth to a series of cases.⁹ In addition to conducting mass surveillance domestically, the US government also tried to garner data that had been stored by US tech companies in other countries. When these tech companies refused to cooperate, controversies between the government and private companies arose.

In 2013, for example, the FBI had required that Microsoft provide a suspect’s emails stored in Microsoft’s Ireland server, a request the company rejected. In *Microsoft Corp. v. United States* (2016), the US Court of Appeals ruled in favour of Microsoft, arguing that the US government could not compel the company to submit its data stored in Ireland, because the Stored Communications Act of 1986 could not be applied extraterritorially. Most notably, the Irish government considered that the US government’s action violated both the European Union’s Data Protection Directive and Ireland’s own data privacy laws. After the Supreme Court took the case, the US government enacted the CLOUD Act (Clarifying Lawful Overseas Use of Data Act), a statutory law that provides a legal basis for the government to access

data stored in foreign jurisdictions under certain conditions. After a warrant was filed according to the Act, the US Supreme Court found the case moot and vacated the lower court decision.

In parallel with the rapid progress of modern technology, the third-party doctrine has encountered some refinement. In *Carpenter v. United States*, the Supreme Court opined that

the fact that the information is held by a third party does not by itself overcome the user's claim to Fourth Amendment protection ... In light of the deeply revealing nature of [cell-site location information, CSLI], its depth, breadth, and comprehensive reach, and the inescapable and automatic nature of its collection, the fact that such information is gathered by a third party does not make it any less deserving of Fourth Amendment protection.¹⁰

That is, even if someone submits certain information to a third party, say, a telecommunications company, they could still have reasonable expectation of privacy in some scenarios. Notably, the Supreme Court emphasised that the CSLI is exceptional and that the third-party doctrine remains good law in general.

Challenges in the Era with Big Data and Artificial Intelligence

From the discussion above, it is clear that traditional arguments can no longer capture new patterns of rights violation and satisfactorily protect privacy in the age of surveillance. In a digital era, moreover, privacy – and information privacy in particular – no longer means seclusion or simply 'leave me alone.' Contrarily, a trust-based, rather than rights-centred, concept of privacy is emerging (Waldman, 2018). Hence, scholars have proposed a variety of new theories, endeavouring to address the privacy concerns. For example, Nissenbaum (2004) argued that the appropriateness of revealing certain information and the distribution of that information dictate whether privacy has been transgressed. Moreover, Solove (2006) provides us with a new taxonomy to understand privacy by distinguishing four groups of activities with privacy implications.

Moreover, the 'Big Brother' that is watching us is not necessarily the state. In fact, tech giants, such as Google, Facebook and Amazon, around the world may control more data than any particular government. Given the data they control, they have become another major threat to our privacy, even though they sometimes defend governmental intrusion on behalf of us, as the aforementioned Microsoft case has demonstrated. In other words, these tech companies have often become part of the problems, rather than part of the solutions. What is thornier is that the information they possess, store and disseminate is usually submitted by ourselves. This may make them more legitimate in accessing, using and processing our data.

To address this new problem, scholars have proposed the concept of ‘Digital Switzerland’, which comprises two ideas: (1) big tech companies are on par with the states, and (2) they are neutral in the sense that they are international in terms of their revenues, markets, users and so on (Eichensehr, 2019). This means that there will be no exceptional treatment for their own countries. Based on these two ideas, the interaction between the governments, the tech companies and the users can be distinctly categorised into three models (Eichensehr, 2019). To illustrate, companies may ally themselves with users to resist governmental intrusion into personal data possessed by the companies, in the name of criminal investigation or national security. Alternatively, users may urge the governments to enact laws that further regulate tech companies because lay people do not have enough bargaining power. Finally, companies may collaborate with governments to suppress certain speech or provide data for criminal investigations. This is particularly so in authoritarian regimes, an issue that will be further discussed later.

Privacy Protections in the European Union

From our previous discussion, it is clear that in the age of globalisation, information privacy is no longer a domestic issue for several reasons. First, a government may request the data stored by foreign companies in foreign jurisdictions for purely domestic concerns. Second, many, if not most, modern big companies do business across national borders. This means that the data they possess could in theory be easily accessed, shared, or processed internationally.

In jurisdictions other than the US, the General Data Protection Regulation (GDPR) of the European Union is probably the most influential law in this regard. The following paragraphs only briefly introduce some pivotal provisions that are closely related to this chapter. The GDPR is now the overarching law in Europe, aiming to protect the privacy right and to facilitate the movement of data throughout the EU. Operating above the level of other laws of the Member States, it was enacted in 2016 and took effect on 25 May 2018. Before that, the basic framework was set out in the Data Protection Directive (95/46/EC), which had been effective since 1995.

One of the GDPR’s core purposes is to strengthen the right to consent and the right to restrict processing personal data, including the right to be forgotten, of natural persons. Specifically, Article 7 of the GDPR explicitly protects the conditions of consent, which, according to Article 4(11), should be ‘freely given, specific, informed, and unambiguous.’ Furthermore, Article 4(4) prescribes ‘profiling’, which means ‘any form of automated processing of personal data consisting of the use of personal data to evaluate certain aspects relating to a natural person’. A data controller should stop profiling when the data subjects oppose it. More importantly, even though the data subjects have consented, they still possess the right to erasure. In this regard, Article 17 clearly grants data subjects the right to erase their personal data when the

data are no longer necessary, when the data have been illegally processed, and when the data subjects withdraw their consent, among other things.

It is worth noting that, according to Article 3 of the GDPR, the territorial scope of the GDPR does not stop at European borders, notwithstanding that it is a European law. In fact, it applies to the processing of personal data outside the EU in some circumstances. For example, non-EU organisations that provide services involving personal data in the EU will be subject to the GDPR, at least in theory. According to Recital 23 of the GDPR,¹¹ therefore, a hotel based only in, say, Japan that advertises in the EU, uses European languages on its website, and accepts euros, may be subject to this provision if it hosts any EU residents. To our knowledge, nevertheless, no multinational corporations that have not registered in any one of the EU member states have been fined because of this provision, partly because they have adjusted their privacy policies accordingly after the promulgation of the GDPR. Moreover, through the supply-chain forces, the GDPR requirements force the organisations that want to do business within the EU to comply with the GDPR (IT Governance Privacy Team, 2016).

As a result, according to Articles 45 and 46 of the GDPR, transferring personal data to international organisations or non-EU/EEA (European Economic Area) countries can only be done when 1) the destination ensures an adequate level of protection recognised by the EU Commission, or when 2) the transfer is subject to appropriate safeguards to protect personal data. Violators will be subject to fines of up to €20 million, or 4 per cent of this total worldwide annual turnover for the preceding financial year.

Given its global reach and prohibitively severe punishment, it should not be surprising that the promulgation of the GDPR has immediately triggered serious responses from the other side of the Atlantic Ocean – that is, the US. This is because the GDPR is arguably stricter than the information privacy law in the US, which is not regarded as meeting the adequacy criteria because of the lack of a national data protection law (IT Governance Privacy Team, 2016). This will force the US tech companies to greatly revise or change their privacy policies, to say the least. As such, the said data-transferring requirements constitute an important issue that may obstruct the EU-US trade relationship.

Indeed, the US and the EU have long disagreed about data retention and access. To address this problem, the US-EU Safe Harbor Framework, according to which many US companies registered with the US Department of Commerce to be given safe harbour from prosecution, was developed to address this issue. Nonetheless, the European Court of Justice declared this framework invalid in October 2015, because it insufficiently complied with the then EU data protection law. Hence, the Safe Harbor Framework was replaced by the EU-US Privacy Shield Framework. On 12 July 2016, the US Secretary of Commerce Penny Pritzker joined European Union Commissioner Věra Jourová to announce the approval of the Privacy Shield Framework as a valid legal mechanism when transferring personal data from the EU to the US. Today, the GDPR has become the *de facto* privacy standard outside Europe.

After the GDPR took effect, tech giants in particular were concerned about its scope and validity. Since its promulgation, the EU has fined Google, Amazon and Facebook billions of dollars in the name of antitrust and breach of privacy protection. It seems that the tension between the EU and these tech companies has escalated in recent years. Most notably, most tech giants in the world, including the GAFA (Google, Apple, Facebook, Amazon) and Microsoft, are American companies. On top of the different regulations across the Atlantic Ocean, moreover, the concept of fundamental rights, including privacy and free speech, sometimes varies significantly between the US and the EU. Both explain the disagreements between these tech companies and the EU in this regard.

Despite the conflicts between the EU and these tech companies, the discord between the US and the EU should not be overstated. In fact, such discord is quite common within the democratic camp, given the economic competition and cultural diversity. The disagreement between the entities on either side of the Atlantic mostly involves the tech companies' operational efficiency and strategies, rather than a power struggle between different ideological camps. The conflict is unlikely to escalate into a full-scale national security issue between the US and the EU, regardless of how vigilant the EU is toward these American tech leviathans. This is mainly because American tech companies are neither affiliated with nor controlled by the US government. In fact, even the US government is investigating these tech giants as well. Sooner or later, these American companies will know where the red line is when more judgments are available. Furthermore, their objective is simply the maximisation of their economic profits.

This is in stark contrast with the US's concern about Chinese companies, where national security is the main worry. In the US–China scenario, the problems caused by Chinese government-controlled tech companies include not only market competition, but, more seriously, the potential sabotage of national security. Given that most if not all Chinese tech giants are directly or indirectly controlled by the CCP, they are often trusted with political missions in addition to their own business agenda. The conflict between the US government and the Chinese tech companies is more like a clash between two sovereign states than a government endeavouring to maintain domestic market competition.

The TikTok/WeChat Controversy

TikTok is a social media platform, upon which users can share their own videos with anyone. It did not become a globally popular app until ByteDance acquired it and merged it with Musical.ly. Since 2017, TikTok has been accused of conducting censorship on a broad swath of political, social and religious topics, including LGBTQ issues, Falun Gong, Tiananmen Square and Tibetan independence, among others. Specifically, some hashtags have been explicitly barred by TikTok, while others are implicitly 'shadow banned',

which means that the posts tagged with these sensitive hashtags could not be searched or found on TikTok by other users, notwithstanding that these posts remain intact on the owners' screen. Therefore, it is more difficult to detect censorship, let alone prevent it.

Worse still, it not only suppresses those videos that criticise the Chinese government, but also promotes those that side with China. Take the Xinjiang 're-education' camp, for example. China has incarcerated millions of Uyghurs in the name of education, cultural assimilation and maintaining social stability. Some were even punished for their religious belief. Even though Xi Jinping has officially admitted (and endorsed) the implementation of such a policy, videos that deny the existence of the camps are more likely to be found from the searches on the platform.¹²

Banning TikTok

In October 2019, US Senator Marco Rubio asked the Committee on Foreign Investment in the United States (CFIUS) to review ByteDance's acquisition of Musical.ly, questioning why TikTok only had a few videos of the Hong Kong protests that had been widely circulated on other social media during and after the Anti-Extradition Law Amendment Bill Movement. In November 2019, the CFIUS started to review the Musical.ly deal because of national security concerns. In response, TikTok claimed that China did not have jurisdiction over the contents of its platform, because TikTok does not operate in China and is not influenced by any foreign government, including the Chinese government. Still, the US government has been increasingly scrutinising TikTok over the personal data it handles, particularly those involving military or intelligence personnel. In December 2019, the US Navy and Army banned the use of TikTok on government-issued phones, because TikTok was conceived as a cybersecurity threat and was flagged for potential security risks.

All these events quickly snowballed into a more comprehensive investigation directed towards TikTok. Invoking his emergency economic power on 6 August 2020, then US President Trump signed an executive order to impose broad sanctions against TikTok unless it was sold in 45 days by ByteDance (Executive Office of the President, Executive Order 13942, 2020). He believed that TikTok might potentially allow China to track the locations of federal employees and contractors, build dossiers of personal information for blackmail, and conduct corporate espionage. Moreover, it 'also reportedly censors content that the CCP deems politically sensitive, such as content concerning protests in Hong Kong and China's treatment of Uyghurs and other Muslim minorities', according to the above executive order. On 14 August, Trump issued a new executive order in which he extended the deadline to 90 days, stating that there is credible evidence that ByteDance could take actions to imperil national security (Executive Office of the President, Order of August 14, 2020).

While TikTok has always insisted that the company has never turned over any data to the Chinese authorities, its parent company, ByteDance, has given in and agreed to divest the US operations of TikTok completely, and several potential buyers have emerged. At first, Microsoft allegedly appeared to be one of them, but it turned out that this was not the case. Other American tech companies, such as Twitter and Oracle, expressed similar interest in acquiring some of TikTok's operations. Meanwhile, TikTok sued the US government, accusing the Trump administration of depriving it of due process and freedom of expression.

In early September 2020, TikTok chose Oracle to be its technology partner in the US, a deal that would make Oracle a 'trusted technology partner' rather than the owner of TikTok. By contrast, ByteDance retained at least majority ownership. Although the terms of the proposed deal would give Oracle full access to TikTok's source code and updates to make sure there were no back doors, national security concerns did not dissipate as long as TikTok's Chinese parent company wielded financial control over the company. At the same time, the Chinese government might not approve of this deal, either. *The Global Times*, China's semi-official newspaper, has claimed that 'China is prepared to prevent Chinese firm TikTok and its advanced technologies from falling into US hands at all costs.' It seemed doubtful that any agreement would be reached. It is worth noting that in August 2020, China adjusted its limitations on exportation, which now includes algorithms.¹³

On 27 September 2020, a federal judge blocked Trump's TikTok ban on the grounds that it might exceed his presidential authority to do so, notwithstanding his broad emergency economic powers with respect to national security matters. In addition, the said deal between TikTok and Oracle as well as Walmart has been under a national security review by the CFIUS. Parallel with this development, the US government also extended the deadline to 27 November. Furthermore, TikTok looked to the US Court of Appeals to review Trump's divestment order and the government's national security review mentioned above. On 29 October, Rubio further proposed the Adversarial Platform Prevention Act, hoping to '[i]ntroduce a set of data protection and censorship related standards and restrictions that must be met by high-risk foreign software'. It defines 'high-risk foreign software' as 'any software owned by an entity organized under the laws of, headquartered in, or whose principal operations are in China, Russia, Venezuela, Cuba, or a country designated as a state sponsor of terrorism, or any software that stores US consumer data in one of those nations'.¹⁴

On 26 November 2020, the US government once again extended the deadline, which should have been 27 November, to 4 December, by which time the deal between TikTok on the one hand, and Oracle and Walmart on the other, was expected to be concluded. Although the deadline had lapsed, further negotiation was ongoing. On 14 December, the Court of Appeals in Washington held a hearing, in which two out of three judges seemed to disagree with the executive order to ban TikTok.

Because of these judicial rulings, the ban on TikTok never really took effect. After his election, President Biden's administration revoked the ban on TikTok on 9 June 2021.¹⁵ Instead, President Biden issued an executive order, which requires the government to evaluate China's threats 'through rigorous, evidence-based analysis' and to 'address any unacceptable or undue risks consistent with overall national security, foreign policy, and economic objectives, including the preservation and demonstration of America's core values and fundamental freedoms'.¹⁶ Specifically, the executive order demands that the Secretary of Commerce, along with other authorities concerned, submit two reports. The first report, which should be submitted within 120 days, should address the harm from the unrestricted sale of, transfer of, or access to United States persons' sensitive data. The second, which should be submitted within 180 days, should recommend executive and legislative actions to address the risk associated with connected software applications that are designed, developed, manufactured, or supplied by persons owned or controlled by, or subject to the jurisdiction or direction of, a foreign adversary.

Banning WeChat

At roughly the same time, the US government banned WeChat on 6 August 2020, on the grounds of 'national emergency with respect to the information and communications technology and services supply chain' (Executive Office of the President, Executive Order 13943, 2020). In the executive order, Trump maintained that 'Like TikTok, WeChat automatically captures vast swaths of information from its users. This data collection threatens to allow the CCP access to Americans' personal and proprietary information.' (Executive Office of the President, Executive Order 13943, 2020).

Owned by another Chinese tech giant, Tencent, WeChat is known as a super all-in-one mobile app that combines a variety of services, such as messaging, social media and payment application, which have been heavily relied upon in the daily lives of most Chinese people. Indeed, it has more than 1 billion users in the world, including an average of 19 million daily active users in the US. Compared with TikTok, it is less popular in the US, but it is still widely used among American Chinese or Americans that do business in China. Not surprisingly, the said ban immediately gave birth to several lawsuits. In September 2020, a federal judge issued an injunction against the executive order, which raised serious concerns regarding the First Amendment. In October, the Ninth Circuit Court of Appeals rejected the request of the Department of Justice that would allow the government to ban WeChat, as the government had not demonstrated an immediate and irreparable injury.

Like the ban on TikTok, the ban on WeChat was revoked in the aforementioned executive order issued on 9 June 2021. This might not, however, mean a retreat by the US. Although the new executive order does not specify any country or tech companies, in practice it has the 'potential for an even broader crackdown on Chinese-owned apps than the Trump administration

orders it replaces by mandating a review of all software applications with potential ties to countries such as China'.¹⁷ To a significant extent, it depends on the suggestions in the two reports that the Secretary of Commerce is to submit in the next few months.

Why do TikTok/WeChat Constitute a National Security Concern?

Although the ban on TikTok and WeChat has been revoked, the danger posed by these information and communications technologies and services has been explicitly recognised in Biden's executive order. One typical way of seeing the importance of TikTok and WeChat is to understand the concept underlying AI. Traditionally, we wrote programs for the computer, and the computer provided feedback with various kinds of data. AI does the reverse: we feed the computer with big data, and the computer provides feedback in the form of a program.¹⁸ In the past, the role of programmers was significant in that the computer only provided computational assistance. In the era of AI, the computer can do much in terms of deep learning, and the role of the programmers is significantly diminishing. For instance, if we feed in a large number of previous game examples of Go, AI then generates for us a playing program named AlphaGo, which can beat almost all human professional players of the Go game. The key parameter that a programmer must tell the computer is: how do we define 'win' in the Go game? Given the specified objective of winning the game, the AI plays the key role and takes care of the rest. Thus, the key to AI design is the *objective function* that we feed into the computer.

In the case of TikTok, the AI determines what video content is to be sent to each individual. It suppresses and promotes certain videos through its 'For You' feed, which provides users with videos to which they do not even subscribe but are recommended by its algorithm. Specifically, the AI in the background analyses the behavioural patterns of individual TikTok users, profiles such patterns of watching previously sent videos, studies the pattern of uploaded videos of each individual, and then sends future videos to users in an 'optimal way', in the sense that some objective function is maximised. If apps such as TikTok and WeChat can collect large amounts of information from US citizens, accurately perform the job of profiling, and then exercise an 'objective' influenced by the CCP, it is not difficult to imagine the national security concerns that may face US officials.

We can use election interference as an example. To many US citizens, the memory of the 2016 presidential-election fake-news attack by Russia is still fresh. In the future, with the help of big data based on personal information, AI will be able to engage in accurate individual profiling, which may facilitate foreign manipulation of domestic electoral results. Another example is the leak of military information from soldiers' video releases. TikTok was banned by the US Navy and Army, because soldiers had uploaded videos recorded inside military facilities, sometimes with their uniforms and identifiable characteristics. In fact, 'one of the most active [subcommunities of

users] consists of young military service members.’ And TikTok gathers the location data when the videos are uploaded in addition to the browsing and search histories,¹⁹ which suggests that the locations of arsenals, barracks, or even warships could be revealed by accident. Moreover, the search histories of American soldiers and lay people may leak sensitive information.

In a public speech on 7 July 2020, the Director of the Federal Bureau of Investigation (FBI) Christopher Wray unmistakably said that China had stolen Americans’ data through a variety of ways, to the extent that the theft had endangered the economic and national security of the US. One of the clandestine efforts he identified was intelligence gathering through social media platforms. Although he did not do a roll call, it is clear that the US government has noticed the potential danger of such platforms. A similar argument has been raised from time to time in official documents of the US. In November 2020, for instance, the Department of State described this issue as one of the elements of China’s challenge in its report (Office of the Secretary of State, 2020).²⁰

In fact, the US is not the only country that has such a worry. Outside of the US, as of 2020, TikTok has been completely banned or intermittently blocked in several Asian countries, such as India, Pakistan, Armenia, Indonesia and Bangladesh, for similar reasons. India, for example, banned TikTok and WeChat because they were ‘prejudicial to the sovereignty and integrity of India, defense of India, security of state and public order’.²¹ Although as of the time of writing this event was still in progress, a question remains: does this contention not exaggerate the danger of the two apps? Why have tech companies affiliated with other countries not received similar challenges? These questions become more puzzling, given that the data-collection and sharing practices of the two platforms are quite standard in the industry. Most social media and e-commerce websites, such as Facebook and Amazon, will recommend posts or books for their users, using roughly similar algorithms. Then, what is wrong with the two Chinese apps?

The answer to the above question is again the *objective function* of the tech companies in question. Given the Chinese government’s access to the data and that the big tech companies have been effectively controlled by the CCP, the CCP has the authority to determine what drives the algorithm. TikTok states in its terms of service that it can share data with its parent company, ByteDance, and ultimately, the Chinese government. With the assistance and development of the AI algorithm, Beijing can use its big data to suppress free speech, manipulate political winds, and threaten any country’s national security. In short, people may not feel threatened that much by AI companies which maximise their profits, such as Facebook and Google, but people do feel threatened by the authoritarian wire-puller behind other AI companies.

Ties Between Chinese Companies and the CCP

Even though TikTok has been banned by the US Department of Defense, the threat to national security has not dissipated. It has simply become more

difficult to detect, as TikTok is popular among not only American armed forces but also lay people. In fact, its downloads have surpassed those of Instagram,²² owned by Facebook, and YouTube, owned by Google. Due to the millions of videos uploaded every day, TikTok has controlled a substantial portion of Americans' daily lives. TikTok officials have always insisted that TikTok stores its data in the US in lieu of China, and TikTok does not send such data to the Chinese government. However, this argument is not plausible because the CCP effectively controls all tech giants in China, both internally and externally.

Internal and External Control of the Business

As we have alluded to in Chapter 1, inside most if not all major corporations, including tech companies, there is a CCP organisation, which serves not only as a whistleblower but also as a policymaker. These CCP committees inside these big companies, be they foreign or domestic, ensure that these companies will faithfully execute the will of the CCP and fulfil their political obligations, sometimes even at the expense of their commercial interests. As a matter of fact, the CCP's reach into private companies, including ByteDance and Tencent, has increased dramatically in the past decade. It has been reported that 'party organizations exist in nearly 70 percent of some 1.86 million privately owned companies.'²³

Externally, all Chinese companies are subject to the national security law as well as other counter-espionage or intelligence laws. To illustrate, in China, privacy is not explicitly protected in the Constitution, although some provisions have enshrined relevant rights. For example, Article 39 of the Chinese Constitution protects people from illegal search and seizure. Article 40 also protects the right of secret correspondence. Nonetheless, the Chinese Constitution is considered as 'perhaps the least important document, in terms of binding force, in the entire Chinese legal system ... The words of the constitution are essentially epiphenomenal.' (Clarke, 2003).

In addition to the inefficacy of constitutional protection, China has enacted many laws that aim to bring these tech giants in line. The most notorious example is the National Security Law, which was enacted in 2015. Article 77 vaguely prescribes that citizens and organisations should report to the authorities concerned any clues, evidence and information that may threaten national security. They should also provide necessary support and assistance to the national security authorities, police authorities and military departments. Article 79 further stipulates that corporations should cooperate with the government and adopt security measures in accordance with the needs of national security. Both provisions may be invoked by the CCP to coerce tech companies into submitting their data. That is, the state does not stand on the opposite side of the corporations, as is the case in democracies. Instead, the Chinese government stands above the private sectors.

In 2016, moreover, China promulgated the Cybersecurity Law, which became effective in June 2017. Its first Article prescribes that the Law is enacted with an eye to defending cyber sovereignty and national security, among other things. In terms of its scope of application, Article 5 of the Cybersecurity Law stipulates that the government shall adopt measures to monitor, prevent and handle cybersecurity risks, and threats from both inside and outside the national border. Article 57 further obligates network operators to stop the dissemination of information prohibited by laws and regulations, and to report to the government when they discover such information. This Article essentially allows the CCP to control the spread of information and garner any data it deems disturbing and dangerous. Not to mention that Article 49 requires network operators to cooperate with the government in conducting surveillance and investigations.

In 2020, China drafted two important laws to further facilitate its access to data stored by Chinese companies or by foreign companies based in China, that is, the Personal Information Protection Law and the Data Security Law.²⁴ After opening for public comment, the Personal Information Protection Law has been in effect since November 2021, while the Data Security Law took effect in September 2021. Not surprisingly, both laws have long-arm jurisdictions when necessary.

International Outreach of Controls

Article 2 of China's Data Security Law stipulates that the government will investigate and punish data activities that infringe upon the national security, public interests, or legitimate interests of citizens and organisations, even if they take place outside China. In other words, China can extend its data governance regime beyond its territory and go after anyone whose data activities may threaten its national security. Article 41 further authorises the authorities concerned to investigate data controllers when they find data activities that are potentially risky to security. More alarmingly, Article 32 prescribes that data controllers should submit their data and cooperate with the authorities concerned when the latter is investigating crime or maintaining national security.

It is unclear how the Chinese government would interpret and invoke these provisions, but one reasonable possibility is that they will give 'Chinese authorities the ability to regulate data controllers, whether they are in China or outside of China, so it has an extra-territorial effect'.²⁵ As such, tech companies that have business in China will be tethered by this law and ultimately be forced to submit their data to the CCP when the data allegedly raise national security concerns.

In a similar vein, Article 3(2) of the Personal Information Protection Law stipulates that the Law is applicable outside China for the purpose of protecting the interests of data subjects in China. Specifically, it will be applied under three circumstances: when organisations provide products or

services to individuals in China, when organisations assess and analyse the behaviour of individuals in China, and when the protection of interests is in accordance with other statutes or regulations. Indeed, its scope of application is broader than the GDPR,²⁶ as foreign entities may be regulated by other Chinese laws and regulations. Of particular note, the draft Personal Information Protection Law deals with, inter alia, cross-border transfers of personal data. Data processors need to meet the standard set by this Law or the approval of the authorities concerned in advance, before they transfer their data to countries other than China.

Undoubtedly, WeChat will be bound by these laws. TikTok, however, may not be under the jurisdictions of these laws, because they regulate those (foreign) tech companies that do business in China, but TikTok is not available in China. Nevertheless, TikTok's parent company, ByteDance, is a Chinese tech company that operates in China and is certainly subject to the said laws. Indeed, Zhang Yiming, the founder of ByteDance, has admitted that his company is willing to cater to the CCP's propaganda and monitor any content that crosses the party's red line. Even though TikTok is formally separated from its Chinese counterpart, Douyin, its algorithm is still developed by ByteDance's China-based engineering teams (Australian Strategic Policy Institute, 2020). As a consequence, there is little difference between TikTok and Douyin in this regard. Admittedly, TikTok has hired many engineers from the US, but they must report to their senior executives in China, who have access to American users' data, including names, birthdays, addresses, emails and passwords, to name just a few (Australian Strategic Policy Institute, 2020). In fact, TikTok has already been accused in California of sending user data to China.²⁷

To sum up, the CCP can effectively control these tech giants both internally and externally. Because of these internal and external controls, it is implausible to claim that China will not have access to data possessed by either WeChat or TikTok.

Fundamental Difference Between Authoritarian and Democratic Countries

Chinese people, in fact, are increasingly dissatisfied with their privacy being violated by these Chinese tech giants. To pacify this unrest, the Chinese government has actually embarked on projects protecting Chinese netizens from commercial data collection to some extent. As mentioned above, China promulgated the Personal Information Protection Law in 2020, which took effect in 2021. Notwithstanding the law, however, the CCP's surveillance activities have only increased rather than decreased in the past decade and it is an unreasonable expectation that the CCP's surveillance will stop in the future because of the law.

Here we should distinguish commercial surveillance from government surveillance. One important feature of China's privacy law is that it 'build[s] consumer trust in a thriving digital economy but does not undermine the government's ability to maintain control'.²⁸ In practice, when the government

does not need the data, commercial surveillance will be restricted in order to facilitate economic development and build up the CCP's reputation. When the state needs the data, however, the protection of privacy will give way to governmental surveillance. In fact, the violation of privacy by tech companies in China should sometimes be attributed to the Chinese government. This is not to say that tech companies have no responsibility at all, but that the boundary between private Chinese tech companies and the Chinese government/CCP is vague and porous in this regard. Many of the distinctions in democracies, such as the line between the government and the ruling party, or the distinction between the state and the private sector, are blurry at best in China.

Indeed, some even go further to claim that 'All Chinese tech companies have deep ties to the Chinese state security apparatus' (Australian Strategic Policy Institute, 2019). Given China's ambition to overtake the US in AI and other cutting-edge technologies, this is understandable because these tech companies have been selected and subsidised by the state, if not owned by the state, to spearhead the development of high technologies. In return, these companies will assist the state in fulfilling its political and military agendas. This symbiotic model has proved to be mutually beneficial to both the state and the companies.

Since much information may be collected internationally, we now turn to discuss the aforementioned concept of a 'Digital Switzerland'. According to Eichensehr, tech giants are both independent and on a par with the states because of their global clout. In the context of information privacy, this suggests that they may be able to protect their users' data from governmental intrusion, even when the requests come from their governments. This may be the case with American tech giants, such as Microsoft, Facebook and Apple, as most of them have resisted governmental mandates that ask for the data they control for use in a criminal investigation.

A 'Digital Switzerland' in China, however, is definitely impossible. The triangular relationship between the states, the companies, and the people that Eichensehr observes in the US does not exist in authoritarian countries. It is not realistic to expect that the Chinese tech companies, such as Tencent, Huawei and Baidu, can stand by Chinese people and resist data requests that come from the Chinese government. No Chinese companies, be they SOE or not, can really reject the government's request, as the CCP itself could participate in the companies' decision-making processes.

Externally, those draconian laws, such as the National Security Law, will make sure that the business tycoons cooperate whenever necessary. With these excuses, it is easy for the Chinese authority to take over a private company that does not cooperate with the government, and examples abound. Under such a shadow, tech companies have no choice but to concede and cooperate. The result is that the CCP has much leeway in requesting information from private tech companies in the name of social stability and national security, given its absolute dominance in China. Instead of forging alliances with the people, Chinese tech giants will always choose the CCP as their ally, if they

have a choice at all. Put briefly, there is little difference between the state and the private sector when the former determines to intervene: the latter only represents the extended hand of the former.

Are Big Data Really Dangerous?

Admittedly, it is one thing that China will have access to the data controlled by all Chinese tech companies sooner or later; it is another thing that these data really pose a serious threat to the national security of the US (or other democratic countries), notwithstanding that it is scary enough that the CCP has access to the data of American people. On this score, some have suggested that the information uploaded to TikTok is truly useless to China, which might have gotten what it wants from the 2015 OPM (Office of Personnel Management) data breach, in which extremely sensitive personal information of millions of American civil servants was stolen.²⁹ Instead, videos uploaded to TikTok are mostly about users' daily lives, their families, pets, and the like. Therefore, the worry that TikTok may use the data to undermine the national security of the US is not justified.

Nonetheless, this argument is not persuasive for several reasons. First, and most obviously, the fact that the OPM hack is more serious in terms of its scope and scale does not justify any underestimation of the potential danger TikTok may bring about. On the contrary, given the lesson of the OPM data breach, we should err on the side of being (over)cautious.

Second, it is possible that people close to high-ranking officials, such as their husbands, daughters, or grandmothers, will use TikTok even if the officials themselves do not, given its popularity among American people. This may raise the risk in two ways, even though the contents of the videos themselves are not sensitive. To begin with, through the videos the users uploaded, the activities and locations of high-ranking officials may be revealed, because TikTok also collects the location data in addition to the video's contents. What is worse, they may use the tablets or mobile phones of these high-ranking officials to do so. This could further put the truly sensitive data stored in these devices at risk.

Third, the rationale of the above-mentioned mosaic theory may warn us against underestimating the danger of ostensibly harmless information. The theory suggests that 'each individual piece of intelligence information, much like a piece of jigsaw puzzle, may aid in piecing together other bits of information.'³⁰ In the context of TikTok, each video, however silly the content is, could be gathered, analysed and eventually provide something that does not appear in the videos at all.

Fourth, even if data controlled by TikTok and WeChat do not put national security at risk immediately, the data they collect today may be a potential threat in the future. The reason is simple: the pace and progress of technological development is beyond imagination. Here the OPM data breach is a good example. So far, there is no evidence (yet) that the data breach of the

OPM has been abused and caused real damage. Nonetheless, this by no means suggests that the data are useless. Once the algorithm becomes more mature as the AI develops, the data that have been breached could be abused and become a real nightmare to the US.

In addition, TikTok may turn its data into information that facilitates the promotion of the CCP's political propaganda. It may portray China as a friend rather than a foe of the US, depict China's military expansion as a 'peaceful rise', and its economic espionage as embracing a free market. This may indirectly affect national security by manipulating public opinion and the elites' perception of the CCP.

As to WeChat, the situation is roughly the same. Tencent 'can and does censor foreign users if Beijing decides special supervision is required' (Australian Strategic Policy Institute, 2020).

Finally, they may damage the economic interests of the US, as users' videos may be taken into consideration when evaluating their social credit, a system that relies on data gathering to tighten social control more efficiently, and has become an important part of the CCP's tech-enhanced authoritarianism. In 2019, ByteDance indicated that the content of livestream videos would be rated and fed into a social credit system (Australian Strategic Policy Institute, 2020). Although the application of the social credit system is limited to China, it is possible that foreign companies operating in the Chinese market will be subjected to the system as well.

In fact, the spillover effect of the social credit system has been pointed out: foreign airlines, for example, have been subject to some sort of credit system (Chen, Lin and Liu, 2018). This applies to foreign airlines' legal representatives, the principal person in charge and other staff with direct responsibilities. In addition to foreign airlines, the extra-territorial reach affects international nongovernmental organisations that register in China. Any activities that are perceived as secessionist or anti-China will be discredited, and this applies to their staff (Chen, Lin and Liu, 2018). From this perspective, it is not impossible that the contents of videos on social media platforms will be used to discredit the uploaders.

In a nutshell, China has spared no efforts in exporting its digital surveillance and techno-authoritarianism, and these Chinese tech giants are extended arms of the Chinese government. It is thus not surprising that the US is proposing the Clean Network program, which according to former US Secretary of State Mike Pompeo, aims to safeguard 'the nation's assets including citizens' privacy and companies' most sensitive information from aggressive intrusions by malign actors, such as the CCP'.³¹ It is unclear whether the idea of decoupling the US economy from China's is feasible both domestically and internationally, but the proposal has resulted in some pushback from China. Wang Yi, China's foreign minister, has proposed to make 'Universally-accepted Global Data Security Rules'³² which would, among other things, require companies to respect local laws and require every participating country to respect the sovereignty, jurisdiction and data-management rights of other countries.

Conclusion

To be fair, there might not have been concrete evidence to prove that either TikTok or WeChat has engaged in sabotage or other espionage behaviour that poses a clear and present danger to the national security of the US. Nonetheless, this does not suggest that the suspicion of TikTok and WeChat is groundless. It would be a fatal mistake to ignore the reach of TikTok and WeChat, given the width and depth of their databases. Speech censorship and content moderation are both real and rampant on both platforms. Even though the data-collection and sharing practices of the two platforms are not uncommon in the industry, their relationship with the CCP is worrisome, given the extremely intense competition between the US and China. With the progress of deep learning and other dimensions of AI, the social media platforms could become the Trojan horses of the technological competition.

Perhaps there is a fundamental conceptual difference between privacy protection and national security protection. For privacy protection, a ‘smoking gun’ is needed, or a specific threat is expected. However, national security is sometimes a one-shot game; the big data collected may just be used once, and once only. In a war, it may be a fatal blow; in an election, it may be fake news spread the night before the voting, and in the banking system, the consequences of one crash may last a very long time. Frequently, it could be too late to reverse the result before one can build the causality link. For national security, the gun is never smoking, until the time comes.

As we discuss the case of TikTok and WeChat, we should not miss the forest for the trees. These two companies are nothing more than the tip of the iceberg. This is why the Biden administration has launched a broader review in the executive order. Search engines, e-commerce companies, airline alliance data exchange, e-payments, GPS using satellites controlled by China, international banking, and 5G or 6G using telecommunications facilities, inter alia, all collect data that are even more extensive than TikTok and WeChat. Even among apps, many other apps, such as mobile payments apps, could be more dangerous because the data they collect are more sensitive. Moreover, they may reach a much wider customer base compared with TikTok, which is most popular among the younger generations. On the one hand, this explains why the Trump administration issued another executive order on 5 January 2021 that banned transactions with eight Chinese apps, including QQ Wallet, WeChat Pay, and most important, Alipay, an app that has been used by 1 billion people every day. On the other hand, it shows that the threat to privacy and national security is omnipresent, not to mention those high-tech gadgets made by Chinese companies, such as Xiaomi, that may include backdoor or other spywares.

In recent years, the trend of ‘the state advances, the private sector retreats’ has escalated to the extent that even Jack Ma, along with other tech tycoons in China, was forced to retire. This is because ‘Xi [Jinping] believes that the Party-state has to maintain tight control over every aspect of the political

and social economic domain in order to ensure economic prosperity as well as political and social stability' (Yu, 2019). Under this pressure, the political atmosphere leaves little room for private companies, let alone SOEs, to reject any orders from the CCP. In this light, every Chinese-controlled company can be seen as an incarnation of the CCP to a considerable extent. This is not to demonise Chinese-owned or controlled companies, which are victims of the authoritarian regime in some sense, but to point out one crucial difference between companies in China and in democracies.

Recently, *The Economist* opined that one achievement of the Trump administration was to correctly recognise the challenge and threat from China.³³ In fact, the Internet has become another 'new field for geopolitical competition', said Geoffrey Gertz, a research associate of the Global Economic Governance Program at the University of Oxford. Although President Biden's national security team may be chiefly composed of former President Obama's staff and be taking a different approach in dealing with China, China's threat seems to be leading to a bipartisan consensus in the US.³⁴

Notes

- 1 'Alibaba's Ma Says Data Resource Is Oil, Water of the Future' (14 October 2015), *Bloomberg News*, www.bloomberg.com/news/articles/2015-10-14/alibaba-s-ma-says-data-resource-is-oil-water-of-the-future.
- 2 It may be useful to have the following informal illustration in mind as we jointly discuss privacy and national security issues. Privacy is about some secrets that individuals do not want others to know, whereas national security is about some secrets that the nation does not want other (especially antagonistic) nations to know. Thus, conceptually, privacy intrusion is very similar to national security offences.
- 3 *Griswold v. Connecticut*, 381 US 479 (1965).
- 4 See *Olmstead v. United States*, 277 US 438 (1928), where the majority opinion stated that wiretapping technology was not physically intrusive, and thus did not constitute a search.
- 5 *Katz v. United States*, 389 US 347 (1967).
- 6 *Smith v. Maryland*, 422 US 735 (1979).
- 7 *United States v. Jones*, 565 US 400 (2012).
- 8 *Halperin v. Central Intelligence Agency*, 629 F.2d 144 (D.C. Cir. 1980).
- 9 These include *Clapper v. Amnesty International USA* (2013), *Klayman v. Obama* (2013), and *American Civil Liberties Union v. Clapper* (2015).
- 10 *Carpenter v. United States*, 585 US ____ (2018).
- 11 Recital 23, EU General Data Protection Regulation, www.privacy-regulation.eu/en/r23.htm.
- 12 For references, see Australian Strategic Policy Institute, 2020.
- 13 Ministry of Commerce, People's Republic of China. *Ministry of Commerce and Ministry of Science and Technology publish the revised Catalogue of Technologies Subject to Export Bans and Restrictions* (30 August 2020), <http://english.mofcom.gov.cn/article/newsrelease/significantnews/202009/20200902998221.shtml>.
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- 18 See www.facebook.com/cateinbigcity/videos/3961569707191616/.
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5 Enacting HFCAA

The Issue of Listed Company Accountability

Background – What is a ‘Public’ Company?¹

The story and the history of the HFCAA are still ongoing. Below is a sketch of what happened in 2020. On 11 May 2020, the US Secretary of Labor Eugene Scalia wrote to the Chairman of the Federal Retirement Thrift Investment Board (FRTIB), the administrating institution of the US Federal pension fund, for the Board’s plan to invest in companies listed in the MSCI All Country World ex USA Investable Market Index,² including approximately \$4.5 billion in Chinese equities.

By attaching a joint letter from the White House national security advisor Robert O’Brien and economic affairs advisor Larry Kudlow, and citing joint statements by the Chairs of the US Securities and Exchange Commission (SEC) and the Public Company Accounting Oversight Board (PCAOB), Secretary Scalia expressed the concerns over the inadequate investors’ disclosure and protection under the Chinese law, and thereby the \$4.5 billion in Chinese equities. He further indicated that due to engaging in activities violating human rights and the US national security and sanction laws, these Chinese investee companies were potentially subject to sanctions or punitive measures, which would undermine their businesses and thus the Board’s investment returns. The Secretary wanted the Board to ‘halt all steps’ associated with such investing, and stated that this was the ‘only course’ to address all the concerns from all levels of the US government.³ On 13 May 2020, the Board announced it was delaying its China investment plan.⁴

The Risk of Investing in Listed Chinese Companies

One week later, on 20 May 2020, the US Senate passed unanimously the Holding Foreign Companies Accountable Act (HFCAA).⁵ The Act was intended to amend the Sarbanes-Oxley Act of 2002 by requiring that all companies whose foreign public accounting firms could not be inspected by the PCAOB for three consecutive years be delisted from the US securities exchanges. The HFCAA also required that the non-compliant companies have to declare that they are not ‘owned or controlled by a governmental entity’.

As will be elaborated below in a section titled ‘The PCAOB Issue: Power and Sovereignty of Auditing’, the PCAOB was designated to audit the public auditors (including US and foreign ones) of the companies listed in the US markets. However, the government of China had long been prohibiting the Chinese audit firms from handing over to the PCAOB their working sheets – that is, the background documents upon which the financial statements were prepared. Although the HFCAA did not single out any particular country, it is widely believed that the target of this legislation was the more than 200 US-listed Chinese companies.

The US administration also attempted to address the PCAOB auditing issue around the same time. On 24 July 2020, the President’s Working Group on Financial Markets (PWG) issued a report, recommending that, as an additional condition on initial and continued listings, the PCAOB should have access to the audit working papers of the issuer’s principal auditors.⁶ The report explicitly listed China as the major concern.

The HFCAA was passed in the House on 2 December 2020⁷ and was signed into law on 20 December of the same year.⁸ The SEC responded by stating that it would consider implementing the HFCAA and the PWG report jointly.⁹

A more dramatic development came from the delisting of ‘Communist Chinese Military Companies’ (CCMCs). On 12 November 2020, the then President Trump issued an Executive Order of Addressing the Threat from Securities Investments that Finance Communist Chinese Military Companies. Effective from 11 January 2021, this order bans any American investment in publicly traded securities of companies identified as CCMCs by the US Department of Defense.¹⁰ As a result of the Order, three Chinese telecom companies (China Mobile, China Unicom Hong Kong and China Telecom) were delisted from the New York Stock Exchange on 11 January 2021.¹¹

From a historical perspective, the facts mentioned above may be seen as a blowing of the trumpet in the US’s ‘financial war’ against China. The apparent purpose of conducting the fight is to prevent US capital from flowing into Chinese companies, especially those listed in the open markets. Occurring at a time when the US–China tension has been high on multiple issues such as the trade war, the accountability for COVID-19, and human rights protection issues in relation to Hong Kong and the Uyghurs, the financial war in general has been viewed as a simple strategic approach that the US has adopted to further counteract China.¹² What is lacking, however, is a discussion about the genuine and systemic risks of investing in Chinese companies, risks that result from the very special eco-political structure in China and the pervasive party-state intervention in the governance of corporates.

In this chapter, we will be looking into this systematic rift between China and the liberal democratic world. The capital markets, either at the global or local level, are founded on a series of corporate governance approaches which are designed to address the ‘agency costs’ arising from the modern corporation structure (Armour et al., 2017). In short, given the separation between

ownership and management, a company is usually run by a subset of ‘insiders’ (that is, major shareholders, the board of directors, or some high-level executive officers), instead of the owners. An insider therefore has the motive to pursue their own interests at the cost of the ‘outside’ owners. This agency cost will be particularly high for publicly listed companies, simply because the stakeholders are possibly millions of general traders. Legal approaches are formulated to regulate the insider’s behaviour, while at the same time maintaining sufficient autonomy for the insider to carry out day-to-day company operations.

The Agency Costs of Chinese Companies

For example, corporate law delegates the commercial decision-making authority to the board of directors or the hired managers, while granting the shareholders the powers to appoint and dismiss the directors so as to ensure that the board of directors is responsive to the shareholders’ interests. The management level of a company may designate certain company information as being confidential for the purpose of competing in the market, while it is required to release certain information to the stakeholders or, in the case of a publicly listed company, to the public, so that the shareholders, bond holders and potential investors are well informed when making decisions related to the company’s affairs.

We argue that these corporate governance approaches, even though they can be seen in the Chinese legislation that has developed over several decades, have been very much eroded under China’s special political and economic structure today. As we can see, since the Chinese government and the CCP are building up a system where the party-state is the actual, final insider of most Chinese companies (state-owned or not), all of the corporate governance norms risk losing this purpose and functions. Put simply, outsiders will find it difficult to know who should be held accountable for the corporate performance, or to whom the company is accountable. The traditional principal-agent problem under democratic capitalism is a bilateral relationship between the company’s managers and public stakeholders. However, this bilateral relationship for Chinese companies is between the CCP and the company’s managers, with the CCP sitting above the managers.

This chapter proceeds as follows. Following the introductory part, the second section will briefly introduce the corporate governance reforms in China since the 1980s. We can see that China has adopted an incremental approach to bring in the Western corporate governance model as a part of its market reform. At the same time, for various reasons, a lot of Chinese companies have been seeking a listing on the foreign stock markets, especially in the US. The third section of this chapter provides an overview of the quarrels between the US and China over the PCAOB’s auditing issue. This long-lasting issue should be seen as an early sign of the potential divergence between the two systems.

The fourth section covers the corporate governance status of Chinese SOEs since 2010. While attempting to copy Singapore's Temasek model, China's State-owned Assets Supervision and Administration Commission (SASAC) appears to be a much more muscular and domineering institution. In the fifth part, we look at the private companies in China, where we will see that the government's intervention, although subtle, is prevalent. Concerns are mounting up along with the enhanced role of the CCP in the corporate governance structure. In the last section, we conclude by highlighting the risk to capital markets imposed by the Chinese party-state extending its command over economic activities.

China's Corporate Governance Reform, 1978–2010

It is fair to say that China built the corporate governance regime from ruins. When it came to power in 1949, the CCP endeavoured to build a centrally planned economy and thus all property and economic activities became state/party-owned. During the following three decades, there were only 'enterprises' but no 'companies' in China, and all the 'enterprises' were SOEs which followed the rules of politics instead of the rules of the market. This remained so until 1978, when Deng Xiaoping initiated the economic reform to make changes to the Chinese business institution possible. As indicated by Guo et al. (2013, p. 258), China adopted gradual and experimental measures for economic reform rather than the 'big-bang' or 'shock-therapy' approach implemented in Eastern Europe.

The early reforms carried out during 1978–92 included measures to separate SOEs from the government, to establish a contract responsibility system, to confer upon SOE managers more autonomy over their business operations, and to allow enterprises to retain certain profits (OECD, 2011, pp. 14–15). The ownership of some enterprises was divided into transferable shares. Shanghai Feilo Acoustics Co. was the first joint-stock company in China. It issued 10,000 shares at 50 RMB per share in 1984, and the shares began to be traded over-the-counter in 1986. Deng Xiaoping gave a stock certificate of Shanghai Feilo Acoustics to John Phelan, the Chairman of the NYSE, as a gift when Phelan visited China.¹³

Establishing Regulatory Rules and Agencies

More significant and substantive reforms came in the 1990s, especially after the time when China's first Company Law was enacted in July 1994. Article 3 of the Law defined two company types: a 'limited liability company' and a 'joint stock limited company'. Both were consistent with the five basic legal characteristics of the business corporation, including legal personality, limited liability, transferable shares, delegated management under a board structure, and investor ownership (Armour et al., 2017), while a joint stock limited company's capital was to be 'divided into equal shares' for better transferability.

As to the governance structure (in the case of a joint-stock company), the 1993 Company Law followed the German model. The shareholders' general meeting was created as the highest decision-making organ of the company (Article 103), while the day-to-day management powers were conferred upon the Board of Directors (Article 11). The Chairman of the Board obtained some unique powers such as being the only legal representative of the company in external activities, and the exclusive authority to convene the meeting of the Board. The Board of Supervisors was created to impose checks and balances on the directors. However, by being distant from the real corporate business, supervisors' functions were limited.

The 1993 Law also contained some transitional features. For example, Article 5 provided that 'A company shall, under the macro-adjustment and control of the State, organise its production and operation independently in accordance with market demand for the purpose of raising economic benefits and labour productivity and maintaining and increasing the value of its assets.' A section was designated for the 'wholly state-owned company' under the 'limited liability company' chapter.

With the establishment of the legal framework, China launched its policy of the 'corporatization of SOEs' with a high degree of prudence. One hundred large central SOEs were selected as priorities in the corporatisation pilot program, followed by some thousands of medium-sized firms being merged with each other or reorganised. As of 1996, approximately 5,800 or 5 per cent of the industrial SOEs were corporatised (World Bank, 1997, p. 24). The primary purpose of the SOE corporatisation was to solicit more capital, and so the securities exchange regime was set up in parallel. The China Securities Regulatory Commission (CSRC) was established in October 1992, and the first Securities Law was passed in 1998.

The demand for capital also led to the first wave of Chinese companies' overseas listings. For example, Tsingtao Brewery became the first SOE to be listed in Hong Kong in July 1993, and Shanghai Petrochemical was also listed on the NYSE in 1993.

In the first decade of the 21st century, the CSRC led the Chinese corporate governance reform by issuing a number of regulations and policies. For example, based on the OECD Principles of Corporate Governance, the CSRC issued the Code of Corporate Governance of Listed Companies in 2002 (OECD, 2011). The Code provided some basic principles of corporate governance, investor protection measures, and the basic code of conduct and professional ethics for corporation insiders.¹⁴

In 2005, the Chinese government published 'Several Opinions of the State Council on Encouraging, Supporting and Guiding the Development of Individual and Private Economy and Other Non-Public Sectors of the Economy',¹⁵ to encourage more private engagement in the growing economy. Corporations' legal systems were also offered guidance to adapt to the policy. Major amendments to the 1993 Corporate Law and the 1998 Securities Law were introduced in 2005, and were considered to be an important step

in the modernisation and Westernisation of the Chinese corporate governance system (Fu, 2016, p. 150). Drawing upon Anglo-American jurisdictions, the revised Company Law rebalanced the powers of the corporate organs, enhanced the legal obligations (in particular, the duty of loyalty and the duty of good care and due diligence) of the corporate insiders, required the appointment of independent directors in listed companies, and created judicial remedies for individual stakeholders.

The revision of the Securities Law improved the institutions in, and administration of, the capital market. In addition, it also strengthened the supervision of listed companies, made issuance examination more transparent, increased the legal responsibilities and rules on the integrity obligations of corporate insiders, and defined the system of civil responsibility to compensate for damage to investors. Following the revision, related agencies made corresponding adjustments to other relevant regulations and criterion documents.

Encouraging Public Listing

Another positive sign of progress during the 2000s was the establishment of the SASAC in 2003. Reporting directly to the State Council, the SASAC was mandated to manage state assets (that is, capital investment in SOEs) and to facilitate the corporatization of SOEs. According to Fu (2016, p. 154), the SASAC took two steps to achieve this. The first was to standardise the Board of Directors of SOEs, which included setting a desirable size and composition of the SOE Board, requiring the setting up of subcommittees under the Board, and rebalancing powers between the chairman and the other positions. The second step was to promote SOEs to be listed in domestic or international stock markets with the hope of improving SOE governance under the pressure of external scrutiny. In addition to improving corporate governance, the policy encouraging public listing also aimed at raising capital, increasing the liquidity of state assets, and building up international reputations for the fast-rising Chinese companies.

All these regulatory and policy developments promoted another wave of Chinese companies' overseas listings. In addition to large SOEs, newly emerging Chinese private companies, especially those in the Internet area, were in dire need of capital that the Chinese capital market could not afford, as the market was still in its infancy and subject to strict government regulations. For example, the Shanghai Stock Exchange imposed an 'IPO moratorium' in 2001 and 2004, respectively, completely banning new listings so as to calm the market and to prevent liquidity from rushing out of existing stocks and into new issues (Asian Corporate Governance Association [ACGA] Alan and Li, 2018, p. 24). The new emerging Chinese companies could thus only seek sources of capital abroad. At the same time, international investors were keen to include the emerging Chinese economy in their portfolios.

Overall, the number and economic weight of these US-listed Chinese companies increased sharply over the ten years between 2000 and 2010. Sina, Sohu and Neteas all became listed on the US Nasdaq in the early 2000s. This frenzy reached its climax in 2007, where 31 Chinese companies completed their US IPOs in that year. The 2008 financial crisis did not result in a large-scale setback for the Chinese economy, and since China was not greatly impacted by the financial tsunami, the 2008 crisis even made the shares of Chinese companies preferable investment targets.

To sum up, at the end of the first decade of the 21st century, China appeared healthy in its corporate governance transition. The legal framework had been established and appeared to match its economic growth. The SOE reforms, while somewhat lagging, continued to progress at a stable pace. The immaturity of its domestic capital market did not prevent ambitious Chinese entrepreneurs from seeking capital in the international pool. In addition to the funds collected, overseas listings also brought an international reputation and new management skills to Chinese companies.

Moving into the second decade of the 21st century, however, several issues arose from the immense overseas listings, and some reversals to the corporate governance reform were floated. The conflicts between the corporate governance in the democratic capital market economies and the Chinese party-state regime quickly became obvious.

The PCAOB Issue: Power and Sovereignty of Auditing

The US adopted the Sarbanes-Oxley Act in 2002 in response to the Enron case and several other capital market scandals. Under the Act, the PCAOB, a nongovernment institution, was created to be a watchdog of the watchdogs. Public accounting firms which audit US-listed companies were obliged to register themselves with the PCAOB, and have their audit worksheets subject to the PCAOB's inspection. This rule applied to accounting firms located in both the US and overseas, as long as their audit clients were publicly listed in the US.

Undoubtedly, the US-listed Chinese companies' auditors were subject to the Sarbanes-Oxley Act. Many of them were the 'Big Four' accounting firms' Chinese branches.¹⁶ They soon found that they were trapped in an international political dilemma: While the US required them to hand over the audit worksheets, China prohibited them from doing so.

CPAs Being Trapped in Political Dilemma

From the outset of when the Sarbanes-Oxley Act came into force, the Chinese government prevented its accounting firms from submitting worksheets to the PCAOB and did not allow the PCAOB to conduct on-the-spot verification in the territory of China.¹⁷ This blocking policy was due to 'national secrecy concerns', at least according to the nominal terms of the codified legislation.

The legal basis of the secrecy prohibitions on audit work papers can be traced back to the 1996 Archives Law of the People's Republic of China, which, in addition to government archives, covers 'historical records in various forms' formed by 'individuals in their political, military, economic, scientific, technological, cultural, religious and other activities' (Article 2). As the number of overseas listings of Chinese companies had grown significantly during 2007–10, it was observed that the Chinese government had progressively tightened its control over audit working papers (Mark, 2014). In 2009, the CSRC, the National Administration of State Secrets Protection (NASSP), and the National Archives Administration issued a joint regulation to prohibit any Chinese companies and accounting firms from providing any 'national secrets' to foreign regulators.¹⁸

The number and size of Chinese companies' US listings kept on growing in the 2010s. Alibaba and JD (a B2C shopping company) both had successful listings in 2014, which drew more investor attention to the rise of Chinese e-commerce industries, and in return drove more Chinese Internet startups to seek a listing in the US. The number of Chinese listings in the US hit a record high of 47 in 2017. Such an upward trend intensified the tension surrounding the PCAOB inspection issue. During 2010–20, numerous bilateral talks were held between the two countries' authorities. Although certain ad hoc solutions were agreed on (for example, the PCAOB may obtain specific audit documents through the Ministry of Foreign Affairs of China),¹⁹ as a whole, the routine PCAOB inspection was still impossible.

China was not the only country to oppose the PCAOB's extraterritorial law enforcement. However, it was the largest stakeholder and the firmest dissenter. According to the PCAOB, 'In the eighteen-month period ended June 30, 2020, 17 PCAOB-registered firms in mainland China and Hong Kong signed audit reports for 202 public companies with a combined global market capitalization (US and non-US exchanges) of approximately \$1.8 trillion. The ten largest of these companies have a combined market capitalization of approximately \$1.3 trillion.'²⁰ An updated list showed that on 1 October 2020, there were 247 companies around the world on which the PCAOB could not conduct its inspections, and 222 of them were located in China and Hong Kong.²¹ The other 'non-cooperative jurisdictions' (NCJs) included France and Belgium, although the PCAOB did announce the achievement of certain agreements with the two European-country governments with regard to its inspection work.²²

As reflected in Secretary Scalia's letter, the HFCAA legislation and the PWG's report mentioned at the beginning of this chapter, the lack of the PCAOB inspection is usually listed as the primary reason underlying the US regulator's concerns over the investment in the Chinese companies. The accounting scandal of Luckin Coffee in April 2020, to some extent, substantiated these concerns.²³

Based on the nature of the debate, the PCAOB issue appears narrower and more specific in content, and is in some sense a more substantiated basis for

the US to take punitive actions (for example, delisting) than other topics, such as human rights violations or national security. In reality, the communication between the US and the Chinese authorities was still ongoing. The enactment of the HFCAA did not really deter the Chinese companies from going to the US capital market, and did not even affect the evaluation of the existing listings. A widely shared anticipation is that the issue will be settled before the three-year grace period lapses.

Failure to Improve Listed Companies' Transparency

However, is the PCAOB inspection merely an issue about the audit gap between the two jurisdictions? This question may deserve some further consideration. A publicly listed company sources its capital from the general public and so its operational information matters to the public interest. That is why corporate and securities laws require listed companies to periodically publish such information. Furthermore, to address the 'agency cost' problem, the laws also require the information to be audited and certified by a third party – that is, public auditors – which in many jurisdictions are served by certified public accountants (CPAs).

To ensure that the numbers in the company books are real, and recorded consistently with laws and Generally Accepted Accounting Principles (GAAP), public auditors usually dig deep into the company's raw data (for example, invoices, stock counting records, etc.) and create various worksheets for reconciliation and adjustment. The audited numbers will then be compiled into financial statements and published. Investors rely on such information to make their investment decisions.

The Enron case revealed another level of risk: public auditors may conspire with their auditing clients. That is why the PCAOB was created to further inspect the auditors' practices. The inspection of the audit work papers is thus crucial to the PCAOB's function. The logic of CPAs is that they are more likely to find financial statements' discrepancies with the help of companies' raw data, and the logic of PCAOB is that they are more likely to find discrepancies that may mislead the general stakeholders with the help of the CPAs' worksheets.

Against this regulatory background, the issue regarding national secrecy is bizarre. Definitely, commercial secrecy is common in the auditors' world, but not national secrecy. A corporate audit involves very limited, if any, national secrets, especially in the private company cases. China's unusually broad national secrecy scope may by assumption reflect the Chinese state's prevailing intervention in corporate affairs. That is, since the state involvement is a general rule in the operation of Chinese companies, all of the information in the companies has the potential of being government-related. In responding to the PCAOB's requirements, the identification of specific pieces of information or even a specific company is too burdensome, and so a total ban is the practical approach.

The above discussion reveals a general issue discussed in the economics literature: the conflicts between internal control and external monitoring. Using business tax evasion as an example, Chen and Chu (2005) showed that, whenever a company wants to have effective internal control, it will have to establish all sorts of tractable and retrievable mechanisms to verify the accuracy and consistency of various information. However, these effective internal-control mechanisms will make the company more transparent not only to company managers, but also to external auditors and agencies, in particular making the company's tax evasion more difficult. The situation in China is similar: if the government or the party does have prevailing interventions in corporate affairs, then these interventions will leave traces and evidence for outsiders to verify. The only way to avoid the side-effects of CCP's authoritarian control is to cut off the information leak to the outside. This is where the accountability problem of Chinese listed companies arises.

The HFCAA's rare requirement that the non-compliant companies declare no government control appears to be based on this assumption. While it is not clear how this provision will be enforced, given the Chinese government's enhanced grip over corporate affairs in recent years, as demonstrated in the following sections, the provision will definitely come under the spotlight in the future.

Continuous SOE Reform, or Reversal?

China's SOE reform speeded up in 2013, when Xi Jinping took the leadership of the country. One of the important reform directions has been to shift from the 'management of assets' to the 'management of capital' in SOE governance.²⁴ Put simply, through the reform, the Chinese government will behave more like a portfolio investor in relation to the SOEs, and will focus on increasing the value of the investment and refrain from engaging in the daily company management.

This capital management-based approach has since been developed into a three-tier corporate control structure. As suggested by Fu (2016, p. 158), the first tier comprises the SASAC as the regulator of state assets; the second tier consists of state holding companies established by the SASAC or transformed from existing SOEs as the investment arm of the SASAC. The SOEs operating the actual businesses and their subsidiaries belong to the third tier.²⁵

It is widely believed that the SASAC's capital management-based approach and the three-tier structure have been very much inspired by Singapore's Temasek model, where Temasek Holdings was established as an investment arm of the government. According to the ACGA, (Allan and Li, 2018, p. 28), the SASAC and Temasek have maintained a long cooperation. Temasek has also provided regular director training courses to SOE managers in China.

After all these reforms, the Chinese SOEs look just like normal companies in any jurisdiction. The SOE may be listed on the exchanges in Shanghai, Hong Kong or New York, and run by a Board of Directors which includes

independent directors. The management level is appointed by the Board. A state holding company behaves as a passive major shareholder, remaining aloof from the operations of the company. According to the US-China Economic and Security Review Commission (2020), 13 Chinese central SOEs were listed in the US markets in October 2020.

The Role of the CCP in Listed Companies

The reality, however, is much more complicated than this. The first and foremost distinction between the Chinese SOEs and their foreign counterparts is the very fact that the Chinese government and the CCP always retain control over the appointment of executive personnel. According to Fu (2016, p. 157), the top three positions (the Chairman of the Board, the Party secretary, and the general manager) of the 53 largest central SOEs were appointed by the CCP's Organisational Department of the Central Party Committee. In the case of the remaining 68 central SOEs, it was always the SASAC that appointed the Party secretary (of the company) and the Chairman (of the Board). The SASAC also appointed other directors, including independent ones, and supervisors, but compared with the Party secretary, other positions were not important.

In practice, the Chinese government and the CCP sometimes exercise their power of appointment in a jaw-dropping way. As McGregor reported (2010, p. 84), in 2004, the CCP Organisational Department decided overnight to rotate the heads of three dominant telecommunications companies (China Mobile, China Unicom and China Telecom, which were just the three being delisted from the NYSE in 2021). Two of the companies were also listed on the New York or Hong Kong stock exchanges. According to McGregor, the rapid expansion of the telecom industries posed a potential threat to the CCP. The rotation was meant to signify the authority of the Party and dissolve any rising power (Wu, 2016, p. 281). In 2009, the CCP again rotated the heads of three state airlines into rival firms.²⁶ In short, these listed companies seemed to be answerable to the CCP, and not to the shareholders. The listed companies' leading positions were just one part of the assignment game of those 'political posts'.

In addition to retaining the power of appointment of key personnel, the Chinese party-state started the 'Party-building' movement along with the 2013 SOE reforms. In fact, Chinese SOEs had long been accommodating the CCP organisation inside their companies, even though its operations were never formally documented, let alone disclosed to the public investors. During the early SOEs' overseas IPOs, investment banks usually chose not to mention this de facto highest decision-making organ (Allen and Li, 2018, p. 23; McGregor, 2010, p. 48).

Party-building vs Corporate Governance

According to Lin and Milhaupt (2020, p. 9), the 2013 'Party-building' initiative was intended to formalise the role of the (Chinese Communist) party

within the companies. The 2015 Guiding Opinions on Deepening State-owned Enterprise Reforms suggested that the ‘Party-building’ would require SOEs to amend the articles of incorporation to specify the leadership of the CCP.²⁷ SOEs were to also establish mechanisms to coordinate the corporate and the Party organisation. ‘In general,’ according to the Guiding Opinions, ‘the positions of the Party secretary and the chairman should be served by the same person.’

The ‘Party-building’ movement can be seen as a counterbalance to ‘the potential loss of party control over the state sector accompanying an increase in private capital investment.’ (Lin and Milhaupt, 2020, p. 9), or the realization of the ‘reform, but without losing Party/state control’ policy (Fu, 2016, p. 159).

It should be noted that the ‘Party-building’ movement is not just an empty policy, for the Chinese party-state has been taking the initiative seriously. For instance, Tianjin Real Estate Development, a local SOE listed in Hong Kong, submitted a proposal to amend its articles of incorporation to align them with the Party-building movement, but failed to garner the necessary two-thirds approval in its shareholders meeting on 6 January 2017. The SASAC thus suspended amendments in SOEs where the state owned less than two-thirds of the shares. In May 2017, Tianjin Real Estate put up the amendment proposal again, and it was passed with a nearly unanimous vote (see Alan and Lin, 2018, p. 47). Something must have happened in the four months between January and May of that year.

Tianjin Real Estate Development’s case also explains the motive underlying the ‘Party-building’ movement. The Hong Kong listing had undermined the state control over the company, and therefore the SASAC stepped in, and cleared the way.

To sum up, since 2013, China’s SOE reform has been moving on dual tracks. On the one hand, the SASAC was designated as a portfolio investor that refrained from meddling in the SOEs operations. On the other hand, the CCP has been enhancing its role within SOEs, maintaining its grip over governance. At first glance, one may wonder whether China is moving in a direction opposite to that of the original corporate reform before 2010. However, if we read the documentation on the CCP’s control over businesses before 2009 (McGregor, 2010), we realise that the 1979–2010 corporate reform had just temporarily gone ‘astray’, and President Xi led it back to the ‘right way’.

Subtle Private Companies in the Future

It has been argued that the private sector and the market economy in China have been the major driving force behind the economic growth over the past 30 years. Today, the private sector accounts for around 70 per cent of the Chinese economy and is the major player in the capital market. Although at first glance, the private companies’ governance is far from the party-state intervention, the risk remains. As observed by Zheng (2017), given

the intertwined nature of the private companies and the government or the CCP, many insider positions in private companies are not based on majority shareholdings. Instead, insiders obtain their controlling position from their ties with the party-state.

The HengFeng Bank case provides an example. The bank is one of the 12 national joint-stock commercial banks in China. In 2013, its largest shareholder was Yantai Blue Sky Investment Development Group Co., Ltd (Blue Sky), a 100 per cent SOE owned by the City of Yantai. Blue Sky controlled 20.55 per cent of HengFeng's shares. The second largest shareholder was the United Overseas Bank of Singapore, which held 14.26 per cent of the shares. The third to fifth largest shareholders were all Chinese private companies. Each held around 7 per cent of the shares.²⁸ The bank was also planning to go to public listing then. In this case, while the government was the largest shareholder, its shareholding was far from constituting a controlling position.

It is thus strange to see that in 2013, Mr Cai was appointed as the director and the chairman of HengFeng Bank because of 'the Party Committee of Yantai City and the City government's decision on recommendation'.²⁹ The appointments did not seem to have gone through the Shareholders' Meeting or the meeting of the Board of Directors, and the private shareholders did not raise any objections. Mr Cai's chairmanship was thus not supported by the majority of the shareholders, but was founded on his close ties with the party-state: he was a member of the Standing Committee of the CCP Yantai Committee and the deputy mayor of the city. In 2017, Mr Cai was arrested due to corruption and sentenced to death in 2020, and HengFeng Bank was also taken over by the central government.³⁰

As Milhaupt and Zheng (2015) argue, the line between SOEs and private companies is blurred in China due to the weak rule of law and other political or economic factors. Thus, equity ownership alone does not reveal the extent to which a given firm is subject to influence by the party-state.

Moreover, the rise of the 'Party-building' movement may worsen the 'Chinese-style insider' problem. As in the case of SOEs, internal CCP organisations have been existing in the context of private companies, but they have been less prevalent, and their function has in general been perceived as being to 'organize social events for Party staff members' (Alan and Lin, 2018, p. 45). The 1993 Corporate Law contained a provision (Article 17) requiring companies to establish a Party organisation based on the Charter of the CCP.³¹ Originally, this provision was not enforced seriously.

Then, a report by the CCP Organisational Department showed that the Party organisation in private companies was underprivileged. The report cited a complaint by one primary CCP cadre that the Party organisation had no money and no power, and not even a chance to speak (McGregor, 2010, p. 217). The 2005 Company Law amended the Party organisation provision by adding a requirement that 'companies shall provide the necessary conditions for the Party organisations to carry out their activities.' However, again this provision was not effectively enforced.

The ‘Party-building’ movement starting in 2013 was primarily focused on SOEs. That is why people were shocked when the CSRC issued the revised Guideline of Corporate Governance of Listed Companies, which explicitly requires all listed companies to set up an internal Party organisation and to provide necessary support to the activities of the organisation (Article 5).³² Actually, one year prior to the revision of the Guideline, the CCP had incorporated the intervention in private companies into its Constitution. Article 33 of the 2017 CCP Constitution provides that

Primary-level Party organisations in non-public sector entities shall implement the Party’s principles and policies, guide and oversee their enterprises’ observance of state laws and regulations, exercise leadership over trade unions, Communist Youth League organisations, and other people’s group organisations, promote unity and cohesion among workers and office staff, safeguard the legitimate rights and interests of all parties, and promote the healthy development of their enterprises.

According to Alan and Lin (2018, p. 45), under the pressure posed by the party-state, Chinese private companies were indeed working on ‘Party-building’. Party organisations in foreign-invested companies more than doubled over the period from 2011 to 2016, from 47,000 to 106,000. Furthermore, around 70 per cent of all foreign-funded firms in China had by then set up Party branches. This is about the same proportion as for China’s private sector, where almost 68 per cent of private businesses had set up Party branches by the end of 2016. Of the above percentage, the denominator includes small and medium-sized businesses; if we focus on influential large businesses, the percentage would be even more shocking.

So far, it is still not clear how exactly the establishment of CCP organisations will interact with the corporate organs. However, the direction of development is worrisome. Moreover, in September 2020, the CCP published the ‘Opinions on Strengthening the United Front Work of the Private Economy in the New Era’,³³ in which it stressed the importance of the Party’s leadership in the private sector and the need to strengthen ‘the ideological and political construction of private economy practitioners’. If we recall the fact that the 2003 Corporate Law was issued following the publication of a general policy encouraging the participation of the private sector, the publication of the Opinions suggests that the policy is moving in the opposite direction.

Conclusions

In this chapter, we have briefly reviewed the corporate reform in China over the past 30 years. We have found that during the first two decades, China was endeavouring to establish a Western-style corporate governance regime to correspond with the liberalisation of the economy. For listed companies in particular, general public investors can hold these companies accountable.

This was a positive development. In the last ten years, however, the reform has moved in the opposite direction, further strengthening the CCP's control over the economy.

To protect general investors, corporate law must be designed to hold company insiders accountable to the outside shareholders. This is particularly important during this era of globalisation, when international companies attempt to raise funds from open financial centres, such as New York, London and Frankfurt. China's party-state regime, however, is eroding this very fundamental axiom of modern corporate law. Needless to say, 'accountable to the (foreign) public' and 'accountable to the CCP' are mutually contradictory notions. All the checks and balances, transparency requirements and remedies under corporate law turn out to be meaningless when the party-state becomes the actual and unanimous company insider.

According to McGregor (2010, Ch. 1), the CEOs of major companies in China must obey instructions from the CCP. Because such party instructions never become public, the corporate governance can never be transparent. This is a fundamental distinction between companies in the democratic regimes and those in China. This distinction resulted in the HFCAA legislation in the US in 2021, and will also possibly lead to similar accountability issues in other international financial markets in the future. Almost by definition, if the CCP is the boss of a listed company in Frankfurt or London, it is hard to believe that this company will be 'transparent' and accountable to the public.

Notes

- 1 Arjun Kharpal, 'Alibaba Shares Dive 7% as Ant Group's Record \$34.5 Billion IPO is Suspended', *CNBC*, 3 November 2020, www.cnbc.com/2020/11/03/ant-group-ipo-in-shanghai-suspended.html.
- 2 MSCI stands for Morgan Stanley Capital International.
- 3 Letter from Eugene Scalia, Secretary of Labor, to Michael Kennedy, Chairman of the FRTIB, https://federalnewsnetwork.com/wp-content/uploads/2020/05/051220_scalia_frtib_letter_FNN.pdf.
- 4 Thrift Savings Plan, 'Board Defers Action on I Fund Transition' (13 May 2020), www.tsp.gov/plan-news/i-fund-transition-defer-2020-05-13/.
- 5 S.945, 116th Congress, 2nd Session (2020).
- 6 PWG on Financial Markets (24 July 2020), 'Report on Protecting United States Investors from Significant Risks from Chinese Companies', *U.S. Department of the Treasury*, <https://home.treasury.gov/system/files/136/PWG-Report-on-Protecting-United-States-Investors-from-Significant-Risks-from-Chinese-Companies.pdf>.
- 7 *Congressional Record*, Vol. 166, No. 203 (2 December 2020), at H6031.
- 8 Public Law No. 116-22 (18 December 2020).
- 9 Chairman Jay Clayton, 'Statement after the Enactment of the Holding Foreign Companies Accountable Act' (18 December 2020), *U.S. Securities and Exchange Commission*, www.sec.gov/news/public-statement/clayton-hfcaa-2020-12.
- 10 White House, 'Executive Order on Addressing the Threat from Securities Investments that Finance Communist Chinese Military Companies' (12 November

- 2020), www.federalregister.gov/documents/2020/11/17/2020-25459/addressing-the-threat-from-securities-investments-that-finance-communist-chinese-military-companies. A list of CCMCs had been initially formulated in June and supplemented in August and December 2020 by the US Department of Defense under Section 1237 of the National Defense Authorization Act for Fiscal Year 1999. See www.defense.gov/Newsroom/Releases/Release/Article/2328894/dod-releases-list-of-additional-companies-in-accordance-with-section-1237-of-fy/. The Office of Foreign Assets Control (OFAC) of the Department of Treasury published a consolidated list of all 35 CCMCs in December. See www.treasury.gov/ofac/downloads/ccmc/ns-ccmc_list.pdf.
- 11 New York Stock Exchange, 'Updated Guidance Related to Compliance with Executive Order 13959 for Issuers of Exchange Traded Products and Closed-End Funds' (7 January 2021), www.nyse.com/trader-update/history#110000332208.
 - 12 For example, see Julian Gewirtz, 'Look Out: Some Chinese Thinkers Are Girding for a "Financial War"', *Politico* (17 December 2019), www.politico.com/news/magazine/2019/12/17/look-out-some-chinese-thinkers-are-girding-for-a-financial-war-086610. See also the editorial 'The Dangers of a US-China Financial War', *Financial Times* (14 May 2020), www.ft.com/content/85bca04a-9529-11ea-abcd-371e24b679ed.
 - 13 Y. Jian, 'China Securities Museum Opens in Century-old Astor House Hotel', *SHINE* (24 December 2018), www.shine.cn/news/metro/1812247124/.
 - 14 China Securities Regulatory Commission, 'Code of Corporate Governance for Listed Companies' (7 January 2001), www.csrc.gov.cn/pub/csrc_en/newsfacts/release/200708/t20070810_69223.html.
 - 15 State Council of China, 'Several Opinions of the State Council on Encouraging, Supporting and Guiding the Development of Individual and Private Economy and Other Non-public Sectors of the Economy' (19 February 2005), www.gov.cn/gongbao/content/2005/content_63162.htm (in Chinese).
 - 16 The big four accounting firms refer to Deloitte, PricewaterhouseCoopers (PwC), KPMG (Klynveld Peat Marwick and Goerdeler, the first characters of the four founders), and Ernst & Young.
 - 17 Chairman Jay Clayton et al., 'Statement on the Vital Role of Audit Quality and Regulatory Access to Audit and Other Information Internationally—Discussion of Current Information Access Challenges with Respect to U.S.-listed Companies with Significant Operations in China', *PCAOB* (7 December 2018), <https://pcaobus.org/News/Speech/Pages/statement-vital-role-audit-quality-regulatory-access-audit-information-internationally.aspx>.
 - 18 China Securities Regulatory Commission, 'Provisions of the China Securities Regulatory Commission, the State Secrecy Bureau and the State Archives Administration on Strengthening Confidentiality and Archives Administration Relating to Overseas Issuance and Listing of Securities' (20 October 2009), www.csrc.gov.cn/pub/newsite/flbg/bmgf/wz/jnss/201012/t20101231_189694.html (in Chinese).
 - 19 PCAOB, 'PCAOB Enters into Enforcement Cooperation Agreement with Chinese Regulators' (24 May 2013), https://pcaobus.org/news-events/news-releases/news-release-detail/pcaob-enters-into-enforcement-cooperation-agreement-with-chinese-regulators_430.
 - 20 PCAOB, 'China-Related Access Challenges' (n.d.), <https://pcaobus.org/International/Pages/China-Related-Access-Challenges.aspx>.

- 21 PCAOB, 'Public Companies that are Audit Clients of PCAOB-registered Firms from Non-U.S. Jurisdictions where the PCAOB is Denied Access to Conduct Inspections' (1 October 2020), <https://pcaobus.org/oversight/international/denied-access-to-inspections>.
 - 22 PCAOB, 'International' (n.d), <https://pcaobus.org/oversight/international>.
 - 23 Luckin Coffee is a Chinese coffeehouse chain with a listing on the Nasdaq since May 2019. In April 2020, the company announced an internal investigation that found its chief operating officer 'fabricated transactions from the second quarter of 2019 to the fourth quarter of 2019', which amounted to around RMB 2.2 billion. See Luckin Coffee, 'Luckin Coffee Announces Formation of Independent Special Committee and Provides Certain Information Related to Ongoing Internal Investigation' (2 April 2020), <https://investor.luckincoffee.com/news-releases/news-release-details/luckin-coffee-announces-formation-independent-special-committee>.
- The U.S.-China Economic and Security Review Commission (2020) indicated that the Luckin Coffee Case illustrated the risk of 'the lack of compliance with international audit inspections'. It pointed out that, 'In presenting information to support its initial public offering, Luckin manipulated critical revenue, operations and customer traffic data. During its IPO, shares traded at \$17 raising \$561 million in capital. Luckin's peak market capitalization was \$12 billion, with shares trading at just over \$50. Within weeks of the disclosure of falsified information, the stock collapsed ultimately leading to losses for investors and its delisting from NASDAQ.'
- 24 Central Committee of the CCP, 'The Decision on Major Issues Concerning Comprehensively Deepening Reforms' (15 November 2013), www.gov.cn/jrzq/2013-11/15/content_2528179.htm.
 - 25 State Council of China, 'Implementation Opinions of the State Council on Advancing the Pilot Program of the Reform of State Capital Investment and Operation Companies' (14 July 2018), www.gov.cn/zhengce/content/2015-11/04/content_10266.htm.
 - 26 A recent instance occurred on 17 January 2020, when a number of positions were rotated among central SOEs and the CCP. Among the shifts, the vice president of China Mobile (Group) was moved to be the general manager of China Telecom (Group), while the China Telecom (Group) vice president was moved to fill the China Mobile (Group) vice president vacancy. A member of the Standing Committee of the Jianxi Provincial CCP Committee was appointed as the chairman of the State Grid Corporation of China, a top-three company of the Forbes 500. The incumbent chairman of the State Grid Corporation was rotated to be the general manager of China Datang Corporation, one of the five major power generation companies in China. See Central News Agency of Taiwan, 'Big Rotate of Leading Positions of China Central SOEs; Many Promotions of Local Party Cadres' (18 January 2020), www.cna.com.tw/news/acn/202001180041.aspx (in Chinese). What should be noted is that the SOEs involved here are all 100 per cent state-owned and so these appointments do not directly give rise to corporate governance problems. However, these 100 per cent SOEs are parent companies of a number of domestic and overseas listed companies, such as China Mobile (a subsidiary of China Mobile (Group)).
 - 27 The CCP and State Council of China, 'Guiding Opinions on Deepening State-owned Enterprise Reforms' (24 August 2015), http://big5.www.gov.cn/gate/big5/www.gov.cn/zhengce/2015-09/13/content_2930440.htm (in Chinese).

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- 29 Ibid., p. 9.
- 30 Reuters Staff, 'China's Hengfeng Bank Ex-chairman Cai Sentenced to Death with Two-year Reprieve' (6 November 2020), *Reuters*, www.reuters.com/article/us-china-crime-hengfeng-idUSKBN27M0LO.
- 31 Article 30 of the Constitution of The Communist Party of China, 'A primary-level Party organization shall be formed in any enterprise, villagers' committee, government organ, school, research institute, subdistrict and community, social organization, company of the People's Liberation Army, and any other primary-level unit where there are three or more full Party members.'
- 32 China Securities Regulatory Commission, 'Code of Corporate Governance for Listed Companies in China' (30 September 2018), www.gov.cn/gongbao/content/2019/content_5363087.htm.
- 33 CCP Central Committee, 'Opinions on Strengthening the United Front Work of the Private Economy in the New Era' (15 September 2020), www.gov.cn/zhengce/2020-09/15/content_5543685.htm.

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6 Antitrust Laws in the Globalisation Era

The Issue of Competitive Neutrality

Introduction

Antitrust law (a term which is used interchangeably with ‘competition law’ in this book) is the legal instrument commonly used to regulate the operation of relevant markets. The vast majority of countries have already enacted their own antitrust legislation. It makes absolute sense for different countries’ antitrust laws to have different regulatory goals to address their own domestic situations, such as promoting efficiency, enhancing consumer welfare, enhancing overall welfare, promoting development, protecting small businesses, protecting exports, supporting industrial policy, or supporting social policies (Bradford et al., 2019). In actual fact, no one can argue that some legislative goals are more appropriate than others, if the respective antitrust laws are enacted and applied to provide a level playing field for all competitors in the relevant markets.

However, it is also true that the sharp differences among countries in terms of their antitrust legislation and practices could artificially create advantages for competitors and undermine the level playing field in the global marketplace. If the disparities due to different antitrust legislation and practices lead to the outcome that the antitrust law in one country (where the antitrust law does not follow the principle of competitive neutrality) is used to take advantage of other countries (where the antitrust law is based on the principle of competitive neutrality), then this should be a matter of serious concern, especially in the globalisation era.

In this chapter, we will briefly introduce the antitrust laws enacted in some selected jurisdictions, including the antitrust legislation in the United States, the European Union and the United Kingdom on the one hand, and China’s Anti-Monopoly Law (AML) on the other hand, to identify their disparities and the sharp contrasts between the legislations in the two different camps. We will then explain that, notwithstanding the natural result that the antitrust laws enacted in different jurisdictions are never identical, they are supposed to follow the principle of competitive neutrality. We will identify the concerns arising from the artificial advantages created by antitrust legislation for their competitors to undermine the level playing field in the global marketplace.

Finally, we will put some thought into possible ways of addressing the competitive neutrality issues arising from the disparities of antitrust legislation, including a possible initiative to formulate a new set of norms to require the enactment and application of antitrust law to be ownership- and nationality-neutral to ensure competitive neutrality and to maintain a level playing field.

Similarities and Disparities Between Antitrust Laws

An Overview of Antitrust Laws and Their Respective Goals

A generally agreed premise is that healthy competition encourages businesses to become more competitive and innovative, and able to offer their goods and services at the favourable prices and qualities. Hence, the core idea of an antitrust policy is to ensure a free and open market for businesses, to expect businesses to act independently of each other, to eliminate undesirable anti-competitive practices and, ultimately, to ensure that consumers enjoy the products and services as the result of better prices and qualities and a broader range to choose from. As pointed out by the United States Federal Trade Commission, ‘Free and open markets are the foundation of a vibrant economy. Aggressive competition among sellers in an open marketplace gives consumers – both individuals and businesses – the benefits of lower prices, higher quality products and services, more choices, and greater innovation.’¹

Broadly speaking, anti-competitive practices include anti-competitive agreements and abuses of a dominant market position. Anti-competitive agreements further include cartels (agreements among competitors to fix prices, rig bids, share markets, or restrict output), collective bargaining and boycotts, exclusive dealings (to restrict the trading counterparts’ ability to decide the content of their deals with others or to choose their business counterparts), and minimum resale prices. They also include concerted practices in which there are either communications or cooperative behaviours between businesses that serve as the basis for them to act in concert and prevent them from acting independently in response to the market conditions. Abuses of a dominant market position include any misuse of market power with the purpose, effect, or likely effect of substantially lessening the competition in a market.²

Antitrust Laws in Jurisdictions with Market-oriented Economies

The majority of countries have their own antitrust legislation. However, the names of these laws vary. For instance, in the US, the three main pieces of antitrust legislation include the Sherman Act of 1890, which outlaws ‘every contract, combination, or conspiracy in restraint of trade,’ and any ‘monopolization, attempted monopolization, or conspiracy or combination to monopolize’; the Federal Trade Commission Act, which bans ‘unfair methods of

competition' and 'unfair or deceptive acts or practices'; and the Clayton Act, which prohibits mergers and acquisitions (M&As) where the effect 'may be substantially to lessen competition, or to tend to create a monopoly', bans certain discriminatory prices, services, and allowances in dealings between merchants (as amended by the Robinson-Patman Act of 1936) and requires companies planning large M&As to notify the government of their plans in advance (as amended by the 1976 Hart-Scott-Rodino Antitrust Improvements Act).³ The United States' antitrust law focuses more on pursuing efficiency to maximise consumer welfare.

In the EU, the antitrust policy is developed from Article 101 (covering horizontal agreements and vertical agreements) and Article 102 (prohibiting businesses with a dominant market position from abusing that position by charging unfair prices, limiting production, or refusing to innovate to the prejudice of consumers) of the Treaty on the Functioning of the European Union (TFEU). Articles 101.1 and 102 of the TFEU read in part:

101.1 The following shall be prohibited as incompatible with the internal market: all agreements ... decisions ... and concerted practices ... which have as their object or effect the prevention, restriction or distortion of competition within the internal market ...

102 Any abuse by one or more undertakings of a dominant position within the internal market or in a substantial part of it shall be prohibited as incompatible with the internal market in so far as it may affect trade between Member States.

The EU's antitrust law also has consumer welfare as its primary objective, although it also has broader goals to foster European integration.

In the UK, the main competition legislation is the Competition Act of 1998, which prohibits agreements between undertakings, decisions by associations of undertakings, or concerted practices which may affect trade within the UK, and have the prevention, restriction, or distortion of competition within the UK as their object or effect, unless they are exempt in accordance with the provisions in the Act. The Act also prohibits abuses in a market of undertakings, including directly or indirectly imposing unfair purchase or selling prices or other unfair trading conditions; limiting production, markets, or technical development to the prejudice of consumers; applying dissimilar conditions to equivalent transactions with other trading parties, thereby placing them at a competitive disadvantage; and making the conclusion of contracts subject to acceptance by the other parties of supplementary obligations which, by their nature or according to commercial usage, have no connection with the subject of the contracts. In addition, merger control is governed by the Enterprise Act 2002, as amended by the Enterprise and Regulatory Reform Act 2013.

Basically, antitrust law in a market-oriented economy is purely designed to correct market distortions. There is no policy goal imposed on antitrust law to

promote distortive effects, such as to provide special advantages to domestic firms or domestic SOEs.

The AML ‘Looks’ Like Antitrust Laws Based on Market-oriented Economies

China adopted its Anti-Monopoly Law in 2007 and brought it into effect in 2008. The AML bears all the ‘hallmarks’ of antitrust laws enacted in most market-oriented economies. The adoption of the AML was therefore welcomed by most countries, mainly because China was perceived ‘like most other countries’ to have ‘recognized that once the market is acknowledged as being the main means by which goods and services are delivered to the community, governments have to ensure that the market works well and this above all requires the market to be competitive or at least not to be hindered by unwarranted anticompetitive structures and behavior’ (Fels, 2012, p. 7).

The AML includes many standard antitrust provisions that can be found in antitrust laws of many other countries. Under the AML, there are four substantive parts, including the prohibition of certain types of monopolistic agreements, with certain specified exemptions (as provided in Chapter 2 of the AML), the prohibition of certain abuses of dominant market position (Chapter 3), establishing a merger review mechanism to address the concentration of firms (Chapter 4), and the prohibition of abuses of government administrative powers which eliminate or restrict competition (Chapter 5).

Notwithstanding the fact that the AML includes many standard antitrust provisions, the stated purposes of the AML provided in Article 1 reveals a clue that the AML is not like the antitrust laws enacted in those jurisdictions where a market-oriented economy prevails. Article 1 of the AML reads: ‘This Law is enacted for the purpose of preventing and restraining monopolistic conducts, protecting fair competition in the market, enhancing economic efficiency, safeguarding the interests of consumers and the public interest and promoting the healthy development of the socialist market economy.’ The elements ‘the public interest’ and ‘the healthy development of the socialist market economy’ are vague and can give the relevant authorities wide discretion in making decisions to distort the market. This will be further elaborated below.

A Level Playing Field and Competitive Neutrality

The Level Playing Field for Global Economic Activities

Under a desirable level playing field in the global marketplace, all countries and firms are supposed to ‘compete on an equal footing to offer consumers everywhere the widest possible choice and the best value for money.’⁴ However, there are various factors affecting the level playing field in international business activities. Examples include border taxes imposed on

imported products, state subsidies and other forms of government support provided for domestic firms, and discriminatory legislation against foreign firms to affect their business activities.

There are also various existing tools to address the distortions to the level playing field caused by these situations. For instance, concerning the border taxes imposed on imported products to affect the foreign products' competitive position in the importing market, the competitive advantage is given by the importing country to its local products by making the imported products more expensive. Such an artificial competitive advantage is addressed by the WTO's tariff negotiations and tariff binding obligations to reduce the trade distortion. Although the removal of tariffs needs to be realised in a progressive manner, at least there are already multilateral schemes to tackle the issue.

As for the subsidies and other forms of support being used by governments to favour particular (or particular groups of) firms, the competitive advantages are given by the subsidising country to its local products by compensating the production costs or by making the products less expensive. The artificial competitive advantages arising from subsidies are addressed by the WTO's rules under the SCM, which makes some subsidies prohibitive and some others actionable. There are many issues concerned with whether the current SCM Agreement is capable of effectively addressing the issues arising from the government subsidies provided to SOEs, and whether SOEs should be considered to be a 'public body' so that the support provided by them is considered as subsidies governed by the SCM Agreement. These are real problems with the SCM Agreement, but are not the main concern of this chapter. More discussion can be found in Chapter 2 of this book.

In addition to this defect, the SCM Agreement only addresses the 'trade aspect' arising from state subsidies. This Agreement does not cover the 'investment aspect' arising from state subsidies. Subsidised capital that is strategically used to acquire firms in another country in important sectors or firms holding important technologies could be a serious threat to the national security of the host state. The issues arising from subsidised capital will be dealt with in Chapter 7 of this book.

In regard to the discriminatory legislation against foreign firms affecting their business activities, international norms are not sufficient either. If a discriminatory piece of legislation is against 'foreign products', the related issues can be addressed by Articles III and X of the GATT 1994. Article III of the GATT 1994 (the national treatment provision) requires that the products of other WTO Members shall be accorded treatment no less favourable than that accorded to like products of national origin in respect of all laws and regulations. In other words, domestic legislation should not be designed or applied so as to discriminate against imported products. Article X requires that WTO Members administer their laws and regulations in a uniform, impartial and reasonable manner. In other words, the application of any domestic legislation should be fair to imported products.

On the other hand, Article XVII of the GATS requires national treatment to be accorded to ‘foreign services and service suppliers’. In this sense, the GATS provides broader protection than the GATT 1994 does, as the GATS may apply to government measures ‘affecting’ foreign firms themselves but not imported goods or services.⁵ However, the GATS’ national treatment obligation is subject to each Member’s specific commitments. In short, no commitment, no obligation (and thus the government may treat foreign firms differently).

In reality, the overall level of national treatment commitment among the WTO Members, especially the developing countries, is limited. It therefore leaves the door open for governments to adopt discriminatory legislation. The insufficient aspect of international norms is that there is no multilateral rule to comprehensively tackle discriminatory legislation or practices against ‘foreign firms’ to ensure a level playing field. This is a matter of particular concern in cases where domestic antitrust legislation is deliberately designed or applied in a discriminatory manner.

Ownership- and Nationality-neutral to Prevent Neutrality Distortion

Anti-competitive practices are measures adopted by individual firms or jointly by two or more firms to create or enhance the difficulty for other firms to enter the market. The fundamental idea of antitrust law is to prevent possible anti-competitive practices so that there will be a level playing field, and to prevent competitive neutrality distortion.

It should be noted that the term ‘competitive neutrality’ is generally used to address the issue that ‘government business activities that are in competition with the private sector should not have a competitive advantage merely by virtue of government ownership and control’ (see Healey, 2014). In this book, however, we use the term based on a broader perspective, to address not only the issue of the competitive advantages granted to SOEs, but also the overall competitive advantages to domestic firms and competitive disadvantages to foreign firms.

In a market-oriented economy, businesses and individuals make their transactions and other economic decisions based on rational commercial considerations, the law of supply and demand, and the interactions between trading counterparts. An economy that operates on a market-oriented basis generally results in better economic outcomes. Within such a market-oriented economy, government interventions are kept at a minimum or at least constrained level. Antitrust laws are basically to ensure that businesses make their transactions and other economic decisions independently of each other and based on commercial considerations so that competition can be enhanced and the market is not distorted.

In order to enhance competition and prevent distortion, an antitrust law should be designed and applied in an ‘ownership- and nationality-neutral’ manner, so that domestic SOEs are also subject to the overall enforcement of

the laws,⁶ and foreign firms are not placed in an artificially disadvantaged position under the law. If an antitrust law is not ‘ownership- and nationality-neutral’, the law not only cannot ensure a level playing field to prevent distortion, but the law itself has already become an ‘anti-competitive’ measure to distort fair competition in the global market. This point will be further elaborated as we introduce the practices in China.

The AML is Distortive Against a Level Playing Field and Competitive Neutrality

The Distortive Aspects of the AML

As mentioned above, China’s AML has many elements which are similar to the antitrust legislation in those jurisdictions where market-oriented economies are in place. However, there are some unique features that make the AML sharply different from other antitrust legislation.

First, as explicitly indicated in Article 1, promoting the ‘socialist market economy’ is one of the AML’s purposes. This is reflected in Deng Xiaoping’s ‘socialism with Chinese characteristics’ reform in the late 1970s. Article 4 further provides that ‘The State shall formulate and implement competition rules which are in accordance with the socialist market economy and improve the macroeconomy and advance a unified, open, competitive and orderly market system.’ The concept of a ‘socialist market economy’ is reflected partly in the wide use of SOEs, which is specifically addressed by the AML. Article 7 states:

Industries controlled by the State-owned economy and relied upon by the national economy and national security or industries implementing exclusive operations and sales in accordance with the law shall be protected by the State to conduct lawful operations by the undertakings. The State shall regulate and control the prices of commodities and services provided by these undertakings and the operation of these undertakings so as to protect the interests of the consumer and facilitate technical progress. The undertakings mentioned above shall operate lawfully, honestly, and faithfully, be strictly self-disciplined, and shall accept public supervision and not use their dominant or exclusive positions to harm the interests of consumers.’

Second, the AML provides a wide spectrum of ‘non-competition factors’ for the authorities to decide whether to allow M&As. Article 27 provides that

The following factors shall be considered in the review of a concentration: (i) the market share in the relevant market of the undertakings concerned and their controlling power over that market; (ii) the degree of concentration in the relevant market; (iii) the influence over access to the

market and technology development; (iv) the influence over consumers and other undertakings; (v) *the influence over national economic development*. [emphasis added]

Article 28 provides in part that ‘However, the anti-monopoly authorities may decide not to prohibit the concentration if the undertakings can prove either that the concentration brings more of a positive effect than a negative effect on competition, or that the concentration is *in harmony with public interests*’ (emphasis added). Two aspects which deserve particular attention include giving the authority discretion to favour domestic firms when foreign companies own technologies which could affect Chinese companies’ effective participation in the competition, and giving exemptions when there are ‘public interests’ involved.⁷

Third, the AML makes the antitrust enforcement a ‘political process’. Article 9 of the AML requires the State Council to establish the Anti-Monopoly Commission (AMC) to organise the investigation, assess the general competition status in the market, and coordinate anti-monopoly enforcement works. The AMC is composed of 19 ministerial- or director-level officials from different ministries or institutions with the responsibility for commerce or industrial policy as its members. As one author commented, ‘The composition of the AMC therefore suggests that enforcement of the AML is in effect a consensus-building process within the Chinese bureaucracy— similar to economic policy- and decision-making in China.’ The involvement of a large number of government actors turns the antitrust enforcement into a ‘political process’ and makes the outcome highly uncertain (Zhang, 2014).

The AML’s Designs for SOEs to Gain Advantages

After some years of implementation, many countries realised that China’s AML was not what they originally and optimistically considered to be, that is, like the antitrust law under the market-oriented economies, when it was first adopted. There are significant overall effects arising from the above-mentioned unique features of the AML, such as the promotion of the socialist market economy, the protection by the state of the industries controlled by the state-owned economy, the influence over access to the market and technology development, and the non-prohibition of concentration for the public interest. Such effects include vastly favouring SOEs, weakening foreign multinational companies (MNCs), and forcing MNCs to transfer their technologies to Chinese firms.

As regards SOEs being singled out by the AML for special protection, it is apparent that China considers that the AML (together with other policies and laws in China) must support its SOEs to compete internationally. In this regard, one author has commented that

SOEs continue to control all important strategic industries in China, such as oil and gas exploration, banking, transportation (including air and rail

transport), telecommunications, and electricity supply. SOEs also have access to easy credit from China's state-owned banks. Many SOEs are now among the largest companies in the world. China's long-term goal is to create SOEs that are 'national champions' that can compete with the world's largest MNCs.

(Chow, 2016, p. 103)

Another author commented that

The SOEs have a built-in preferential position that can manifest itself in two ways. Either they can get gentle slaps on the wrist for offenses that prompt far harsher sanctions against private companies, especially foreign companies who are either suppliers or competitors with SOEs, or the SOEs could prod Chinese anti-monopoly enforcement authorities to take actions against their foreign competitors. The AML can all too easily function as a new form of protectionism by virtue of its differential application to foreign firms vis-à-vis SOEs doing business in China.

(Epstein, 2014, p. 4)

AML's Designs are Disadvantageous to MNCs

As for the AML being used to weaken foreign MNCs and to force MNCs to transfer their technologies to Chinese firms, an author has commented that it could be China's intention to have such undesirable results and seeks to remind us that there is a need to address such a problem:

China seems intent on using the AML to strengthen the position of SOEs and to weaken the position of MNCs, their main competitors in China as well as on the worldwide stage. MNCs are viewed as possessing 'huge advantages in technology, scale, capital, etc. It is easy for them to gain a competitive edge, even monopoly positions, in the market. ... There is a need to impose countermeasures to regulate multinationals' anticompetitive conduct. There is a general sense in the state-controlled media that greedy MNCs are able to take advantage in many different areas of China's still developing legal and economic system to gain various advantages over Chinese companies.

(Chow, 2016, p. 103)

What is worse is the 'oft-concealed enforcement practices' by government agencies, which are given very broad inspection and investigation powers under Article 39 of the AML that allows them to enter the business premises of business operators under investigation, or to force the investigated business operators to respond to various forms of interrogation 'to explain the relevant conditions' to the public authorities. The government authorities also have

broad powers to examine, duplicate, seize and examine all business documents, bank records and accounts. The only ineffective procedural ‘protection’ for the investigated firms in Article 39 is a ‘written report’ to be submitted to the chief person-in-charge of the anti-monopoly authority (Epstein, 2014). Other allegations of the Chinese practice deviating from the basic requirements of due process and transparency include ‘pressure to ‘admit guilt’ without the ability to see and respond to evidence; inability to have appropriate legal representation present for unannounced, on-site investigations known as ‘dawn raids’; inability to have appropriate legal representation in ongoing proceedings; insufficient transparency during competition reviews; insufficient transparency in publishing case decisions’.⁸

In addition, the prolonged merger notification procedure also greatly hinders MNCs’ international merger activities. In this regard, it was remarked by one author that:

the merger notification process in China is notoriously protracted. Large multinational companies increasingly find that their merger transactions are held up by Anti-Monopoly Bureau of the Ministry of Commerce’s (MOFCOM) clearance decisions. For instance, in Google/Motorola, China was the last jurisdiction to clear the transaction. In fact, MOFCOM did not clear the transaction until the last day of the statutory review period. In some cases where merging parties had not reached a desirable outcome with MOFCOM, the agency required or encouraged the parties to withdraw their filings and re-file. Thus, in some cases, MOFCOM’s review time significantly exceeds the statutory review period.

(Zhang, 2014, pp. 681–682)

Helping Giant SOEs to Meet the National Outward Looking Policy

In the early 2000s, China launched its ‘going global policy’ to encourage its firms to engage in global competition (Bellabona and Spigarelli, 2007, p. 93). In 2015, China announced its ‘Vision and Actions on Jointly Building the Silk Road Economic Belt and 21st-Century Maritime Silk Road’ (the Belt-and-Road Initiative)⁹ to intensively engage in economic activities throughout a very large portion of the whole of Asia and Europe. In addition, as cited, Xi Jinping declared in 2016 that ‘SOEs should become “stronger, better, and bigger.”’ Also, in 2017, China amended its Constitution to assert that the CCP ‘plays a leadership role’ in the firms’ decision making.¹⁰

The AML definitely must give way to the national policies of ‘going global’ and the ‘Belt-and-Road Initiative’, the constitutional requirement of the CCP’s leadership role in the firms’ decision making, and Xi Jinping’s declaration requiring SOEs to become ‘stronger, better, and bigger’, although, as reported by some articles,¹¹ the AML still randomly applies to SOEs.

In recent decades, China has established an enormous number of SOEs, many of which are listed in the Fortune Global 500 (FG500). It has been calculated that

Chinese SOEs represent a significant portion of FG500 SOEs. In particular, in 2000, there were 27 FG500 SOEs, of which 9 were from China; in 2017, 75 of the 102 FG500 SOEs were from China ... In addition to the 75 giant SOEs in [the] FG500, there are more than 150,000 SOEs in China. China's SOE sector has grown significantly (in the number and total market capitalization of SOEs) since China's launch of [the] marketization reform in 1978.

(Lin et al., 2020, pp. 31–32)

At the present time, 'Three of the world's five largest companies are Chinese (Sinopec Group, State Grid, and China National Petroleum). China's largest SOEs hold dominant market positions in many of the most critical and strategic industries, from energy to shipping to rare earths.'¹² China's SOEs have exerted great influence in the global market. In actual fact, it is the Chinese government which strongly and effectively utilises its political control over its SOEs.

There is no other country in the world that is so intensively establishing SOEs to engage in domestic and international economic activities. This also gives China a strategic competitive advantage in other countries, where SOEs do not exist for most economic activities or do not play such an enormous role in the marketplace. The lenient rules for SOEs and outward-looking firms provide advantageous conditions for them. The AML's role in nurturing giant Chinese SOEs and other private companies to engage in competition in the international marketplace is extremely harmful to the level playing field.

An Example: Huawei vs Qualcomm

The well-known initiative *Made in China 2025* was announced in 2015, in which ten industries were identified and expected to become world leading players by 2025. Among these ten industries, the one which raised the most attention and controversy is perhaps telecommunications, the main reason being that most of the US entity list sanctions announced in 2018–20 were targeted at the telecommunications industry. The US sanction targets include ZTE, Huawei and the many suppliers which provide components and parts to these companies. As for the future telecommunications industries such as those engaged in 5G or 6G and consumer electronics, the leading Chinese company generally identified is Huawei.¹³ However, if one tries to identify a US or European company that can be treated as a typical telecommunications company, it would be very difficult, because the telecommunications industry is very much fragmented. In the US, for instance, if we really try to identify just one company which is competing with Huawei, it may be Qualcomm.

Qualcomm is basically a chip design and software company, which is only a small part of the telecommunications picture. These chips and software are installed in cell site stations, terminal devices such as smart phones or pad computers, and Wi-Fis and satellites in order to facilitate communication. In the US or even in the US plus the EU, the terminal devices, satellites, consumer electronics and site stations are made by different companies, such as Apple, Motorola, Ericsson, Orbital Sciences Corporation (OSC), etc. Furthermore, none of the above companies provides a ‘cloud’ service. However, when we enter into the official website of Huawei, we see that ‘it has everything’ (see Figure 6.1), from chips design to chip manufacturing, to terminal devices to clouds and data management, and to site station products. In the telecommunications industry, this is a typical example of complex vertical and horizontal integration.

Theoretically, vertical and horizontal integration as in Huawei’s case may increase operational efficiency. For instance, whether or not a new design

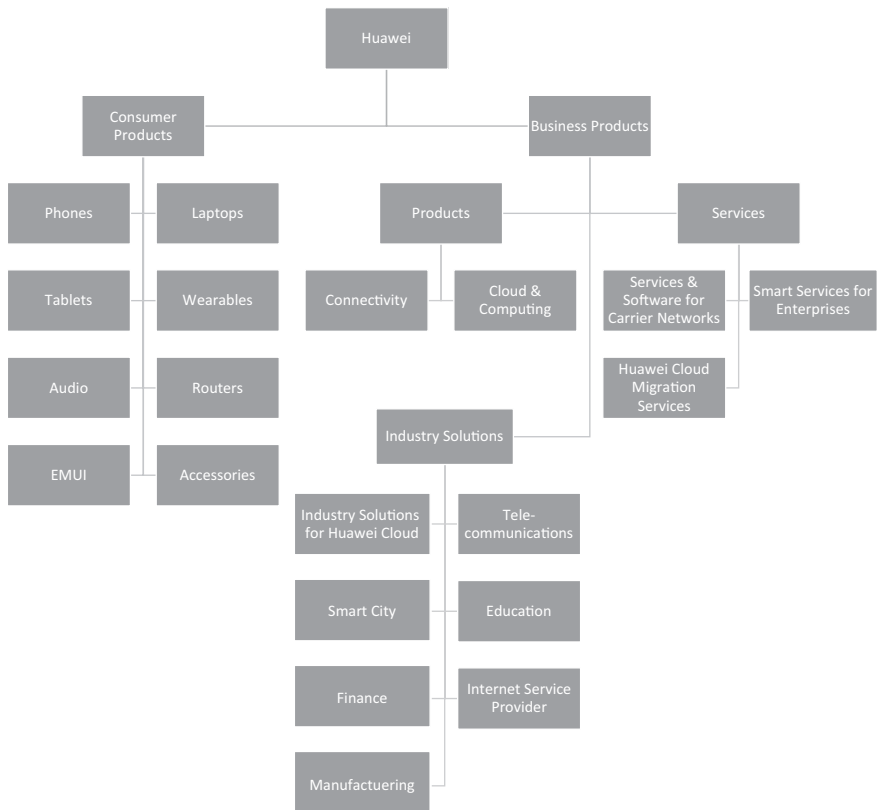


Figure 6.1 Horizontal and vertical ICT map of Huawei

Source: www.huawei.com/en/

of chip has any bugs can be easily tested in-house by integrated companies in the same group, and these bugs can be effectively fed back, modified and removed more efficiently. Furthermore, these integrations may also put the devices and site-station products in a better position to compete with other companies' (domestic or foreign) products, which goes against the spirit of the antitrust law.

In the US, for instance, it would be unimaginable if Qualcomm, Apple, Motorola and OSC were to merge. In fact, even if they were to file an application to the FTC, it would be unimaginable for the FTC to approve the case. In practice, leaving aside the possible national security issues, the 5G or 6G competition is indeed a competition between a consolidated Huawei and at least half a dozen separate US and European companies. This example illustrates the unlevel playing field if we do not consider the international competition aspects of antitrust laws. In short, the US or EU antitrust laws simply cannot deal with the unfair competition from a foreign company which has been highly (both vertically and horizontally) integrated. As long as the telecommunications service is something the US and EU have agreed to open to other countries, Huawei's competition attempts cannot be denied, according to the WTO rules.

In recent years, there have been more and more criticisms concerning the monopoly position of the largest three US companies related to the Internet, namely, Facebook, Google and Amazon. According to Wu (2018), a business that is too big would damage the checks and balances, and the political balance in general. Ironically, even if many people challenge Amazon's monopoly position, for example, we have heard relatively few challenges concerning the threat from Alibaba, even though it is an even stronger and bigger monopoly in the field of e-commerce. Many scholars still treat the antitrust law as a *domestic* law, neglecting the aspects of international competition neutrality.

Needing a New Framework to Ensure Competitive Neutrality

The Adverse Effect of Anti-competitive AML

Antitrust law is supposed to be designed and used to remove undesirable anti-competitive practices. However, when an antitrust law itself has become anti-competitive, the consequences are multiple. From the perspective of competitors, the obvious consequences are that Chinese SOEs are protected and given artificial competitive advantages, and that foreign firms are discriminated against. From the states' perspective, the antitrust law with its anti-competitive nature is also against the economic interests of other countries. They definitely need to consider adopting an approach to avoid or to deter such legislation or its resulting adverse effects.

One commentator has warned of the likely risk of other countries adopting a similar legislative approach to counteract or retaliate against the abusive legislation:

The differential enforcement of the anti-monopoly laws poses major dangers in this regard, for the same laws that protect against anticompetitive practices are all too often used to achieve the very abuse that they are intended to guard against. Commissioner Ohlhausen bluntly puts the point: ‘Critics claim that China is using its antitrust law to promote industrial policy.’ The unfortunate situation in China is but one example of that dangerous set of practices, which unchecked could spread to other countries, motivated either by imitating what China has done or retaliating against its abuses. The risk is that the disease can spread all too easily. Other nations can protest against these practices.

(Epstein, 2014, p. 9)

To ‘Retroactively’ Correct ‘Non-neutrality’?

As explained above, the distortion of competitive neutrality comes as the result of many forms of government intervention, such as import tariffs, state subsidies and domestic discriminatory laws. There have been some mechanisms to address these issues, although the mechanisms are not complete or sufficiently adequate. However, ways to improve the mechanisms are at least discussed. For instance, there have been discussions at the WTO on how to reform the SCM Agreement to address the subsidies provided to SOEs and those provided by SOEs. However, there has not been any discussion on how to systemically address the anti-competitive effect arising from the anti-trust law itself.

There are different options for countries to consider. One option is to adopt practices similar to China’s approach in its AML, so as to equalise the competitive positions of their firms. However, such a ‘clawback’ approach is undesirable, because it would fundamentally change the nature and the original function of the antitrust laws to enhance and ensure competition. Such ‘clawback’ legislation is also against the spirit of a market-oriented economy.

Another option is to have a new set of international norms to govern anti-trust laws. In the past, there were discussions on whether and how to include WTO rules to address anti-competitive practices. For instance, the Doha Ministerial Declaration of 2001 includes a statement in paragraph 23 indicating that

Recognizing the case for a multilateral framework to enhance the contribution of competition policy to international trade and development, and the need for enhanced technical assistance and capacity-building in this area as referred to in paragraph 24, we agree that negotiations will take place after the Fifth Session of the Ministerial Conference on the basis of a decision to be taken, by explicit consensus, at that session on modalities of negotiations.¹⁴

After the Doha Declaration, WTO Members set up a working group to clarify 'core principles including transparency, non-discrimination and procedural fairness, and provisions on "hardcore" cartels (i.e., groups of companies that secretly fix prices); ways of handling voluntary cooperation on competition policy among WTO Member governments; support for progressive reinforcement of competition institutions in developing countries through capacity building.'¹⁵ More specifically, there were discussions on a possible WTO agreement on competition policy.¹⁶ However, later on, Members were not in agreement on continuing the negotiation. The idea of concluding a multilateral agreement under the WTO ended without any concrete results.

We are not arguing for a resumption of a round of negotiations on concluding a multilateral agreement on competition policy to address various kinds of private anti-competitive practices. Instead, we bring the distortive effect arising from the antitrust law itself to the awareness of countries and suggest having a collective effort to address the issue. We also expect that certain appropriate proposals at a multilateral level that are aimed at concluding a set of norms will be initiated.

Such a set of multilateral norms should require the antitrust laws of the parties to be of competitive neutrality, and can be established either within or outside the WTO. However, according to the experience of the past 26 years of the WTO's history, the difficulty in negotiating and concluding a new agreement under the WTO is extremely high. Hence, preferably, such multilateral norms can be established outside the WTO. The specific issues to be covered should include the rules to require the enactment and application of antitrust laws to be ownership- and nationality-neutral to ensure competitive neutrality and to maintain a level playing field for domestic and foreign firms. The multilateral agreement should include as many countries as possible as its parties, so that most countries follow the principle of competitive neutrality in regard to its antitrust laws and so that there will be minimum leaks arising from countries not joining the agreement.

Furthermore, there could be collective measures included in the multilateral agreement to address the issues arising from the non-participation of those countries whose antitrust laws are themselves anti-competitive in the sense that they discriminate against foreign firms and benefit local SOEs.

Finally, we also suggest that the above-mentioned multilateral rules should 'retroactively' address the 'already existing' unfairness caused by the competitive 'non-neutrality' of an antitrust law. Because, even if there is a set of multilateral norms to require the antitrust laws of the parties to be of competitive neutrality, such norms are merely to ensure the level playing field 'prospectively', so that in future, anti-competitive laws will be of competitive neutrality. It would not be able to correct the existing unfairness already caused by the non-neutrality of antitrust law, if no specific scheme is created to address such a problem. As we indicate in this chapter, for instance, Huawei has already completed its horizontal and vertical mergers and has become an uncontrollable gigantic company able to compete in telecom services internationally.

The result is partly contributed to by the inactive enforcement of the AML in China for the purpose of preventing a monopoly in the market. If there is a set of norms to require competitive neutrality of antitrust law and, even if China is bound by the new set of norms, Huawei will still enjoy such unfair advantages arising from the previous non-neutral nature of the AML. We argue that there must be a mechanism to be built into the suggested multilateral rules so as to ‘retroactively’ correct the structural and systemic unfairness caused by the previous competitive ‘non-neutrality’ of antitrust laws of some countries.

Conclusions

In this era of globalisation, not only are firms competing against one another, but domestic legislation also greatly affects the business activities of domestic and foreign firms. It is a matter of course that countries will have their own sovereign rights to decide the content of their own legislation. However, there is also a serious concern arising from countries enacting their legislation in such a way to systemically favour domestic firms and discriminate against foreign ones.

It is especially undesirable for a country to use its antitrust law to grant substantial favours to domestic firms, or to take advantage of firms from other countries resulting in a distortion of competition in the global marketplace. The reason is abundantly clear. Antitrust law is supposed to ensure proper competition. It is not a legitimate tool to give special advantages to domestic firms or SOEs, or to discriminate against or take advantage of foreign firms.

We argue that there is a need to establish a set of multilateral norms to require the enactment and application of antitrust law to be ownership- and nationality-neutral to ensure competitive neutrality and to maintain a level playing field for domestic and foreign firms. And, very importantly, in the suggested multilateral rules, there must be a mechanism to ‘retroactively’ correct the structural and systemic unfairness caused by the previous competitive ‘non-neutrality’ of antitrust laws. We hope that our argument will serve as a basis for such an initiative.

Notes

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- 2 For a general explanation, see the Australian Competition and Consumer Commission (ACCC) ‘Anti-competitive Behavior’, www.accc.gov.au/business/anti-competitive-behaviour.
- 3 FTC, ‘The Antitrust Laws’, www.ftc.gov/tips-advice/competition-guidance/guide-antitrust-laws/antitrust-laws.
- 4 OECD, ‘Levelling the Playing Field: Competition in the Global Market Needs to be Fair’, see www.oecd.org/trade/topics/levelling-the-playing-field/.

- 5 According to the WTO jurisprudence, the GATS and the GATT 1994 are not mutually exclusive. Thus, a government measure subject to the GATT 1994 because of affecting imported goods may also be subject to the GATS because of affecting supply or distribution services. See WTO Appellate Body Report, *EC – Bananas III*, para. 221.
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7 Cross-border M&A Control

The Issue of State Capitalism

Introduction

A foreign firm entering a market of the host state often takes the form of a cross-border M&A. In addition to the purpose of entering the new market, foreign acquiring firms might also have the objectives of gaining access to local resources, absorbing a competitor, or having access to important technologies and know-how that the target company holds.¹

M&As conducted by foreign firms could be subject to the control of anti-trust laws and foreign investment reviews. Traditionally, antitrust controls and investment reviews in relation to cross-border M&As have focused on the market structure and competition effect. However, over the past several decades, China has adopted state-controlled capitalism and used its SOEs and massive subsidisation policies as tools to venture abroad to acquire firms in critical sectors or ones that hold critical technologies. As for the host states, such kinds of M&A activity challenge the regulatory architecture of many countries and give rise to national security concerns. The national security anxieties resulting from transnational acquisitions have existed for decades (Malawer, 2006), but recently they have emerged as more critical issues.

Antitrust laws and foreign investment reviews were designed based on the principles of the market-oriented economy. They are not sufficient to effectively address security concerns if no new mechanisms are introduced. Although in the past, the relevant issues emerged from time to time in anti-trust law enforcement, national security was not considered within a systemic framework; it was touched upon only sporadically. However, in those sporadic cases, the national security concerns that emerged were quite different from the national security concerns that many countries are facing today.

The current issues involve the systemic impacts that certain foreign investment has on the operation of a market-oriented economy (for example, a country's massive use of SOEs and government-subsidised capital to acquire businesses in other countries) and the fundamental adverse implications of foreign investment for an open economic and social system (for example, the foreign investor or its home-state government's exerting stringent control over the media sectors). The concerns are thus becoming more serious than those

that emerged in earlier times. The new trend of regulatory reform is to incorporate the national security elements in the cross-border investment review. The question remains whether merely introducing the national security elements in the foreign investment review is sufficient, or whether the same elements should also be systematically considered under the antitrust law enforcement. In this regard, one of the fundamental issues here is to address ‘the conflict between the demand of national security and an open foreign investment policy’ (Byrne, 2006, p. 850).

To address these issues, this chapter will discuss how FDIs are different for authoritarian and democratic countries; why foreign SOEs, foreign subsidised capital, foreign ownership of certain sectors, and foreign control of certain new technologies pose threats to national security when the related foreign firms engage in cross-border M&As; why and to what extent the antitrust and investment laws have a critical role to play in addressing such threats, and what would be the complementary roles between the two laws.

It should be noted that we do not promote any transformation of anti-trust law into national security legislation. In actual fact, antitrust law needs to continue to serve its purpose as an economic norm to prevent distortive market effects arising from cross-border M&As and ensuring that they have a positive contribution to the economy. However, we argue that, due to the massive use of SOEs, the overwhelming subsidisation of capital by some countries in acquiring foreign enterprises, the possibility of critical sectors and new technologies being acquired by hostile state governments under the mask of commercial activities, and the fact that failure to minimise the related national security risks could have a huge and adverse security-related economic impact, there are good reasons for incorporating a systematic analysis of national security within the M&A review.

National Security Risks due to Differences in Economic Structure

Party-state Capitalism vs Market-oriented Economy

There are various types of economic systems. Many countries have their economic operations based on the concept of a free market. In these countries, the market functions based on supply and demand. Although governments still intervene to a certain extent, they tend to refrain from systematically interfering with the operation of the market. In such markets, antitrust law is essential to ensure that there is sufficient competition and a fair opportunity to make transactions. On the other hand, certain other countries adopt a command economic system to control a significant portion of resources and sectors. Many such controls are exercised through their SOEs.

Take China as an example: It was mentioned in Chapter 6 that, in recent decades, China has established an enormous number of internationally huge SOEs. We will argue that this strategy has fundamentally and structurally changed the function of the market-oriented economy, with the target or

effect of permanently dominating the market. This is one of the reasons why the other countries are becoming increasingly worried about the potential national security threats caused thereby. The following comment highlights the problem very clearly:²

The U.S. government, for the most part, does not hold an ownership stake, either wholly or strategically, in technologies, industries, infrastructure, resources, and businesses of national importance. U.S. telecommunications infrastructure is entirely in private-sector hands; our leading defense and aerospace suppliers are publicly traded entities, but not 'public' entities in the sense of government ownership; and even our 'national laboratories' are federally funded but managed principally by private-sector contractors.

This model served the United States well from the stance of innovation and economic efficiency, and, following the U.S. success, many Western democracies also privatized many of their national assets starting in the 1980s and continuing into the twenty-first century...

A strong and dynamic economy based on private market ownership and principles has given us great power in the world and fueled the technological innovation that ensures the dominance of U.S. forces on the battlefield. This state of affairs, however, also leaves the United States potentially more vulnerable than other countries, because our national security, both defensively and offensively, is inherently more dependent on the private sector ... Hence, the federal government has the need and the desire to regulate commercial activities related to national security.

The National Security Issue Deserves More Attention

The result of better economic outcomes for the market-oriented economy is based on the premise that there are no adverse external interventions in the market's operation. If an adverse external intervention has caused a national security threat and no mechanisms are built up in response to such a threat, a market-oriented economy could become vulnerable and be subject to the undesirable distortions caused by massive foreign subsidised capital, foreign governments' control of the merged firms, or their control over critical IP rights and technologies. It is thus essential to introduce a stronger national security screening mechanism under the legal regimes of antitrust and M&A laws.

The need for having a stronger national security screening mechanism to address cross-border M&As was not very well recognised until recent years. The reasons for the delayed recognition of national security issues are multiple. One of the reasons comes from the late awareness that China itself is acting as a party-state capitalist and it continues to develop its enormous capitalistic capabilities. Another reason is that foreign investments (either through M&A activity or other methods) were traditionally made from

friendly, developed countries; however, in the globalisation era, investments may come from everywhere.

As an OECD report indicates:

Most countries have had some arrangement to review, assess, and address potential national security risks arising from specific FDI for decades. However, as international investment largely took place among allies and the ratio of FDI to GDP was much lower than it is today – only 7% in 1990 against around 40% now – the amount of FDI covered by such mechanisms was likely relatively low. Correspondingly, the mechanisms enjoyed limited attention in most countries.³

Another reason why the concern is more complicated today is that, in the globalisation era, some international standards commonly applied to all countries are becoming more and more important. Take telecommunications as an example. The 5G or 6G standard involves core chip designs and base-station facilities. Once a particular standard is adopted, the chips and facilities are needed globally, and this implies huge profits. Moreover, the future of telecommunications is related to accurate satellite positioning, and hence possible espionage and the guidance of missiles could be involved. The national security concerns here are evident.

Apparently, the need to take national security considerations into screening international M&A activities deserves much more attention at the present time. Nevertheless, it must be noted that building a stronger national security review mechanism in the investment review of cross-border M&As does not imply that the market economy should be modified. On the contrary, the market-oriented economy could be strengthened by preventing the fundamental threats to the system arising from foreign cross-border M&As, especially those generated by party-state capitalism.

Different FDI Patterns – Some Examples

It may be useful to use some real-life examples to highlight the patterns of FDI sourced from China vs those from democratic countries. We will pick several large-amount FDI from democratic countries to China, and also introduce the huge FDI from China to the rest of the world: the Belt and Road Initiative (BRI). From these examples we can identify vivid distinctions in the FDI strategies.

Investments from Democratic Countries to China⁴

Starting from the most recent ones, TSMC, the largest chip manufacturer in the world and based in Taiwan, announced in 2021 that they would invest \$21 billion to expand capacity in their factory in Nanjing, China. The Nanjing factory uses 12 nanometre (nm) technology to produce chips mainly for

moving vehicles, whereas the most advanced technology adopted in Taiwan is 3 nm used in advanced ICT products.

Audi, the largest German-based car manufacturer in China, announced in the beginning of 2021 that they would partner with the FAW group of China to produce electric vehicles. The project was expected to enter into production by 2024. This project was the first one where a foreign company (instead of the Chinese partner) held the majority shares (60 per cent) in a joint venture. Other than Audi, the other two German car brands, Mercedes Benz and BMW, also heavily invested in China, despite rising US–China tension. The three largest German car manufacturers had 30–40 per cent of their total world sales in China, and that was why their investments in China had always been so intensive. As well as German cars, Toyota of Japan also cooperated with the FAW group at Tianjin, and invested \$1.2 billion in electric vehicle production.

Investments from the US to China have evidently slowed down in recent years, especially after the US–China tension escalated during the Trump administration. A recent major investment by General Electric was announced in 2010, stating that the US manufacturer would invest \$2 billion in China to boost innovation and customer services. Ford Motor's recent investment was made in 2015, targeting at building R&D capacity. Tesla announced in 2018 that they planned to invest \$5 billion to build their Chinese factory. Tesla said that this plan was in response to the trade war between the US and China; the FDI in China should save Tesla from the cost of escalating tariffs that China imposed on American goods.

In general, FDIs by companies based in Europe are less affected by the recent US–China tension. The car industry is an extreme example, because China is the largest vehicle market in the world, and also because European mechanical engineering is less dependent on US technologies. The car industry is also less involved with sensitive information technologies, which are dominated by US companies. This basically describes the industry spectrum of FDIs in China.

China's Belt and Road Initiative⁵

The most typical outbound FDI sourced from China is, undoubtedly, the BRI. The project was announced in 2013 by Xi Jinping, and thereafter promoted by Premier Li Keqiang. The 'Belt' refers to the Silk Road economic belt, and the 'Road' is short for the 21st-century Maritime Silk Road, covering the Indo-Pacific sea routes.

Excluding China, this project was estimated to need \$900 billion for infrastructure investment in Asia each year, much more substantial than the above-mentioned inbound FDIs from democratic countries. The financing of BRI was mainly through the Asian Infrastructure Investment Bank (AIIB). As of 2015, China had announced that infrastructure-related projects costing over \$160 billion are undergoing planning or construction. The infrastructure

items being invested include ports, skyscrapers, railroads, roads, airports, dams, coal-fired power stations, railroad tunnels, and also ‘soft’ infrastructure, such as the coordination of talent pools and technology databases. Soon after the BRI’s launching, China has become the largest bilateral lender in the world. Loans are usually backed by collaterals, such as rights to a mine, a port, or money.

The leader of this project is Zhang Gaoli, who is also a member of the seven-member CCP Politburo Standing Committee, and four other political heavyweights were named deputy leaders. The steering committee reports directly to the State Council of China. The project was scheduled to end by 2049, which will coincide with the centennial anniversary of the People’s Republic of China’s founding.

BRI has had some relatively more enthusiastic participants in the world. In Asia, these include Kazakhstan, Pakistan, Indonesia, Singapore, Laos, Malaysia and Myanmar. In Africa, they include Kenya, Nigeria, South Sudan, Ethiopia, Egypt, Tanzania, Ghana and Zambia. In Europe, Italy and Greece are heavily involved. Moreover, Russia and China initiated the Ice Silk Road project, which promotes development in the Arctic, and complements the original BRI.

Comparisons of Investment Patterns

From the above descriptions, we can identify the difference in FDI patterns from democratic countries to China vs those outbound from China. First, in democratic countries, since their businesses are mostly private, the corresponding FDI decisions are business ones: companies try to minimise costs or maximise revenues through such foreign investments. For China, however, since most SOEs are controlled or owned by the party or the government, the orientation of FDIs are mostly political. The BRI is an extreme example, with the investment amount astonishingly huge, and all decisions and steering are politically oriented and controlled.

Second, FDIs from democratic countries to China are often for specific economic purposes, such as expanding factory capacity or enhancing R&D activities; whereas FDIs sourced from China are mostly investments in infrastructures, which individual businesses would not do. Indeed, China’s BRI plays the role of the World Bank or International Monetary Fund, which facilitate the infrastructure construction of various countries; the term ‘Belt and Road’ is just a geographic description of the fund-receiving countries in the world. The difference between BRI and the conventional World Bank loans is about the collaterals behind such loans. Under BRI, collaterals sometimes (when loans cannot be repaid) ended up with seizing the assets of the target country, such as the Hambantota Port as a Chinese colony.

Third, using the framework of Christensen et al. (2019), we can say that most FDIs from democratic countries to China are *demand-pulled*, meaning that such investments are driven by real business demand; whereas most

China-sourced FDIs are *supply-pushed*, meaning that such infrastructure investments are top-down from the government. In the former case, the government's role is limited to requested supports from the investors, but in the latter case, the bilateral governments (China and the host country) are the key players, and the demand of the infrastructure is at most an estimate figure. Because such investments are top-down, citizens and private sectors are less involved and informed. Domestic oppositions sometimes end up in the form of antagonism against foreign countries.

Fourth, because the steering committee and leaders of the BRI are all political heavyweights, it is hard to imagine that these investments do not have political objectives. Thus, BRI has aroused opposition in the US, India and several other sensitive countries. From the viewpoint of some US scholars, BRI is a debt trap and evidence of neo-imperialism, in an attempt to extend the influence of China. Viewing investments as a political initiative that extends China's diplomatic influence causes fundamental doubts on such projects. This is also why review of FDIs, especially those sourced from China, as we will describe below, is receiving more and more political scrutiny.

National Security Risks Related to M&As

Countries might have different criteria for identifying a national security risk. In this part of the chapter, other than the above-mentioned commercial areas that are directly and evidently related to national security (such as 5G or 6G), we identify four main areas where the risks to national security interests could become serious in relation to foreign cross-border M&As. They include the overwhelming subsidisation of capital, the massive and strategic use of SOEs in acquiring enterprises, the possibility of critical sectors of infrastructure, and the new technologies being acquired by unfriendly foreign firms. These four areas of concern are interrelated. They are intertwined with each other in that foreign SOEs could receive massive government subsidies, which vastly empower them to acquire critical sectors of infrastructures and even critical technologies.

China's SOEs as a National Security Issue

As a matter of fact, SOEs are created and maintained in both market economies and non-market economies. SOEs per se are not a problem. Many countries create SOEs to provide public goods (such as national defence) and merit goods (such as public health and education), or to increase their population's access to public services. The creation of SOEs could also occur because the relevant markets are imperfect or unable to accomplish critical societal needs.⁶ These are all positive purposes for establishing SOEs.

However, when the overseas activities of a country's SOEs are designed and implemented in a systemic and large-scale manner at a national level, they could become a concern. As explained above, China has established an

enormous number of SOEs not only to engage in domestic economic activities, but also to enter into global markets. This leads to a fundamental and structural change in the function of the market-oriented economy. State capitalism has been the policy of China. For instance, Chinese Premier Li Keqiang's 2018 announcement of the *Made in China 2025* initiative is designed to help many Chinese SOEs compete more effectively in overseas markets as well as improve their high-end export capability.⁷

State capitalism has given rise to wide-ranging concerns. The SOEs' massive-scale, outward-looking approach is exemplified in an OECD report, which states:

Overseas activities may be formulated at national level. One example is China where an internationalization of the SOE sector mostly through foreign investment was codified in the Central Committee's 'Go Global Strategy' in 2000. In addition to the acquisition of resources, central goals of this strategy include (1) the building of competitiveness through acquisitions of high technology, know-how and brand power; and (2) an increasing reliance on overseas production, as opposed to direct exports.⁸

A foreign government which systematically and massively promotes its SOEs to engage in acquisitions and investment is actually using the market position created by its SOEs in the host state to unfairly engage on a huge scale in competition in the market-oriented economy, and even to control or dominate critical sectors or technologies. Such a strategic and overwhelming outward-looking approach for SOEs cannot be overlooked by the legal regimes of countries which are based on market-oriented economies.

In addition, foreign SOEs' investments and operations in a market also involve national security concerns, because SOEs often behave as instruments of the non-economic policies of their host state. A US congressional record of a public hearing in 2012 stated before the US-China Economic and Security Review Commission:

Because SOEs 'often behave as instruments of Chinese foreign policy,' SOE investments and operations in the US market also raise national security and other strategic concerns. The primary motive of SOEs often is not merely economic, but rather to further the objectives of the government, whether it be to obtain advanced technologies, secure access to raw materials, maximize production output or achieve geopolitical influence.⁹

State-subsidised Capital as a National Security Issue

A government subsidy is generally understood as a financial contribution by a government or any public body, where a government practice involves a direct transfer of funds (for example, in the form of grants, loans, or an equity infusion), potential direct transfers of funds or liabilities (for example,

loan guarantees), where government revenue that is otherwise due is foregone or not collected (for example, fiscal incentives such as tax credits), where a government provides goods or services other than general infrastructure, or purchases goods, or where a government makes payments to a funding mechanism, or entrusts or directs a private body to carry out one or more of the above functions.¹⁰

The issue of foreign government subsidies is closely linked with SOEs. For instance, China's SOEs or state-controlled enterprises are reported to have received massive government subsidies.¹¹ A national security threat could arise from both the foreign government's control of such enterprises and the foreign government's subsidies to them. However, it is also possible for massive subsidies to be provided to private firms to enable these private-but-state-backed firms to unfairly enter another country's market, to acquire local enterprises and to create national security concerns.

Government subsidies can be used not only to support the sales of goods or services to distort international trade in goods or services (the distortion of which is supposed to be dealt with by the subsidies rules under the SCM of the WTO), but also to strategically facilitate acquisitions of important businesses or to distort the support of the operation of a business in the host state. From the perspective of individual businesses, government subsidies definitely distort investment decisions. Local, smaller companies cannot compete with heavily subsidised foreign companies and can be pushed out of the market. From a broader perspective, a foreign government's massive subsidies can also generate unwanted change in the overall structure of the economy and affect the market's operation. Such concerns are not merely about the threat to the fairness of individual transactions, but also about an overall threat to the market-oriented economy caused by the foreign subsidised capital.

The issue of subsidised capital emerged in the early 2000s, when 'American tensions flared in the context of the bid by China National Overseas Oil Corporation (CNOOC) for the US oil firm Unocal. Chevron, the rival and ultimately successful bidder, contended that subsidized finance supplied by the Chinese government enabled CNOOC to offer an artificially high price for Unocal.'¹² In addition, for instance, the above-mentioned US congressional record of a public hearing in 2012 before the US-China Economic and Security Review Commission stated:

China's renewable energy sector demonstrates the effects of state capitalism in China and Chinese government subsidies at work. Over the past decade, China's solar industry has expanded at a phenomenal rate. In 2008, 'China became the largest producer of solar panel cells in the world, shipping ... roughly one-third of worldwide total [solar] cell shipments.' China remained the world's largest producer of solar cells in 2009 and 2010 and captured more than half of the global cell market for the first time in 2010. This rapid and unprecedented expansion was the direct result of the Chinese government's support for its solar energy

industry, including its granting of an extraordinary range and amount of subsidies to the industry.¹³

In recent years, the EU has started considering the regulatory gap with respect to ‘Foreign subsidies distorting the internal market regarding the general market operation of economic operators active in the EU; acquisitions of EU undertakings; public procurement procedures; and Foreign subsidies in the context of access to EU funding.’¹⁴ This shows that foreign subsidies are one of the key factors which could give rise to national security concerns.

Foreign Ownership/Control of Certain Sectors as a National Security Issue

The sectors which involve high degrees of national security interest include certain national infrastructure sectors, the sectors having access to sensitive information, and the media sectors which are critical in protecting freedom and pluralism, among others.

As for the infrastructure sectors, if we take the UK as an example, there are 13 national infrastructure sectors being named as involving national security interests, including chemicals, civil nuclear, communications, defence, emergency services, energy, finance, food, government, health, space, transport and water.¹⁵ The UK considers that increased access (to businesses, physical assets, people, operations, or data) and the ability to engage in espionage; greater opportunities to engage in disruptive or destructive actions or an increase in the impact of such actions, and the ability to exploit an investment to dictate or alter services or to utilise ownership or control as an inappropriate leverage in other negotiations – all pose national security risks. National security risks are likely to arise in the above sectors in relation to the ownership or control by foreign investors.¹⁶

Regarding the sectors having access to sensitive information, different countries might have different categories of information which are treated as sensitive. If we take the US as an example, the Committee on Foreign Investment (CFIUS) has treated sensitive personal data as a strategic asset under the federal regulations since 15 October 2020. Actually, in the past, CFIUS had already blocked the acquisition of MoneyGram International (which was a US money transfer provider) by Ant Financial (which was a Chinese financial technology company and Alibaba affiliate) ‘reportedly due to concerns that Ant Financial may provide the Chinese government access to Americans’ financial data through MoneyGram’.¹⁷ The new rule vests greater power in CFIUS to exercise the authority ‘to suspend, modify or prohibit foreign investments in the US’ or ‘to force divestitures for completed transactions’ where there may be a risk to national security, such as a risk arising from the access to sensitive data, including location data, biometric data, health data and financial data.¹⁸

For many countries, transmission channels are also a critical sector from a national security perspective. An OECD report explains the evolution of

transmission channels that are considered as being closely related to the national security threat:

Traditional concerns focused primarily on *sabotage* and *espionage* – especially when defense assets and strategically located real estate were the main assets of concern. Later, with the possibility of foreigners acquiring critical infrastructure assets after privatization, the possibility that infrastructure’s *availability* could be *disrupted* or *withheld* became apparent as a further risk. Some countries are concerned about the risk of supply disruption in sectors beyond critical infrastructure. *Leakage* of information, know-how, or sensitive personal data or the *malicious manipulation and alteration* of such data corresponds to the recent inclusion in security-related policy of technology assets and enterprises that hold or generate personal data. [emphasis added]¹⁹

In relation to the media sectors, these sectors are critically important in protecting freedom and pluralism in democratic countries. There is a common view in democratic societies that ‘Media freedom and pluralism are at the heart of any democracy. They embrace editorial independence, the free flow of ideas, and public access to a wide range of information sources and views.’²⁰ ‘Without the right to freedom of expression, and free, independent and pluralistic media, there is no true democracy.’²¹ Hence, there is a high national security interest involved in ensuring and protecting the freedom and pluralism of the media sectors. It would be very undesirable to see certain media being acquired by a foreign government-controlled firm in such a way that the media and its views are directly or indirectly influenced or controlled by a foreign government and that the freedom and pluralism of the media are threatened.

Sometimes a sector itself (such as the wind-farm sector) might not seem to be sensitive to national security, but when it is combined with other factors, such a sector could become highly sensitive from a national security perspective. An example of this concern arising from a foreign wind-farm investment combined with a sensitive site location is the Chinese wind-farm project, Blue Hills Wind, in Texas. This wind-farm site is near the Laughlin Air Force Base in Del Rio, Texas, which had started training Air Force pilots as early as 1943 and is now the US Air Force’s largest pilot training base. Because this Air Force base is so critical to US national security and the wind farm owned by a Chinese-controlled company will be so close to the Air Force base making it very easy to collect data from the site, it has been described as ‘the greatest national security concern’.²² This investment, despite the fact that it has passed its CFIUS review, has even triggered the enactment of the ‘Lone Star Infrastructure Protection Act’ of Texas, which bans all contracts in Texas that have been concluded with businesses or government officials from China, Iran, North Korea, Russia, or a country designated by the Texas governor as a threat to critical infrastructure.²³

Foreign Ownership/Control of Critical Technologies as a National Security Issue

Many intellectual property rights form the basis of critical technologies. Hence, IP and such technologies are interconnected. They will be dealt with together for the purpose of the discussion in this chapter.

IP rights and technologies can be obtained by foreign companies through a variety of ways, including through a sale or licensing of an IP right and through M&As with the company holding the rights, among other measures. In other words, an M&A is a frequently used method for foreign firms to acquire critical IP rights and technologies.

IP rights and technologies, such as advanced, dual use (that is, civilian and military use) and network technologies, can be extremely valuable and sometimes even critical to ensure national security. In recent years, IP rights and technologies have been receiving more and more attention from the national security perspective. One example is the 5G business, which enables lower latency and allows wider connections of more devices simultaneously with smart devices. From the US's perspective, there are two major reasons why 5G is of high national security importance, namely, 'the role that 5G will play in global economic growth and technological development' and its importance 'to protect U.S. and allied networks against cyber threats' by 'reducing the vulnerabilities of communications networks in the United States and in allied countries to cyber espionage by China and other competitor nations.'²⁴

For example, Motor Sich is a Ukrainian company producing aero engines for aircraft vehicles of various applications, including helicopters.²⁵ Its engines can be used for civilian, cargo and military aircraft, and even for cruise missiles and drones. It was reported that China started making an effort to acquire Motor Sich in 2009. After that, some Chinese companies continued buying up shares of the company for the purpose of gaining access to the 'strategically important dual-use technologies critical for the Chinese military'. In 2016, Chinese entities purchased from the company's chairperson of the Board of Directors 56 per cent of the company's shares, and it was reported in 2021 that about 75 per cent of the company's shares are held under Chinese ownership.²⁶ As a result, the largest shareholder of Motor Sich is the Chinese company Skyrizon Aviation, which is controlled by Chinese tycoon Wang Jing, who is reported as being 'renowned for spearheading China's global ventures into developing countries with high levels of corruption and strong military connections to China', and who gave an oath 'to serve Chinese national interests and use his company to expand China's global influence.'²⁷

The US Department of Commerce (USDOC) announced on 14 January 2021 that Skyrizon Aviation has 'significant ties to the PRC and the People's Liberation Army' and is 'actively seeking to acquire IP and technology to advance key military capabilities that threaten US national security, including the capability to develop, produce, or maintain military items, such as aircraft engines, satellites, and cruise missiles.' The USDOC further states that

‘Skyrizon’s predatory investments and technology acquisitions in Ukraine represent an unacceptable risk of diversion to military end use in the PRC.’²⁸ As reported:

Skyrizon is said to have transferred technology from the Ukrainian plant to a newly built China-based facility. However, Skyrizon’s attempted acquisition of Motor Sich was not approved by Ukrainian regulators. In 2017, a Ukrainian court froze shares in Motor Sich after the country’s SBU state security service launched legal action and an investigation, citing a ‘conspiracy’ by ‘foreign companies ... to move the company’s assets and production capacity outside Ukraine’.²⁹

In March 2021, for national security reasons, Ukraine decided to nationalise Motor Sich in order to prevent the Chinese takeover of the company.³⁰

Another example is the technologies in artificial intelligence. AI is about building smart machines to perform certain tasks which may exceed human intelligence. AI is a recently and rapidly developing field of technology and has highly significant implications for, or potential impacts on, national security. For instance, a report prepared by the US Congressional Research Service cites a case where AI was the reason to block a potential cross-border M&A:

In particular, the Chinese government is reported to be aggressively pursuing AI investments in the United States. Amid growing scrutiny of transactions involving Chinese firms in the semiconductor industry, in September 2017 President Trump, following the recommendation of the CFIUS, blocked a Chinese firm from acquiring Lattice Semiconductor, a U.S. company that manufactures chips that are a critical design element for AI technology.

The report also explains that AI could involve a national security threat because of its susceptibility to possible malicious manipulation: ‘AI algorithms are vulnerable to bias, theft, and manipulation, particularly if the training data set is not adequately curated or protected.’ In addition, AI’s wide range of uses also make it of high national security importance: ‘AI is being incorporated into a number of other intelligences, surveillance, and reconnaissance applications, as well as in logistics, cyberspace operations, information operations, command and control, semiautonomous and autonomous vehicles, and lethal autonomous weapon systems.’³¹ As regards the issue of AI in social media, please see the further discussion in Chapter 4.

Many critical technologies are owned by large, often listed, firms. However, as reported by the OECD, ‘companies that produce advanced technology or hold and generate sensitive personal information may be small, non-listed, sometimes young and little known, and produce items whose national security relevance is not obvious at first.’³² Therefore, even if critical technologies are

involved, the sizes of firms being acquired by foreign firms or the value of the M&A might not be scrutinised.

Antitrust Law and the Role of National Security

In addition to different approaches for the purpose of identifying national security risks, countries also adopt different mechanisms and approaches to manage national security risks arising from foreign M&A activity. In theory, M&As allow merged businesses to operate in a more efficient manner so that the businesses may enhance their competition and benefit the consumers. However, in practice, it is possible that some M&A transactions may ‘change market dynamics in ways that can lead to higher prices, fewer or lower-quality goods or services, or less innovation’.³³ Hence, regulating undesirable M&As is also an important component of antitrust laws. In the following discussion, we use some jurisdictions as examples to illustrate the use or non-use of antitrust laws to address national security threats.

Antitrust Laws Governing M&As

The fundamental idea of antitrust law – to prevent possible anti-competitive practices – applies to M&A activities. The main ideas include requiring the intended M&A transaction (which exceeds certain thresholds) to be referred to the competition authority in order to alert the authority of the possible concerns over anti-competitive behaviour arising from the M&A, and requiring the competition authority to screen the transaction based on the possible consequential effects of undesirably reducing competition in the market.

The mandatory pre-merger notification requirement will be triggered if the deal exceeds certain notification thresholds. The parameters and sizes of the threshold for transactions vary from country to country. The parameters of the thresholds for merger notification include turnover, assets, market share and transaction value.³⁴ The M&A transaction will have to wait for a period of time, or for clearance from the competition authority before closing the transaction.

When screening an M&A transaction, the competition authorities usually make an assessment of the likelihood of substantially preventing or lessening competition in the relevant market, or creating, maintaining, or enhancing the ability of the company (after completing the M&A) to exercise market power. Take the United States Horizontal Merger Guidelines’ explanation as an example for illustration:³⁵

The unifying theme of these Guidelines is that mergers should not be permitted to create, enhance, or entrench market power or to facilitate its exercise. ... these Guidelines generally refer to all of these effects as enhancing market power. A merger enhances market power if it is likely

to encourage one or more firms to raise price, reduce output, diminish innovation, or otherwise harm customers as a result of diminished competitive constraints or incentives. In evaluating how a merger will likely change a firm's behavior, the Agencies focus primarily on how the merger affects conduct that would be most profitable for the firm ... A merger also can enhance market power by increasing the risk of coordinated, accommodating, or interdependent behavior among rivals.

Sporadic Consideration of National Security in Traditional Antitrust Practices

Since the core content of antitrust policy is to eliminate anti-competitive practices and to ensure that consumers enjoy better prices and qualities and broader choices, 'national security' per se has not been a factor to be brought into the operation of antitrust law for 'systemic' analysis in most jurisdictions. However, in actual fact, national security has randomly emerged as an issue in the enforcement of antitrust law in several previous antitrust cases.

Take the US's antitrust law enforcement as an example: As early as the First World War, there had already been an antitrust case in which the national interest issue was raised and antitrust law was used to address such an issue. During the war, in 1916, German agents conspired to 'disrupt the flow of war material from US factories to countries fighting against Germany', 'by disrupting war industry labour (in which they actually succeeded) and planning to bomb factories and transportation facilities' (O'Keefe, 2020). Since conspiracy between firms could be against the antitrust law, the issue fell within the realm of antitrust law enforcement. The US government successfully prosecuted the conspirators based on Section 1 of the Sherman Act for the reason that such acts were restraining trade (O'Keefe, 2020). Although there was a competition consideration involved in the decision, national security was in fact the key factor that affected the prosecution of the conspirators.

In 1940, an American firm (Bausch & Lomb) and a German firm (Carl Zeiss) formed a cartel by entering into agreements in which Bausch & Lomb was prevented from selling its products outside the US without the prior consent of Carl Zeiss, and Carl Zeiss was also prevented from selling its products within the US without the consent of Bausch & Lomb. Furthermore, inventions resulting from their joint efforts had to be patented in the US by Bausch & Lomb for the account of Carl Zeiss. The US government alleged that their agreements had adversely affected war production. The US government was concerned about the American firm's ability to use patent rights to military optical instruments. The case was settled through a consent decree, which indicated that the sole ownership of these patents obtained by Bausch & Lomb for the account of Carl Zeiss was vested in the American firm (O'Keefe, 2020). The result satisfactorily addressed the national security concern.

There were other cases where national security was a reason being cited to support not enforcing the antitrust law. For instance, in 1943, the leader of the Antitrust Division of the Department of Justice intended to prosecute railroad companies for price-fixing. The attempt was abandoned because the Roosevelt administration considered that the enforcement of the antitrust law was against the national security priority (O’Keefe, 2020). In 1974, there was a case about an antitrust break-up of the Bell telephone system. The then Secretary of Defense asserted that the AT&T network was the most important communication network that the United States had to service its strategic systems in the country, and that it was essential that the US keep together this communications network that it had to rely on. However, the Department of Justice was still able to pursue the case (O’Keefe, 2020).

From the late 1980s into the 1990s, due to the end of the Cold War and the resulting excess capacity, there were many M&As in the defence industry in the United States. Some people worried about the situation in which the enforcement of antitrust law based on economic analysis could undermine national security. However, the Defense Science Board considered that the antitrust enforcement and review of M&A activity was in the public interest to protect the Department of Defense against the risk of firms’ enhanced market power (O’Keefe, 2020).

Although the issues of national security emerged from time to time in previous antitrust law-enforcement activities, the consideration of national security did not take place within a systemic framework. It was touched upon sporadically. In addition, the national security concerns that emerged in previous cases were quite different from the national security concerns that many countries are facing today. The current issues involve huge impacts on the operation of the market-oriented economy (such as a foreign country’s massive use of SOEs and subsidised capital to acquire domestic businesses), or fundamental adverse implications for an open economic system and society (such as foreign firms or governments dominating the key sectors or controlling the media sectors). The concerns are now much more serious than what emerged in earlier times.

Security Considerations in Antitrust Laws – Country Perspectives

The US Perspective – No Systemic Consideration of National Security

More and more countries are facing national security risks. However, there are different views as to the proper ways to address them. In this regard, there is the issue of whether national security should be left to national security laws to address, or whether it should be introduced into the operation of antitrust laws. One view is that antitrust law should not be used as a tool to address national security issues; instead, such issues should be addressed by other laws.

On this issue, the US has tended to avoid using antitrust laws to systematically address national security issues. In 2020, the US FTC Commissioner Noah Joshua Phillips stated:

As the U.S. Constitution itself makes clear, there is no responsibility more essential for a government than the protection of its citizens ... like other non-competition considerations, antitrust is an imperfect tool. And, when it comes to national security, the U.S. government has other tools. We have, for example, separate and distinct systems requiring mergers to be notified to one set of enforcers who monitor antitrust concerns and to another set of government officials responsible for national security review ... The CFIUS is authorized to review national security implications of certain cross-border transactions. Note that CFIUS is not an antitrust tool, but a national security one. The U.S. government has other tools beyond CFIUS to address national security risks in the private sector. On 6 August 2020, the then President Trump signed an executive order banning China's TikTok and WeChat services from mobile app stores in the U.S. The order relied upon the International Emergency Economic Powers Act and the National Emergencies Act.³⁶

However, some other countries have started considering using antitrust law to handle national security concerns. The EU and UK's recent developments in this regard are the apparent examples.

The EU and the UK – Systemic Consideration of National Security is in Place

As for the EU's use of antitrust law to control M&A activity, pre-notification for mergers meeting certain thresholds is required. The threshold parameter is turnover value:

Mergers not meeting EC (European Commission) thresholds may still be subject to review by the competition authorities of an individual Member State. While the EC does not conduct its own national security review, several provisions in the TFEU and EC Merger Regulation contemplate Member States conducting such a review. Within the TFEU, Article 346 provides two blanket exceptions to all provisions of the treaty ... Second, Article 346(1)(b) of the TFEU provides: any Member State may take such measures as it considers necessary for the protection of the essential interests of its security which are connected with the production of or trade in arms, munitions and war materiel; such measures shall not adversely affect the conditions of competition in the common market regarding products which are not intended for specifically military purposes.³⁷

In the UK, as mentioned above, merger control is governed by the Enterprise Act 2002, as amended by the Enterprise and Regulatory Reform Act 2013. The Enterprise Act 2002 provides that a proposed merger will be determined on the basis of whether the merger has resulted (or may be expected to result) in a ‘substantial lessening of competition’ in the relevant market, with the condition that the Secretary of State may intervene in the assessment of the Competition and Markets Authority (CMA) on public interest grounds, including national security, media plurality and broadcasting standards, and the stability of the UK financial system.³⁸

A Loophole in the Existing Considerations

In regard to the regulations of the above-mentioned countries, the M&A-related terms such as agreements, decisions, practices and merger notifications are the subjects to be regulated or monitored. However, what if such terms are past-tense, in the sense that they have been completed in foreign countries? For instance, if Qualcomm (a chip design and software company) is considering merging with Intel (a chip manufacturing company) or Apple (a mobile phone company which is the downstream user of the chip), this typical vertical integration case will have to be reviewed by the US FTC. The review criteria may include whether such vertical integration would compromise fair competition, and whether consumer choices would be affected. For a Chinese company such as Huawei, as we illustrated in Chapter 6, which has completed its vertical integration (of chip design, chip manufacturing, telecommunication facilities, and mobile phones) in China, it may be more effective in competing in the US, and more capable of providing attractive packages to US consumers.

In the above-mentioned example, since the M&A activity was completed in China, there are no agreements or decisions or practices to be reviewed by the FTC. The target of the competition between Qualcomm and Huawei is 5G or 6G, which does affect consumer benefits, but the competition scenario involves global markets, and is much broader than the US market. This is a pure ‘fair competition’ topic of concern, and there is no M&A decision to be reviewed by CFIUS. If the US has a national security concern over Huawei, it can only ban it from entering the US market. This, however, violates the US WTO accession promises for telecommunications services, unless Article XIV of the GATS (the national security exception) is applied.

Systemic National Security Considerations Under Investment Law

US Investment Review to Consider National Security

The US established the interagency CFIUS pursuant to Section 721 of the Defense Production Act of 1950, as amended, to review investment transactions in the United States involving foreign companies in order to

determine whether such transactions raise national security concerns. In 2018, the Foreign Investment Risk Review Modernization Act (FIRMA) was enacted to strengthen and modernise CFIUS to ‘address national security concerns more effectively, including by broadening the authorities of the President and CFIUS to review and to take action to address any national security concerns arising from certain non-controlling investments.’³⁹

Institutionally, CFIUS is chaired by the US Department of the Treasury. Members with voting power are from various departments and offices (State, Defense, Commerce, Justice, Homeland Security, Energy, Science and Technology Policy of the White House; and the USTR). Other government agencies, such as the Department of Labor and the Office of the Director of National Intelligence, are required to produce a threat assessment in relation to any transaction under scrutiny. Hence, CFIUS conducts an ‘all-of-government’ review to ensure that the reviewed transaction is evaluated from a comprehensive perspective.

The powers of CFIUS are broad. It can approve a transaction with or without conditions. In the case where CFIUS determines that the only adequate and appropriate resolution is to block a given transaction, it will send the matter to the White House for the final decision. No judicial review of CFIUS’ national security decisions is available. Very importantly, there is no statute of limitations on the authority of CFIUS to review already completed transactions: ‘Unlike the competition reviews under the Hart–Scott–Rodino statute, there is no transaction size threshold for CFIUS jurisdiction ... It has the authority to take actions, however, only if the transaction under review “threatens to impair the national security.” And, therefore, the vast majority of foreign M&A is not filed with CFIUS.’⁴⁰

EU Investment Review to Consider National Security

In recent years, the EU and its Member States have increased their concerns ‘that predatory investors might take advantage of vulnerable businesses through heavily discounted takeovers.’ For instance, in Spain, ‘Further measures restrict FDI by investors (i) controlled by the government of a non-EU member state, (ii) that already have invested in the strategic industries of another member state, and (iii) that have been brought before administrative or judicial proceedings for illegal conduct.’ ‘Germany is considering further support measures to protect its national industrial sector, including restricting FDI in significant assets such as automotive companies’, while Italy stressed that ‘the government would not allow Italy ‘to become someone’s shopping territory.’⁴¹

The EC issued its ‘Guidance to the Member States concerning FDI and free movement of capital from third countries, and the protection of Europe’s strategic assets, ahead of the application of Regulation (EU) 2019/452 (FDI

Screening Regulation),⁴² in which the Commission called upon its Member States to:

Make full use already now of its FDI screening mechanisms to take fully into account the risks to critical health infrastructures, supply of critical inputs, and other critical sectors as envisaged in the EU legal framework;

For those Member States that currently do not have a screening mechanism, or whose screening mechanisms do not cover all relevant transactions, to set up a full-fledged screening mechanism and in the meantime to use all other available options to address cases where the acquisition or control of a particular business, infrastructure or technology would create a risk to security or public order in the EU, including a risk to critical health infrastructures and supply of critical inputs.

Under the EU's FDI screening regulation, EU Member States are advised to adopt needed measures to prevent any foreign investor from acquiring or taking control of a local company if it would result in a security threat. The relevant factors to decide a security threat include 'the effects on critical infrastructure, technologies (including key enabling technologies) and inputs which are essential for security or the maintenance of public order, the disruption, failure, loss or destruction of which would have a significant impact in a Member State or in the Union'.⁴³

UK Investment Review to Consider National Security

In 2020, the National Security and Investment Bill 2019–21 was presented in the House of Commons of the UK to significantly strengthen the government's powers to investigate and prohibit M&A transactions on national security grounds. The Bill introduces a new regime for reviewing and intervening in M&A transactions that might raise national security concerns to 'enable the Secretary of State to "call in" acquisitions of sensitive entities and assets ("trigger events") to undertake a national security assessment. This can happen up to 5 years after a trigger event has taken place'; to 'establish a requirement for proposed acquirers of sensitive entities and assets to seek authorization and to obtain approval from the Secretary of State before completing their acquisition'; to 'create a voluntary notification system to encourage notifications from parties who consider that their trigger event may raise national security concerns', and to 'create a power to impose remedies to address risks to national security, sanctions for non-compliance with the regime and mechanisms for legal challenge'.⁴⁴

Under the new rule, the number of M&A transactions to be assessed for national security concerns will be increased and a qualifying M&A without clearance could be subject to serious remedial consequences, including civil and criminal penalties.

Regulatory Gaps to be Closed

From the above explanation, it is clear that the national security consideration has been incorporated into the foreign investment reviews of M&As in many jurisdictions. However, it is not consistently incorporated in the antitrust review of cross-border M&As. We argue that certain gaps exist in the following situations in antitrust law enforcement and investment reviews of cross-border M&As in order to address national security issues. The gaps that need to be closed include: (1) not properly identifying the scope of national security arising from cross-border M&As; (2) not introducing the national security element into the antitrust review of M&As; (3) not introducing the national security element into the foreign investment review mechanism, and (4) not considering the nature of national security for the purpose of setting forth the thresholds for the intervention into M&A transactions.

In terms of identifying the scope of national security arising from cross-border M&As, this chapter has explained that a foreign country's systemic use of SOEs and government-subsidised capital to acquire domestic businesses and the foreign acquisition of key sectors or technologies are the main concerns. For those countries where these national security elements have not been incorporated into their legislation, there are good reasons for them to do so.

In the following discussions, we will focus on introducing the national security element to the antitrust review of M&As, on introducing the national security element to the foreign investment review mechanism, and on considering the nature of national security for the purpose of setting forth the thresholds for intervention in M&A transactions.

Introducing National Security Into the Antitrust Assessment of M&As

As explained above, not all antitrust laws systemically incorporate national security elements into their assessment of M&A transactions. We argue that regulating M&As which involve 'possible adverse security-related economic impacts' is also the task of antitrust law.

First, according to the United States' experience of antitrust law enforcement, the antitrust authority actually cannot avoid 'sporadically' addressing national security issues. If the antitrust authority must address the issue sporadically, why not address the issue in a systemic manner so that national security can be more comprehensively assessed?

Second, SOEs, foreign subsidised capital, and foreign ownership or control of critical sectors or technologies could result in direct or indirect control by foreign governments, create structural market distortions and have a substantial impact on the operation of the market-oriented economy, while also having fundamental adverse implications for an open economic system and society. Antitrust law is designed to ensure that there will be sufficient competition and fair transactions, that economic operations will be based on

the concept of a free market, and that the market is functioning based on supply and demand. Hence, antitrust law is suitable for addressing the related security issue. If an external adverse intervention has caused a national security threat and if there is no permanent mechanism built into the legal regimes, a market-oriented economy could become vulnerable and be subject to undesirable distortions by massive foreign subsidised capital, or by control by unfriendly foreign governments of merged firms or their control of critical IP rights and technologies. It is thus essential to introduce a stronger national security screening mechanism to be built into the legal regimes of antitrust law on cross-border M&As.

Third, although there might already be certain foreign investment screening mechanisms to review national security risks, their functions and remedies are not the same as the functions and remedies of antitrust law. Antitrust law has to do with addressing market distortion and ensuring sufficiently fair competition, whereas investment screening involves various considerations which might not have anything to do with the market's operation, including political considerations. Their respective tools and remedial measures are not the same. For instance, there are pre-merger notification requirements in most antitrust laws. This allows the antitrust authorities to receive information in advance and to decide whether to intervene. By contrast, not all investment review legislation adopts similar notification requirements. In addition, for instance, under antitrust law, possible remedial measures include forcing a divestiture or reversal of integration that had already taken place, requiring access to the infrastructure, or requiring licensing or disclosure of interfaces in relation to the infrastructure owned by the merging firm, whereas the main tool arising from an investment review is a denial of investment. Therefore, practically, it is still meaningful to have both antitrust law and the law governing the investment review addressing national security issues from different perspectives.

Introducing National Security into the Investment Review of Cross-border M&As

On the issue of introducing the national security element into the foreign investment review mechanism, this chapter has explained that more and more countries have modified their investment legislation to address such an issue.

One aspect that needs to be further considered is that although national security is of utmost importance, it should not be used in an abusive manner, with the result that decisions made by the investment authority become purely political. Certain mechanisms are necessary in order to ensure that a decision to deny a proposed cross-border M&A is made on the grounds of security concerns.

For example, as mentioned above, the participants in CFIUS decisions are from multiple government agencies of the US. Consideration from various perspectives can thus take place to prevent obvious bias. The formation of a decision-making institution is a good example for other countries to consider.

However, the participation of an antitrust authority in the process that can provide views from the perspective of the structure and operation of the market-oriented economy is lacking. We are not suggesting that the decision-making body with regard to the foreign investment review should include the antitrust authority among the decision-making members. However, an independent assessment of the potential structural implications arising from a cross-border M&A should be of importance to the investment review authority in understanding the nature and seriousness of the proposed transaction from a market perspective.

Changing the Intervention Thresholds for the Antitrust Law and Investment Review

There should be a proper balance between the need for stricter control to ensure the protection of the national security interest and, at the same time, the need to avoid too-rigid regulations that would suffocate active business activities. In order to alert relevant government agencies to be aware of the M&A transactions that might involve national security concerns, most countries have set up certain thresholds for the purpose of determining whether the merging firm needs to report to the authorities in charge of the antitrust law or investment review, whether the M&A transaction will have to wait for a period of time or for clearance from the competition authority before closing the transaction, and whether the authorities will intervene in the transaction.

The most commonly adopted threshold is the size of the transactions or turnovers. Threshold sizes vary from country to country. However, in considering possible security concerns, countries have begun to lower their thresholds. For instance, in 2018, the UK lowered the ‘turnover threshold’ for its intervention from £70 million to £1 million, and set forth a target of a 25 per cent or greater share of supply to be the ‘share of supply threshold’.⁴⁵ Lowering the threshold is useful to ensure that possible security concerns are identified.

In addition to using the quantitative criteria (such as the size of transactions or turnovers) to decide whether the merging firm has a notification obligation, it is also necessary to use the qualitative criteria to decide the notification obligation. The qualitative criteria must be based on the facts that the acquiring firm is a foreign country’s SOE, that the acquiring firm uses a foreign government’s subsidised capital, and that the proposed M&A involves certain designated sectors or critical technologies.

Conclusions

As an OECD report indicates:

Over the past decades, many countries have eliminated most barriers to trans-border capital flows and have abolished exceptions to national

treatment for the acquisition, ownership, or operation of assets by foreigners. This openness has created economic opportunities for home and host economies and for enterprises. With these opportunities came occasional risks, including for the recipient country's essential security.⁴⁶

There is an apparent need to systematically introduce the national security element into the antitrust law enforcement and investment review in order to close the regulatory gaps and to respond to the immense adverse effect arising from foreign SOEs, foreign subsidised capital, and foreign ownership or control of critical sectors and technologies. In the end, the balance between free trans-border capital flows and national security concerns hinges upon the mutual 'trust' between the countries involved. As illustrated in the case of the Blue Hill Wind Farm discussed earlier, when an investment is administered by a foreign SOE with an opaque collectivist agenda, then the national security concern will eventually dominate, and the benefits from the international free flow of capital, if any, will be sacrificed.

Notes

- 1 OECD, 'Acquisition- and Ownership-related Policies to Safeguard Essential Security Interests: Current and Emerging Trends, Observed Designs, and Policy Practice in 62 Economies' (May 2020), 24, www.oecd.org/investment/OECD-Acquisition-ownership-policies-security-May2020.pdf.
- 2 David Fagan and Brian Williams, 'Intersection of National Security with M&A: The Committee on Foreign Investment in the United States', *Tax Executive* (30 January 2020), <https://taxexecutive.org/intersection-of-national-security-with-ma-the-committee-on-foreign-investment-in-the-united-states/>.
- 3 OECD, 'Acquisition- and Ownership-related Policies ...', 6.
- 4 The examples raised can be found in various news releases. Li Fusheng, 'Audi to Hold Majority Stake in New Chinese JV', *China Daily* (19 January 2021), www.chinadaily.com.cn/a/202101/19/WS60066835a31024ad0baa3903.html; 'Toyota Plans New \$1.2 Billion EV Plant in Tianjin with FAW: Document', *Reuters* (29 February 2020), www.reuters.com/article/us-toyota-china-electric-idUSKBN20N0H0; 'Taiwan Approves TSMC's Plan to Expand Nanjing, China Chipmaking Plant', *Focus Taiwan* (30 July 2021), <https://focustaiwan.tw/business/202107300008>.
- 5 References in this subsection include the following: World Bank, 'Belt and Road Initiative' (29 March 2018); Ming Wan, *The Asian Infrastructure Investment Bank: The Construction of Power and the Struggle for the East Asian International Order* (Palgrave Macmillan 2016), 70; Scott Wingo, 'How Will China Respond When Low-income Countries Can't Pay Their Debts?', *Washington Post* (30 June 2020); American Enterprise Institute and Heritage Foundation, 'Global China Investment Tracker', aei.org/china-global-investment-tracker/; Nilanthi Samaranayake, 'Chinese Belt and Road Investment Isn't All Bad—or Good', *Foreign Policy* (2 March 2021), <https://foreignpolicy.com/2021/03/02/sri-lanka-china-bri-investment-debt-trap/>; Maria Abi-Habib, 'How China Got Sri Lanka to Cough Up a Port', *New York Times* (25 June 2018).

- 6 PWC, 'State-Owned Enterprises: Catalysts for Public Value Creation?' (April 2015), 14, www.pwc.com/gr/en/publications/assets/state-owned-enterprises-catalysts-for-public-value-creation.pdf.
- 7 Ibid.
- 8 OECD, 'SOEs Operating Abroad: An Application of the OECD Guidelines on Corporate Governance of State-Owned Enterprises to the Cross-border Operations of SOEs', 3, www.oecd.org/corporate/ca/corporategovernanceofstate-ownedenterprises/44215438.pdf.
- 9 *Chinese State-Owned and State-Controlled Enterprises: Hearing Before the U.S. China Economic and Security Review Commission*, 112th Cong. 84 (2012), www.uscc.gov/sites/default/files/transcripts/2.15.12HearingTranscript.pdf.
- 10 Agreement on Subsidies and Countervailing Measures (Marrakesh, Morocco, 15 April 1994), Marrakesh Agreement Establishing the World Trade Organization, Annex 1A, THE LEGAL TEXTS: THE RESULTS OF THE URUGUAY ROUND OF MULTILATERAL TRADE NEGOTIATIONS 231 (1999), 1869 U.N.T.S. 14, Art 1.
- 11 US-China Business Council, *Member Survey* (August 2019), www.uschina.org/sites/default/files/member_survey_2019_-_en_0.pdf.
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8 Forced Technology Transfer

The Issue of Insufficient Remedies

Introduction

Forced technology transfer has become an increasingly high-profile and controversial issue in international commerce. In view of the US–China conflicts which we referred to in Chapter 2,¹ it is fair to say that FTT was one of the most important factors contributing to the trade war between the US and China in 2018. Certainly, FTT could be brought to the WTO’s Dispute Settlement Body (DSB) because it can be argued that there has been a violation of the TRIPS Agreement or Member accession protocols. It could also be a breach of the respective country’s IP laws. Nevertheless, neither the WTO nor China’s domestic judicial system has been able to properly tackle this issue for decades. In fact, only a few cases have been formally filed and even fewer have been settled or decided because of the nature of this controversy.²

This chapter provides a bird’s-eye view of FTT in China, including its origins, forms and effects. Furthermore, it analyses why FTT may continue, even after the tentative deal reached by the US and China in early 2021. Our main point is that many disputes in international trade involve a governmental role. In the case of FTT, whether a technology transfer is ‘forced’ depends on how the scenario is interpreted. Very often, a fair interpretation of the contract (between the foreign company and the local one) hinges upon the neutrality and independence of the government (and the judicial system). Knowing that the CCP ‘controls’ both the executive and the judicial systems in China, it is a reasonable prediction that the problem of FTT will not be solved in the foreseeable future. In fact, there are many other similar situations involving serious unfairness in remedial mechanisms. In this chapter, we use FTT as an example to illustrate our point.

The reason why we choose FTT as an example is that it has the following features. First, unlike the cases of other WTO rules, FTT usually involves the *ex ante* collaboration between a local business (say, in Country A) and a foreign company (affiliated with, say, Country B) which owns the technology. This collaboration often takes the form of a joint venture of business, with terms signed in a mutually agreed contract. When there is a controversy over FTT, the foreign firm could in theory choose to terminate the collaboration

contract, and bring the case to a remedial mechanism, either within country A or at the WTO. For other violations of the WTO rules, such as inappropriate government subsidies violating the SCM Agreement, any local collaborators of the foreign company do not exist. As we will explain, the above-mentioned *ex ante* collaboration is key to making the FTT controversy difficult to address.

The second feature of FTT is the vested interest of the foreign company. Given the existing collaboration and investment of the foreign company, formally bringing a complaint to the WTO would change the original collaboration between the two companies into a scenario of dispute settlement between two governments. This often implies the formal termination of the original collaboration, and hence forgoing the company's vested interest. In practice, the authoritarian administration has hundreds of ways of harassing the foreign company which filed the complaint. However, if the mutual collaboration is to come to an end after the formal complaint is made, very few companies would be willing to spend much resource to go through the dispute settlement procedure, since there is no longer any expected future interest. This may be the reason why very few cases have been brought to the WTO dispute settlement system. Therefore, if there is any alternative procedure to find a solution, the foreign company would prefer to pursue that alternative option.

Third, the above-mentioned alternative procedure usually involves the government or court system in Country A. For instance, the local business may exaggerate the terms of the original collaborating contract, which makes the foreign company feels it is being treated unfairly and being 'forced'. Whether the interpretation of the contract is fair or not often needs to be determined by the administrative or judicial system. Thus, for the case of a possible FTT, the fairness of the administrative or judicial system is crucial. Without an independent and fair government system, the foreign company can hardly find justice or remedies in any FTT cases.

Defining Forced Technology Transfers

Before discussing the concept of FTT, we should understand that technology transfers take place in almost all developing countries at different times. Indeed, some already-developed states, such as South Korea, have applied the same strategy along their development route. In fact, technology transfer was first discussed publicly and internationally at the United Nations in 1961 (Tundang, 2020). Since then, it has been widely used as a method to help developing countries build their domestic industries, and is often believed to be one of the most efficient ways for poorer countries to garner technologies in order to upgrade their domestic industries and foster economic growth (Cheng, 2012).

To a certain degree, international technology transfer from rich countries to poor ones is economically desirable for the technology-exporting

companies. This is because their supply chains can obtain cheaper and safer components more efficiently, thereby improving their products' competitiveness. Ultimately, technology transfer could be beneficial to the whole world because more products become affordable to more people. In a nutshell, developing countries modernise their economies, while developed states expand their access to more markets through technology transfer – a symbiotic relationship between the donor and the recipient countries. This is the win-win ideal situation.

Why, then, does international technology transfer give rise to many concerns and disputes that possibly contribute to a trade war? In addition to all its side-effects in development theory, such as entrenched dependency, the crucial reason behind the concerns is that technology transfer can take place in many forms, and some of them involve coercion and intimidation. To be more specific, policies regarding international technology transfer can be arranged in various ways, depending on how 'voluntarily' foreign companies transfer their technologies to their collaborators in developing countries. Some policies are commonplace around the globe and normally do not raise concerns, such as those policies that aim to enhance the absorptive capacity of technology transfer. However, others give rise to suspicion, such as those that demand that data be stored locally, or the disclosure of the source code. Between these two poles, some transfers fall within a grey area, such as those involving opaque administrative requirements (Andrenelli et al., 2019). For instance, China's Technology Import and Export Regulation (TIER) has long been criticised as overly ambiguous and burdensome.

To decide whether a technology transfer mode has been abused and has become an FTT, Andrenelli et al. (2019) suggested several factors to be considered: 1) whether it is essentially a quid pro quo between access to the domestic market and technology transfer; 2) whether there is discrimination against foreign companies; 3) whether the process and requirements to access the market are transparent, and 4) the role of the authorities (including the local and central administrations and judicial courts). These four factors are interrelated and, when working together, the combination may render one ostensibly normal measure closer to FTT. For instance, if the process of applying for approval is not transparent and unreasonably time-consuming, foreign companies may have no choice but to transfer their technology in exchange for the permission, even if the authorities remain completely silent on the surface, or even if the statutes do not explicitly ask for FTT as a quid pro quo.

FTT in China: Approach, Controversy and Effect

As we mentioned above, the pivotal issue is not the technology transfer per se; rather, it is the fact that foreign companies are being compelled to do so. Let us explain how FTT is implemented in China.

Mandatory Joint Venture

In China, international technology transfer usually takes place alongside foreign direct investment (FDI), which can take at least six different forms, including an equity joint venture (JV), contractual JV, a wholly foreign-owned enterprise (WFOE), shareholding enterprises, joint exploration and others. Nevertheless, not all business sectors are available for all forms of FDI in China. To protect its own industries, China has prohibited WFOEs from competing with local companies in some sectors of its market (Lovely and Huang, 2018). For example, the *Catalogue of Industries for Guiding Foreign Investment 2017*, published by the Chinese government, has required that foreign companies find a Chinese partner and form a JV in order to invest in certain industries. While this list has been revised repeatedly in order to meet the needs of China's economic development strategies, the point is clear: in order to enter some sectors of the Chinese market, foreign companies are often compelled to find a local collaborator, as we mentioned in the first section of this chapter.

By using the above regulation as leverage, Chinese companies have gained huge bargaining power over foreign companies and could take advantage of this in the process of negotiation and cooperation. It has been observed that business sectors, such as the automotive, aviation and high-speed rail sectors (Carbaugh and Wassell, 2019), have faced a serious threat of FTT.

Furthermore, Chinese laws often prescribe that the local partners in JVs formed in this way must control no less than 50 per cent of the shares. Under such JVs, the IP rights, or the know-how, are to be shared between partners, which makes it difficult to prevent the leakage of technologies. As such, technologies possessed by foreign companies are unlikely to remain secret from the Chinese partners (Lee, 2020a; Prud'homme and Zhang, 2019).

Technology 'Sharing' with JV Partners

Because of this type of 'involuntary sharing' of technology information, FTT has emerged as a major controversy between China and other economically developed democracies, such as the US, the EU and Japan.³ But this is actually not something new in China. Since the 'Reform and Opening Up' in the late 1970s, China has adopted this strategy, which later became known as the 'market for technology', meaning that China opened up the largest and fastest growing market in the world in exchange for technology transfers from foreign companies. This was not an issue when China was a 'horse-and-cart economy', without a technology infrastructure.⁴ Nowadays, however, China has not only become the world's second largest economy, but has also become a powerful contender in many domains of cutting-edge technology. The rising strength that has accompanied China's economic transformation has rendered FTT a sensitive issue in international relations. After all, if some of China's

modern technologies have been based on previous FTTs, the technology competition today would be unfair in the first place.

Most notably, FTT has long been outlawed by international trade and investment agreements despite its prevalence. In fact, when China became a Member of the WTO in 2001, it had promised in its Accession Protocol that FTT would no longer be allowed in China.⁵ Notwithstanding that China did revise several laws and prohibit FTT after joining the WTO, it has been widely reported that FTT has continued to be an issue for more than two decades.⁶ Some even go further and contend that ‘It is an open secret – not even a secret – that FTT is part of the price of doing business in China.’⁷

Some may wonder why foreign companies did not make loud complaints about FTT. One crucial reason is the lure of dealing with the largest market in the world. Few foreign companies can resist the pressure of their shareholders, some of whom may have already been Chinese (companies) for a long time. Moreover, foreign companies usually do not have any choice because they face a sort of prisoner’s dilemma: If they refuse to acquiesce and transfer their technology to Chinese companies, other foreign companies in the same sector might do so instead. Therefore, even if all foreign companies know that it is better in the long run for all of them to jointly resist the temptation of FTT, some companies will be mesmerised by the short-term interests at the expense of other foreign companies (Branstetter, 2018).

As a result, foreign companies might choose their second-best strategy and the Chinese government will still achieve its purpose by taking advantage of the potential economic conflicts between foreign companies. Even worse, once the partnership breaks up or comes to an end, former local Chinese partners may become the most competitive opponents in the global market, using the technology previously obtained from FTT. That is, the partnership may back-fire in the long run. Finally, once the JV is formed and the operation starts, the foreign companies face a huge sunk cost to stop the JV, if they decide not to follow the FTT request. The dynamics are always to the benefit of the Chinese local company.

Admittedly, foreign companies receive a royalty when they transfer the technology or IP. However, the royalty rates in licensing or other forms of compensation are usually below-market (Prud’homme et al., 2018, p. 161). Moreover, it should be emphasised that although technology transfer seems to be a transaction between foreign and Chinese companies, it is usually done with the (implied) consent of the CCP,⁸ partly because in every big company, there is a CCP committee, whose members serve not only as whistleblowers but also as policymakers. From this perspective, it is unconvincing to argue that FTT is purely a transaction between private entities, even though sometimes it ostensibly appears to be so.

Once a JV is formed, it also creates an opportunity to compel foreign companies to disclose other technologies, IP, or trade secrets that are not necessary or even irrelevant with approval. It has been alleged that Chinese officials

may deliberately leak information to other Chinese companies in exchange for personal interests.⁹ This kind of FTT may be worse, as the foreign companies do not even know when their technologies have been transferred, which companies have transferred these technologies, and the identity of the recipient companies – that is, potential competitors in the future.

Fifteen Shades of FTT

In addition to the above-mentioned approaches, China has used other means to force or induce technology transfer. According to the US White House Office of Trade and Manufacturing Policy, there are 15 instruments of coercion that the Chinese government has applied, including discriminatory patent and other IP rights restrictions, data localisation, government procurement, forced research and development (R&D) localisation, the CCP's intervention in corporate governance, and so on.¹⁰ China does not apply all of the aforementioned means to all foreign companies at the same time. For example, R&D localisation is widely used in the wind-energy industry and automotive industry, while mandatory JV is usually applied to civil aircraft and parts.¹¹

Moreover, both the White House and the USTR have repeatedly referred to government-sponsored IP and technology theft (either physical or cyber-enabled), trademark violations, and other concerns that arguably violate WTO norms and clearly breach China's own commitments that promise to eradicate FTT and unfair competition.¹² For example, in cooperation with the US Department of Homeland Security and other organisations, the US telecommunications company Verizon has reported that 96 per cent of espionage cases were attributable to actors in China, although the identity of these actors could not be discerned (Verizon, 2013). Furthermore, a US Department of Defense organisation maintained that 'The scale of the espionage ... continues to increase' in 2018 (Brown and Singh, 2018, p. 17). More importantly, because such economic espionage is either supported or conducted by the Chinese government,¹³ it is difficult for foreign private businesses to pursue an investigation.

It has also been suggested that different FTT strategies exert different levels of leverage over foreign companies. Some are more likely to lead foreign companies to give in than others. Prud'homme et al. (2018) argued that FTT is more likely to succeed in China when seven conditions are met: (1) when there is strong state support for the development of certain types of industries; (2) oligopolistic competition; (3) other policies closely complementing FTT policies; (4) high technological uncertainty; (5) policy modes of operation offering basic appropriability and tailored to the industrial structure; (6) avoidance of reform by the state, and (7) stringent policy compliance mechanisms. Prud'homme and colleagues also suggested that in the absence of these conditions, FTT may only spur limited technology transfer, or that the transferred technologies may not be cutting-edge ones.

Special Features of Certain Industries

To name a few examples, in the high-speed rail industry, China once relied on Japan, France and Germany to build its domestic high-speed rail system (Prud'homme et al., 2018, p. 158). Today, China has become the largest exporter of high-speed rail after two decades of 'accumulated' technologies. Moreover, China relied on Spain and other foreign companies in 2005 to manufacture, assemble and install its wind turbines in China and required these companies to do so inside China. In less than two decades, Chinese companies have come to account for the lion's share of the domestic market for wind turbines.¹⁴

Nonetheless, not all FTTs are fruitful. In the semiconductor industry, for instance, China still lags behind. It is said that the FTT may even entrench the dependency of Chinese companies.¹⁵ That is, Chinese companies have become more reliant on their foreign partners, rather than developing their own technology in the long run.

Finally, although FTT has been cited as one of the most troubling issues when conducting business in China, some companies maintain that they are not particularly worried about that. General Electric (GE), for example, indicated that the competition from China will not be imminent, even though Chinese companies have obtained its advanced technologies through the establishment of JVs. According to GE, Chinese companies have not exhibited the skill of innovation when working with the advanced technology; that is, once they have become familiar with the transferred technology, they produce nothing new, and in the meantime, GE has developed more advanced technology (Carbaugh and Wassel, 2019, p. 318). In this story, Chinese companies will always be chasing and asking for another round of cooperation. Boeing, one of the leaders in the aircraft production industry, made similar statements, suggesting that maintenance is key in the aircraft industry. Even if Boeing shares aircraft production technology with its Chinese partners, the latter still need to rely on foreign companies for logistic support (Carbaugh and Wassel, 2019, pp. 318–319).

Overall, however, in terms of the macroeconomic impact, studies have suggested that FTT has 'undercut the national welfare of the United States even if it is profitable for US investors' (Sykes, 2021). This is because even if American investors could benefit from China's market in the short term, FTT may render other American companies less competitive not only in China but worldwide, resulting in a level of damage that has not been properly assessed. The result is that FTT 'transfers surplus from the United States to China' (Sykes, 2021), and the surplus thus transferred is astronomical.¹⁶

International Treaties That Regulate FTT

First, it should be noted that WTO rules do not regulate technology transfer directly (Qin, 2019). However, FTT often constitutes market access barriers

and leads to discriminatory treatment of foreign investors. For instance, the dispute *China – Intellectual Property Rights II* that the US brought up in 2018 against China was primarily based on the violation of national treatment, not FTT itself (see Note 20). The only exception that clearly prohibits FTT is China's Accession Protocol, in which China committed to prohibiting FTT that it had practiced for decades before joining the WTO.

In a sense, FTT can be understood as a government subsidy to local companies and a tax imposed on foreign companies (Carbaugh and Wassel, 2019). From this perspective, it is regulated by several agreements that include national treatment provisions which require all WTO Members not to discriminate against foreign companies. Related rules include the TRIPS, the GATS, and the Agreement on Trade-Related Investment Measures (TRIMS).

If the technology that is forced to be transferred involves IP, FTT may be regulated by the TRIPS Agreement, which facilitates cross-border technology transfer by strengthening the protection of IP rights. Although the TRIPS Agreement does not clearly mention FTT, Article 3 of the TRIPS Agreement obligates all WTO Members to 'accord to the nationals of other Members treatment no less favorable than that it accords to its own nationals with regard to the protection of IP'. Article 28 required exclusive rights to be conferred upon a patent owner to prevent third parties from 'making, using, offering for sale, selling, or importing for these purposes' the product subject to the patent or the product obtained from the process subject to the patent. A patent owner 'shall also have the right to assign, or transfer by succession, the patent and to conclude licensing contracts'. Article 39 further protects trade secrets by stipulating that 'Natural and legal persons shall have the possibility of preventing information lawfully within their control from being disclosed to, acquired by, or used by others without their consent in a manner contrary to honest commercial practices.' Finally, Articles 41–61 require Members to provide a series of enforcement mechanisms that could be invoked to deter FTT. Among these enforcement mechanisms, two are particularly noteworthy: violators could be requested to compensate the right holders for damages and other expenses, and the judicial authorities might order the goods that cause the damage to be destroyed.

In a similar vein, the GATS also emphasises market-access commitments and national treatment. In focusing on investment issues, Article XVI stipulates that each WTO member 'shall accord services and service suppliers of any other Member treatment no less favorable than that provided for under the terms, limitations and conditions agreed and specified in its Schedule'. Additionally, Article XVII provides that 'each Member shall accord to services and service suppliers of any other Member, in respect of all measures affecting the supply of services, treatment no less favorable than that it accords to its own like services and service suppliers.' These two provisions in the GATS might provide a certain extent of protection for foreign companies from FTT, although it does not clearly mention FTT either. Of particular

note, the GATS has not been able to comprehensively constrain China as well, because it merely regulates the service sector.

Furthermore, the TRIMS, which concentrates on trade-related performance requirements, also has some provisions regarding remedies against certain types of FTT. For instance, Article 2 states that ‘Without prejudice to other rights and obligations under GATT 1994, no Member shall apply any TRIM that is inconsistent with the provisions of Article III or Article XI of GATT 1994.’ Furthermore, Articles III and XI of the GATT 1994 respectively prescribe national treatment on internal taxation and regulation, and the general elimination of quantitative restrictions. Both may address issues regarding FTT in certain contexts.

In addition to the three WTO rules mentioned above, China has committed to prohibit the use of technology transfer as a quid pro quo for the access to its domestic market since it became a member of the WTO in 2001. Specifically, China made a binding commitment that, among other things, it would not condition performance requirements on technology transfer.¹⁷ In addition, in the Working Party Report, China agreed that it would ‘ensure national and most-favored-nation treatment to foreign right-holders regarding all intellectual property rights across the board in compliance with the TRIPS Agreement’.¹⁸

From the above brief introduction, it should be fair to argue that FTT violates several WTO rules. Indeed, the problem is never the lack of relevant international and domestic regulations. On the contrary, the problem is in the difficulty of proving the existence of FTT. In the TRIPS Agreement, TRIMS, or GATS, the key idea is national treatment. The term used is ‘no less favorable’. In reality, it is rather difficult for anyone to argue that a long-delayed review or a very detailed scrutiny is for the sake of ‘thoroughness’, or is actually providing ‘less favorable’ treatment.

Remedies to FTT (and the Lack Thereof) in China

Bring a Case to the WTO?

In 2018, both the US and the EU brought complaints pertaining to China’s TIER and its FTT practice. The US requested consultations with China (DS542) on 23 March 2018, claiming that certain of China’s measures concerning IP rights violated the TRIPS Agreement.¹⁹ The EU, Ukraine, Saudi Arabia and Taiwan requested to join in the consultations. The DSB established a panel in November 2018. Nonetheless, the proceedings were suspended later, and no substantive decision has been made to date.²⁰

Likewise, on 1 June 2018, the EU requested consultations with China (DS549), claiming that China had violated several provisions of the TRIPS Agreement, GATT 1994, and its own Protocol of Accession. Japan, the US and Taiwan requested to join in the consultations.²¹ The EU replaced and revised this request on 8 January 2019. In this revised version, the EU detailed

how China had violated a variety of WTO Agreements by applying its JV-related regulations and statutes, including but not limited to the Catalogue of Industries for Guiding Foreign Investment (2017), the Foreign Trade Law of the People's Republic of China, the Anti-Monopoly Law, and 'unpublished measures through which China imposes and administers the above restrictions'.²² Taiwan, Japan and the US requested joining in the consultations again in mid-January 2019. The EU had not requested the establishment of a panel.

The WTO dispute resolution mechanism has not been able to address China's FTT, partly because Chinese laws on the surface do not have clauses on FTT. Admittedly, the jurisprudence of the WTO recognises unwritten practices as a basis for a claim (Brum, 2019). However, foreign companies are usually unwilling to accuse their Chinese collaborators, fearing that they will encounter more serious political and economic countermeasures in the future.

Foreign companies have faced informal pressures when doing business in China. Many foreign companies that are involved in FTT are usually MNCs. This means that they may have other businesses and assets in China, in addition to the business in which the FTT is taking place. Therefore, offending their Chinese partners may put their other business activities in China in jeopardy. It simply does not pay for them to bring a case against a Chinese business, which could have a high degree of support from the CCP, if the foreign company faces the prospect of terminating all businesses in China. As a result, foreign companies either refuse the unequal offer, or do not transfer their technologies in the first place. This explains why there are very few relevant cases so far that involve technology transfer in the WTO jurisprudence (Tundang, 2020).

Alternatively, there is another option for the foreign company: to file a civil-suit complaint in China, with the hope of legally refusing the FTT request from the Chinese JV partner, and retaining whatever contract conditions were originally perceived by the foreign company. If this is successful through a Chinese court ruling, then it seems to be nothing more than a civil case between two businesses, and will not escalate to the stage of a WTO dispute, where two countries are confronting each other.

The Judicial System in China

In China, the judicial system comprises four levels: basic courts, intermediate courts, provincial high courts, and the Supreme People's Court (SPC). In addition to these four tranches, China also established multi-provincial circuit courts in 2015 and specialised commercial courts in 2018 (deLisle, 2019). The establishment of a comprehensive court system, nonetheless, does not imply a separation of powers between the judicial and the executive branches. Instead, given its exponential economic growth in the past three decades, China has become more confident in its own political system, which emphasises, among other things, 'the rule of law with Chinese characteristics'.

For example, the president of the SPC had publicly averred that China does not need judicial independence, which is ‘the “trap” of “Western” ideology’.²³ Admittedly, judicial reform is ongoing, but the aim is not to enhance judicial independence. On the contrary, it endeavours to further weaponise the judiciary in order to meet the needs of the central government, rather than those of the people. Therefore, when speaking of the judiciary in China, it is crucial to bear in mind that the judiciary is *not* a coordinating branch amongst the three. Instead, it is a supporting institute of the government.

Furthermore, given that China is a party-state, in which the CCP has firm control of every apparatus of the government, the impact of the CCP upon the judiciary cannot be overlooked. In China, there is neither parliamentary sovereignty nor judicial supremacy, but only party supremacy. This can be seen from the fact that every president of the SPC has been charged by the CCP with overseeing lower courts since 1954 (Zhu, 2010). Furthermore, the presidents of each of the lower courts shoulder similar responsibilities within their respective jurisdictions. In other words, the CCP has planned to ‘equip courts with cadres’ for decades (CCP Central Committee Office, 1985 (Li, 2012). Since the leader of the party will simultaneously be the president of the court at the same level (Li, 2012), the judiciary is usually the CCP’s mouthpiece. Accordingly, decisions by the SPC may simply reiterate what has already been decided by the Central Committee of the CCP (Zhu, 2010).

Against this background, it should not be surprising that corruption in China’s judicial system is often reported or widely disseminated. Although it has rarely been documented officially, empirical studies have pointed out that corruption ‘exists in all main court divisions where key functional judicial power is exercised [and] occurs at almost all levels of the judicial system, involving all types of judges, regardless of their rank, level of education, and income’ (Li, 2010). Indeed, it is an institutionalised activity that should be attributed to the mechanisms of the judiciary (Li, 2012). To be more specific, it has been argued that judicial decision-making processes in China are influenced by four categories of factors (Ng and He, 2017): administrative, political, social and economic factors, each of which can easily lead to irregularities.

To be specific, administrative factors include the hierarchical and bureaucratic control over the judges. In China, there is little difference between courts and other government bureaus in terms of the decision-making process (Ng and He, 2017). Decisions made by junior judges are subject to the supervision of senior judges and the presidents of the courts. These presidents, who control the promotion and discipline of other judges in their respective courts, are not only CCP members but also party secretaries. As if this is not absurd enough, the president of a court does not necessarily need to have had a law degree for an extensive period in the past. Although to some extent this has been changed after the revision of China’s Judges Law in 2019, legal expertise is still not the first priority when court presidents are chosen. Instead, political savvy and absolute loyalty may be prioritised. Furthermore, another

institutional design that aims to limit judicial independence is the adjudication committee inside each court. This committee is composed of the president, vice presidents, division heads, and, most importantly, the disciplinary inspector of the local party, a representative of the party leader who has the final say in a court (Ng and He, 2017).

On top of the administrative factor that hinders judicial independence, courts are treated as political weapons of the CCP. In other words, there is no separation of powers in China, and courts of different levels are subordinate to the people's congress of the same level, which in turn is controlled by the CCP. This is reflected in the political manipulation of the judicial docket. The political branches, or the CCP, can require the judiciary to shoulder some responsibilities that are completely outside the scope of judicial power on the one hand, but can dissuade the judiciary from hearing certain cases because of their political sensitivity, on the other. All these extrajudicial activities are overseen by the CCP's Political and Legal Committees (PLC), which are instituted at each level of the judiciary.

The social factor refers to the undue influences that come from a judge's professional or personal networks. In China, where relationship, or in Chinese, *guanxi*, is particularly important in every domain of daily life, judges could easily consider their relationships with the parties in addition to the merits in the cases. What is worse, the social ties may extend indefinitely. Occasionally, judges not only consider their own social ties, but also those of their superiors (Li, 2012). They may do so voluntarily or involuntarily, given that their superiors control their discipline and promotion.

Finally, judicial independence is heavily restricted in China, because the funding of courts comes from local governments of the same level, rather than the central government. Although China has become the second biggest economy on earth, the degree of economic development actually varies dramatically from one region to another. Some coastal areas are incredibly affluent, while inland counties are miserably poor. To wit, court budgets depend on the vibrancy of the local economy to a considerable extent. When funding is not sufficient, courts must fund themselves. This inevitably results in judicial dependence on local governments and ultimately, to local protectionism.

Filing a Suit in China?

Empirical studies have suggested that local courts favour local firms. As suggested, some of this favouritism might be to secure the revenue of local governments; because doing so will indirectly increase the funding of local courts. Due to this peculiar situation, judicial independence could be even more severely lacking in places where there is little foreign investment (Wang, 2015).

Of particular note, the four factors mentioned above (administrative, political, social and economic) that influence judicial decisions are mutually reinforcing. For example, the hierarchal supervision may aggravate the undue

influence that comes from the social ties, because it is difficult to reject a ‘plea’ from someone who can supervise you.

For example, take the trademark lawsuit of Michael Jordan, undoubtedly the most famous basketball player in the world. A Chinese company established Qiaodan Sports, which is the Chinese translation of ‘Jordan’ in 2000, and used a trademark that was misleadingly similar to Nike’s Air Jordan brand: the silhouette of Michael Jordan. Since its inauguration, Qiaodan Sports has opened 6,000 stores across China and registered about 200 trademarks.²⁴ After Jordan discovered this in 2012, he filed 80 lawsuits against Qiaodan in China. The Chinese lower courts accepted Qiaodan’s argument that ‘Jordan’ is a common English-language last name and does not refer to Michael Jordan specifically. After appealing, the SPC finally delivered a decision partly in favour of Jordan in 2020²⁵ and ruled that Qiaodan should pay Jordan US\$46,000 in compensation, a trivial amount compared with how much Qiaodan had earned and how much Jordan had lost in the two decades of this trademark’s usage in China.

Another case is McDonald’s, which sued a Chinese company, Wonderful, over a trademark dispute. Registered in 2001, the logo of Wonderful is a ‘W’, which rotates McDonald’s iconic ‘M’ in yellow with a red background without changing the radian, scale and other designs.²⁶ In 2011, Beijing’s First Intermediate People’s Court rejected McDonald’s claim that Wonderful’s logo closely resembled that of McDonald’s. McDonald’s appealed, only to be rejected again in 2013.²⁷

Another possibility is that the foreign company may face suits and harassments other than the original dispute. Companies that failed to comply with (say, FTT) requests of the local government may face harassments such as tax evasion checks, emission violation charges, antitrust suits, labour rights complaints, and factory fire-alarm checks, etc. All these harassments may be linked to the only real problem that is involved.²⁸

In summary, if a foreign company affiliated with country A faces the threat of FTT from China and has experienced some harassment, should it comply with country A and let it file a complaint with the WTO? Definitely not, because doing so would make it impossible for all this company’s businesses in China to operate in the future. Should it bring a civil suit against this unfair treatment in China? Definitely not, because a possibly unfair judicial system might not be considered as an effective way to remedy the FTT, as the cases of Jordan and McDonald’s have taught us. Should it withdraw from the market in China? If it does so, it may incur a huge sunk cost already invested in China. This is indeed the paradox faced by many foreign investors.

Reforms and Changes Under the Xi Regime

To be fair, China has made some progress in judicial independence, at least in the commercial area. The Chinese government has recently pushed forward a series of judicial reforms, though whether, or to what extent, this effort will

be translated into reality remains questionable. For example, the Chinese government has vowed to make the judicial system more transparent, at least in certain areas, such as the number and identity of judges and the rulings in certain categories (Finder, 2019). Nonetheless, data that are perceived by the CCP as dangerous to social stability (such as cases regarding the death penalty and judicial discipline) are still kept confidential. Moreover, the SPC has claimed that the funding of lower courts by the central government will be increased, so that lower courts need not rely on local governments (too much). While this may help the central government to combat local protectionism and reduce the economic control that local governments impose on local courts, it also means that the administrative and political control over the judiciary will be increased as well. For these reasons, foreign companies may prefer to solve FTT issues informally or under the table, in order not to waste their efforts and money on litigation.

Moreover, the role of the SOEs in China makes the problem of FTT even thornier. Whether an SOE is bound by WTO agreements and rules or not remains a conundrum. Arguably, SOEs are not directly regulated by WTO agreements, which constrain the Members. Even though China promised that its SOEs would engage in fair competition and fulfil the non-discriminatory obligation in its Accession Protocol, those provisions have rarely been invoked in any disputes between WTO Members (Yang, 2018; Zhou et al., 2019). Studies have maintained that '[Many GATT rules] restrict the types of regulations which governments can impose on international traders, but do not purport to regulate the traders themselves. If the government is the trader or controls the trader, the rules may be ineffective since decisions ostensibly made independently by the trader may in fact reflect actions of the government' (Jackson et al., 2008, p. 444).

This is definitely a loophole, because SOEs in China are essentially the extended arms of the CCP. In China, both the distinction between the government and the party, and the line between the state and the private sector, are non-transparent and unpredictable. Indeed, even private companies sometimes would have to shoulder political missions in addition to their own business agenda, because many of them might be firmly controlled by the government. This is because 'Xi [Jinping] believes that the Party-state has to maintain tight control over every aspect of the political and social economic domain in order to ensure economic prosperity as well as political and social stability' (Yu, 2019). Let alone the SOE.

Moreover, given the policy of 'state enterprises advance, the private sectors retreat', SOEs have gradually come to control the Chinese market that was once dominated by private companies (Economy, 2018). This means that foreign companies may have even less choice for their JV candidates other than SOEs, when they want to conduct business in China in the future. As a consequence, technology transfer that takes place under this scenario is more likely to be forced by the SOEs, or by the CCP behind them. Nonetheless, the WTO regime is unable to effectively regulate these SOEs that are acting

under the guise of the freedom of contract. The WTO'S inefficacy in this regard has partly resulted in the 2020 Joint Statement of the Trilateral Meeting of the Trade Ministers of Japan, the United States and the European Union.²⁹ In this Joint Statement, the three economic superpowers have urged the importance of market-oriented conditions and the reform of the WTO. Although the Statement did not explicitly blame China, it is clear that all the announcements target Beijing.

Genuine Reforms?

Given that FTT has been heavily criticised by many, if not all, foreign companies that do business in China, the CCP has taken several measures in response to these critiques. On the one hand, it consistently denies any accusation that there is FTT in China, claiming that all foreign companies 'voluntarily' transfer their technologies and trade secrets to Chinese companies. According to its argument, the government never intervenes, let alone coerces foreign companies to submit. Instead, the Chinese laws clearly prohibit FTT and honour the spirit of the WTO rules. That is, foreign companies are well protected in China and all technology transfer is made with freedom of contract.

On the other hand, the CCP has also planned to amend some statutes and regulations to pacify the growing tension between China and foreign companies and countries. For example, in 2019, China revised its Anti-Unfair Competition Law, to which it added Article 32, which stipulates the burden-shifting in civil proceedings regarding the theft of trade secrets. As mentioned above, it is extremely difficult for a right-holder to prove that the accused has stolen trade secrets, know-how, and other IP from it. This provision requires the accused to disclose or prove that it has not conducted any misappropriation, and might enhance the protection of foreign companies that are reluctant to bring their cases to Chinese courts because of the burden of proof.

Additionally, China has planned to relax the restrictions on ownership in certain sectors by removing the aforementioned 50 per cent cap restriction by 2022. Moreover, it has promulgated the Foreign Investment Law, which took effect on 1 January 2020, several articles of which aim to protect foreign companies' rights and interests and further open the domestic market. For instance, Article 4 of the Foreign Investment Law maintains that the treatment given to foreign investors during the investment-access stage cannot be lower than that given to their domestic counterparts. Article 22 stipulates that the state should protect the IP rights of foreign investors and enterprises; Article 23 provides the government officials' confidentiality obligation and protects the trade secrets from being disclosed by government officers. Furthermore, Article 39 provides the legal liability for violating the said two provisions. In addition, Article 5 of China's Administrative License Law also stipulates the confidentiality obligation of administrative agencies, and Article 32 clearly prohibits FTT.

On top of these domestic legal changes, China also reached agreement with the US and promulgated the so-called ‘Phase One’ trade deal in January 2020. By concentrating mostly on FTT issues, the deal emphasises market terms, voluntary transfer, and governmental non-intervention.³⁰ In Chapter 2 of the Agreement which regulates technology transfer, Article 2.2 unmistakably provides that ‘Neither Party shall require or pressure persons of the other Party to transfer technology to its persons in relation to acquisitions, joint ventures, or other investment transactions.’ Article 2.3 tackles the problem of administrative and licensing requirements, and Article 2.4 deals with due process and transparency. All are pressing problems that need to be addressed immediately. All these send a message to foreign countries and their investors that China has no intention of forcing technology transfer and that China will adhere to fair trade practices. Moreover, this could be seen as an olive branch in the US–China trade war, which may shift the blame to the US if the trade war continues.

Notwithstanding these legal reforms, people in the fields of international commerce and IP law generally remain sceptical about their effects (Lee, 2020b). As mentioned above, FTT had been legally prohibited even before the said legal reforms. To a considerable extent, these legal changes simply reiterate the *de jure* status quo. The real problem is not the lack of regulations, but the Chinese government’s determination to implement these laws. It is widely known that the gap between law in books and law in action is wide in China. As we mentioned, one of the reasons for spawning FTT is the difficulty in proving its existence. Moreover, the wording of these laws is vague, which ‘allow[s] Beijing to technically comply with its international commitments and project an image of openness, while continuing to maintain its influence over foreign investors for use as potential leverage in its broader dealings with other state players’ (Kim and Lo, 2019).

In fact, some surveys seem to undergird this scepticism. For example, the European Chamber in its latest Business Confidence Survey 2020 pointed out that

16% of respondents report having felt compelled to transfer technology in order to maintain market access, down four percentage points from 2019. However, a sectoral breakdown shows that industries have very different technology transfer experiences. In industries like medical devices, aerospace and aviation, and environment—the crown jewels of European innovation—nearly a third of members report having been compelled to transfer technology in order to maintain market access. Furthermore, of the respondents that felt compelled to do so, 36% report that the incident happened less than a year ago.³¹

Likewise, the USTR has contended in its 2020 Report to Congress on China’s WTO Compliance that ‘It is more difficult to assess China’s implementation of the chapter on technology transfer, as these commitments extend to

informal, unwritten measures that China takes to force or pressure foreign companies to transfer their technology to Chinese entities.³²

Although these surveys have been conducted during the process of China's legal reform, it is unclear to what extent the situation will change in the near future. Conceptually, technology advantage is a life-and-death issue to the CCP, because the legitimacy of its dictatorship depends largely on prosperous economic growth.³³ If the economic development remains stagnant or even encounters recession, the CCP's reign and authority might encounter a serious challenge.

The CCP clearly and correctly knows that its current successful low- and middle-tech manufacturing economy is not enough to maintain the exponential economic growth it has showcased to both international and internal audiences in the past three decades. Vietnam, India and Bangladesh have gradually become the default choice for low-cost, offshore manufacturing. Therefore, China must shift to a sophisticated high-tech economy, by cajoling, co-opting, and often coercing Western and Japanese businesses (Hout and Ghemawat, 2010). Furthermore, the 'cajoling, co-opting, and often coercing' strategy explains why FTT will not disappear (Hout and Ghemawat, 2010).

This is concretely reflected in the *Made in China 2025* – the national strategic plan to develop China's manufacturing sector, particularly in those areas involving cutting-edge technology. Announced in 2015, this project identifies ten priority sectors, including but not limited to information technology, numerical control tools and robotics, and new materials. The end goal is the transformation of China's economy by making China not only self-sufficient, but also the leader in these high-tech fields. Since its promulgation, it has spawned many controversies and complaints as to the approaches adopted to implement this goal.

For example, the EU Chamber of Commerce has clearly condemned FTT as one of the strategies that China has adopted to achieve this *Made in China 2025* project, in addition to restrictions of market access and other tools (EU Chamber of Commerce in China, 2017, p. 56; Gordon and Milhaupt, 2019). In a nutshell, we should be cautious about the progress of China's legal reform that it claims will eliminate FTT. Instead of relying on China's lip-service reforms, democracies need to discover other ways out of the legal labyrinth.

What Could be the Solutions?

As mentioned above, FTT is not a problem unique to China – it is prevalent in all developing countries around the world. What makes FTTs in China different from that in other jurisdictions is China's global economic clout and political ambitions (Mastel, 1996). It has become a bipartisan consensus in the US that some countermeasures must be adopted. The disagreement between the political parties concerns what exactly these countermeasures should be.

It has been suggested that indiscriminate tit-for-tat tariffs may not be the best solution, as these are too broad to be truly effective. Instead, a better solution should be more narrowly tailored. That is, the solutions should

target those industries that are either most vulnerable to FTTs, or encounter FTTs most frequently. This is because FTT does not occur in every industry; nor is FTT equally effective in every industry. In particular, some cutting-edge technologies are closely related to national security, including but not limited to information technology, aerospace engineering, and new energy. In fact, it has been reported that ‘The Pentagon and defense companies have long maintained that the theft of IP has permitted China to create systems that closely resemble US weapons such as helicopters, armed drones, and jet fighters’ (Carbaugh and Wassel, 2019).

More importantly, the solution should be multinational rather than unilateral. Because of the prisoner’s dilemma, the FTT has a spillover effect that is detrimental to both the ‘donor companies’ and other foreign companies in the same fields. Hence, it is suggested that the US and other traditional democratic countries with advanced technology industries, such as Germany and Japan, should form an alliance because they also face similar problems (Carbaugh and Wassel, 2019; Branstetter, 2018). The said trilateral Joint Statement is a good start (see Note 3), but is far from being enough. Other measures might also be needed. Combining these two strategies might be able to ensure China to modify its policy.

Certainly, there are some difficulties that should be resolved in advance. Indeed, the prisoner’s dilemma is a dilemma because it is individually rational for each country to comply with China unilaterally. As mentioned above, foreign companies are reluctant to publicly disclose the oppression they have experienced. This might not be a problem because they might be more willing to report to their own governments, at least anonymously. But the second issue involves coordination: some foreign countries may want to free-ride or piggyback on others’ resistance because the US is one of the few democracies, perhaps the only one, that can stand toe-to-toe with China. Smaller countries may have concerns that China will seek retribution not only economically but also diplomatically, a risk they simply cannot take. To solve this problem, technology-holding countries might need to jointly address the FTT issue.

Finally, the WTO jurisprudence has recognised unwritten practices as a basis for a claim (Brum, 2019). This suggests that, even without a textual violation, it is still possible for the WTO panel to find that China is in breach of provisions of the TRIPS and other Agreements regarding honest commercial practice, transparency, and so on. It is reasonable to assume that the more that the coordinated works are put together, the more likely it will be able to substantiate the argument.

Notes

- 1 See *China and the World Trade Organization – Communication from China*, WT/GC/W/749 (23 July 2018).
- 2 See for instance, *China – Certain Measures Concerning the Protection of Intellectual Property Rights* (‘China – Intellectual Property Rights II’) – Request

for the establishment of a panel by the United States, WT/DS542/8 (18 October 2018), in which the US brought up the case against China concerning the latter's forced technology transfers. The complaint was about China's possible violation of Article 3.1 (national treatment) and Article 28 (patent rights) of the TRIPS Agreement. The case was later dropped because China amended its domestic law. The details will be discussed in a later section.

- 3 'Joint Statement of the Trilateral Meeting of the Trade Ministers of the United States, European Union, and Japan' (14 January 2020), https://trade.ec.europa.eu/doclib/docs/2020/january/tradoc_158567.pdf.
- 4 D. Dodwell, 'China is on the Path to Global Technology Dominance', *South China Morning Post* (24 March 2017), www.scmp.com/business/global-economy/article/2081771/be-afraid-china-path-global-technology-dominance.
- 5 *Protocol on The Accession of The People's Republic of China*, WT/L/432 (December 10 2001), www.worldtradelaw.net/misc/ChinaAccessionProtocol.pdf.download.
- 6 F. Abbott, 'Inside Views: US Section 301, China, and Technology Transfer: Law and Its Limitations Revisited (Again)', *Intellectual Property Watch* (6 June 2018), www.ip-watch.org/2018/06/07/us-section-301-china-technology-transfer-law-limitations-revisited/.
- 7 S. Babones, 'The Trade War That Wasn't: Tit-For-Tat Tariffs Are Unlikely to Have Any Real Effect', *Forbes* (5 April 2018), www.forbes.com/sites/salvatorebabones/2018/04/05/the-trade-war-that-wasnt-tit-for-tat-tariffs-are-unlikely-to-have-any-real-effect/?sh=52895c733ec5.
- 8 White House Office of Trade and Manufacturing Policy, *How China's Economic Aggression Threatens the Technologies and Intellectual Property of the United States and the World* (June 2018), 2–4, <https://trumpwhitehouse.archives.gov/wp-content/uploads/2018/06/FINAL-China-Technology-Report-6.18.18-PDF.pdf>. USTR, *2019 Report to Congress on China's WTO Compliance* (March 2020), 12–13, https://ustr.gov/sites/default/files/2019_Report_on_China%E2%80%99s_WTO_Compliance.pdf.
- 9 USTR, *Findings of The Investigation into China's Acts, Policies, and Practices Related to the Technology Transfer* (22 March 2018), <https://ustr.gov/sites/default/files/Section%20301%20FINAL.PDF>.
- 10 White House Office of Trade and Manufacturing Policy, *How China's Economic Aggression Threatens ...*, 6.
- 11 United States International Trade Commission, *China: Effects of Intellectual Property Infringement and Indigenous Innovation Policies on the U.S. Economy* (May 2011), www.usitc.gov/publications/332/pub4226.pdf.
- 12 White House Office of Trade and Manufacturing Policy, *How China's Economic Aggression Threatens ...*, 2–4; USTR, *2019 Report to Congress on China's WTO Compliance*, 33.
- 13 USTR, *2019 Report to Congress on China's WTO Compliance*, 13.
- 14 Keith Bradsher, 'How China Obtains American Trade Secrets', *New York Times* (15 January 2020), www.nytimes.com/2020/01/15/business/china-technology-transfer.html.
- 15 Yu Zhou, 'U.S. Trade Negotiators Want to End China's Forced Tech Transfers. That Could Backfire', *Washington Post* (28 January 2019), www.washingtonpost.com/news/monkey-cage/wp/2019/01/28/u-s-trade-negotiators-want-to-end-chinas-forced-tech-transfers-that-could-backfire/.

- 16 Occasionally, FTT could spawn a counterproductive effect, meaning that it might discourage technology transfer that would have occurred without the pressure. Studies have suggested that owing to China's poor protection of intellectual property rights, foreign companies would 'avoid fully integrating with the Chinese operation' and usually transfer at least five-year-old or soon-to-be obsolete technologies (Schiappacasse, 2004).
- 17 *Protocol on the Accession of The People's Republic of China*, WT/L/432 (10 December 2001), www.worldtradelaw.net/misc/ChinaAccessionProtocol.pdf.download.
- 18 *Report of the Working Party on the Accession of China*, WT/ACC/CHN/49 (1 October 2001), <https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=Q:WT/ACC/CHN49.pdf&Open=True>.
- 19 *China – Certain Measures Concerning the Protection of Intellectual Property Rights*
- 20 Neither the US or China have ever explained to the public the reason for suspension of the panel proceeding. A plausible guess is that China had removed certain disputed legal provisions in March and May 2019, including Article 24(3) (requiring foreign licensors to indemnify Chinese licensees for all liabilities for infringement resulting from the use of the transferred technology), Article 27 (requiring that any improvements in imported technology belong to the party making the improvement), Article 29 (prohibiting a license contract from restricting a Chinese party from improving the technology or from using the improved technology) of the TIER and Article 43(4) of the Regulations for the Implementation of the Law of the People's Republic of China on Chinese Foreign Equity Joint Ventures (providing a Chinese JV party with the right to continue to use technology transferred after the expiration of the transfer contract). The US suspension request made in June 2019 might have also been in response to the upcoming Trump-Xi talk during the 28 June G20 Summit.
- 21 *China – Certain Measures on the Transfer of Technology – Request for consultations by the European Union*, WT/DS549/1 (6 June 2018), <https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:WT/DS/549-1.pdf&Open=True>.
- 22 Ibid.
- 23 Michael Forsythe, 'China's Chief Justice Rejects an Independent Judiciary, and Reformers Wince', *New York Times* (18 January 2017), www.nytimes.com/2017/01/18/world/asia/china-chief-justice-courts-zhou-qiang.html?smid=fb-share&fbclid=IwAR26YbirMgxtRPXdeDLCsJFzZp0IUphjOYrunoHVsJtCjKdEQDjMx10-R_A.
- 24 R. Davis, 'Michael Jordan Awarded \$46,000 for "Emotional Damages" but Not Full Rights to His Own Name in China', *Variety* (31 December 2020), <https://variety.com/2020/sports/news/michael-jordan-china-trademark-lawsuit-qiaodan-1234877353/>.
- 25 Supreme People's Court (2018) Administrative Retrial No 32.
- 26 Y. Yunting, 'How Could McDonald's Beat Free Rider of Trademark in China?', *Bridge IP Commentary* (23 October 2011), www.chinaiplawyer.com/administrative-litigation/.
- 27 Xinhua Net, 'McDonald's sued Trademark Appeal Board and was Rejected', *Sina News* (4 September 2013), <http://finance.sina.com.cn/chanjing/gsnews/20130904/172316664285.shtml>.
- 28 We do have evidence supporting our statement here, although the company involved does not want to have its case made public.

- 29 'Joint Statement of the Trilateral Meeting ...'.
- 30 *Economic and Trade Agreement Between the Government of the United States of America and the Government of the People's Republic Of China* (January 2020), https://ustr.gov/sites/default/files/files/agreements/phase%20one%20agreement/Economic_And_Trade_Agreement_Between_The_United_States_And_China_Text.pdf.
- 31 European Union Chamber of Commerce in China, *European Business in China Business Confidence Survey 2020* (10 June 2020), www.europeanchamber.com.cn/en/publications-archive/774/European_Business_in_China_Business_Confidence_Survey_2020.
- 32 USTR, *2020 Report to Congress on China's WTO Compliance* (January 2021), <https://ustr.gov/sites/default/files/files/reports/2020/2020USTRReportCongressChinaWTOCompliance.pdf>.
- 33 T. Ferguson, 'China's Communist Party and Lost Legitimacy', *Forbes* (29 June 2012), www.forbes.com/sites/timferguson/2012/06/29/chinas-communist-party-and-lost-legitimacy/?sh=dc4b393676da.

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9 How to Dance with the Dragon?

Chinese-style Globalisation?

China's extremely rapid economic growth in the past three decades has made it a unique economic development model (for some people, even a role model) to study. We have seen many books and papers discussing 'Chinese-style capitalism' or 'socialist market economy'. Most of the previous analyses along these lines has been trying to understand how, in the process of its development, China has resolved its various domestic problems; how it has transformed its traditional agricultural and banking sectors; what role the state-owned enterprises have played in the process of development; how China has implemented the principle of 'crossing the river by feeling the stones', and how its economic planning mechanism has been intertwined with the market mechanism. In a word, previous studies have focused mainly on the various *internal* operations of China's unique economic system.

The purpose of this monograph is very different: here we study the *external* interactions between China and the rest of the world. Specifically, we analyse how China's unique economic structure has fit into the long-evolving global economy. Common sense tells us that the term 'peoples' republic' for a communist regime usually does not have any democratic implications in terms of the word 'republic'. Hence, we would not be surprised if a socialist market economy were to be very different from a market economy. This implies that China's economic regime may have some external interface with the liberal market economies in the world. Evidently, we need both internal and external angles to have a full picture. Therefore, this book endeavours to complement previous studies on China's economy.

In 2020, China (as the complaining party) lost the dispute over its market economy status in the WTO, allowing the dispute to lapse.¹ Though the WTO panel report was never made available to the public, it is reasonable to infer that in its interim report, the panel had touched upon the most sensitive part – that is, China's unique economic system was considered to be that of a non-market species. In previous chapters, we have explained why various aspects of the Chinese system may cause conflicts with democratic regimes.

As we come to the end of this book, we would like to put aside such detailed conflicts, and get back to the macro picture of ‘globalisation’.

If we search through dictionaries, we find that the term ‘globalisation’ can be defined as something like ‘the speeding up of movements and exchanges (of human beings, goods, services, capital, technologies or cultural practices) all over the planet’. One effect of globalisation is that it ‘promotes and increases interactions between different regions and populations around the globe’.

If we look at the *entities* of the above-mentioned increasing interactions, we would agree that these entities often refer to individuals, private businesses, or social groups, such as environmental protection activists or human rights NGOs. In these definitions, the entities being referred to usually do not include political ones. Indeed, if political entities were to be included, then globalisation would have been an element for centuries, and we should not have had to invent such a new term in the first place.

For instance, Alexander the Great conquered a large area of Egypt, southern Europe, western Asia and India; his kingdom was a global one, nearly 3,000 years ago. The Mongolian empire, the Napoleonic empire, the British and Spanish colonial empires in the American and African continents, and even NATO formed in 1949, are all examples of the globalisation of *political* entities and activities. To paraphrase Thomas Friedman (2005), as far as the military force exercised by political entities has been of concern, *the world has already been flat* for centuries. The fact that we usually do not connect such political empires with the word ‘globalisation’ suggests that the entities involved in the globalisation in the 21st century are expected to be individuals and private businesses.

Conceptually, we also have good reason not to connect globalisation with political entities. Globalisation is often associated with the advancement of transportation technologies and the Internet, which have significantly reduced the cost of communication and of travelling around the world. For powerful empire builders in ancient times, however, long-distance travelling was never a hurdle that could interfere with their strong political will. As such, the advancement of transportation technologies and the Internet has dramatically lowered the cost of communication and travelling for many that originally did not have that capacity. The activities referred to in the globalisation era are mostly economic, cultural and educational, etc.; this is why we see ‘goods, services, capital, technologies or cultural practices’ in the previously mentioned definition of globalisation. In discussions on globalisation that we know of, there are no references to any political activities implemented by any governments.

From this, we can see why many democratic countries have felt uncomfortable with the rise of China over the past three decades: China has been disturbing and distorting the very definition of globalisation.

The Internal/External Conflicts of China’s Globalisation

The key feature of China’s global activities in the past couple of decades has been that almost all the activities that China has been engaged in can be

connected to some political entities and usually have some political purposes. Based on the doctrine of collectivism, by definition most things in China are to be determined collectively. In practice, however, these collective decisions are actually made by the Party-state. Let us review some examples that are either well-known or have already been mentioned in previous chapters.

- 1) As for educational and cultural interactions, such activities in democratic countries are always initiated and implemented by private entities. However, since 2004, China has established roughly 550 Confucius Institutes and more than 1,190 tutoring classes around the world,² all being supported by the Chinese government. Because such cultural and educational institutes have often been found to have collective national agendas, many countries in the world have begun to ban their operation. When cultural and educational activities are covered and supported by a political entity under the ideology of collectivism, it is more like the cultural infiltration of an empire, rather than a true globalisation activity.
- 2) There are hundreds of app companies and dozens of social media businesses around the world, with each of them independently trying to find new business opportunities. However, under China's collectivism, all high-tech giants in China are connected to the CCP. China's software firms may design products such as TikTok and WeChat, which have algorithms that have been alleged to have a hidden agenda (such as avoiding the circulation of politically sensitive videos) set by the underlying political entity. If entertainment programs circulating internationally are screened and penetrated according to a collective will, it is more closely resembles the spreading of an empire's ideology, rather than a true globalisation activity.
- 3) In 2021, some businesses started to ban the use of cotton planted in Xinjiang. This was due to the 're-education camps' for the Uyghurs in Xinjiang, a situation which many democratic countries believe to be close to an ethnic 'cleansing'. Many international boycotts similar to that against Xinjiang cotton production have previously been held, for example, one that boycotted the production of diamonds after the film *Blood Diamond* publicised the exploitation of workers in Sierra Leone. This time, however, the business boycotts have been counteracted by China. In retaliation, the Chinese government has identified several businesses and banned their operations in China. Using the Swedish fast-fashion company H&M as an example, one week after H&M was identified as a target, its garments were unavailable on nearly all of China's e-commerce apps, and its stores have disappeared from smartphone maps. People in democratic countries may wonder how could a physical store disappear from the map?³ Well, it just can easily happen in China. Under the ideology of collectivism, it is the Chinese government that coordinates and determines how the market works. This is a case of how a Chinese political entity embroils itself with the activities of international businesses.

To sum up, the traditional wisdom of globalisation refers to the increasing interactions of individuals and private businesses due to the convenience of transportation or electronic transmission, whereas China has its own definition of globalisation. Under collectivism and authoritarian control, China's version of globalisation refers to the concerted export of ideologies (through education and cultural activities), politically determined market access (to friendly businesses), and the imposition of its ideology and CCP supremacy in apps and social media. To China, globalisation does not mean 'China goes to the globe.' Instead, taking advantage of its enormous economic and political clout, China sees globalisation as an opportunity to bring every other country in the world into line. We may refer to this as the *Chinese backdoor listing of globalisation*.

In 2001, China joined the WTO, and was able to enjoy the 'flat' world market with few trade barriers. China has greatly benefited from globalisation. It has increased its GDP roughly 40 times in 30 years, and since 2010, has become the world's second largest economy. In 2004, it began to establish the Confucius Institutes worldwide, to export national propaganda. In 2010, China sanctioned the importation of Norwegian salmon, to punish the latter's awarding the Nobel Peace Prize to China's opposition protestor, Liu Xiaobo. This is the most famous example of China as a political entity imposing sanctions on economic exports. After 2010, China's politically backed economic interference steadily increased, and we started to see frequent boycotts and sanctions in response to all kinds of events that China claimed had impacted its 'national feeling'. Under collectivism and authoritarian control, these boycotts gradually became routine and systemic rather than incidental and sporadic, and the scale and strength of such sanctions have been increasing. The 2021 Xinjiang cotton event made the democratic world step back and think: if China was to shape a new globalisation order, how different would the world be?

Here we clearly see the fundamental conflicts between an internal collectivist regime and an external globalisation environment. As we have argued in Chapter 5, if Chinese companies want to participate in the global financial market and obtain global funds, then they must face the accountability requests of global investors. However, this is inconsistent with the CCP's logic of internal control under collectivism. We have also demonstrated that in democratic market economies the government always remains at an arm's-length distance from private businesses, but this globally common distance between private and public entities is fundamentally inconsistent with China's collectivism. In short, China faces a systemic paradox between internal authoritarian controls and merging into the external global market.⁴ This internal/external paradox has become more and more severe in recent years.

In various chapters of this book, we have elaborated many types of economic conflicts between China and democratic countries. These conflicts, as we have shown, are rooted in the political regime differences: democratic

countries and authoritarian countries have distinct political ideologies, constitutional designs, administrative regimes, power structures, and most importantly their respective *values*, which are behind all such differences. We argue that it is the value-based institutional differences that eventually drive the economic conflicts, and therefore such conflicts are somewhat inevitable.

Value Differences Between Authoritarianism and a Liberal Market Economy

A good example of the economic differences caused by value distinctions has been made by the American economic historian and 1993 Nobel Prize winner Robert Fogel. In his book *Time on the Cross* (1974), co-authored with Stanley Engerman, the authors argued that the US system of slavery in the South was profitable for slave owners, because they could organise the cotton plantation labour input to maximise their profits, and the physical fatigue or work enthusiasm of slaves was not the concern of slave owners. Due to the economies of scale of labour on cotton plantations, Southern slave farms were more productive per unit of labour than Northern farms. The implication from this is that slavery in the American South was not quickly going away on its own (as it had in some historical instances such as in ancient Rome) because, despite its exploitative nature, slavery was profitable and productive for slave owners.

Whether a slavery system is allowed in a society or not is of course a matter of *value*. If we treat human dignity and individual freedom as values that transcend production efficiency, then we should abolish the slavery system, no matter how economically efficient it is, compared with the society that does not allow it. Thus, it is the *value* that determines the relative efficiency of cotton plantations between North and South.

The above story is very much like the scenario in all our previous chapters. For instance, e-commerce platforms in democratic countries are less competitive than Alibaba in China, because the latter has benefited from China's Internet blocking, which results in asymmetric market access to the advantage of Alibaba. Democratic e-commerce platforms are less competitive simply because they cannot be effectively reached by Chinese citizens. As another example, Chinese high-tech businesses are more competitive because they receive staggeringly high government subsidies from initiatives such as *Made in China 2025*. Democratic regimes cannot implement these 'competitive strategies' such as Internet blocking or unconstrained government subsidies, because they respect the value of free information flow and the laws regulating the use of governmental resources.

The more important message that Fogel and Engerman delivered was that, because the slavery system was profitable, 'it would not quickly go away.' This seems to drive us to a disturbing conclusion. If China has benefited from its Internet blocking, overwhelming government subsidies, and widespread use of artificial intelligence to intrude into individual privacy, then these measures are unlikely to disappear soon.

This leaves us in a quandary about reaching a conclusion concerning our analysis in the previous chapters. For example, toward the end of Chapter 8, we argue that, to solve the problem of forced technology transfers (FTT), the US should not act unilaterally; it should form a coalition with its democratic allies, such as the EU and Japan, and force China to stop its practices of FTT. Similarly, to force China to allow the symmetric market access of e-commerce, a strong coalition of democratic countries is also important, as we have shown in Chapter 3. This coalition proposal is partly based on the intrinsic property of the prisoner's dilemma, where only the coordinated actions of democratic countries would be able to counteract the individual interests of a single country. Although coordinated moves are better than unilateral ones, we are however not sure how China would react to the democratic coalition. Even if democratic countries are coordinated and form a coalition, why would China concede? In sum, if it took a long time for the South to completely abolish slavery after the US Civil War, on what basis we can expect China to change in the near future, even if democratic regimes jointly demand that it do so?

The above query leads to a deeper question. The 2018 and 2020 trilateral statements drawn up by the US/EU/Japan were delivered to the WTO. The WTO is an international economic organisation honouring rule-based operations, and the rules enactments are based on consensus rather than majority voting. The above-mentioned trilateral statements declared that some existing WTO rules on government subsidies are unfair and outdated, and cannot deal with the high-tech industries with increasing returns. If, however, China took advantage of these old-rule loopholes and benefited in the past, how can people expect China to concede when facing these trilateral statements? Without China's joining the consensus, how would any amendment to the existing WTO rules be possible?

Differences Between Now and the Containment Period

It is also important to note that the world today is different from what it was during the Period of Containment, when the NATO coalition led by the US was confronted with the Warsaw Pact led by the then Soviet Union. The key difference between the situation now and that of 70 years ago, in our opinion, is the highly integrated global economic system in the 21st century. In the 1950s, the world was far from being globalised; there was no complicated global supply chain, and China and the Soviet Union together only constituted a very small portion of the world market. In the 2020s, however, China is arguably the largest market and factory in the world.

Take the car industry as an example: The total number of car sales in China in the year 2019 was more than 21 million; the second largest market was the US, which had 17 million car sales. Of the 21 million cars sold in China, nearly 5.2 million were German cars, whereas the total car sales in Germany amounted to only 3.6 million in the same year.⁵ In terms of the cars manufactured in China, those made by the two most famous German luxury

brands, Mercedes Benz and BMW, far exceeded 1 million in 2020. Given these data, it is of no surprise that democratic regimes (such as Germany) face a prisoner's dilemma as they deal with China.

The cover story of the 20 March 2021 issue of *The Economist* magazine was 'The Brutal Reality of Dealing With China'. The car industry is just one of many examples of democratic countries' vested interests in China. *The Economist* listed several other examples, including the banking industry of the UK, the tech companies of the US, luxury goods from France, and the mining industry of Australia. Indeed, we face an epoch-defining contest between autocracy and liberal values, but the vested interests turn out to be the confounding factors.⁶

Given that the global economy has become highly integrated with China, both in terms of production line and marketing channels, we understand that it is very difficult for the world to even consider decoupling itself from China, especially in terms of physical economic manufacturing and distribution, as what happened during the Cold War and Containment period. Given this reality, what can we do instead?

In sum, we have the following list of interwoven problems:

- The world economy is highly globalised and more integrated now than it was in the Cold War period. This globalised and integrated world economy is currently governed by the WTO rules, most of which were framed 27 years ago, when there were no prevalent high-tech industries with increasing returns, there was no Internet, and there were no such widespread superficially private businesses which are actually tightly controlled by the authoritarian government (such as the Chinese government), or a political party (such as the CCP). Thus, the current WTO rules have evident loopholes.
- Authoritarian countries are better at taking advantage of the above-mentioned loopholes in the WTO rules, thereby making the competition between businesses from democratic countries and those from authoritarian countries intrinsically unfair.
- Rule-amendment in the WTO requires a consensus. However, those countries that take advantages under the current rules have no intention of joining the consensus to modify the rules. This makes the amendment of rules toward a reasonable direction extremely difficult.
- The global liberal market economy's operation is in fundamental conflict with the collectivist ideology and China's authoritarian control. This is why China has tried to resist some global transparency requirements. Moreover, China's various business, cultural and academic outreaches have often had a collective agenda in the background mandated by the authoritarian regime, which has also given rise to serious defensive measures in democratic countries.
- Because China is an important market as well as having a factory-of-the-world economy, a coalition of democratic countries is necessary to avoid

the prisoner's dilemma. Such a dilemma has been worsened by the huge vested interests that many democratic countries have already acquired in China. Even with a strong coalition, for example, in the form of US/EU/Japan trilateral statements, there is no guarantee that China would concede, or that the unfair situation could be turned around.

- The advantage of authoritarian countries under the current WTO rules is reflected in the very fact that those autocratic regimes can ignore some very fundamental values of democracy. Democratic countries treat these values as being transcendent and thus unconditionally important. These values could result in business operations being restricted in democratic countries, and accordingly, the competition efficiency of businesses could be lowered.
- It is political regimes that shape the economic strategies of authoritarian countries, and not the other way around. As such, any coalition in democratic countries will be effective only if this coalition can change or challenge authoritarian political regimes.

If democratic countries accept the above observations/presumptions, then they must find better strategies to make their coalition effective. Borrowing the term from *The Economist*, on what theme should the *epoch-defining contest* between democratic and authoritarian regimes be based?

Identifying the Democratic Values Behind Coalition Strategies

In our previous chapters, we have mentioned several conflicts between democratic and authoritarian regimes, including e-commerce, government subsidies, privacy intrusion using Big Data, listed company accountability, antitrust across countries, national security caused by FDI's, and forced technology transfers. We can ask ourselves a question. If a coalition is formed among democratic countries in one or several of the areas mentioned above, which one(s) will give rise to the strongest pressure on the authoritarian country? If we can identify the most effective threat to the authoritarian political regime, then we are most likely to be able to change the unfair scenario. The 2018 trilateral statement focuses on government subsidies, and the 2020 trilateral statement focuses on forced technology transfers. In our opinion, neither of these two touches upon the fundamental political regimes of authoritarian countries. Therefore, our prediction is that they are unlikely to be effective.

Alternatively, we may think of a coalition based not on the previously mentioned conflicts per se, but based on value-rooted constitutional differences which are behind such conflicts. A typical example may be the democratic values behind the 'Clean Network' initiative.⁷ This is a US government-led effort to address the long-term threat to data privacy, human rights, security and principled collaboration, which were proposed to the free world by malign authoritarian actors. The result has been an alliance of democratic countries and the companies therein, based on democratic values. The Clean Network

is intended to implement internationally accepted digital trust standards across a coalition of partners, with trustable telecommunications equipment vendors.

In December 2020, the United States announced that more than 60 nations, representing more than two-thirds of the world's gross domestic product, and 180 telecom companies, had publicly committed to the principles of the Clean Network. This alliance of democratic regimes included 27 of the 30 NATO members; 26 of the 27 EU members; 31 of the 37 OECD nations; 11 of the 12 Three Seas nations, as well as Japan, Israel, Australia, Singapore, Taiwan, Canada, Vietnam, India and New Zealand.

The key difference between the Clean Network initiative and the earlier cited trilateral statements is that the former is founded on the very fundamental democratic values of privacy, human rights and security. Although the target companies may indeed include Huawei, that target is chosen because Huawei and its satellite firms (perhaps together with TikTok and WeChat) are companies that have been identified as being tied to the CCP to harm democratic values such as privacy, security and human rights. Thus, the Clean Network initiative is a coalition based on democratic values, and not solely on the competitive fairness of a particular branch of the telecommunications industry per se.

As we know, the Clean Network attempts to establish a collaboration among democratic countries, to ensure three things: 1) at the individual level, privacy is well-respected; 2) at the business level, intellectual property rights and know-how are well-protected, and 3) at the country level, national security is well-warranted. To have these goals achieved in the Internet era, we need to have telecommunications facilities provided by trusted vendors, to ensure that software and app companies are not controlled by malign government entities, and to see that Big Data generated through various services are not handled with a hidden agenda dictated by a government not accountable to the public. Indeed, respecting privacy and protecting intellectual property rights are democratic values, and we need national security to protect such values. As such, the Clean Network initiative has aligned different aspects of Internet-related businesses with some common democratic values.

Can We Expect China to Rectify Itself in the WTO?

The Clean Network initiative also reveals that we cannot rely on the WTO to deal with the issue of an 'unclean' network. The so-called 'cleanliness' of the Internet is indeed a feature of cyberspace, in contrast to the physical space in the traditional international trade of goods. As we mentioned in Chapter 3, although Internet blocking has resulted in some obvious asymmetric market access and trade barriers, it is difficult to establish a case of a cyberspace trade barrier in the WTO. Similarly, other than observing the fact that some Chinese propaganda has been sent to users and some anti-Chinese videos have not been circulated, it is difficult to prove whether there is country-backed

artificial intelligence behind TikTok. Even in the case of the US 2016 presidential election, it is practically impossible to prove that the election result was affected by ‘fake news’.

In short, it is rather difficult to define the integrity of the Internet, not to mention enacting some rules to regulate members. Thus, if we expect to establish new WTO rules to deal with these cyberspace interferences, we may either never reach a constructive consensus, or we will reach a consensus with extremely low ambition. In a sense, the Clean Network initiative (together with its future new details) can be seen as a new WTO coalition, which may be referred to as ‘E-WTO’, to deal with Internet-related trades, such as those related to telecommunications facilities vendors and some services that use the Internet.

Establishing new E-WTO rules does not, however, mean that the old WTO rules should be disbanded. The E-WTO rules will, nonetheless, bring some pressure to bear on the old ones. Take the e-commerce plurilateral initiative as an example: whenever new rules are established, they will exert pressure on the old institution. If the old e-commerce initiative continues to have a low degree of ambition, then following the view expressed by China that the Internet is a purely sovereign issue that the WTO cannot deal with, the democratic countries can engage in telecommunications and e-commerce using the consensus formed in the new E-WTO. This will make the old WTO dysfunctional in cyberspace-related trades. Alternatively, the old WTO will feel the pressure from a truly level playing field, and will be able to enact more ambitious and fairer rules. In this case, the new E-WTO may play a relatively minor role.

We can extend the analysis further along the lines of ‘protecting democratic values’. It is fair to say that equal opportunity and fairness are also very important democratic values. In international trade, the very reason behind the rules on government subsidies, tariffs and quotas, national treatment, and intellectual property rights is indeed to guarantee ‘a level playing field’. As we mentioned in Chapter 2, some of these rules have now turned out to be unfair due to some newly arising factors, and some members have large numbers of government-in-essence ‘private businesses’. Conceptually, we should modify the out-of-date rules in order to restore the level playing field. If, however, some WTO Members which are benefiting from, and taking advantage of, such unfair old rules resist the revision of the rules toward fairness, then the WTO cannot do much about it. This constitutes a systematically unfair competition scenario, violating another democratic value of ours: fairness.

Perhaps the only breakthrough direction is to form a new coalition among truly liberal market economies, with a possible name being the ‘Fair Competition’ initiative. This coalition would welcome truly believing market-economic members to join, and require that members provide absolutely transparent government subsidy figures, ensure that operation of SOEs are market-oriented, and also guarantee that no governmental instructions regarding business objectives are allowed. Conceptually, only democratic countries governed by the rule of law and healthy checks and balances could

possibly fulfil such criteria. We may refer to this coalition as a fair-competition WTO, or 'F-WTO'. Presumably, the old WTO rules could be easily revised in a better and fairer direction in this F-WTO.

To sum up, it is our belief that, under the current consensus resolution system, it is difficult to reform the WTO to make it a better and fairer organisation. The E-WTO or F-WTO may play the roles of 'parallel competing institutions', which are competing with the old WTO, thereby allowing for the possibility of replacing the old one. Only through these new parallel organisations can the existing unfair rules and the uneven playing field under the old WTO be really changed.

Conceptually, the E-WTO or F-WTO is like an FTA signed by some plurilateral, possibly democratic, liberal market-economy members. The only difference is that the traditional FTA must be consistent with Article XXIV of the GATT, whereas the new E-WTO or F-WTO does not need to be. This incompatibility may be dealt with in three possible ways, as explained by the following:

- The new E-WTO, for instance, may be a part of the Trade in Services Agreement (TiSA), and the telecommunications or e-commerce activities covered may be claimed to be the early-harvest items.
- The new E-WTO may be treated just like a new FTA with all the queries and complaints from the old-WTO members being handled by the dispute settlement system of the WTO. If the dispute settlement system is not working, then there is nothing that the old WTO can do.
- The E-WTO and F-WTO are jointly the foundations of a potentially new WTO, with the rules being modified and strengthened in particular to prevent unfair play by party/state authoritarian regimes, such as China.

Reconstructing a Set of Fair World Trade/Economic Rules

The usual economics textbooks tell us that, when John Maynard Keynes wrote the *General Theory of Employment, Interest and Money* (1936), he was trying to construct a new economic theory, replacing the old doctrine of Say's Law, which stated that 'supply creates its own demand.' Keynes found that the economic system at the time of the Great Depression was fundamentally different from what the Classical school of economics described, and that this was the reason why we had persistent long-term unemployment all over the world in the 1930s. Keynes then emphasised the role of demand-side government expenditure, particularly during periods of deep depression, to rectify the dysfunctional (supply-side) economy at that time.

In our opinion, the world trade system today faces problems that are pretty much similar to those at that time. The purpose of the GATT, the GATS and the Marrakesh Agreement was to lower various kinds of trade barriers and to form a level playing field, so that the benefits of open and free trade could be facilitated, and that an efficient international division of labour could be

achieved among true market economies. Decades after the above-mentioned agreements, the world has changed significantly, but the WTO has been unable to modify its rules accordingly.

After 2001, when China, the world's largest authoritarian country, joined the WTO, the definition of a market economy became gradually blurred, and until recently, the WTO's Members did not realise that the old WTO rules were unable to discipline the prevalent SOEs in China, which are businesses with unprecedented influence from their governments. With the accession of China, economic competition under the old WTO rules has not only been unfair, but also to the benefit of the authoritarian party/state regime. Worse still, Internet blocking, the privacy intrusion and the threat to national security through Big Data and artificial intelligence have all been escalating.

A well-known quotation by Keynes was that 'the power of vested interest is vastly exaggerated.' This seems to have been echoed by the paradox described in *The Economist* magazine cited above. Many countries in the world are unable to face or deal with the evident unfairness created by the authoritarian party/state system of China, mainly because of the vested interests in China's market. The Trump administration did try to rectify the unfairness, but it applied a unilateral force with little in terms of a coalition with democratic allies, and the then President Trump did not spell out the fundamental unfairness and value differences between the US and China. Without specifying the unfairness and the embedded failed democratic values, it is easy for the US–China conflicts to degenerate to a Thucydides trap: a power struggle between two big countries with no right or wrong.

This book has attempted to spell out that differences in values, and establish theoretical explanations for what lies behind the US-China trade and economic conflicts. It is our belief that, only with the clarification of such ideas, can we make the right choice regarding 'the brutal reality of dealing with China' (the cover title of *The Economist*). Despite having only very briefly discussed the strategic coalitions, this book has concrete ideas concerning what a fair and free economic operation should be. After all, as Keynes finally said: 'It is ideas, not vested interests, which are dangerous for good or evil' (Keynes, 1936). We have tried our best to clarify the ideas – the good ones – and we hope that they will be dangerous to the authoritarian states.

Notes

- 1 B. Baschuk, 'China Loses Landmark WTO Dispute Against EU' *Bloomberg* (16 June 2020), www.bloomberg.com/news/articles/2020-06-16/not-with-a-bang-china-loses-landmark-wto-dispute-against-eu.
- 2 G. Sands, 'Are Confucius Institutes in the US Really Necessary?' *The Diplomat* (20 February 2021), <https://thediplomat.com/2021/02/are-confucius-institutes-in-the-us-really-necessary/>.
- 3 See *The Economist*, 3–9 April 2021 issue.
- 4 See Chen and Chu (2005) for a more detailed theoretical discussion.

- 5 These statistics can be easily obtained from Google search. Although different sources provide slightly different numbers, they are roughly consistent with each other. We skip the detailed references here to avoid including the minor differences in numbers.
- 6 See *The Economist*, 20–26 March 2021 issue, 7.
- 7 M.R. Pompeo, ‘The Clean Network’, US Department of State, <https://2017-2021.state.gov/the-clean-network/index.html>.

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Index

- 5G 12, 18, 26, 118, 127, 135
6G 12, 18, 82, 116, 127, 141
- Aaronson, S. A. 41–42
ACCC 121n2
access asymmetry 36, 45, 50, 51–53, 59
Acemoglu, D. 3, 9, 11–12
ACGA 92, 96
Administrative License Law 165
agency costs 88–89
Ahmed, U. 54
AI 10, 63, 66–67, 74–75, 81–82, 136
Alito, S. 65
AlphaGo 74
Amazon 8–10, 22, 50, 70, 118
AMC 113
American Civil Liberties Union v. Clapper 83n9
AML 106, 109, 112–16, 118–19, 121
Andrenelli, A. 153
Ant Financial 133
Anti-Extradition Law Amendment Bill Movement 71
antitrust law 106–9, 111–13, 118–21, 124–25, 137–40, 144–47; *see also* competition law
Anti-Unfair Competition Law 165
Appellate Body 32, 38, 122
Apple 70, 79, 117–18, 141
Arab Spring 9, 45
Arab Spring 9, 45
Archives Law 94
Armour, J. 88, 90
Australia 2, 43, 48–49, 49, 51–52, 59, 179, 181
Australian Strategic Policy Institute 78–79, 81
authoritarianism 13, 15–16, 81, 177
- Babones, S. 169n7
Beijing 48–50
Bellabona, P. 115
Belt-and-Road Initiative 115, 122, 127–29, 147
big data 10, 67, 74–75, 80, 180–81
Bolton, J. 1
Brandeis, L. 64
Branstetter, L. G. 155, 168
Brazil 48–49, 49, 51–52, 59
Brown, M. 156
Brum, J. 160, 168
ByteDance 62, 70–72, 75–76, 78, 81
- C. Biancotti, Peterson Institute for International Economics 84n19
Canada 43, 48–49, 49, 51–52, 57, 59, 181
Canada – FIT 38n15
Carbaugh, B. 154, 157–58, 168
Carpenter v. United States 67, 83n10
Catalogue of Industries for Guiding Foreign Investment 154, 160
CCMCs 88
CCP 5, 8–9, 75–79, 97–100
CFIUS 71–72, 133–34, 136, 140–42
Chen, Y. J., Lin, C. F., and Liu, H. W. 81
Cheng, T. K. 152
Children's Online Privacy Protection Act 62
China – Publications and Audiovisual Products 53
China's Company Law 90–92, 99
Chinese listed companies 96
Chow, D. C. K. 114
Chu, C. Y. C. 5, 96
clawback 119
Clayton Act 108
Clean Network 81, 180–82
CLOUD Act 66

- CMA 141
 CNOOC 132
 Code of Corporate Governance of Listed Companies 91
 Competition Act 108
 competition law 106
 competitive neutrality 111
 consumer welfare 108
 corporate governance 87–89, 90
 CPC 54
 CPTPP 33, 57
 cross-border data-flow 41–42, 44, 58
 CSRC 91, 94, 100
 CTB 21, 35–36, 55–57
 Cybersecurity Law 77

 data fusion 66
 Data Protection Directive 66, 68
 Data Security Law 77
 Defense Production Act 141
 deLisle, J. 160
 Diamond, L. 3
 digital mercantilism 41, 46–47, 53–54, 57
 Digital Switzerland 68, 79
 digital trade 42, 46, 58; *see also*
 e-commerce
 Dispute Settlement Body(DSB) 151, 159
 Doha Declaration 120
 Douglas, W. O. 64
 Douyin 62, 78
 DSB 151, 159

 e-commerce 35–37, 42–44, 47, 53–58, 182–83
EC – Bananas III 122n5
 Economic and Security Review Commission 97, 131–32
 Economy, E. 164
 efficiency-driven 11
 Eichensehr, K. E. 68, 79
 Enterprise Act 108, 141
 Enterprise and Regulatory Reform Act 108, 141
 entity list 2, 16n5, 116
 Ericsson 117
 European Union's Data Protection Directive 66
 EU-US Privacy Shield Framework 69
 extractive 11–12

 Facebook 47, 67, 70, 75–76, 79, 118
 factor-driven 11
 Fagan, D. 147n2, 150n40

 FDI 125, 127–30, 142–43, 154; review 56, 126–27, 141–43, 145–46
 FDI Screening Regulation 143
 Federal Trade Commission 107
 Federal Trade Commission Act 107
 Finder, S. 164
 FIRRMA 142
 forced technology transfer 151–68
 Foreign Investment Law 165
 Foreign Trade Law of the People's Republic of China 160
 Fortune Global 500 (FG500) 116
 Fourth Amendment 64–65, 67
 France 43, 48–49, 49–50, 51–52, 59, 94, 157, 179
 Freedom House 8, 36, 47
 FRTIB 87
 FTT 18, 151, 153–54, 156–60, 163, 165–68
 Fu, J. 91, 96–98

 GAAP 95
 GATS 14, 53–54, 111, 141, 158–59
 GATT 5, 19–21, 28–29, 54–55, 110–11, 159
 GDPR 68–70, 78
 Germany 2, 24–25, 43, 48–49, 49, 51–52, 59, 138, 142, 157, 168, 178–79
 GFW 42, 45–47, 50, 53, 58–59
 Ghemawat, P. 167
 globalization 63, 106–21, 173–76
 Google 47, 67, 70, 75–76, 115, 118
 Gordon, J. N. 167
 government subsidies 5, 21, 23, 25, 27–33, 132, 177–78
 GreatFire.org 47, 49
Griswold v. Connecticut 64

Halperin v. Central Intelligence Agency 83n8, 84n30
 Hamilton, I. A. 84n22
 harmful information 45–46, 51
 Harrell, P. 149n24
 Hart–Scott–Rodino 108, 142
 Hart-Scott-Rodino Antitrust Improvements Act 108
 He, X. 161–62
 HengFeng Bank 99
 HFCAA 87–88, 96
 HFCE 43–44, 53
 Hong Kong 2, 62, 71, 88, 91, 94, 96–98
 Hout, T. 167
 Huawei 1–2, 116–21, 141, 181

- Hufbauer, G. 148n12
 human rights 13–14, 180–81
 Hurska, A. 149n26, 149n27
- inclusive 11–12
 increasing returns 19–22, 27–30, 36–37, 178–79
 India 48–49, 49, 51–52, 58, 59, 62, 75, 130, 167, 181
 Indonesia 11, 48–49, 49, 51–52, 59, 75, 129
 Inglehart, R. 3–4
 innovation-driven 12
 intellectual property theft 4, 18, 156, 168
 International Emergency Economic Powers Act 140
 Internet Blocking 8, 41, 47, 55, 177, 184
 internet censorship 41
 internet control 14, 55–56
 investment review 124–27, 141–47
 IP rights 126, 135, 154, 156, 158–59, 165
 IT Governance Privacy Team 69
- Jackson, J. H. 164
 Japan 1–2, 14, 25, 30, 43, 44, 48–49, 49, 51–52, 59, 128, 154, 157, 159–60, 165, 168, 178, 180–81
 John, J. 148n17
 joint stock limited company 90
 Jones, D. 122n7
 JSI 58–59
- Katz v. United States* 65
 Kerr, O. S. 65
 Kim, D. 166
 King, K. 148n17
Klayman v. Obama 83n9
 knowledge-based economy 5, 19–21, 29
 knowledge-based industries 36
 Korea 43, 48–49, 49–50, 51–52, 59, 152
- Lee, J. 154, 166
 Lee, P. C. 5
 level playing field 109, 112
 LGBTQ 70
 Li Keqiang 128, 131
 Li, L. 161
 liberal market economy 179
 limited liability company 90–91
 Lin, LY-H 97–98
 Linz, J. 3
 Lipset, S.M. 3
 Lo, I. 166
- M&A 108, 124–27, 130, 135–37, 139–46;
 control 108, 124–26, 140–41
Made in China 2025 18, 22–28, 131, 167
 Malawer, S. S. 124
 Malaysia 48–49, 49–50, 51–52, 59, 129
 Manak, I. 58
 market access 28, 42, 52–4, 157–58, 166–67, 176–78
 market-oriented economy 108–9, 111, 124–27, 131–32, 144–46
 Marrakesh Agreement 5, 13, 19–20, 30, 183
 Martina, M. 84n23
 Mastel, G. 167
 McDonald's 163
 McGregor, R. 5–6, 97–99, 101
 Meng Wanzhou. 1–2
 Mexico 48–49, 49, 51–52, 57, 59
 Michael Jordan 163
Microsoft Corp. v. United States 66
 Milhaupt, C. J. 97–99, 167
 Mizulin, N. 148n14
 MNCs 113–15, 160
 mode-3 44, 50
 Moll, T. 148n12
 mosaic theory 65–66, 80
 Motor Sich 135–36
 Motorola 115, 117–18
 MSCI 87, 101n2
 Musical.ly 62, 70–71
- narrow corridor 9
 NASSP 94
 National Archives Administration 94
 National Emergencies Act 140
 national security 74, 125–27, 130–31, 133, 135, 137–45
 National Security and Investment Bill 143
 National Security Law 76, 79
 nationality-neutral 111–12, 120–21
 Negroponte, N. 34, 56, 63
 Ng, K. H. 161–62
 Nissenbaum, H. 67
- O'Keefe, C. 138–39
 OECD 90–91, 127, 131, 133, 136, 146
 Olearchyk, R. 149n29
Olmstead v. United States 83n4
 OPM 80–81
 OSC 117–18

- PACOB 87–89, 93–95
 Party-building 97–100
 party-state capitalism 125, 127
 per-click waiting time 48–52
 Personal Information Protection Law 77–78
 Phillips, N. J. 140, 149n36
 PLC 162
 plurilateral e-commerce negotiations 42
 Pompeo, M. R. 81
 Pozen, D. E. 65
 PPF 11–12
 President's Council of Advisors on Science and Technology 66
 privacy 62–70, 78–79, 82, 181
 profiling 56, 63, 65–66, 74–75
 property doctrine 64–65
 Prosser, W. L. 64
 protectionism 113, 162
 Protocol on the Accession of the PRC 155, 158, 159, 164
 Prud'homme, D. 154–57
 public observation doctrine 65
 PWG 88, 94

 Qiaodan Sports 163
 Qin, J. Y. 157

 Rajan, R. G. 9
 relationship between economic development and democracy 3–4, 7–8
 right to consent 68
 right to restrict processing 68
 Robinson, J. 3, 9, 11–12
 Robinson-Patman Act 108
 Rubini, L. 148n12
 rule of law 7; in China 160–62

 Safe Harbor Framework 69
 Sarbanes-Oxley Act 87, 93
 SASAC 90, 92, 96–98
 Schlabach, G.R. 66
 SCM 14, 28, 33, 110, 119
 SEC 87–88
 Securities Law 91–92
 Segalis, B. 148n17
 Shanghai 48–49, 49, 51, 96
 Shenzhen 48–49, 49, 51
 Sherman Act 107, 138
 Singapore 4, 7, 34, 48–49, 49, 51–52, 59, 90, 96, 129
 Singh, P. 156

Smith v. Maryland 83n6
 social credit system 10, 81
 socialist market economy 8–9, 15, 109, 112–13, 173
 SOE 30–34, 90–93, 96–100, 113–16, 130–31, 164
 Solove, D. J. 67
 Sotomayor, S. 65
 SPC 160–61, 163–64
 Spigarelli, F. 115
 state capitalism 26, 131–32
 state-subsidized capital 131
 subsidies 4–5, 21–25, 27–28, 31, 131–33
 surveillance 65, 66–67, 78–79
 capitalism 62
 Switzerland 48–49, 49, 51–52, 59
 Sykes, A. O. 157

 Taiwan 25, 48–49, 49, 51–52, 59, 127–28, 159–60, 181
 Taylor, P. 122n11
 Temasek 90, 96
 Tencent 73, 76, 79, 81
 TFEU 108, 140
 third-party doctrine 65, 67
 Thucydides trap 15–16, 184
 Tiananmen Square 8, 45, 50, 70
 Tibet riots 45
 TIER 153, 159
 TikTok 62–63, 70–76, 78, 80–82, 140, 175
 TRIMS 158–59
 TRIPS 14, 151, 158–59, 168
 Trump 1–2, 18–19, 71–73, 140, 184
 Tundang, R. E. 152, 160
 Turkey 48–49, 49, 51–52, 59

 U.S. 19, 23, 43, 48–49, 49, 51–52, 59, 64, 139, 141
 U.S.-China Economic and Security Review Commission 97, 131–32
US – Gambling 53
 UK 2, 43, 48–49, 49, 51–52, 59, 108, 133, 140–41, 143, 146, 179
 United States Horizontal Merger Guidelines 137
United States v. Jones 65, 83n7
 UNSD 54
 URL 49
 Uruguay Round 54
 USDOC 135
 USMCA 57–58

- USTR 4, 18–20, 36, 156, 166
Uyghur “re-education” camp 45
Uyghurs 71, 88, 175
- Verizon 156
VPN 8
- Waldman, A. E. 67
Wang Yi. 81
Wang, Y. 162
Warren, S. 64
Wassell, C. 154
WebPageTest.org 49
WebSitePulse.com 49
WeChat 62, 73–75, 78, 80–82, 140, 175
Welzel, C. 3–4
WFOE 154
Williams, B. 147n2, 150n40
Wonderful 163
- WTO 20–21, 27, 34–7, 53–7, 159–60,
181–84; dispute settlement 13–14, 19,
151–152, 183
Wu, T. 41, 54, 118
- Xi Jinping 15, 71, 96, 115, 128
Xinjiang 71, 175
- Yang, M. 84n25
Yang, X. 164
Yu, H. 83, 164
- Zhang, A. H. 113, 115
Zhang, T. 154
Zheng, W. 99
Zhou, Y. 164
Zhu, S. 161
ZTE 1, 116
Zuboff, S. 62